# LECTURE NOTES FOR STAT652

CUN-HUI ZHANG

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# 1. Theory of Point Estimation

1.1. **Statistical decision problems.** Suppose $X$ is observed as a random element in a sample space $\mathscr{X}$ from a measurable space endowed with a family of probability measures $\{\mathbb{P}_\theta, \theta \in \Theta\}$, with an unknown parameter $\theta$ in a certain parameter space $\Theta$. Unless otherwise stated, here and in the sequel we take the sample space $\mathscr{X}$ as $\mathbb{R}^n$ or a Borel subset of it. In statistics, the probability family $\{\mathbb{P}_\theta, \theta \in \Theta\}$ is referred to as a parametric model, especially when $\Theta$ is finite-dimensional and the dependence of $\mathbb{P}_\theta$ on $\theta$ is "smooth". When $X$ is endowed with joint densities $p_n(x;\theta)$ with respect to a sigma-finite measure $\nu_n(dx)$ in $\mathscr{X}$, the statistical model can be equivalently described by the density family $\{p_n(x;\theta), \theta \in \Theta\}$.

Let $\mathbb{E}_\theta$ be the expectation under $\mathbb{P}_\theta$. Given a loss function $L(\theta, a)$ for $\theta \in \Theta$ and $a$ in an action space $\mathscr{A}$, a decision rule $\delta = \delta(x)$ is a mapping from $\mathscr{X}$ to $\mathscr{A}$, a randomized decision rule is a random $\delta(x) \in \mathscr{A}$ with a known conditional distribution given $X = x$, and the risk (function) of the (randomized) decision rule is defined as

$$R(\theta, \delta) = \mathbb{E}_\theta\big[L(\theta, \delta(X))\big], \ \theta \in \Theta. \tag{1.1}$$

Unless otherwise stated, measurability is assumed for all spaces, sets and functions under consideration.

In the point estimation theory, we are interested in the estimation of a (possibly vector-valued) function $\tau(\theta)$ of the parameter $\theta$, including the case of $\tau(\theta) = \theta$, so that we assume

$$L(\theta, a) \geqslant 0 \ \forall(\theta, a), \quad L(\theta, \tau(\theta)) = 0.$$

We shall focus on the squared Euclidean loss as a primary example,

$$L(\theta, a) = \|a - \tau(\theta)\|_2^2,$$

so that the risk $R(\theta, \delta)$ is the mean squared error (MSE). Let $b(\theta) = \mathbb{E}_\theta[\delta(X)] - \tau(\theta)$ be the bias of the estimator. The MSE is the sum of the variance and the squared bias,

$$\mathbb{E}\big[\|\delta(X) - \tau(\theta)\|_2^2\big] = \mathrm{trace}\big(\mathrm{Cov}_\theta\big(\delta(X)\big)\big) + \|b(\theta)\|_2^2.$$

1.1.1. *Sufficiency and completeness.* Statistical decision problems can be sometimes simplified by restricting the decision rules to those depending only on sufficient statistics.

A function $T = T(X)$ of the data is sufficient in the statistical model $\{\mathbb{P}_\theta, \theta \in \Theta\}$ if the conditional probability given $T$ can be written as $\mathbb{P}_\theta\{X \in A|T\} = H(A|T)$ almost surely for all $A \subseteq \mathscr{X}$ and $\theta \in \Theta$ for a family of distributions $H(dx|t)$ not depending on $\theta$.

There is no loss of statistical efficiency if we confine ourself to randomized decision rules depending only on sufficient statistics. Given $T = t$ with a sufficient $T = T(X)$, one may generate $X'$ according to the conditional distribution $H(dx|t)$ and apply the decision rule $\delta(X')$. It can be readily seen from the definition of sufficiency that $\delta(X')$ has the same distribution as $\delta(X)$ under $\mathbb{P}_\theta$ for all $\theta \in \Theta$.

If the distribution of $X$ has density $p_n(x;\theta)$ under $\mathbb{P}_\theta$ for all $\theta \in \Theta$, then $T(X)$ is sufficient iff $p_n(x;\theta) = g(T(x);\theta)h(x)$ for some nonnegative functions $g(t;\theta)$ and $h(x)$. This is called the factorization criterion for sufficiency.

In point estimation, there is no loss of statistical efficiency under convex loss if we confine ourself to nonrandomized decision rules depending only on sufficient statistics.

**Theorem 1.1** (Rao-Blackwell)**.** *Let $T = T(X)$ be a sufficient statistic and $L(\theta, a)$ be a loss function. Suppose $L(\theta, a)$ is convex in $a$ and $\delta$ is a decision rule with $R(\theta, \delta) < \infty$. Let $\overline{\delta}(T) = \mathbb{E}[\delta(X)|T]$ be the conditional expectation of $\delta$ given $T$. Then,*

$$R(\theta, \overline{\delta}) \leqslant R(\theta, \delta), \quad \forall\, \theta \in \Theta,$$

*and for each $\theta \in \Theta$ the inequality is strict if and only if (iff) $\mathbb{E}_\theta[\mathrm{Var}(L(\theta, \delta(X))|T)] > 0$.*

Proof. Due to the convexity of the loss function, Jensen's inequality provides

$$R(\theta, \overline{\delta}) = \mathbb{E}_\theta\big[L(\theta, \mathbb{E}_\theta[\delta(X)|T])\big] \leqslant \mathbb{E}_\theta\big[\mathbb{E}_\theta\big[L(\theta, \delta(X))\big|T\big]\big] = R(\theta, \delta).$$

Due to the sufficiency of $T$, $\overline{\delta}$ can be used as it does not depend on $\theta$. □

Statistical decision problems can be sometimes simplified via the completeness of certain statistics. A statistic $T = T(X)$ is complete in the model $\{\mathbb{P}_\theta, \theta \in \Theta\}$ if $\mathbb{E}_\theta[h(T)] = 0\; \forall\, \theta \in \Theta$ implies $\mathbb{P}_\theta\{h(T) = 0\} = 1\, \forall\, \theta \in \Theta$. Thus, $\mathbb{E}_\theta[\delta_1(T)] = \mathbb{E}_\theta[\delta_2(T)]\, \forall \theta \in \Theta$ implies $\delta_1(T) = \delta_2(T)$ almost surely when $T$ is complete.

A statistical model is given by an exponential family of densities $\{p_n(x; \theta), \theta \in \Theta\}$ with respect to $\nu_n(dx)$ if $p_n(x; \theta) \propto \exp[\eta(\theta)^\top T(x)]\, \forall \theta \in \Theta$ for a function $T(\cdot) \in \mathbb{R}^d$. In this case,

$$p_n(x; \theta) = \exp[\eta(\theta)^\top T(x) - \psi_n(\eta(\theta))] \quad \forall \theta \in \Theta, \tag{1.2}$$

where $\psi_n(\eta) = \log \int \exp[\eta^\top T(x)]\nu_n(dx)$. In (1.2), $\eta(\theta) \in \mathbb{R}^d$ is called the natural parameter and $\{\eta(\theta) : \theta \in \Theta\}$ the natural parameter space. Taking a linear mapping to a lower dimensional space if necessary, we assume without loss of generality the linear span of the natural parameter space is $\mathbb{R}^d$ and $T(X)$ does not live in a hyperplane in $\mathbb{R}^d$. Often, it is convenient to set $\theta$ itself as the natural parameter so that

$$p_n(x; \theta) = \exp[\theta^\top T(x) - \psi_n(\theta)] \quad \forall \theta \in \Theta. \tag{1.3}$$

The exponential family in (1.3) can be viewed as a sub-model to its extension

$$p_n(x; \theta) = \exp[\theta^\top T(x) - \psi_n(\theta)] \quad \forall \theta \in \Theta^*, \tag{1.4}$$

with $\Theta^* = \{\theta \in \mathbb{R}^d : \int \exp[\theta^\top T(x)]\nu_n(dx) < \infty\}$. It follows from Hölder's inequality that $\Theta^*$ is a convex set and $\psi_n(\theta) = \log \int \exp[\theta^\top T(x)]\nu_n(dx)$ is a convex function in $\Theta^*$. Moreover, $\psi_n(\cdot)$ is analytic in the interior of $\Theta^*$, and

$$\mathbb{E}_\theta[T(X)] = \nabla \psi_n(\theta),\;\; \mathrm{Cov}_\theta(T(X), T(X)) = \nabla^{\otimes 2}\psi_n(\theta)\; \forall \theta \in \Theta^*, \tag{1.5}$$

where $\nabla$ is the gradient operator and $\nabla^{\otimes 2}$ the Hessian operator. In the exponential family (1.3), $T(X)$ is sufficient, and $T(X)$ is complete when $\Theta$ has an nonempty interior.

1.1.2. *Some lower bounds.* General risk minimization in the statistical decision theory is not a perfectly defined problem without additional provisions. Consider the estimation of $\tau(\theta) = \theta$ as a typical example. Given a point $\theta^* \in \Theta$, the trivial estimator $\delta = \tau(\theta^*)$ gives the minimum MSE $R(\theta, \delta) = 0$ at $\theta = \theta^*$. However, the trivial estimator does not extract any information from the observed data as it does not depend $X$. Thus, it is expected to perform poorly when the true parameter $\theta$ turns out to be "significantly" different from $\theta^*$. In general, it is typically impossible to achieve zero MSE simultaneously at two distinct parameter points, say $\theta_1$ and $\theta_2$, so that there must be some lower bound on the bias and variance of any estimator at $\theta_1$ and $\theta_2$. The following proposition, due to Hammersley [Ham50] and Chapman-Robbins [CR51], gives a qualitative statement of this phenomenon.

**Theorem 1.2.** *Let $p_n(x;\theta)$ be the joint density of $X$ with respect to a sigma-finite measure $\nu_n(\cdot)$. Let $\mu(\theta) = \mathbb{E}_\theta[\delta(X)] \in \mathbb{R}$ be the mean of a real-valued estimator $\delta$. Suppose $p_n(x;\theta_k)$ are absolutely continuous with respect to a density $p_n(x)$, $p_n(\cdot;\theta_k) \ll p_n(\cdot), k = 1, 2$. Then,*

$$\big|\mu(\theta_2) - \mu(\theta_1)\big|^2 \leqslant \mathrm{Var}(\delta(X)) \int \left(\frac{p_n(x;\theta_2)}{p_n(x)} - \frac{p_n(x;\theta_1)}{p_n(x)}\right)^2 p_n(x)\nu_n(dx),$$

*where the variance is computed under $p_n(x)$. In particular, if $p_n(\cdot;\theta_2) \ll p_n(\cdot;\theta_1)$, then*

$$\big|\mu(\theta_2) - \mu(\theta_1)\big|^2 \leqslant \mathrm{Var}_{\theta_1}(\delta(X))\mathrm{Var}_{\theta_1}\big(p_n(X;\theta_2)/p_n(X;\theta_1)\big).$$

Proof. Let $\mu = \int \delta(x)p_n(x)\nu_n(dx)$. We have

$$\mu(\theta_2) - \mu(\theta_1) = \int (\delta(x) - \mu)\left(\frac{p_n(x;\theta_2)}{p_n(x)} - \frac{p_n(x;\theta_1)}{p_n(x)}\right)p_n(x)\nu_n(dx).$$

The conclusions follow from an application of the Cauchy-Schwarz inequality to the integral on the right-hand side. □

When the bias of an estimator $\delta$ is zero at both $\theta_1$ and $\theta_2$ in the estimation of $\tau(\theta)$, Theorem 1.2 provides a lower bound for the variance of the estimator at $\theta_1$ when $\tau(\theta_2) \neq \tau(\theta_1)$ and $p_n(\cdot;\theta_2) \ll p_n(\cdot;\theta_1)$. In general, Theorem 1.2 provides a lower bound for the mean squared error for any estimator when $\tau(\theta_1) \neq \tau(\theta_2)$. However, for a fixed sample size, such lower bounds are in general not attainable by any estimator if we simultaneously consider three or more parameter points.

Optimal point estimation can be achieved for a fixed sample when regularity conditions are imposed such as unbiasedness or equivariance under a suitable class of transformations. Optimality can be also achieved in terms of a single criterion such as the Bayes risk

$$R(\Pi, \delta) = \int_\Theta R(\theta, \delta)\Pi(d\theta), \tag{1.6}$$

where $\Pi(d\theta)$ is a prior distribution on $\Theta$, or the maximum risk

$$R_{\max}(\delta) = \sup_{\theta\in\Theta} R(\theta, \delta). \tag{1.7}$$

The Bayes approach provides an explicit lower bound for the weighted risk at two points.

**Proposition 1.1.** *Let $p_n(x;\theta)$ be the joint density of $X$ with respect to a sigma-finite measure $\nu_n(\cdot)$. Then, under the squared-error loss $L(\theta, a) = \|a - \tau(\theta)\|_2^2$, for $\pi_1 \wedge \pi_2 > 0$*

$$\min_\delta \big\{\pi_1 R(\theta_1, \delta) + \pi_2 R(\theta_2, \delta)\big\} = \|\tau(\theta_2) - \tau(\theta_1)\|_2^2 \int \frac{\pi_1 p_n(x;\theta_1)\pi_2 p_n(x;\theta_2)}{\pi_1 p_n(x;\theta_1) + \pi_2 p_n(x;\theta_2)}\nu_n(dx).$$

Proof. With $w_k(x) = \pi_k p_n(x;\theta_k)/\sum_{j=1}^2 \pi_j p_n(x;\theta_j)$ and $\delta(x) = \sum_{k=1}^2 \tau(\theta_k)w_k(x)$,

$$\begin{aligned}&\textstyle\int \sum_{k=1}^2 \|\delta(x) - \tau(\theta_k)\|_2^2 \pi_k p_n(x;\theta_k)\nu_n(dx)\\ &= \|\tau(\theta_2) - \tau(\theta_1)\|_2^2 \textstyle\int w_1(x)w_2(x) \sum_{k=1}^2 \pi_k p_n(x;\theta_k)\nu_n(dx)\end{aligned}$$

is the minimum of the weighted risk $\pi_1 R(\theta_1, \delta) + \pi_2 R(\theta_2, \delta)$ □

1.2. **Unbiased point estimation.** An estimator $\delta(X)$ of $\tau(\theta)$ is unbiased if

$$\mathbb{E}_\theta[\delta(X)] = \tau(\theta)\ \forall\, \theta \in \Theta. \tag{1.8}$$

Unbiased estimation is not guaranteed to be feasible in general. For example, when $X$ is a binomial$(n,\theta)$ variable with $\Theta = (0,1)$, $\mathbb{E}_\theta[\delta(X)]$ is always a polynomial of no more degree than $n$, so that there exists uniquely an unbiased estimator iff $\tau(\theta)$ is such a polynomial.

An estimator $\delta$ of $\tau(\theta)$ is locally minimum variance unbiased (LMVU) at a parameter point $\theta_0$ if it is unbiased and

$$\mathbb{E}_{\theta_0}\big[\|\delta(X) - \tau(\theta_0)\|_2^2\big] = \inf_{\text{unbiased } \delta'} \mathbb{E}_{\theta_0}\big[\|\delta'(X) - \tau(\theta_0)\|_2^2\big] < \infty. \tag{1.9}$$

The estimator is uniformly minimum variance unbiased (UMVU) if it is unbiased and

$$\mathbb{E}_\theta\big[\|\delta(X) - \tau(\theta)\|_2^2\big] = \inf_{\text{unbiased } \delta'} \mathbb{E}_\theta\big[\|\delta'(X) - \tau(\theta)\|_2^2\big] < \infty \quad \forall\, \theta \in \Theta. \tag{1.10}$$

While the UMVU estimator (if exists) is optimal among unbiased estimators under the MSE, one may prefer a suitable biased alternative. In some reasonable cases such as Example 1.1 below, there exist estimators with smaller MSE than the UMVU estimator for all $\theta \in \Theta$.

The following theorem gives a mathematical solution to the LMVU and UMVU problems. Because $\delta(X)$ is LMVU for $\tau(\theta)$ iff each element of $\delta(X)$ is LMVU for the corresponding element of $\tau$, it suffices to consider real-valued $\tau(\theta)$.

**Theorem 1.3.** *Consider the square loss $|a - \tau(\theta)|^2$ with $\tau(\theta) \in \mathbb{R}$. Let $\mathscr{U}$ be the class of all statistics $U \in \mathbb{R}$ with $\mathbb{E}_\theta[U] = 0$ for all $\theta \in \Theta$. Let $\mathscr{U}_\theta = \{U \in \mathscr{U} : \mathbb{E}_\theta[|U|^2] < \infty\}$. Let $\delta$ be an unbiased estimator of $\tau(\theta)$. Then, $\delta$ is LMVU at $\theta = \theta_0$ iff*

$$\mathbb{E}_{\theta_0}[|\delta(X)|^2] < \infty,\ \mathbb{E}_{\theta_0}\big[U\delta(X)\big] = 0 \quad \forall\, U \in \mathscr{U}_{\theta_0}, \tag{1.11}$$

*and $\delta$ is UMVU iff (1.11) holds for all $\theta_0 \in \Theta$. If $\mathbb{E}_{\theta_0}[|\delta(X)|^2] < \infty$, then $\delta - U_{\theta_0}$ is LMVU at $\theta = \theta_0$, where $U_\theta$ is the $L_2(\mathbb{P}_\theta)$ projection of $\delta$ to $\mathscr{U}_\theta$. If $\mathbb{E}_\theta[|\delta(X)|^2] < \infty$ and $\mathbb{P}_{\theta_0}\{X \in A\} = 0$ implies $\mathbb{P}_\theta\{X \in A\} = 0$ for all $\theta \in \Theta$, then the UMVU estimator exists iff $\mathbb{P}_\theta\{U_\theta = U_{\theta_0}\} = 1$ for all $\theta$ and $\theta_0$ in $\Theta$, iff $\delta(X) - U_{\theta_0}$ is the unique UMVU estimator.*

Proof. Suppose $\mathbb{E}_{\theta_0}[|\delta(X)|^2] + \mathbb{E}_{\theta_0}[|\delta'(X)|^2] < \infty$ where $\delta'$ is also an unbiased estimator. We have $U = \delta' - \delta \in \mathscr{U}_{\theta_0}$. If $\delta$ is LMVU at $\theta_0$, then

$$\mathbb{E}_{\theta_0}[U\delta] = (d/dt)\mathbb{E}_{\theta_0}\big[|\delta(X) - tU - \tau(\theta)|^2\big]\Big|_{t=0} = 0.$$

If $\mathbb{E}_{\theta_0}[U\delta] = 0$, then $\mathbb{E}_{\theta_0}[|\delta(X) - \tau(\theta)|^2] \leqslant \mathbb{E}_{\theta_0}[|\delta'(X) - \tau(\theta)|^2]$. This gives the LMVU and UMVU criteria. Moreover, as $\mathbb{E}_{\theta_0}[U(\delta - U_{\theta_0})] = 0$ for all $U \in \mathscr{U}_{\theta_0}$, $\delta - U_{\theta_0}$ is LMVU at $\theta_0$. Suppose $\mathbb{P}_{\theta_0}\{X \in A\} = 0$ implies $\mathbb{P}_\theta\{X \in A\} = 0$ for all $\theta \in \Theta$. If $\mathbb{P}_\theta\{U_\theta = U_{\theta_0}\} = 1\,\forall\theta$, then $\delta - U_{\theta_0}$ is LMVU under $\mathbb{P}_\theta$ for all $\theta$ and thus is UMVU. Finally, if $\delta^*$ is UMVU, then $\mathbb{P}_\theta\{\delta^* - \delta = U_\theta\} = 1$ for all $\theta \in \Theta$, so that $\mathbb{P}_\theta\{U_\theta = \delta^* - \delta = U_{\theta_0}\} = 1$ for all $\theta \in \Theta$. □

While the solution of the LMVU and UMVU problems in Theorem 1.3 are quite complete mathematically, one may have difficulties to find an unbiased estimator, to identify the subspace $\mathscr{U}$, or to compute $U_\theta$. In the discrete case where $|\mathscr{X}| < \infty$, $\mathscr{U}$ can be explicitly identified as the subspace of all vectors in the $L_2(\nu_n)$ orthogonal complement of the linear span of the vectors $p_n(\cdot;\theta) : \mathscr{X} \to \mathbb{R}$, $\theta \in \Theta$. When a complete sufficient statistic exists, the Rao-Blackwell theorem provides an explicit general solution as follows for convex loss.

**Theorem 1.4.** *Consider the estimation of $\tau(\theta)$ under a convex loss $L(\theta, a)$. Suppose there exists an unbiased estimator $\delta$ with finite risk $R(\theta, \delta)$ for all $\theta \in \Theta$. Suppose there exists a complete sufficient statistic $T = T(X)$. Then, there exists uniquely a uniformly minimum risk unbiased estimator $\overline{\delta}$ satisfying*

$$R(\theta, \overline{\delta}) \leqslant R(\theta, \delta') \quad \forall\, \theta \in \Theta \tag{1.12}$$

*for all unbiased estimators $\delta'$. Moreover, $\overline{\delta}$ can be taken as $\overline{\delta} = \mathbb{E}[\delta|T]$.*

Proof. Let $\overline{\delta} = \mathbb{E}[\delta|T]$ and $\overline{\delta}' = \mathbb{E}[\delta'|T]$ with unbiased $\delta$ and $\delta'$. Then, $\mathbb{P}_\theta\{\overline{\delta} = \overline{\delta}'\} = 1\,\forall\theta$ by the completeness of $T$ so that $R(\theta, \overline{\delta}) = R(\theta, \overline{\delta}') \leqslant R(\theta, \delta')\,\forall\theta$ by Rao-Blackwell. □

It follows from Theorem 1.4 that under the existence condition for a complete sufficient statistic $T = T(X)$, the UMVU estimator exists in the estimation of $\tau(\theta)$ iff

$$\mathbb{E}_\theta\big[\,\overline{\delta}(T)\big] = \tau(\theta)\ \forall\,\theta \in \Theta$$

for some $\overline{\delta}(\cdot)$ with finite second moment. This is the case when the model is given by the exponential family (1.3) with an nonempty interior of the natural parameter space.

**Example 1.1.** *Suppose iid $X_1, \ldots, X_n$ are observed from $N(\mu, \sigma)$ with unknown $\mu \in \mathbb{R}$ and $\sigma > 0$. The sample mean $\overline{X}$ and sample variance $\widehat{\sigma}^2 = \sum_{i=1}^n (X_i - \overline{X})^2/(n-1)$ are jointly complete sufficient. Moreover, they are respectively UMVU estimators of $\mu$ and $\sigma^2$. However, for $\widetilde{\sigma}^2 = \sum_{i=1}^n (X_i - \overline{X})^2/(n+1)$ as the minimum risk equivariant estimator discussed Section 1.4,*

$$\mathbb{E}\Big[\big(\widetilde{\sigma}^2 - \sigma^2\big)^2\Big] = \frac{2\sigma^4}{n+1} < \mathbb{E}\Big[\big(\widehat{\sigma}^2 - \sigma^2\big)^2\Big] = \frac{2\sigma^4}{n-1},$$

*so that the UMVU estimator $\widehat{\sigma}^2$ is "inadmissible" under the squared-error loss.*

**Example 1.2.** *Suppose iid $X_1, \ldots, X_n$ are observed from an unknown common density with respect to the Lebesgue measure. Then, $\{X_i, i \in [n]\}$ as a set of vectors is complete sufficient and the UMVU estimator of a real-valued $\tau$ exists iff $\tau = \mathbb{E}[h(X_1, \ldots, X_k)]$ for some $k \leqslant n$ and $h$ with finite variance. In this case, the UMVU is given by the $U$-statistic*

$$\widehat{\tau} = \sum_{1 \leqslant i \neq \cdots \neq i_k \leqslant n} \frac{h(X_{i_1}, \ldots, X_{i_k})}{n(n-1)\cdots(n-k+1)}.$$

*The variance of $\widehat{\tau}$ can be computed via Hoeffding's decomposition discussed in Section 1.3.*

In many cases where the use of UMVU estimator is not feasible or desirable, one may use estimators whose risk is close to an information lower bound. In the rest of the section we consider a density family $\{p_n(x;\theta), \theta \in \Theta\}$ with $\Theta \in \mathbb{R}^d$. Suppose a score function $s_n(x;\theta)$ is well defined in the sense of

$$s_n(x;\theta) = (\partial/\partial\theta)\log p_n(x;\theta) \in \mathbb{R}^d, \quad \mathbb{E}_\theta\big[s_n(x;\theta)\big] = 0, \quad \mathbb{E}_\theta\big[\|s_n(x;\theta)\|_2^2\big] < \infty.$$

The Fisher information (matrix) is defined as

$$I(\theta) = \mathbb{E}_\theta[s_n(X;\theta)s_n(X;\theta)^\top] \in \mathbb{R}^{d\times d}. \tag{1.13}$$

Consider the estimation of $\tau(\theta) \in \mathbb{R}$ in a small neighborhood of $\theta_0 \in \Theta \subseteq \mathbb{R}^d$. Consider fixed $\dot\tau_0 \in \mathbb{R}^d$ and curves $h_t \to h_0 \in \mathbb{R}^d$ with $\theta_0 + th_t \in \Theta_0$ such that

$$h_0^\top \dot\tau_0 = \lim_{t\to 0+} \big(\tau(\theta_0 + th_t) - \tau(\theta_0)\big)/t, \tag{1.14}$$

$$\mathbb{E}_{\theta_0}\big[g(X)h_0^\top s_n(X;\theta_0)\big] = \lim_{t\to 0+} \int g(x)\frac{p_n(x;\theta_0+th_t) - p_n(x;\theta_0)}{t}\nu_n(dx)$$

for all functions $g$ with $\mathbb{E}_{\theta_0}[g^2(X)] < \infty$. The following theorem is a version of the Cramér-Rao inequality.

**Theorem 1.5.** *Let $\mathbb{H}_0 \subseteq \mathbb{R}^d$ be the collection of $h_0 \neq 0$ satisfying* (1.14) *and $\overline{\mathbb{H}}_0$ be the linear span of $\mathbb{H}_0$. Suppose the fisher information $I(\theta_0)$ is invertible in $\overline{\mathbb{H}}_0$. Consider an estimator $\delta(X) \in \mathbb{R}$ with mean $\mu(\theta) = \mathbb{E}_\theta[\delta(X)]$ and some $\dot\mu_0 \in \mathbb{H}$ satisfying*

$$h_0^\top \dot\mu_0 \quad = \quad \lim_{t\to 0+} \big(\mu(\theta_0 + th_t) - \mu(\theta_0)\big)/t \tag{1.15}$$

*for the curves in* (1.14) *generating $\mathbb{H}_0$. Let $Q_0$ be the orthogonal projection to $\overline{\mathbb{H}}_0$. Then,*

$$\mathrm{Var}_{\theta_0}(\delta(X)) \geqslant \dot\mu_0^\top Q_0 I^{-1}(\theta_0) Q_0 \dot\mu_0, \tag{1.16}$$

*and the equality holds iff $\delta(x) - \mu(\theta_0) = a_0\dot\mu_0^\top Q_0 I^{-1}(\theta_0)Q_0 s_n(x;\theta_0)$ for some constant $a_0 \in \mathbb{R}$. In particular, if $\delta(X)$ is unbiased then* (1.16) *holds with $\dot\mu_0 = \dot\tau_0$.*

Proof. As in the proof Theorem 1.2,

$$\big(\mu(\theta_0 + th_t) - \mu(\theta_0)\big)/t = \int \delta(x)\frac{p_n(x;\theta_0+th_t) - p_n(x;\theta_0)}{t}\nu_n(dx).$$

Taking the limit as $t \to 0+$, we find by conditions (1.14) and (1.15) that

$$h^\top \dot\mu_0 = \mathbb{E}_{\theta_0}\big[\delta(X)h^\top s_n(X;\theta_0)\big]$$

for all $h \in \mathbb{H}_0$ and then by linear extension to all $h \in \overline{\mathbb{H}}_0$. It follows by Cauchy-Schwarz that

$$\mathrm{Var}_{\theta_0}(\delta(X)) \geqslant \big(h^\top\dot\mu_0\big)^2/\mathbb{E}_{\theta_0}\big[\big(h^\top s_n(X;\theta_0)\big)^2\big] = \big(h^\top\dot\mu_0\big)^2/\big(h^\top I(\theta_0)h\big),\ \forall h \in \overline{\mathbb{H}}_0.$$

The above lower bound is maximized at $h_* = Q_0 I^{-1}(\theta_0)Q_0\dot\mu_0$, yielding (1.16). The equality attains in (1.16) iff $\delta(x) - \mu(\theta_0) = a_0 h_*^\top s_n(x;\theta_0)$ for some constant $a_0 \in \mathbb{R}$. □

**Example 1.3.** *For the exponential family* (1.3) *with open $\Theta \subset \mathbb{R}^d$,*

$$s_n(x;\theta) = T(x) - \dot\psi_n(\theta), \quad \mathbb{E}_\theta[T] = \dot\psi(\theta), \quad I(\theta) = \ddot\psi_n(\theta) = \mathrm{Cov}_\theta(T,T),$$

*where $\dot\psi_n$ and $\ddot\psi_n$ are respectively the gradient and Hessian of $\psi_n$. Because $T(X)$ does not live in a hyperplane in $\mathbb{R}^d$, $v^\top I(\theta)v = \mathrm{Var}_\theta(v^\top(T - T'))/2 > 0$ for $v \neq 0$ where $T'$ is an independent copy of $T$. The invertibility of $I(\theta)$ follows. In the estimation of $\tau(\theta) = \mathbb{E}_\theta\big[\delta(T(X))\big] \in \mathbb{R}$, $\delta(T(X))$ is UMVU as $T$ is complete and sufficient. Moreover, $\delta(T(X))$ attains the Cramér-Rao lower bound iff $\delta(x) - \tau(\theta) \propto \dot\tau(\theta)^\top\ddot\psi^{-1}(\theta)(T(x) - \dot\psi(\theta))$ iff $\tau(\theta)$ is a linear functional of $\dot\psi(\theta)$.*

1.3. **$U$-statistics.** In Example 1.2, UMVU estimators are shown to be $U$-statistics in a nonparametric setting. Here we discuss properties of $U$-statistics.

For positive integers $n \geqslant m$, the projection of functions $h_m : \mathbb{R}^m \to \mathbb{R}$ to the space of all permutation invariant functions of $x_{1:n} = (x_1, \ldots, x_n)^\top \in \mathbb{R}^n$ is

$$U_{n,m}(x_{1:n}; h_m) = \sum_{1\leqslant i_1 \neq \cdots \neq i_m \leqslant n} \frac{h_m(x_{i_1}, \ldots, x_{i_m})}{n(n-1)\cdots(n-m+1)}.$$

$U$-statistics are formed by applying this projection to data $X_{1:n} = (X_1, \ldots, X_n)^\top \in \mathbb{R}^n$,

$$U_{n,m} = U_{n,m}(h_m) = U_{n,m}(X_{1:n}; h_m).$$

Here $h_m$ is called a kernel. If $U_{n,m}(X_{1:n}; h_m)$ cannot be written as $U_{n,k}(X_{1:n}; h_k)$ for some $k < m$ and $h_k : \mathbb{R}^k \to \mathbb{R}$, $m$ is the order of the $U$-statistic $U_{n,m}(X_{1:n}; h_m)$. Let $h_m^s(x_{1:m}) = U_{m,m}(x_{1:m}; h_m)$ as the symmetrization of $h_m$. The $U$-statistics can be written as

$$U_{n,m} = U_{n,m}(h_m) = U_{n,m}(h_m^s) = \binom{n}{m}^{-1} \sum_{1\leqslant i_1 < \cdots < i_m \leqslant n} h_m^s(X_{i_1}, \ldots, X_{i_m}). \tag{1.17}$$

For $m > k \geqslant 0$, a kernel $h_m$ is order-$k$ degenerate if

$$\mathbb{E}\big[h^s(X_{1:m})\big| X_{1:k} = x_{1:k}\big] = 0 \quad \forall x_1, \ldots, x_k,$$

and $h_m$ is (completely) degenerate if it is order-$(m-1)$ degenerate.

Suppose $X_i$ are iid and $\mathbb{E}[|h_m^s(X_{1:m})|] < \infty$. Hoeffding's decomposition [Hoe48] writes

$$U_{n,m}(h_m) = \overline{h}_0 + mU_{n,1}(\overline{h}_1) + \cdots + U_{n,m}(\overline{h}_m) = \sum_{k=0}^{m} \binom{m}{k} U_{n,k}(\overline{h}_k), \tag{1.18}$$

where $\overline{h}_0 = \mathbb{E}[h_m^s(X_{1:m})]$ and $\overline{h}_k$ are completely degenerate order-$k$ kernels given by

$$\overline{h}_k(x_{1:k}) = \mathbb{E}\left[h_m^s(X_{1:m}) - \sum_{\ell=0}^{k-1} \binom{m}{\ell} U_{m,\ell}(\overline{h}_\ell) \middle| X_{1:k} = x_{1:k}\right],\ k \in [m].$$

This can be seen as follows. Let $\Delta_k = h_m^s(X_{1:m}) - \sum_{\ell=0}^{k-1}\binom{m}{\ell} U_{m,\ell}(\overline{h}_\ell)$. As $\Delta_1 = h_m^s(X_{1:m}) - \overline{h}_0$, $\mathbb{E}[\overline{h}_1(X_1)] = 0$, so that $\overline{h}_1$ is order-0 degenerate. If $\overline{h}_k$ is degenerate for a $k \in [1, m)$,

$$\begin{aligned} &\mathbb{E}\big[\overline{h}_{k+1}(X_{1:(k+1)})\big| X_{1:k} = x_{1:k}\big] \\ = \ &\mathbb{E}\left[\Delta_k - \sum_{1\leqslant i_1<\cdots<i_k\leqslant m} \overline{h}_k(X_{i_1}, \ldots, X_{i_k}) \middle| X_{1:k} = x_{1:k}\right] \\ = \ &0 \end{aligned}$$

due to $\mathbb{E}\big[\overline{h}_k(X_{i_1}, \ldots, X_{i_k})\big| X_{1:k} = x_{1:k}\big] = 0$ for $i_k > k$.

$U$-statistics of a more general form is

$$U_{n,m} = \sum_{1\leqslant i_1\neq\cdots\neq i_m\leqslant n} h_{n,m,i_1,\ldots,i_m}(X_{i_1}, \ldots, X_{i_m}),$$

for which Hoeffding's decomposition can still be carried out when $X_i$ are independent but not necessarily identically distributed.

**Theorem 1.6.** *Suppose $X_1, X_2, \ldots$ are iid random variables and $\mathbb{E}[|h_m^s(X_{1:m})|] < \infty$. Then, the U-statistics $U_{n,m}(h_m)$ in (1.17) form a reverse martingale and*

$$\mathbb{P}\Big\{U_{n,m}(h_m) \to \mathbb{E}\big[h_m(X_{1:m})\big]\Big\} = 1.$$

*Suppose further that $\mathrm{Var}(h_m^s(X_{1:m})) < \infty$. Then, the summands $\overline{h}_k(X_{i_1}, \ldots, X_{i_k}), 1 \leqslant i_1 < \cdots < i_k \leqslant n, 1 \leqslant k \leqslant m$, in Hoeffding's decomposition (1.18) are all uncorrelated,*

$$\mathrm{Var}\big(U_{n,m}(h_m)\big) = \sum_{k=1}^{m} \binom{m}{k}^2 \binom{n}{k}^{-1} \mathrm{Var}\big(\overline{h}_k(X_{1:k})\big), \tag{1.19}$$

*and with the quantile function $Q(t)$ of $X_1$ and a Brownian motion process $W(t)$ in $[0,1]$*

$$\begin{aligned} &\Big(\tbinom{n}{k}^{1/2} U_{n,k}(\overline{h}_k), k = 1, \ldots, m\Big) \\ \rightsquigarrow\ &\bigg(\sqrt{k!}\int_{0 \leqslant t_1 < \cdots < t_k \leqslant 1} \overline{h}_k\big(Q(t_1), \ldots, Q(t_k)\big) W(dt_1) \cdots W(dt_k), k = 1, \ldots, m\bigg). \end{aligned} \tag{1.20}$$

*Moreover, for $\overline{h}_k(t_1, \ldots, t_k) = \prod_{j=1}^k \phi(t_j)$ with $\int \phi(Q(t))dt = 0$,*

$$k! \int_{0 \leqslant t_1 < \cdots < t_k \leqslant 1} \overline{h}_k\big(Q(t_1), \ldots, Q(t_k)\big) W(dt_1) \cdots W(dt_k) = H_k\bigg(\int \phi(Q(t))W(dt)\bigg)$$

*with the Hermite polynomial $H_k(t) = (-1)^k e^{t^2/2}(d/dt)^k e^{-t^2/2}$.*

The right-hand side of (1.20) is defined as Itô's integral, and is a more explicit version of the expressions in [RV80, DM83]. As stochastic processes indexed by $\overline{h}_k$, the two sides in (1.20) are $L_2(\mathbb{P})$ isometric. In fact, the components on the right-hand side of (1.20) are uncorrelated with

$$\mathbb{E}\Bigg[\bigg(\sqrt{k!}\int_{0 \leqslant t_1 < \cdots < t_k \leqslant 1} \overline{h}_k(Q(t_1), \ldots, Q(t_k)) W(dt_1) \cdots W(dt_k)\bigg)^2\Bigg] = \mathrm{Var}\big(\overline{h}_k(X_{1:k})\big). \tag{1.21}$$

Proof of Theorem 1.6. As $\overline{h}_k$ are completely degenerate, for $k \in [m]$

$$\mathbb{E}\big[\overline{h}_k(X_{i_1}, \ldots, X_{i_k})\overline{h}_\ell(X_{j_1}, \ldots, X_{j_\ell})\big] = \mathrm{Var}\big(\overline{h}_k(X_{1:k})\big) I\{(i_1, \ldots, i_k) = (j_1, \ldots, j_\ell)\}$$

for all $1 \leqslant i_1 < \cdots < i_k \leqslant n$ and $1 \leqslant j_1 < \cdots < j_\ell \leqslant n$, which implies (1.19). For (1.20) it suffices to consider iid uniform $X_i$ in $[0,1]$ as we may write $X_i = Q(X_i')$ with iid uniform $X_i'$ in $[0,1]$. Moreover, it suffices to consider $h(x_1, \ldots, x_m) = \prod_{j=1}^m \phi_j(x_j)$ with $\mathbb{E}[\phi_j(X_1)] = 0$ and $\mathbb{E}[\phi_j(X_1)\phi_k(X_1)] \in \{0,1\}$ due to the linearity in $\overline{h}_k$ and the isometry of the two sides in (1.20). Because $\int_0^1 g(t)(W(dt))^k = I\{k=2\}\int_0^1 g(t)dt$ for $k \geqslant 2$,

$$\begin{aligned} &\bigg(n^{-k/2}\sum_{i=1}^{n}\prod_{\ell=1}^{k} \phi_{j_\ell}(X_i), 1 \leqslant j_1 < \cdots < j_k \leqslant m, k \leqslant m\bigg) \\ \rightsquigarrow\ &\bigg(\int_0^1 \bigg(\prod_{\ell=1}^{k}\phi_{j_\ell}(t)\bigg)(W(dt))^k, 1 \leqslant j_1 < \cdots < j_k \leqslant m, k \leqslant m\bigg). \end{aligned}$$

The central limit theorem (1.20) then follows from the continuous mapping theorem. □

1.4. **Invariant decision problems.** Consider the location family as a primary example to illustrate the idea before discussion of equivariance in general. In a location family

$$X = (X_1, \ldots, X_n)^\top \sim p_n(x - \theta\, 1_n) = p_n(x_1 - \theta, \ldots, x_n - \theta) \tag{1.22}$$

for all $x = (x_1, \ldots, x_n)^\top \in \mathbb{R}^n$ with an unknown $\theta \in \mathbb{R}$ and a known joint density $p_n(x)$ with respect to the Lebesgue measure in $\mathbb{R}^n$. An estimator $\delta(x)$ is location equivariant if

$$\delta(x + c\, 1_n) = \delta(x_1 + c, \ldots, x_n + c) = \delta(x) + c \quad \forall c \in \mathbb{R},\ x \in \mathbb{R}^n. \tag{1.23}$$

Consider the estimation of $\theta$ in (1.22) under a loss function of the form $L(\theta, a) = \rho(a - \theta)$. For location equivariant estimators $\delta$, the risk

$$R(\theta, \delta) = \mathbb{E}_\theta\big[\rho(\delta(X) - \theta)\big] = \mathbb{E}_\theta\big[\rho(\delta(X - \theta\, 1_n))\big] = \mathbb{E}_0\big[\rho(\delta(X))\big] \tag{1.24}$$

is a constant not depending on $\theta$. Thus, under the above provisions, the statistical decision problem can be solved by minimizing $\mathbb{E}_0[\rho(\delta(X))]$ as a single criterion.

**Theorem 1.7.** *Suppose $X = (X_1, \ldots, X_n)^\top$ is observed from a location family as in* (1.22). *Let $\delta_0(x)$ be a location equivariant decision rule, $Y = X - X_1\, 1_n$, and*

$$\delta^*(X) = \delta_0(X) - c^*(Y), \quad c^*(Y) = \mathrm{argmin}_c\, \mathbb{E}_0\big[\rho(\delta_0(X) - c)\big|Y\big]. \tag{1.25}$$

*Then, $\delta^*(X)$ is a location equivariant estimator of $\theta$ and*

$$R(\theta, \delta^*) = \mathbb{E}_\theta\big[\rho(\delta^*(X) - \theta)\big] \leqslant \mathbb{E}_\theta\big[\rho(\delta(X) - \theta)\big] = R(\theta, \delta) \tag{1.26}$$

*for all $\theta \in \mathbb{R}$ and location equivariant estimators $\delta$. In particular, for the squared loss*

$$\delta^*(X) = \frac{\int u p_n(X - u 1_n) du}{\int p_n(X - u 1_n) du} = \frac{\int u p_n(X_1 - u, \ldots, X_n - u) du}{\int p_n(X_1 - u, \ldots, X_n - u) du}. \tag{1.27}$$

The estimator (1.27) is known as Pitman's estimator [Pit39].

Proof. For any equivariant estimator $\delta(X)$, we write $\delta(X) = \delta_0(X) - c(Y)$, so that

$$\mathbb{E}_0\big[\rho(\delta(X))\big|Y\big] = \mathbb{E}_0\big[\rho(\delta_0(X) - c(Y))\big|Y\big] \geqslant \mathbb{E}_0\big[\rho(\delta_0(X) - c^*(Y))\big|Y\big] = \mathbb{E}_0\big[\rho(\delta^*(X))\big|Y\big]$$

in (1.24) by the optimality of $c^*(Y)$. For the squared loss and $\delta_0(X) = X_1$,

$$\delta^*(X) = X_1 - \mathbb{E}[X_1|Y] = X_1 - \frac{\int x_1 p_n(Y + x_1 1_n) dx_1}{\int p_n(Y + x_1 1_n) dx_1} = \frac{\int u p_n(X - u 1_n) du}{\int p_n(X - u 1_n) du}$$

with $u = X_1 - x_1$, as the joint density of $X_1$ and $Y_{2:n}$ at $\theta = 0$ is $p_n(y + x_1 1_n)$, $y_1 = 0$. $\square$

The above development can be extended to more general invariant statistical decision problems. A statistical model $\{\mathbb{P}_\theta, \theta \in \Theta\}$ with observation $X$ in a sample space $\mathscr{X}$ is invariant under a bijection $g$ in $\mathscr{X}$ if there exists a mapping $\overline{g}$ in $\Theta$ such that

$$\mathbb{P}_\theta\big\{gX \in B\big\} = \mathbb{P}_{\overline{g}\theta}\big\{X \in B\big\}, \quad \forall\, \theta \in \Theta,\ B \subset \mathscr{X}. \tag{1.28}$$

Let $G$ be an algebraic group of such transformations $g$ in $\mathscr{X}$ satisfying (1.28). Suppose throughout that the distribution of $X$ under $\mathbb{P}_\theta$ are distinct from each other in $\Theta$. Because the inverse of $g$ gives the inverse of $\overline{g}$, the collection $\overline{G}$ of the transformations $\overline{g}$ induced by $g \in G$ through (1.28) also forms a group. The group $\overline{G}$ is transitive if $\{\overline{g}\theta_0 : \overline{g} \in \overline{G}\} = \Theta$

for each $\theta_0 \in \Theta$. In an invariant statistical model, a decision problem with loss $L(\theta, a)$ is invariant if for each $\overline{g} \in \overline{G}$ there exists a transformation $g^*$ in the action space $\mathscr{A}$ such that

$$L(\overline{g}\theta, g^* a) = L(\theta, a) \quad \forall\, \theta \in \Theta,\ a \in \mathscr{A}. \tag{1.29}$$

In such an invariant statistical decision problem, a decision rule $\delta$ is equivariant if

$$\delta(gx) = g^*\delta(x), \quad \forall\, g \in G,\ x \in \mathscr{X}. \tag{1.30}$$

Suppose throughout that $L(\theta, a) \neq L(\theta, a')$ for at least one $\theta \in \Theta$ when $a \neq a'$, so that the collection $G^*$ of $g^*$ induced by $\overline{g} \in \overline{G}$ through (1.29) also forms a group. If $G^*$ is commutative and $\delta(X)$ is equivariant, then $\delta(gX) = g^*\delta(X)$ is also equivariant. An estimator $\delta^*(X)$ is minimum risk equivariant (MRE) if it is equivariant and

$$R(\theta, \delta^*) = \inf\big\{R(\theta, \delta) : \delta(gX) = g^*\delta(X)\ \forall g \in G\big\}. \tag{1.31}$$

The following lemma asserts that in an invariant decision problem, an equivariant decision rule has constant risk in the orbit $\{\theta = \overline{g}\theta_0, \overline{g} \in \overline{G}\}$ for each $\theta_0 \in \Theta$.

**Lemma 1.1.** *In an invariant decision problem, the risk of equivariant decision rules satisfy $R(\theta, \delta) = R(\overline{g}\theta, \delta)$ for all $\theta \in \Theta$ and $\overline{g} \in \overline{G}$. Consequently, $R(\theta, \delta)$ does not depend on $\theta$ when $\overline{G}$ is transitive in $\Theta$.*

Proof. We have $R(\theta, \delta) = \mathbb{E}_\theta[L(\overline{g}\theta, g^*\delta(X))]$ by the invariance of the loss function, so that $R(\theta, \delta) = \mathbb{E}_\theta[L(\overline{g}\theta, \delta(gX))]$ by the equivariance of $\delta(X)$. It then follows from the invariance of the model that $R(\theta, \delta) = \mathbb{E}_{\overline{g}\theta}[L(\overline{g}\theta, \delta(X))] = R(\overline{g}\theta, \delta)$. $\square$

In an invariant decision problem, a statistician adheres to the principle of equivariance if they confine their choice of decision rules to equivariant ones. Lemma 1.1 implies that under the principle of equivariance, the risk minimization in (1.31) is a single-criterion constrained optimization problem when $\overline{G}$ is transitive in $\Theta$.

We first consider a direct extension from the location model. A function $Z(\cdot)$ is invariant in the sample space $\mathscr{X}$ under $G$ if $Z$ is measurable and $Z(gx) = Z(x)$ for all $g \in G$ and $x \in \mathscr{X}$, and $Z(\cdot)$ is maximal invariant in $\mathscr{X}$ if it is invariant in $\mathscr{X}$ and $Z(x') = Z(x)$ implies $x' = gx$ for some $g \in G$ for all pairs $\{x', x\}$ in $\mathscr{X}$. A maximal invariant $Z(x)$ in $\mathscr{X}$ is a measurable representation of the quotient space $\mathscr{X}/G$ under the relationship $x' = gx$ for some $g \in G$. It is typically a natural projection in practical examples.

**Theorem 1.8.** *Suppose $X \in \mathscr{X}$ is observed in an invariant decision problem with loss $L(\theta, a)$ and groups of transformations $G$ on $\mathscr{X}$, $\overline{G}$ on $\Theta$, and $G^*$ on $\mathscr{A}$. Let $\delta_0(x; w)$ be a class of equivariant decision rules indexed by $w \in \mathscr{W}$. Suppose all equivariant decision rules are of the form $\delta_0(X; w(Z))$ for some invariant $Z(\cdot)$ in $\mathscr{X}$. Let $\theta_0 \in \Theta$ and*

$$\delta^*(X) = \delta_0(X; w^*(Z(X))), \quad w^*(z) = \text{argmin}_{w \in \mathscr{W}}\, \mathbb{E}_{\theta_0}\big[L(\theta_0, \delta_0(X; w))\big|Z = z\big]. \tag{1.32}$$

*Suppose $\overline{G}$ is transitive and $\delta^*(X)$ is measurable. Then, $\delta^*(X)$ is MRE as in* (1.31).

Theorem 1.8 provides extensions of Theorem 1.7 to the scale and location-scale models in examples below. Bayesian approaches can be also taken to construct MRE estimators.

Proof. As $\delta^*(gx) = \delta_0(gx; w^*(Z(gx))) = g^*\delta_0(x; w^*(Z(x))) = g^*\delta^*(x)$, $\delta^*(x)$ is equivariant in $\mathscr{X}$. As $\mathbb{E}_{\theta_0}[L(\theta_0, \delta_0(X; w(Z)))|Z = z] \geqslant \mathbb{E}_{\theta_0}[L(\theta_0, \delta_0(X; w^*(z)))|Z = z]$, $R(\theta_0, \delta) \geqslant R(\theta_0, \delta^*)$. As $\overline{G}$ is transitive in $\Theta$, the conclusion follows from Lemma 1.1. $\square$

Here is a general scheme of constructing $\delta_0(x,w)$ in the presence of an equivariant mapping from $\mathscr{X}$ to $G$. A cross-sectional mapping is an invariant $Z : \mathscr{X} \to \mathscr{X}$ with $Z(x) \in \{gx : g \in G\}$. Due to the invariance of $Z(\cdot)$, it is a measurable way of choosing exactly one representative member in each orbit $\{gx : g \in G\}$ in $\mathscr{X}$.

**Proposition 1.2.** *Let $Z(x) = (h(x))^{-1}x$ with an equivariant mapping $h : \mathscr{X} \to G$. Then,*

$$x = h(x)Z(x), \quad Z(gx) = Z(x), \quad \forall x \in \mathscr{X}, g \in G,$$

*so that $Z(x)$ is cross-sectional. Moreover, given an equivariant decision rule $\delta_0(\cdot)$, all equivariant rules $\delta(x)$ can be written as*

$$\delta(x) = \delta_0\big(h(x)w(Z(x))\big)$$

*for some $w(z) = w(\delta, z) \in \mathscr{X}$, provided that $G^*$ is transitive.*

Proof. For any $g \in G$, the equivariance of $h(x)$ gives $Z(gx) = (gh(x))^{-1}gx = Z(x)$. Let $\delta_0(\cdot)$ be a fixed equivariant rule. For any equivariant $\delta(\cdot)$ and $z = Z(x)$, $\delta(z) = g^*_{\delta,z}\delta_0(z)$ for some $g^*_{\delta,z} \in G^*$ due to the transitivity of $G^*$. Let $w(z) = w(\delta, z) = g_{\delta,z}z$. For any $x \in \mathscr{X}$,

$$\delta(x) = (h(x))^*\delta(Z(x)) = (h(x))^*\delta_0(w(\delta, Z(x))) = \delta_0(h(x)w(Z(x))).$$

The equivariance of $\delta_0(h(x)w)$ follows from the equivariance of $\delta_0(\cdot)$ and $h(\cdot)$. $\square$

**Example 1.4.** *In a scale model, the joint density of the observation is given by*

$$X \sim p_n(x;\theta) = p_n(x/\theta)/\theta^n,\ x \in \mathbb{R}^n,\ \theta > 0$$

*with respect to $dx$. The group $G$ is given by $g_b x \to bx$, $b > 0$, and $\overline{G}$ by $\overline{g}_b\theta = b\theta$. Let $\mathscr{X} = \{x \in \mathbb{R}^n : x_1 \neq 0\}$. We have $g\mathscr{X} = \mathscr{X}$ for all $g$ and $\mathbb{P}_\theta\{X \in \mathscr{X}\} = 1$ for all $\theta$. The function $h(x) = g_{|x_1|}$ is equivariant $\mathscr{X} \to G$, $Z(x) = x/|x_1|$ is cross-sectional, and $x = h(x)Z(x)$. Let $r > 0$. For the estimation of $\tau(\theta) = \theta^r$, the squared relative error $L(\theta, a) = |a/\theta^r - 1|^2$ is invariant with $g^*_b a = b^r a$. It turns out that $\overline{G}$ and $G^*$ are transitive, $|X_1|^r$ is an equivariant rule, all equivariant rules are of the form $\delta(X) = w(Z)|X_1|^r$, and*

$$w^*(z) = \underset{w>0}{\arg\min}\, \mathbb{E}_1\Big[\big(w|X_1|^r - 1\big)^2\Big|Z = z\Big] = \frac{\mathbb{E}_1\big[|X_1|^r\big|Z = z\big]}{\mathbb{E}_1\big[|X_1|^{2r}\big|Z = z\big]}.$$

*As the joint density of $T = |X_1|$ and $Z$ is $t^{n-1}p_n(tz)$ at $\theta = 1$ with respect to the Lebesgue measure for $t$ and $z_{2:n}$ and counting measure for $z_1 \in \{\pm 1\}$, Pitman's MRE estimator is*

$$\widehat{\tau} = w^*(Z)|X_1|^r = \frac{|X_1|^r \int_0^\infty t^{r+n-1}p_n(tZ)dt}{\int_0^\infty t^{2r+n-1}p_n(tZ)dt} = \frac{\int_0^\infty v^{r+n-1}p_n(vX)dv}{\int_0^\infty v^{2r+n-1}p_n(vX)dv}, \quad v = \frac{t}{|X_1|}.$$

**Example 1.5.** *Consider the estimation of the scale parameter $\sigma$ in the location-scale model,*

$$X \sim p_n(x;\theta) = p_n((x-\mu 1_n)/\sigma)/\sigma^n, \quad x \in \mathbb{R}^n,\ \theta = (\mu,\sigma)^\top,\ \mu \in \mathbb{R},\ \sigma > 0,$$

*with respect to $dx$. The group $G$ is given by $g_{b,c}(x) = bx + c1_n$, $x \in \mathbb{R}^n$, $b > 0$, $c \in \mathbb{R}$, and $\overline{G}$ by $\overline{g}_{b,c}\theta = (b\mu + c, b\sigma)$. Let $\mathscr{X} = \{x : x_1 \neq x_2\}$. We have $g\mathscr{X} = \mathscr{X}$ for all $g$ and $\mathbb{P}_\theta\{X \in \mathscr{X}\} = 1$ for all $\theta$. The function $Y(x) = x - x_1 1_n$ is scale equivariant and maximal location invariant $\mathscr{X} \to \mathscr{X}$, $h(x) = g_{|Y_2(x)|,x_1}$ is equivariant $\mathscr{X} \to G$, $Z(x) = Y(x)/|Y_2(x)|$ is cross-sectional, and $x = h(x)Z(x)$. Let $r > 0$. For the estimation of $\tau(\theta) = \sigma^r$, $L(\theta,a) = |a/\sigma^r - 1|^2$ is invariant with $g^*_{b,c}a = b^r a$. It turns out that $\overline{G}$ and $G^*$ are transitive, $|Y_2|^r$ is equivariant $\mathscr{X} \to \mathscr{A}$, equivariant rules are of the form $\delta(X) = w(Z)|Y_2|^r$, and*

$$w^*(z) = \operatorname*{arg\,min}_{w>0} \mathbb{E}_{0,1}\big[\big(w|Y_2|^r - 1\big)^2\big|Z = z\big] = \frac{\mathbb{E}_{0,1}\big[|Y_2|^r\big|Z = z\big]}{\mathbb{E}_{0,1}\big[|Y_2|^{2r}\big|Z = z\big]}.$$

*The MRE estimator $w^*(Z)|Y_2|^r$ is a function of $Y_{2:n}$, and $Y_{2:n}$ is an observation from a scale model with the density family $\{\overline{p}_n(y_{2:n}/\sigma)/\sigma^{n-1}, \sigma > 0\}$ in $\mathbb{R}^{n-1}$, where $\overline{p}_n(y_{2:n}) = \int p_n\big(y_{1:n} + u1_n\big)du$ with $y_1 = 0$. As in Example 1.4, Pitman's MRE estimator is*

$$\widehat{\tau} = \frac{\int_0^\infty v^{r+n-2}\overline{p}_n(vY_{2:n})dv}{\int_0^\infty v^{2r+n-2}\overline{p}_n(vY_{2:n})dv} = \frac{\int_0^\infty v^{r+n-2}\int p_n\big(vX + u1_n\big)dudv}{\int_0^\infty v^{2r+n-2}\int p_n\big(vX + u1_n\big)dudv}.$$

**Example 1.6.** *Consider the estimation of the location parameter $\mu$ in the location-scale model in Example 1.5, with $\theta = (\mu,\sigma)^\top$ and the same transformation groups $G$ and $\overline{G}$. Define $Y(x) \in \mathscr{X}$, $h(x) \in G$ and $Z(x) = Y(x)/|Y_2(x)| \in \mathscr{X}$ as in Example 1.5. The loss function $L(\theta,a) = |a-\mu|^2/\sigma^2$ is invariant with $g^*_{b,c}a = ba + c$. It turns out that $\overline{G}$ and $G^*$ are both transitive. Given equivariant rules $\delta_0(x)$ for $\mu$ and $\delta_1(y)$ for $\sigma$, equivariant rules $\delta(x)$ for $\mu$ must be of the form $\delta(x) = \delta_0(x; w(Z(x)))$ with $\delta_0(x;w) = \delta_0(x) - w\delta_1(Y(x))$, so that the MRE is given by*

$$\delta^*(X) = \delta_0(X) - w^*(Z)\delta_1(Y)$$

*with the optimal*

$$w^*(z) = \text{argmin}_{w>0}\mathbb{E}_{0,1}[(\delta_0(X) - w\delta_1(Y))^2|Z = z].$$

*More explicitly,*

$$\widehat{\mu} = \delta^*(X) = \frac{\int_0^\infty \int tv^{n+1}p_n(v(X - t1_n))dtdv}{\int_0^\infty \int v^{n+1}p_n(v(X - t1_n))dtdv}.$$

*For $X = (X_1,\ldots,X_n)^\top$ with iid $X_i \sim N(\mu,\sigma^2)$, $\delta_0(X) = \overline{X}$ is independent of $Y$ and $Z$, so that $\overline{X}$ is the MRE of $\mu$.*

**Example 1.7.** *In the linear model, we observe $X \in \mathbb{R}^n$ and deterministic $M \in \mathbb{R}^{n\times d}$ with*

$$X = M\beta + \sigma\varepsilon, \quad \varepsilon \sim p_n(\cdot) \text{ with respect to the Lebesgua measure in } \mathbb{R}^n. \tag{1.33}$$

*Assume* $\mathrm{rank}(M) = d$. *Let $\mu = M\beta$ and $\mathbb{H}_0$ be the range of $M$. We write* (1.33) *as*

$$p_n(x;\theta) = p_n((x-\mu)/\sigma)/\sigma^n, \ x \in \mathbb{R}^n, \ \theta = (\mu^\top, \sigma)^\top, \ \mu \in \mathbb{H}_0, \ \sigma > 0. \tag{1.34}$$

*This includes the location-scale model in Example 1.6 as a special case with one-dimensional $\mathbb{H}_0 = \{c1_n, c \in \mathbb{R}\}$. Let $P_0$ be the orthogonal projection to $\mathbb{H}_0$ in $\mathbb{R}^n$, $P_0^\perp = I_n - P_0$, and $r > 0$. In the estimation of $\tau(\theta) = \sigma^r$ under the loss function $(a/\sigma^r - 1)^2$, the MRE estimator $\widehat{\tau}$ is based on $Y = P_0^\perp X$. The density of $Y$ is $\overline{p}_n(y/\sigma)/\sigma^{n-d} = \int_{\mathbb{H}_0} p_n\big(y/\sigma + u\big)du/\sigma^{n-d}$ with respect to the Lebesgue measure in the $(n-d)$-dimensional $P_0^\perp\mathbb{R}^n$, so that by Example 1.4*

$$\widehat{\tau} = \frac{\int_0^\infty v^{r+n-d-1}\int_{\mathbb{H}_0} p_n\big(vX+u\big)dudv}{\int_0^\infty v^{2r+n-d-1}\int_{\mathbb{H}_0} p_n\big(vX+u\big)dudv}.$$

*For the estimation of $\mu \in \mathbb{H}_0$ under the loss function $\|a-\mu\|_2^2/\sigma^2$, the MRE estimator is*

$$\widehat{\mu} = \frac{\int_0^\infty \int_{\mathbb{H}_0} tv^{n+1}p_n(v(X-t))dtdv}{\int_0^\infty \int_{\mathbb{H}_0} v^{n+1}p_n(v(X-t))dtdv}.$$

*For $\varepsilon \sim N(0, I_n)$, $\widehat{\mu} = P_0X$ is the MLE due to the independence of $P_0X$ and $P_0^\perp X$.*

Let $T(X)$ be a sufficient statistic and $\delta(X)$ an equivariant estimator. Let $\mathscr{F}_0$ be the sigma-field generated by $T(X)$. For equivariant $\delta$, $\mathbb{E}[\delta(gX)|\mathscr{F}_0] = g^*\mathbb{E}[\delta(X)|\mathscr{F}_0]$, so that the estimator $\mathbb{E}[\delta(X)|\mathscr{F}_0]$ is still equivariant. Thus, by Rao-Blackwell, for invariant statistical decision problems with convex loss functions, it suffices to consider deterministic equivariant rules depending on sufficient statistics.

**Example 1.8.** *Suppose $X = (X_1, \ldots, X_n)^\top \in \mathbb{R}^{n\times d}$ is observed with iid $X_i \sim N(\mu, \Sigma)$, with parameter $\theta = (\mu, \Sigma)$ and sample size $n > d$. In this model $\overline{X} = \sum_{i=1}^n X_i/n$ and $S_n = (X - \overline{X}1_n)^\top(X - \overline{X}1_n)$ are sufficient. Let $G$ be the group of bijections $g_{B,c}X = XB^\top + 1_nc^\top$ with invertible $B \in \mathbb{R}^{d\times d}$ and $c \in \mathbb{R}^d$. The group $\overline{G}$ is given by $\overline{g}_{B,c}\theta = (B\mu + c, B\Sigma B^\top)$. In the estimation of $\tau(\theta) = \Sigma$, consider as in the examples below loss functions $L(\theta, a)$ convex in $a$ and invariant with transformations $g^*_{B,c}a = BaB^\top$ in the action space. In such decision problems, an equivariant estimator of $\Sigma$ based on the sufficient statistics must be a function of $S_n$. Moreover, an equivariant estimator $\delta(S_n)$ must satisfy $\delta(S_n) = S_n^{1/2}\delta(I_d)S_n^{1/2}$. As $\delta(I_d) = \delta(UI_dU^\top) = U\delta(I_d)U^\top$ for all orthonormal $U \in \mathbb{R}^{d\times d}$, $\delta(I_d) = wI_d$ for some $w > 0$. It follows that the MRE estimator is given by*

$$\delta(S_n) = w^*S_n, \quad w^* = \arg\min_{w>0} \mathbb{E}_{(0,I_d)}\big[L((0,I_d), wS_n)\big].$$

*Let $\lambda_j$ be the eigenvalues of $\Sigma^{-1/2}a\Sigma^{-1/2}$. For the loss function*

$$L_1((\mu,\Sigma),a) = \mathrm{trace}\big(\Sigma^{-1}(a-\Sigma)\Sigma^{-1}(a-\Sigma)\big) = \textstyle\sum_{j=1}^d(\lambda_j - 1)^2,$$

*the optimal constant factor is $w_1^* = 1/(n+d)$. For the loss function*

$$L_2((\mu,\Sigma),a) = \mathrm{trace}\big(a\Sigma^{-1}\big) - \log\det\big(a\Sigma^{-1}\big) - d = \textstyle\sum_{j=1}^d\big(\lambda_j - 1 - \log\lambda_j\big).$$

*the optimal constant factor is $w_2^* = 1/(n-1)$.*

### 1.5. **Bayes methods.**

1.5.1. *The Bayes approach.* In a Bayes approach to the statistical decision problem, the unknown parameter $\theta$ is treated as random itself with a distribution, called the prior and denoted by $\Pi$, in the parameter space $\Theta$, and the objective is to minimize the Bayes risk

$$R(\Pi,\delta) = \int_\Theta R(\theta,\delta)\Pi(d\theta) = \int_\Theta\int_{\mathscr{X}} L(\theta,\delta(x))\mathbb{P}\{X\in dx|\theta\}\Pi(d\theta)$$

in (1.6). Here $\mathbb{P}\{X\in dx|\theta\}$ is used to denote the probability family to emphasis that $\theta$ is treated as random. Suppose the statistical model is given by the conditional density $p_n(x|\theta), \theta\in\Theta$, with respect to a sigma-finite measure $\nu_n(dx)$ in $\mathscr{X}$. The posterior (distribution) and posterior risk of taking action $a\in\mathscr{A}$ are respectively given by

$$\Pi_n(d\theta|x) = \frac{p_n(x|\theta)\Pi(d\theta)}{p_n(x;\Pi)},\ R(a|x) = R_\Pi(a|x) = \int_\Theta L(\theta,a)\Pi_n(d\theta|x), \tag{1.35}$$

where $p_n(x;\Pi) = \int_\Theta p_n(x|\theta)\Pi(d\theta)$ is the marginal density of $X$ under prior $\Pi$. If $T(X)$ is sufficient, then the posterior depends on $x$ only through $T(x)$ by the factorization criterion. When the prior has a density $\pi(\theta)$, the posterior (density) is given by

$$\pi_n(\theta|x) = p_n(x|\theta)\pi(\theta)/p_n(x;\Pi). \tag{1.36}$$

The Bayes risk of a decision rule $\delta$ is the integration of its posterior risk

$$R(\Pi,\delta) = \int_{\mathscr{X}} R\big(\delta(x)\big|x\big)p_n(x;\Pi)\nu_n(dx),$$

and the Bayes rule, denoted by $\delta^*$, is the minimizer of the posterior risk

$$\delta^*(x) = \delta^*_\Pi(x) = \arg\min_{a\in\mathscr{A}} R_\Pi(a|x). \tag{1.37}$$

It follows that when $\delta^*$ is well defined as in (1.37), the Bayes rule is optimal,

$$R(\Pi,\delta^*) = \min_{\delta\in\mathscr{D}} R(\Pi,\delta),$$

where $\mathscr{D}$ is the class of all decision rules. We assume throughout the Bayes solution exists.

For example, in the estimation of $\tau(\theta)$ under the squared loss $L(\theta,a) = \|a-\tau(\theta)\|_2^2$,

$$\delta^*(x) = \int \tau(\theta)\Pi_n(d\theta|x),\quad R_\Pi(\delta^*|x) = \int \big\|\delta^*(x)-\tau(\theta)\big\|_2^2\Pi_n(d\theta|x),$$

are respectively the posterior mean and variance. For the estimation of $\tau(\theta) = \theta\in\mathbb{R}$ under the absolute loss $L(\theta,a) = |a-\theta|$, the Bayes rule $\delta^*(x)$ is the posterior median. One may also take the point of view that the objective of the Bayes analysis is to characterize or compute the posterior. This would provide justification for the estimation of $\theta$ by the posterior mode as the maximizer of the posterior density.

The prior has the interpretation as a probabilistic summary of the available knowledge about the parameter $\theta$ before observing $X$. This interpretation is coherent as the posterior can be then treated as the new prior summarizing the aggregated knowledge about $\theta$ after observing $X$: If new data $Y = y$ becomes available and $Y$ is independent of $X$ given $\theta$,

$$\Pi_n(d\theta|x,y) = \frac{p_n(y|\theta)\Pi_n(d\theta|x)}{\int_\Theta p_n(y|\theta)\Pi_n(d\theta|x)}. \tag{1.38}$$

In the Bayes approach, the prior is a probability in the parameter space, $\int_\Theta \Pi(d\theta) = 1$. In a generalized Bayes approach, one may also take a sigma-finite measure with $\Pi(\Theta) = \infty$ as an improper prior as long as the posterior, posterior risk and Bayes rule are still well defined by (1.35), (1.36) and (1.37).

One may chose a non-informative prior to express no preference among the probabilities in the statistical model. Jeffreys [Jef46] reasoned that such a prior should be "flat" in the space of probabilities, and thus not dependent on a specific parametrization of the model. This leads to Jeffreys' prior

$$\pi(\theta) \propto \sqrt{\det(I(\theta))},$$

where $I(\theta)$ is the Fisher information matrix in (1.13). Indeed, for differentiable invertible functions $\tau = \tau(\theta)$, $I(\tau) = J_\tau^\top I(\theta(\tau)) J_\tau$ with the Jacobian $J_\tau = (\partial\theta/\partial\tau_1, \ldots, \partial\theta/\partial\tau_d) = \nabla\theta^\top$, so that the prior $\widetilde{\pi}(\tau) = \pi(\theta(\tau))|\det(J(\tau))| \propto \sqrt{\det(I(\tau))}$ induced from Jeffrey's prior under parametrization $\theta$ matches Jeffreys' prior under parametrization $\tau$. In this sense, Jefferys' prior is invariant. However, Jeffreys' prior is not guaranteed to be proper. For example, in the Gaussian scale model $N(0, \theta^2)$, $I(\theta) = 1/(2\theta^2)$, so that $\int_0^\infty \sqrt{I(\theta)}d\theta = \infty$.

When the density belongs to an exponential family, conjugate priors provide explicit Bayes estimators of the mean as a linear interpolation of the MLE and the prior estimator corresponding to the prior mode. Let $X = (X_1, \ldots, X_n)^\top$ with iid observations $X_i$ from density $\exp[\theta^\top x_1 - \psi(\theta)]$ with respect to $\nu(dx_1)$. The joint density of $X$ is

$$p_n(x; \theta) = \exp[\theta^\top T(x) - n\psi(\theta)]$$

with sufficient statistic $T(X) = \sum_{i=1}^n X_i$ and with respect to the product measure $\nu_n(dx) = \prod_{i=1}^n \nu(dx_i)$. For such an exponential family, the conjugate prior is given by

$$\pi(\theta|k, \mu) \propto \exp[\theta^\top \mu - k\psi(\theta)]$$

with respect to the Lebesgue measure $d\theta$. Under this prior, the Bayes estimator is

$$\mathbb{E}\big[\psi'(\theta)\big|T(X)\big] = \frac{T(X) + \mu}{n + k} = \left(\frac{n}{n+k}\right)\overline{X} + \left(1 - \frac{n}{n+k}\right)\frac{\mu}{k},$$

where $\overline{X}$ is the MLE for $\psi'(\theta)$ and $\mu/k = \psi'(\text{prior mode})$.

**Example 1.9.** *For the binomial $p_n(x; \widehat{\eta}) \propto \theta^x(1-\theta)^{n-x}$, the conjugate prior is given by the beta density $\pi(\theta|k, \mu) \propto \theta^{\mu-1}(1-\theta)^{k-\mu-1}$ with $\mu > 0$ and $k - \mu > 0$ in this parametrization. With the Fisher information $I(\theta) = n/(\theta(1-\theta))$, $\pi(\theta|1, 1/2)$ is Jeffreys' prior.*

The prior may involve some parameter $\gamma$, say $\Pi(d\theta|\gamma)$, as in the conjugate prior above with $\gamma = (k, \mu)$. Instead of picking a predetermined value of $\gamma$, one may also treat $\gamma$ as random in a hierarchical Bayes approach with a joint prior $\Pi(d\theta|\gamma)\Pi(d\gamma)$. In this case, the Gibbs sampler [GS90] can be used to compute the posterior given $X = x$ as follows,

$$\theta^{(k)} \sim \Pi(d\theta|x, \gamma^{(k-1)}), \quad \gamma^{(k)} \sim \Pi(d\gamma|x, \theta^{(k)}), \quad k = 1, 2, \ldots$$
$$\Pi(\theta \in B|x) = \lim_{K\to\infty} K^{-1} \textstyle\sum_{k=1}^K I\{\theta^{(k)} \in B\},$$
$$\mathbb{E}_\Pi\big[\theta\big|X = x\big] = \lim_{K\to\infty} K^{-1} \textstyle\sum_{k=1}^K \theta^{(k)},$$

under proper conditions (e.g. irreducibility and aperiodicity) for the convergence to a unique stationary distribution in Markov chain Monte Carlo (MCMC) [Tan93].

1.5.2. *Gaussian processes.* In a Gaussian model with a centered Gaussian prior,

$$X|\theta \sim p_n(\cdot|\theta) \sim N\big(M\theta, \Sigma\big), \quad \theta \sim \pi(\theta) \sim N\big(0, \Sigma_{\text{prior}}\big), \tag{1.39}$$

with known $M \in \mathbb{R}^{n\times d}$, $\Sigma \in \mathbb{R}^{n\times n}$, and $\Sigma_{\text{prior}} \in \mathbb{R}^{d\times d}$. Here $\pi(\cdot)$ denotes the density of the Gaussian prior $\Pi$, and the prior is centered for simplicity as the mean $\mathbb{E}_\Pi[\theta]$ can be simply subtracted in the uncentered case. Suppose $\text{rank}(\Sigma) = n$ and $\text{rank}(\Sigma_{\text{prior}}) = d$. Because $(X^\top, \theta^\top)^\top$ is jointly Gaussian, the posterior is also Gaussian and given by

$$\begin{aligned} \pi_n(\theta|x) &\sim N\big(\mu_{\text{post}}(x), \Sigma_{\text{post}}\big), \\ \Sigma_{\text{post}} &= (M^\top\Sigma^{-1}M + \Sigma_{\text{prior}}^{-1})^{-1}, \\ \mu_{\text{post}}(x) &= \mathbb{E}_\Pi[\theta|X = x] = \Sigma_{\text{post}}M^\top\Sigma^{-1}x, \end{aligned} \tag{1.40}$$

where the formulas for $\Sigma_{\text{post}}$ and $\mu_{\text{post}}(x)$ are directly given by completing the square in

$$-2\log\big(p_n(x|\theta)\pi(\theta)\big) \;=\; \big(x - M\theta\big)^\top\Sigma^{-1}\big(x - M\theta\big) + (\theta - \mu_{\text{prior}})^\top\Sigma_{\text{prior}}^{-1}(\theta - \mu_{\text{prior}}) + C.$$

Alternatively and equivalently, the least squares formula gives

$$\begin{aligned} \mu_{\text{post}}(x) &= \text{Cov}(\theta, X)\text{Cov}(X, X)^{-1}x = \Sigma_{\text{prior}}M^\top(M\Sigma_{\text{prior}}M^\top + \Sigma)^{-1}x, \\ \Sigma_{\text{post}} &= \Sigma_{\text{prior}} - \Sigma_{\text{prior}}M^\top(M\Sigma_{\text{prior}}M^\top + \Sigma)^{-1}M\Sigma_{\text{prior}}. \end{aligned} \tag{1.41}$$

The following examples demonstrate the Gaussian process method as useful tools for statistical inference as well as the computation of quadratic penalized least squares.

**Example 1.10** (Ridge regression)**.** *In ridge regression with response $X$ and design $M$,*

$$\widehat{\theta} = \mathop{\arg\min}_{\theta\in\mathbb{R}^d}\big\{\|X - M\theta\|_2^2 + \lambda\|\theta\|_2^2\big\}. \tag{1.42}$$

*Without further assumption,* (1.42) *can be viewed as an $\ell_2$ penalized least squares estimator. Because the Gaussian mean is also the mode, $\widehat{\theta}$ has the Bayesian interpretation as the posterior mean in* (1.40) *with $\Sigma = \sigma^2 I_n$ and $\Sigma_{\text{prior}} = (\sigma^2/\lambda)I_d$ in* (1.39)*. It follows that*

$$\begin{aligned} \widehat{\theta} = \mu_{\text{post}}(X) &= (M^\top M + \lambda I_d)^{-1}M^\top X, \\ \Sigma_{\text{post}} &= \sigma^2(M^\top M + \lambda I_d)^{-1}. \end{aligned}$$

**Example 1.11** (Generalized Bayes)**.** *When the improper prior for $\theta$ is taken as the Lebesgue measure in the linear model, $X|\theta \sim N(M\theta, \Sigma)$, the posterior is given by*

$$\pi_n(\theta|x) = \frac{\exp[-(x - M\theta)^\top\Sigma^{-1}(x - M\theta)/2]}{\int\exp[-(x - M\theta)^\top\Sigma^{-1}(x - M\theta)/2]d\theta} \sim N(\mu_{\text{post}}, \Sigma_{\text{post}})$$

*with $\Sigma_{\text{prior}}^{-1} = 0$ in* (1.40)*, provided* $\text{rank}(\Sigma) = \text{rank}(M) = d$*. It follows that*

$$\mu_{\text{post}} = (M^\top\Sigma^{-1}M)^{-1}M^\top\Sigma^{-1}x, \quad \Sigma_{\text{post}} = (M^\top\Sigma^{-1}M)^{-1},$$

*respectively the weighted least squares estimator of $\theta$ (without penalty) and its variance when $\theta$ is treated as deterministic. For $\Sigma = I_n$, this matches Example 1.10 with $\lambda = 0$.*

In what follows, we consider a few examples where the prior is infinite-dimensional. As in the case of ridge regression, the posterior mean is analytically identical to a quadratic penalized least squares estimator in these examples.

**Example 1.12** (Smoothing spline)**.** *Suppose* $(u_i, X_i)^\top$ *are observed with*

$$X_i = \theta(u_i) + \varepsilon_i,\ 1 \leqslant i \leqslant n,$$

*where* $\theta(\cdot)$ *is an unknown smooth function in* $[0,1]$*,* $u_i \in [0,1]$ *are deterministic distinct design points and* $\varepsilon_i$ *are iid* $N(0,\sigma^2)$ *given* $\theta(\cdot)$*. Let* $\alpha \in \mathbb{N}_+$*. Denote by* $f^{(\alpha)}(\cdot)$ *the* $\alpha$*-th derivative of* $f(\cdot)$*, and* $f^{(-\alpha)}(t) = \int_0^t f^{(1-\alpha)}(u)du$ *as the* $\alpha$*-th integration with* $f^{(0)} = f$*. In this nonparametric regression problem, the smoothing spline estimator is defined as*

$$\widehat{\theta} = \mathrm{argmin}_{\theta(\cdot)}\big\{\|X_{1:n} - \theta(u_{1:n})\|_2^2 + \lambda\|\theta^{(\alpha)}\|^2_{L_2[0,1]}\big\}, \tag{1.43}$$

*where* $f(u_{1:n}) = (f(u_1),\ldots,f(u_n))^\top \in \mathbb{R}^n$ *for all functions* $f(\cdot)$*. As* $\theta^{(\alpha)} \in L_2[0,1]$ *and* $\{\sqrt{2}\sin(k\pi u), k \geqslant 1\}$ *is an orthonormal basis in* $L_2[0,1]$*,* $\theta(u)$ *can be written as*

$$\theta(u) = \textstyle\sum_{j=0}^{\alpha-1}\beta_j u^j + \sum_{k=1}^\infty (k\pi)^{-\alpha}\xi_k\sqrt{2}\sin^{(-\alpha)}(\pi k u),\ 0 \leqslant u \leqslant 1, \tag{1.44}$$

*with* $\|\theta^{(\alpha)}\|^2_{L_2[0,1]} = \sum_{k=1}^\infty \xi_k^2$ *and unknown* $\beta_j$ *and* $\xi_k$*. Let* $\beta = (\beta_0,\ldots,\beta_{\alpha-1})^\top$ *and*

$$\big(\widehat{\beta}^{(d)\top}, \widehat{\xi}^{(d)\top}_{1:d}\big)^\top = \underset{b\in\mathbb{R}^\alpha, v\in\mathbb{R}^d}{\arg\min}\Big(\Big\|X_{1:n} - M_{n\times\alpha}b - \Psi_{n\times d}v\Big\|_2^2 + \lambda\|v\|_2^2\Big\} \tag{1.45}$$

*with* $M_{n\times\alpha} = (u^j_{1:n}, j = 0,\ldots,\alpha-1)$ *and* $\Psi_{n\times d} = ((k\pi)^{-\alpha}\sqrt{2}\sin^{(-\alpha)}(k\pi u_{1:n}), k = 1,\ldots,d)$*. It follows from the* $\widehat{\theta}$ *version of the Fourier expansion formula* (1.44) *that*

$$\widehat{\theta}(u) = \lim_{d\to\infty}\Big(\textstyle\sum_{j=0}^{\alpha-1}\widehat{\beta}^{(d)}_j u^j + \sum_{k=1}^d (k\pi)^{-\alpha}\widehat{\xi}^{(d)}_k\sqrt{2}\sin^{(-\alpha)}(\pi k u)\Big). \tag{1.46}$$

*Because* (1.45) *can be viewed as generalized Bayes in a Gaussian model,* (1.40) *provides*

$$\big(\widehat{\beta}^{(d)\top}, \widehat{\xi}^{(d)\top}_{1:d}\big)^\top = \begin{pmatrix} M^\top_{n\times\alpha}M_{n\times\alpha} & M^\top_{n\times\alpha}\Psi_{n\times d} \\ \Psi^\top_{n\times d}M_{n\times\alpha} & \Psi^\top_{n\times d}\Psi_{n\times d} + \lambda I_d\end{pmatrix}^{-1}(M_{n\times\alpha}, \Psi_{n\times d})^\top X_{1:n}.$$

*Here the prior* $\Pi$ *is given by the product measure with* $\xi_k \sim N(0,\sigma^2/\lambda)$ *and the Lebesgue measure as improper prior for each* $\beta_j$*. The induced prior on* $\theta^{(\alpha)}$ *is also improper because* $\mathbb{P}_\Pi\big\{\|\theta^{(\alpha)}\|^2_{L_2[0,1]} = \infty\big\} = 1$*. However, the prior for the integrated Fourier series* $\theta_F(t) = \theta(u) - \sum_{j=0}^{\alpha-1}\beta_j t^j$ *is proper as the process* $\{(\lambda^{1/2}/\sigma)\theta_F(t), 0 \leqslant t \leqslant 1\}$ *is a Brownian motion for* $\alpha = 1$ *and its integral* $W^{(1-\alpha)}(t)$ *for* $\alpha > 1$*. This expression of the prior for* $\theta_F(t)$ *gives*

$$\widehat{\theta}(v_{1:d}) = \mathbb{E}_\Pi[\theta(v_{1:d})|X_{1:n}] = \textstyle\sum_{j=0}^{\alpha-1}\widehat{\beta}_j v^j_{1:d} + \widehat{\theta}_F(v_{1:d}) \tag{1.47}$$

*for all* $d \geqslant n$ *and distinct* $v_j \in [0,1]$ *with* $v_{1:n} = u_{1:n}$*, and*

$$\big(\widehat{\beta}^\top, \widehat{\theta}^\top_F(u_{1:n})\big)^\top = \begin{pmatrix} M^\top_{n\times\alpha}M_{n\times\alpha} & M^\top_{n\times\alpha}I_{n\times d} \\ I^\top_{n\times d}M_{n\times\alpha} & I^\top_{n\times d}I_{n\times d} + \lambda\big(\Sigma_W^{(1-\alpha)}(v_{1:d})\big)^{-1}\end{pmatrix}^{-1}(M_{n\times\alpha}, I_{n\times d})^\top X_{1:n},$$

*where* $I_{n\times d} = (I_n, 0_{n\times(d-n)})$ *and* $\Sigma_W^{(1-\alpha)}(v_{1:d}) = \mathrm{Cov}(W^{(1-\alpha)}(v_{1:d}), W^{(1-\alpha)}(v_{1:d}))$*. Note that* $X_{1:n} = M_{n\times\alpha}\beta + I_{n\times d}\theta(v_{1:d}) + \varepsilon_{1:n}$*, so that* $M = (M_{n\times\alpha}, I_{n\times d})$ *in* (1.40)*. A justification to call the estimator smoothing spline is that the covariance of* $W^{(1-\alpha)}(u)$ *and* $W^{(1-\alpha)}(v)$*, as a function of* $(u,v)^\top \in [0,1]^2$*, is a bivariate spline of degree* $\alpha$*.*

**Example 1.13** (Kriging). *Gaussian process regression is commonly used to analyze spatial data observed from different locations in a manifold. The method is also called Kriging or Wiener-Kolmogorov prediction. Consider the example where the manifold is taken as the unit sphere $\mathbb{S}^2$ in $\mathbb{R}^3$, e.g. a model for the surface of the planet Earth in geostatistics. Suppose independent $(u_i, X_i)^\top$ are observed with*

$$X_i = \theta(u_i) + \varepsilon_i,\ \ i = 1, \ldots, n,$$

*where $\theta(\cdot)$ is an unknown smooth function in $\mathbb{S}^2$, $u_i \in \mathbb{S}^2$ are deterministic distinct design points and $\varepsilon_i$ are iid $N(0, \sigma^2)$ given $\theta(\cdot)$. Given a Gaussian process prior for $\theta(\cdot)$, the Bayes inference about $\theta(\cdot)$ can be carried out via (1.40) as in Example 1.12. For any $v_k \in \mathbb{S}^2$, $k = 1, \ldots, d$ with $d \geqslant n$ and $v_{1:n} = u_{1:n}$, the posterior of $\theta(v_{1:d})$ is Gaussian with*

$$\begin{aligned} \mathbb{E}_\Pi\big[\theta(v_{1:d})\big|X_{1:n}\big] &= \Sigma_{\text{prior}}(v_{1:d})I_{n\times d}^\top(\Sigma_{\text{prior}}(u_{1:n}) + \sigma^2 I_n)^{-1}X_{1:n}, \\ \Sigma_{\text{post}}(v_{1:d}) &= \big(\sigma^{-2}I_{n\times d}^\top I_{n\times d} + \Sigma_{\text{prior}}^{-1}(v_{1:d})\big)^{-1}, \end{aligned} \tag{1.48}$$

*where $I_{n\times d} = (I_n, 0) \in \mathbb{R}^{n\times d}$, and $\Sigma_{\text{prior}}(v_{1:n})$ and $\Sigma_{\text{post}}(v_{1:n})$ are respectively the prior and posterior covariance matrices of $\theta(v_{1:d})$.*

*In applications where no additional domain input is used to construct the prior, it is natural to require $\theta(\cdot)$ to have a stationary prior distribution under rotation and reflection transformations in $\mathbb{S}^2$. In such stationary cases, the prior covariance of $\theta(u)$ and $\theta(v)$ is a function of the geodesic distance between $u$ and $v$ in $\mathbb{S}^2$,*

$$\mathrm{Cov}_\Pi\big(\theta(u), \theta(v)\big) = g\big(d(u, v)\big), \quad d(u, v) = \arccos\big(u^\top v\big) \in [0, \pi].$$

*A continuous function $g : [0, \pi] \to \mathbb{R}$ is said to be nonnegative-definite (positive-definite) iff the covariance matrix $\big(g\big(d(v_i, v_j)\big)\big)_{d\times d}$ is nonnegative-definite (positive-definite) for all $d \geqslant 2$ and $v_i \in \mathbb{S}^2$. Of course, we must use a nonnegative-definite $g(\cdot)$ to construct a stationary Gaussian prior for $\theta(\cdot)$ in $\mathbb{S}^2$, and we would prefer to use a positive-definite $g$. For example, $g(t) = \cos(t)$ is nonnegative-definite but not positive-definite because the rank of the resulting $\Sigma_{\text{prior}}(v_{1:d}) = v_{1:d}v_{1:d}^\top$ is no greater than 3. Let $P_k$ denote the Legendre polynomial of degree $k$. By* [Sch42] *a function $g$ is nonnegative-definite iff*

$$g(t) = \sum_{k=1}^\infty a_k P_k(\cos(t)), \quad a_k \geqslant 0\,\forall\, k \geqslant 1,\ \sum_{k=1}^\infty a_k < \infty.$$

*Moreover, $g(t)$ is positive-definite if in addition $a_k > 0$ for all $k$* [XC92]. *The above results hold for domains $\mathbb{S}^m$ when $P_k$ are taken as the Chebyshev polynomials $P_k(\cos(t)) = \cos(kt)$ for $m = 1$ and in general the Gegenbauer polynomials.*

**Example 1.14** (White noise). *In the white noise model, a Gaussian process*

$$X(t) = \int_0^t \theta(u)du + \sigma_n W(t),\ \ 0 \leqslant t \leqslant 1,$$

*is observed where $\theta(\cdot)$ is an unknown function, $\sigma_n > 0$ is a known constant and $W(\cdot)$ is a standard Brownian motion process given $\theta(\cdot)$. The white noise model can be viewed as the limit of nonparametric regression with uniform design in $[0, 1]$, and a special case of the Gaussian functional regression below.*

**Example 1.15** (Functional regression). *In functional data, functions are observed and unknown parameters can be also functions. Consider as an example Gaussian functional regression in which a stochastic process $\{X(t), 0 \leqslant t \leqslant 1\}$ is observed in the model*

$$X(t) - X(0) = \int_0^t (K\theta)(u)du + \sigma_n W(t),\ 0 \leqslant t \leqslant 1,$$

*where $\theta(t)$ is an unknown function with a Gaussian prior, $W(t)$ is a Brownian motion process independent of $\theta(t)$, and $K : f \to Kf$, called the design operator, is a known linear operator in $L_2[0,1]$. In a more general setting*

$$X(t) - X(0) = (J\theta)(t) + \sigma_n W(t), 0 \leqslant t \leqslant 1,$$

*the operator $K$ can be identified as $(K\theta)(t) = (d/dt)(J\theta)(t)$. In the differential form, the model is written as*

$$dX(t) = (K\theta)(t) + \sigma_n dW(t).$$

*Suppose the operator $K$ has a discrete (functional) SVD given by*

$$(Kf)(u) = \sum_{k=0}^{\infty} \lambda_k \phi_k(u) \int_0^1 \psi_k(v) f(v)dv$$

*with $\lambda_k \geqslant 0$ and some (complete) orthonormal bases $\{\phi_k, k \geqslant 0\}$ and $\{\psi_k, k \geqslant 0\}$ in $L_2[0,1]$. As in the matrix case, the basis elements $\phi_k$ and $\psi_k$ are respectively called left- and right-singular functions. The Brownian motion can be written as*

$$W(t) = \sum_{k=0}^{\infty} \varepsilon_k \int_0^t \phi_k(u)du \quad \text{with iid } \varepsilon_k \sim N(0,1).$$

*For simplicity, we may conveniently take the prior of $\theta$ as a Gaussian process of the form*

$$\theta(t) = \sum_{k=0}^{\infty} \beta_k \xi_k \psi_k(t)$$

*with $\beta_k \geqslant 0$ and iid standard Gaussian $\{\xi_k, k \geqslant 0\}$ independent of $\{\varepsilon_k, k \geqslant 0\}$. In this case,*

$$X(t) - X(0) = \textstyle\int_0^t \sum_{k=0}^{\infty} \lambda_k \beta_k \xi_k \phi_k(u)du + \sigma_n W(t),$$

*so that the observation can be transformed into a Gaussian sequence model with observations*

$$Z_k = \int_0^1 \phi_k(t)dX(t) = \lambda_k \beta_k \xi_k + \sigma_n \varepsilon_k, \quad k = 0, 1, \ldots \tag{1.49}$$

*Consequently, the Bayes estimator is given by the shrinkage estimator*

$$\widehat{\theta}(t) = \sum_{k=0}^{\infty} \beta_k \widehat{\xi}_k \psi_k(t), \quad \widehat{\xi}_k = \frac{\lambda_k \beta_k Z_k}{(\lambda_k \beta_k)^2 + \sigma_n^2}.$$

*Given a Gaussian regression problem with design operator $K$, the main difficulty with the above analysis is to solve the SVD problem for $K$. If an analytical solution is not feasible, numerical methods can be used to discretize the problem into finite-dimensional linear regression. In the white noise model, $K$ is the identity operator with $\lambda_k = 1$, corresponding to linear regression with orthonormal designs.*

1.5.3. *Bayes estimation and unbiasedness.* When $\theta$ is treated as random, unbiasedness becomes $\mathbb{E}[\delta(X)|\theta] = \tau(\theta)$ in the estimation of a function $\tau(\theta)$. In general, Bayes rules are biased, or equivalently unbiased estimators are suboptimal in the Bayes setting.

**Proposition 1.3.** *No unbiased estimator $\delta(X)$ for $\tau(\theta)$ can be a Bayes solution under the squared $\ell_2$ loss unless $\mathbb{E}_\Pi\big[\|\delta(X) - \tau(\theta)\|_2^2\big] = 0$.*

Proof. For unbiased Bayes rules, $\mathbb{E}_\Pi[\delta(X)^\top \tau(\theta)] = \mathbb{E}_\Pi[\|\delta(X)\|_2^2] = \mathbb{E}_\Pi[\|\tau(\theta)\|_2^2]$. $\square$

In the estimation of the natural parameter under the squared $\ell_2$ loss in an exponential family with a continuous multivariate sufficient statistic $T(X)$, the Bayes rule is given by Tweedie's formula [Rob56, Efr03] in terms of the marginal density of $T(X)$. Suppose

$$T(X) \sim \overline{p}_n(t|\theta)dt = \exp\big[\theta^\top t - \psi_n(\theta)\big]\overline{h}_n(t)dt,\ t \in \mathbb{R}^d. \tag{1.50}$$

Let $\delta_0(t) = -\nabla \log(\overline{h}_n(t))$ with the gradient operator $\nabla$. Tweedie's formula writes

$$\delta^*(t) = \mathbb{E}_\Pi\big[\theta\big|T(X) = t\big] = \frac{\int_\Theta \theta e^{\theta^\top t - \psi_n(\theta)}\Pi(d\theta)}{\int_\Theta e^{\theta^\top t - \psi_n(\theta)}\Pi(d\theta)} = \delta_0(t) + \nabla \log\big(\overline{p}_n(t;\Pi)\big) \tag{1.51}$$

for the Bayes estimator $\delta^*(T(X))$, where $\overline{p}_n(t;\Pi) = \int \exp\big[\theta^\top t - \psi_n(\theta)\big]\overline{h}_n(t)\Pi(d\theta)$ is the marginal density of $T(X)$ with respect to $dt$. For differentiable mappings $f : \mathbb{R}^d \to \mathbb{R}^d$ satisfying proper integrability conditions, Stein's identity [Ste81, Hud78] provides

$$\mathbb{E}_\theta\big[(\delta_0(T(X)) - \theta)^\top f(T(X))\big] = \int \big(\delta_0(t) - \theta\big)^\top f(t)\overline{p}_n(t|\theta)dt = \int(\mathrm{div} f)(t)\overline{p}_n(t|\theta)dt, \tag{1.52}$$

where $\mathrm{div} f = \mathrm{trace}(\nabla f)$ is the divergence of $f$. This can be seen as integration by parts. With constant $f \in \mathbb{R}^d$, it follows that $\delta_0(T(X))$ is the UMVU estimator of $\theta$ by Theorem 1.4.

**Theorem 1.9.** *Let $\delta_0(t) = -\nabla \log(\overline{h}_n(t))$ and $\delta^*(t) = \delta_0(t) + \nabla \log\big(\overline{p}_n(t;\Pi)\big)$ be respectively the UMVU and Bayes estimators of $\theta$ under the squared $\ell_2$ loss in the exponential family* (1.50). *Then, the posterior risk of the Bayes estimator $\delta^*$ is given by*

$$\mathbb{E}_\Pi\big[\big\|\theta - \delta^*(T(X))\big\|_2^2\big|T(X) = t\big] = \Delta \log\big(\overline{p}_n(t;\Pi)/\overline{h}_n(t)\big) \tag{1.53}$$

*with the Laplace operator $\Delta = \nabla^\top\nabla$. Moreover, the risk function of $\delta_0$ is given by*

$$R(\theta, \delta_0) = \mathbb{E}_\theta\big[-\Delta \log \overline{h}_n(T(X))\big] \tag{1.54}$$

*and the risk function of $\delta^*$ is given by*

$$R(\theta, \delta^*) = R(\theta, \delta_0) + \mathbb{E}_\theta\big[2\Delta \log\big(\overline{p}_n(t;\Pi)\big) + \big\|\nabla \log\big(\overline{p}_n(t;\Pi)\big)\big\|_2^2\big]. \tag{1.55}$$

Proof. As $R(\theta, \delta_0) = \mathbb{E}_\theta\big[(\delta_0(T(X)) - \theta)^\top \delta_0(T(X))\big]$, (1.54) follows from (1.52). In the expansion of $\mathbb{E}_\theta\big[\|\delta_0(t) - \theta + \nabla \log(\overline{p}_n(t;\Pi))\|_2^2\big]$, the cross-product term is given by (1.52),

$$2\int \big(\delta_0(t) - \theta\big)^\top \big(\nabla \log\big(\overline{p}_n(t;\Pi)\big)\big)\overline{p}_n(t|\theta)dt = 2\int \big(\Delta \log\big(\overline{p}_n(t;\Pi)\big)\big)\overline{p}_n(t|\theta)dt.$$

This gives (1.55). Finally, (1.53) can be seen as the differentiation of (1.51). $\square$

When a prior $\Pi$ can be chosen to satisfy $\mathbb{E}_\theta\big[2\Delta \log\big(\overline{p}_n(t;\Pi)\big) + \big\|\nabla \log\big(\overline{p}_n(t;\Pi)\big)\big\|_2^2\big] \leqslant 0$ for all $\theta$, the Bayes estimator has smaller risk than the UMVU estimator for all $\theta$. In this case, the UMVU estimator is *inadmissible*. For the estimation of the Gaussian mean of dimension 5 or greater, such a proper prior was constructed in [Str71]; See the discussion on empirical Bayes and admissibility.

1.5.4. *Bayes equivariant rules.* Consider an invariant decision problem with observation $X \in \mathscr{X}$, probability family $\{\mathbb{P}_\theta, \theta \in \Theta\}$, loss function $L(\theta, a)$ on $\Theta \times \mathscr{A}$, and transformation groups $G$ on $\mathscr{X}$, $\overline{G}$ on $\Theta$ and $G^*$ on $\mathscr{A}$. When $\overline{G}$ is transitive, the risk function $R(\theta, \delta)$ is constant by Lemma 1.1 for equivariant decision rules, so that any equivariant Bayes rule, if exists, is MRE. Assume $\overline{G}$ is transitive in the rest of this subsection.

We shall prove that the Bayes rule with an invariant prior is equivariant and thus MRE. A (possibly improper) prior $H$ in $\Theta$ is invariant under transformations in $\overline{G}$ if

$$\int_\Theta f(\overline{g}\theta)H(d\theta) = \int_\Theta f(\theta)H(d\theta), \quad \forall\, \overline{g} \in \overline{G},\ f \in L_1(H). \tag{1.56}$$

A measure $H_\ell$ in the group $\overline{G}$ is left invariant if

$$\int_{\overline{G}} f(\overline{g}\overline{h})H_\ell(d\overline{h}) = \int_{\overline{G}} f(\overline{h})H_\ell(d\overline{h}), \quad \forall\, \overline{g} \in \overline{G},\ f \in L_1(H_\ell).$$

Such $H_\ell$ is called the left Haar measure. When the left Haar measure exists in $\overline{G}$, an invariant prior in $\Theta$ can be induced through a natural mapping $\overline{h} \to \theta = \overline{h}\theta_0$,

$$\int_\Theta f(\theta)H(d\theta) = \int_{\overline{G}} f(\overline{h}\theta_0)H_\ell(d\overline{h}),$$

with a fixed $\theta_0 \in \Theta$. The invariance of the induced measure $H$ can be seen from

$$\int_\Theta f(\overline{g}\theta)H(d\theta) = \int_{\overline{G}} f(\overline{g}\overline{h}\theta_0)H_\ell(d\overline{h}) = \int_{\overline{G}} f(\overline{h}\theta_0)H_\ell(d\overline{h}), \quad \forall\, \overline{g} \in \overline{G},\ f \in L_1(H).$$

Similarly, the prior $H$ can be induced from the left Haar measure on $G$ through $h \to \overline{h}\theta_0$.

The (left and right) Haar measures exist on topological groups. Here we say that a group is topological if the following conditions hold: (i) The group is endowed with a topology, a collection of open subsets containing the entire group and the empty set and closed under the open operations of union and finite intersection; (ii) The topology is Hausdorff in the sense of open separation of any two points in the group; (iii) Each point in the group has a compact neighborhood; (iv) The topology can be generated by a countable sub-collection of open sets through the open operations. Under these conditions, the Haar measure is sigma-finite, Borel, and unique up to a constant factor. When the group is compact, the Haar measure is finite and will be taken as the Haar probability hereafter.

We say that a decision rule $\delta$ is almost equivariant under a prior $\Pi$ if

$$\mathbb{P}_\Pi\{\delta(gx) = g^*\delta(x)\} = 1\ \forall g \in G.$$

Suppose a proper invariant prior $H$ exists. The risk of the Bayes rule $\delta^*$ must satisfy

$$\int_\Theta R(\theta, \delta^*)H(d\theta) = \int_\Theta \mathbb{E}\big[L(\overline{g}\theta, \delta^*(x))\big|\overline{g}\theta\big]H(d\theta) = \int_\Theta \mathbb{E}\big[L(\theta, (\overline{g}^{-1})^*\delta^*(gx))\big|\theta\big]H(d\theta).$$

Thus, the Bayes rule is almost equivariant if it is unique. However, the above argument has two drawbacks. Firstly, almost equivariance does not imply equivariance which requires $\delta(gx) = g^*\delta(x)\ \forall\, g \in G, x \in \mathscr{X}$. Secondly, as $\int_\Theta R(\theta, \delta^*)H(d\theta) = \infty$ for improper priors, the argument heavily relies on the existence of a proper invariant prior, which essentially means the compactness of the group $\overline{G}$.

In the rest of this subsection we describe a remedy to the above issues. Suppose the statistical model is given by a density family $\{p_n(x|\theta), \theta \in \Theta\}$ with respect to a sigma-finite measure $\nu_n(dx)$ in the sample space satisfying the following condition,

$$\int_{\mathscr{X}} f(gx)\nu_n(dx) = J_g \int_{\mathscr{X}} f(x)\nu_n(dx), \quad \forall\, f \in L_1(\nu_n), \tag{1.57}$$

i.e. the Jacobian $J_g = \nu_n(dg^{-1}x)/\nu_n(dx)$ is a constant. Take a fixed $\theta_0 \in \Theta$ and define

$$\begin{aligned} &\overline{p}_n(x|\theta) = J_h p_n(h^{-1}x|\theta_0),\ \theta = \overline{h}\theta_0,\ h \in G, \\ &\overline{R}(a|x) = \frac{\int L(\theta, a)\overline{p}_n(x|\theta)H(d\theta)}{\int \overline{p}_n(x|\theta)H(d\theta)}, \\ &\overline{\delta}(x) = \arg\min_{a\in\mathscr{A}} \overline{R}(a|x). \end{aligned} \tag{1.58}$$

**Theorem 1.10.** *Suppose an invariant model is given by a density family $\{p_n(x|\theta), \theta \in \Theta\}$ with respect to a sigma-finite measure $\nu_n$ satisfying* (1.57)*. Suppose $\overline{\delta}(x)$ is uniquely defined in* (1.58) *with an invariant sigma-finite measure $H$ in $\Theta$. Then, $\overline{\delta}(x)$ is an equivariant generalized Bayes rule. If in addition $H$ is a proper prior, then $\delta(X)$ is MRE.*

Proof. It follows from (1.57) and the invariance of $p_n(x|\theta)$ that for $\theta = \overline{h}\theta_0$,

$$\int f(x)p_n(hx|\theta)\nu_n(dx) = J_h \int f(h^{-1}x)p_n(x|\overline{h}\theta_0)\nu_n(dx) = J_h \int f(x)p_n(x|\theta_0)\nu_n(dx).$$

Because $\overline{p}_n(hx|\theta) = J_h p_n(x|\theta_0)$ by (1.58) and $\nu_n$ is sigma-finite, $p_n(hx|\theta) = \overline{p}_n(hx|\theta)$ almost everywhere in $\nu_n$. It follows that the statistical model is also given by $\{\overline{p}_n(x|\theta), \theta \in \Theta\}$ by the invertibility of $h \in G$, and $\overline{\delta}(x)$ is the generalized Bayes rule. Moreover, for any $g \in G$ and $\theta = \overline{h}\theta_0$,

$$\overline{p}_n(gx|\overline{g}\theta) = \overline{p}_n(gx|\overline{g}\overline{h}\theta_0) = J_{gh}p_n(h^{-1}x|\theta_0) = (J_{gh}/J_h)\overline{p}_n(x|\theta).$$

Thus, by the definition of the posterior risk $\overline{R}(a|x)$ and the invariance of $H$,

$$\overline{R}(a|x) = \frac{\int L(\overline{g}\theta, g^*a)p_n(gx|\overline{g}\theta)H(d\theta)}{\int p_n(gx|\overline{g}\theta)H(d\theta)} = \frac{\int L(\theta, g^*a)p_n(gx|\theta)H(d\theta)}{\int p_n(gx|\theta)H(d\theta)} = R(g^*a|gx).$$

Because $g^*$ are surjections due to their invertibility in $\mathscr{A}$, $\overline{\delta}(gx) = g^*\delta(x)$ when $\overline{\delta}(x)$ is uniquely defined. $\square$

1.6. **Minimaxity and admissibility.**

1.6.1. *Minimax theory.* In a statistical decision problem, a decision rule $\delta^*_{\rm mm}$ is said to be minimax if it minimizes the maximum risk in (1.7):

$$\sup_{\theta\in\Theta} R(\theta,\delta^*_{\rm mm}) = R_{\rm mm},\quad R_{\rm mm} = \inf_{\delta\in\mathscr{D}} R_{\max}(\delta),\quad R_{\max}(\delta) = \sup_{\theta\in\Theta} R(\theta,\delta). \tag{1.59}$$

Because the Bayes risk $R(\Pi,\delta)$ in (1.6) is no greater than the maximum risk $R_{\max}(\delta)$, a Bayes rule $\delta^*_\Pi$ with prior $\Pi$ is minimax when its Bayes and maximum risks agree.

**Theorem 1.11.** *Suppose the Bayes rule $\delta^*_\Pi$ under a prior $\Pi$ satisfies*

$$R(\Pi,\delta^*_\Pi) = \int_\Theta R(\theta,\delta^*_\Pi)\Pi(d\theta) = R_{\max}(\delta^*_\Pi). \tag{1.60}$$

*Then, $\delta^*_\Pi$ is minimax, and $\delta^*_\Pi$ is the unique minimax rule when the Bayes solution with the prior $\Pi$ is unique. Moreover, the prior $\Pi$ is the least favorite in the sense that*

$$R(\Pi,\delta^*_\Pi) = \sup_{\Pi'\in\mathscr{P}} R(\Pi',\delta^*_G),$$

*where $\mathscr{P}$ is the set of all priors in $\Theta$.*

**Example 1.16.** *In the binomial case $X\sim \mathrm{bin}(n,\theta)$ with the beta$(a,b)$ prior for $\theta$, the Bayes estimator of $\theta$ is $\delta^*_{a,b}=\delta^*_{a,b}(X)=(a+X)/(a+b+n)$, and its risk function is*

$$R(\theta,\delta^*_{a,b}) = \frac{n\theta(1-\theta)+(a(1-\theta)-b\theta)^2}{(a+b+n)^2}.$$

*For $a=b=n^{1/2}/2$, $R(\theta,\delta^*_{a,b}) = 1/\{4(1+n^{1/2})^2\}\ \forall\theta$, so that $\delta^*_{a,b}$ is minimax. Clearly the MLE $\delta_{\rm ml}=X/n$ is not minimax as $R(\theta,\delta_{\rm ml})=\theta(1-\theta)/n$. However, the minimax estimator has smaller risk than the MLE only in a very small interval,*

$$|\theta-1/2|\leqslant \{n^{1/2}/2+1/4\}^{1/2}/(1+n^{1/2})\approx n^{-1/4}/\sqrt{2}.$$

Because $R_{\max}(\delta)=\sup_{\Pi\in\mathscr{P}} R(\Pi,\delta)$, Theorem 1.11 relies on the minimax theorem

$$\min_{\delta\in\mathscr{D}}\sup_{\Pi\in\mathscr{P}} R(\Pi,\delta) = \max_{\Pi\in\mathscr{P}}\inf_{\delta\in\mathscr{D}} R(\Pi,\delta), \tag{1.61}$$

attained at a Bayes rule with respect to a least favorable prior. However, as (1.59) just calls for a decision rule $\delta^*_{\rm mm}$, it suffices to impose the weaker version of the condition

$$\min_{\delta\in\mathscr{D}}\sup_{\Pi\in\mathscr{P}} R(\Pi,\delta) = \sup_{\Pi\in\mathscr{P}}\inf_{\delta\in\mathscr{D}} R(\Pi,\delta). \tag{1.62}$$

**Theorem 1.12.** *Suppose that for some decision rule $\delta^*$ and a sequence of priors $\Pi_k$*

$$R_{\max}(\delta^*) = \lim_{k\to\infty}\inf_{\delta\in\mathscr{D}} R(\Pi_k,\delta).$$

*Then, $\delta^*$ is minimax.*

PROOF. For any decision rule $\delta$, $R_{\max}(\delta)\geqslant \inf_{\delta\in\mathscr{D}} R(\Pi_k,\delta)$. $\square$.

**Example 1.17.** *For the estimation of mean $\theta \in \mathbb{R}^d$ under squared $\ell_2$ loss with observation $X \sim N(\theta, I_d)$. The risk of the MLE $\delta_{\rm ml} = X$ is $R(\theta, \delta_{\rm ml}) = d$. For the Gaussian prior $\Pi : \theta \sim N(0, \gamma I_d)$, the Bayes risk is $d\gamma/(1+\gamma) = d + o(1)$ by (1.40) as $\gamma \to \infty$, so that $\delta_{\rm ml}$ is minimax and $R_{\rm mm} = d$. However, $\delta_{\rm ml}$ is not the unique minimax rule for $d \geqslant 3$, and some proper Bayes estimator is also minimax for $d \geqslant 5$.*

A drawback of Theorem 1.12 is the lack of a criterion for the uniqueness of the minimax solution. Both (1.61) and (1.62) are called minimax theorems. In general, the minimax theorem holds for the risk $R(\Pi, \delta)$ as a bivariate function of inputs $\Pi \in \mathscr{P}$ and $\delta \in \mathscr{D}$ if

$$R_{\rm mm} = \inf_{\delta \in \mathscr{D}} \sup_{\Pi \in \mathscr{P}} R(\Pi, \delta) = \sup_{\Pi \in \mathscr{P}} \inf_{\delta \in \mathscr{D}} R(\Pi, \delta). \tag{1.63}$$

This minimax theorem, if holds, guarantees "$\epsilon_1$-minimax" rules are "$\epsilon_2$-Bayes" for all $\epsilon_2 > \epsilon_1 > 0$, but not the existence or uniqueness of the minimax solution. The value of $R_{\rm mm}$ provides a benchmark to measure the performance of decision rules. In more complex statistical decision problems where an explicit optimal solution is not easy to formulate or solve, minimax risk is commonly used to characterize the optimal rate of convergence. Sufficient conditions are provided in the celebrated minimax theorem of von Neumann for (1.61) and in more general versions such as Sion's minimax theorem for (1.62) and (1.63). Here we provide a set of sufficient conditions to directly guarantee condition (1.60).

**Theorem 1.13.** *Suppose $\Theta$ is a compact Hausdorff space. Let $\mathscr{C}_0(\Theta)$ be the space of all continuous functions on $\Theta$, $\mathscr{D}_0$ be a collection of randomized decision rules, and $\mathscr{S}_0 = \{R(\theta, \delta) : \delta \in \mathscr{D}_0\}$ be the corresponding collection of risk functions. Suppose $\mathscr{D}_0$ contains at least one Bayes solution for each prior $\Pi \in \mathscr{P}$ and $\mathscr{S}_0$ is a convex subset of $\mathscr{C}_0(\Theta)$. Suppose $R_0 = \min\{R_{\max}(\delta) : \delta \in \mathscr{D}_0\}$ is attained by a certain $\delta^* \in \mathscr{D}_0$, $R_{\max}(\delta^*) = R_0$. Then, $\delta^*$ is a Bayes rule for a proper prior $\Pi$, and condition (1.60) of Theorem 1.11 holds for the Bayes rule $\delta^*_\Pi = \delta^*$. Consequently, the Bayes rule $\delta^*$ is minimax.*

Proof. Let $f_0$ be the constant function $f(\theta) = R_0$ and $\mathscr{S}_1$ be the convex hull of $\mathscr{S}_0 \cup \{f_0\}$. Let $\mathscr{Q}_0 = \{f \in \mathscr{C}_0 : \max_\theta f(\theta) < R_0\}$ and $\overline{\mathscr{Q}}_0 = \{f \in \mathscr{C}_0 : \max_\theta f(\theta) \leqslant R_0\}$ as the closure of $\mathscr{Q}_0$. Because $\mathscr{Q}_0$ and $\mathscr{L}_1$ are convex and $\mathscr{Q}_0 \cap \mathscr{S}_1 = \varnothing$ and $\{\delta^*, f_0\} \in \overline{\mathscr{Q}}_0 \cap \mathscr{S}_1$, the Hahn-Banach separation and Riesz-Markov representation theorems provide

$$\int_\Theta f(\theta)\mu_0(d\theta) \leqslant R_0\mu_0(\Theta) = \int_\Theta R(\theta, \delta^*)\mu_0(d\theta) \leqslant \int_\Theta R(\theta, \delta)\mu_0(d\theta)$$

for all $f \in \overline{\mathscr{Q}}_0$, all $\delta \in \mathscr{D}_0$, and a finite signed measure $\mu_0$. For any continuous positive $h(\theta)$, the above inequality with $f(\theta) = R_0 - h(\theta) \in \mathscr{Q}_0$ yields $\int_\Theta h(\theta)\mu_0(d\theta) \geqslant 0$. It follows that $\mu_0$ is a finite positive measure, and $\Pi(d\theta) = \mu_0(d\theta)/\mu_0(\Theta)$ is a proper prior. Let $\delta_\Pi$ be a Bayes solution in $\mathscr{D}_0$ for the prior $\Pi$, we have $R(\Pi, \delta^*) \leqslant R(\Pi, \delta_\Pi)$ so that $\delta^*$ is also a Bayes rule for the prior $\Pi$. The conclusion follows as $R(\Pi, \delta^*) = R_0 = R_{\max}(\delta^*)$. □

A fundamental theorem of statistical decision theory is the assertion that the minimax rules are Bayes or generalized Bayes rules. Theorem 1.13 provides a set of sufficient conditions for this assertion. A main drawback of the theorem is the requirement of the continuity of the risk functions of Bayes rules under a compact topology in $\Theta$. It does not cover Example 1.17 for $d = 1, 2$ and other cases where the minimax risk is attained by generalized Bayes rules but not proper Bayes rules.

1.6.2. *Complete class theory.* As the performance of (possibly randomized) decision rules is measured by the risk function, the following terminologies are natural and reasonable. A decision rule $\delta_1$ is said to be *as good as* another decision rule $\delta_2$ if $R(\theta, \delta_1) \leqslant R(\theta, \delta_2)$ for all $\theta \in \Theta$, $\delta_1$ is *better than* $\delta_2$ if $\delta_1$ is as good as $\delta_2$ and $R(\theta, \delta_1) < R(\theta, \delta_2)$ for some $\theta \in \Theta$, and $\delta_1$ is *equivalent* to $\delta_2$ if $R(\theta, \delta_1) = R(\theta, \delta_2)$ for all $\theta \in \Theta$. A decision rule $\delta_0$ is *admissible* if no rule $\delta$ is better than $\delta_0$, and a decision rule $\delta_0$ is *inadmissible* if it is not admissible.

A collection $\mathscr{D}_0$ of decision rules is *(essentially) complete* if some $\delta_1 \in \mathscr{D}_0$ is (as good as) better than $\delta_2$ for every rule $\delta_2 \notin \mathscr{D}_0$. The collection $\mathscr{D}_0$ is *minimum (essentially) complete* if it is (essentially) complete and no proper sub-collection of it is (essentially) complete. Thus, the minimum complete class, if exists, is the collection of all admissible rules.

Let $\mathscr{D}_0$ be a collection of randomized decision rules, and $\mathscr{S}_0 = \{R(\theta, \delta) : \delta \in \mathscr{D}_0\}$ the corresponding risk set. Then, the convex hull of $\mathscr{S}_0$ is also composed of risk functions of randomized decision rules. This can be seen as follows. For $\delta_0$ and $\delta_1$ in $\mathscr{D}_0$ and random variable $U \sim \text{unif}[0, 1]$ independent of $\{\delta_0, \delta_1\}$, $R(\theta, \delta_t) = tR(\theta, \delta_1) + (1-t)R(\theta, \delta_0)$, $0 \leqslant t \leqslant 1$, where $\delta_t = \delta_1$ when $U \leqslant t$ and $\delta_t = \delta_0$ when $U > t$. For example, $\mathscr{S}_0$ can be taken as the convex hull of the risk set of all Bayes and minimax rules in Theorem 1.13.

A second fundamental theorem of statistical decision theory is the assertion that all admissible decision rules are Bayes or possibly generalized Bayes rules. This assertion holds in the following setting similar to the setting of Theorem 1.13.

**Theorem 1.14.** *Suppose $\Theta$ is a compact Hausdorff space. Suppose the risk functions are continuous in $\Theta$ for all admissible decision rules and at least one Bayes rule for each proper prior. Then, every admissible rule is a Bayes rule for a certain proper prior. If the minimum complete class exists, then the set of Bayes rules is complete.*

Proof. Let $\mathscr{C}_0(\Theta)$ be the space of all continuous functions on $\Theta$, $\mathscr{D}_0$ be the collection of all randomized admissible or Bayes decision rules with continuous risk functions, $\mathscr{S}_0 = \{R(\theta, \delta) : \delta \in \mathscr{D}_0\}$ the corresponding risk set, and $\mathscr{S}_1$ the convex hull of $\mathscr{S}_0$ in $\mathscr{C}_0(\Theta)$. Let $\delta^*$ be an admissible rule. Define

$$\mathscr{Q}_0 = \Big\{ f \in \mathscr{C}_0(\Theta) : f(\theta) \leqslant R(\theta, \delta^*)\ \forall \theta \in \Theta,\ \max_{\theta \in \Theta} \big( R(\theta, \delta^*) - f(\theta) \big) > 0 \Big\},$$

and $\overline{\mathscr{Q}}_0$ as the closure of $\mathscr{Q}_0$ under the $\|\cdot\|_\infty$ norm. Because $\delta^*$ is admissible, $\mathscr{Q}_0 \cap \mathscr{S}_1 = \varnothing$. Because $\mathscr{Q}_0$ and $\mathscr{L}_1$ are convex and $\delta^* \in \overline{\mathscr{Q}}_0 \cap \mathscr{S}_1$, the Hahn-Banach separation and Riesz-Markov representation theorems provide a finite signed measure $\mu_0$ satisfying

$$\int_\Theta f(\theta) \mu_0(d\theta) \leqslant \int_\Theta R(\theta, \delta^*) \mu_0(d\theta) \leqslant \int_\Theta R(\theta, \delta) \mu_0(d\theta)$$

for all $f \in \mathscr{Q}_0$ and $\delta \in \mathscr{D}_0$. For any continuous positive $h(\theta)$, the above inequality with $f(\theta) = R(\theta, \delta^*) - h(\theta) \in \mathscr{Q}_0$ yields $\int_\Theta h(\theta)\mu_0(d\theta) \geqslant 0$. It follows that $\mu_0$ is a finite positive measure, and $\Pi(d\theta) = \mu_0(d\theta)/\mu_0(\Theta)$ is a proper prior. Let $\delta_\Pi$ be a Bayes solution in $\mathscr{D}_0$ for the prior $\Pi$, we have $R(\Pi, \delta^*) \leqslant R(\Pi, \delta_\Pi)$ so that $\delta^*$ is also a Bayes rule for the prior $\Pi$. The conclusion follows. $\square$

Theorems 1.13 and 1.14 are called fundamental theorems of statistical decision theory because they identify minimax and admissible rules as Bayes rules. Between the two, Theorem 1.14 renders stronger support for the Bayes approach. Again, a main drawback of the theorem is the requirement of the continuity of the risk functions under a compact topology in $\Theta$ for all admissible rules and at least one Bayes rule for each prior.

The existence of a minimum complete class essentially requires that the convex hull $\mathscr{S}_1$ of the risk set of a complete class $\mathscr{D}_0$ be closed from below. Let $\overline{\mathscr{S}_1}$ be the closure of $\mathscr{S}_1$ and $\mathscr{Q}(f) = \{h : h(\theta) \leqslant f(\theta) \; \forall\, \theta \in \Theta\}$. The lower boundary of $\mathscr{S}_1$ is composed of functions $f_- : \Theta \to [0, \infty)$ satisfying $\mathscr{Q}(f_-) \cap \overline{\mathscr{S}_1} = \{f_-\}$. The risk set $\mathscr{S}_1$ is closed from below if it contains its lower boundary.

**Lemma 1.2.** *Let $\mathscr{D}_0$ be an (essentially) complete class of randomized decision rules, and $\mathscr{S}_0 = \{R(\cdot, \delta) : \delta \in \mathscr{D}_0\}$ be the corresponding risk set. Let $\mathscr{B}_-$ be the collection of functions $f_-(\cdot)$ satisfying the following two conditions: (i) $R(\theta, \delta_{k-1}) \geqslant R(\theta, \delta_k) \to f_-(\theta) \,\forall\, \theta \in \Theta$ for a monotone sequence of rules $\delta_k$ in $\mathscr{D}_0$ with $\delta_k$ as good as $\delta_{k-1}$; (ii) $\mathscr{Q}(f_-) \cap \mathscr{S}_0 \subseteq \{f_-\}$. If $\mathscr{B}_- \subseteq \mathscr{S}_0$, then $\mathscr{B}_-$ is minimum (essentially) complete.*

Proof. Consider the partial ordering defined by $f_1(\theta) \leqslant f_2(\theta) \,\forall\, \theta$. Let $\delta_0 \in \mathscr{D}_0$ and $f_0(\cdot) = R(\cdot, \delta_0)$. Let $f_-(\cdot)$ be the limit of a monotone sequence as in condition (i) in the risk set $\mathscr{Q}(f_0) \cap \mathscr{S}_0$. If $f_- \in \mathscr{B}_-$, then $f_- \in \mathscr{Q}(f_0) \cap \mathscr{S}_0$ is a lower bound of the sequence $R(\cdot, \delta_k)$. Otherwise $R(\cdot, \delta_0') \in \mathscr{Q}(f_-) \cap \mathscr{S}_0 \subseteq \mathscr{Q}(f_0) \cap \mathscr{S}_0$ for some $\delta_0' \in \mathscr{D}_0$, so that $R(\cdot, \delta_0')$ is a lower bound of the sequence. Consequently, by Zorn's lemma, $\mathscr{Q}(f_0) \cap \mathscr{S}_0$ contains a minimum element $f_-^* = R(\cdot, \delta^*)$ with $\mathscr{Q}(f_-^*) \cap \mathscr{Q}(f_0) \cap \mathscr{S}_0 = \{f_-^*\}$. Because $\delta^*$ is admissible and as good as $\delta_0$, $\mathscr{D}_0$ is minimum (essentially) complete. □

Let $\mathscr{C}_0(\Theta)$ be equipped with the $\|\cdot\|_\infty$ norm. In the proofs of Theorems 1.13 and 1.14, the Hahn-Banach theorem asserts the separation of disjoint convex sets by bounded linear functionals in $\mathscr{C}_0(\Theta)$, and the Riesz-Markov representation theorem identifies such bounded linear functionals to finite measures. It is possible to extend this proof to sigma-compact $\Theta$, but then the separating functional will be given by sigma-finite measures in general, leading to the inclusion of generalized Bayes rules in the complete class.

The following theorem provides the admissibility of unique Bayes rules.

**Theorem 1.15.** *Let $\delta_\Pi^*$ denote Bayes rules under a proper prior $\Pi$ with finite risk $R(\Pi, \delta_\Pi^*)$.*
*(i) The Bayes solutions $\delta_\Pi^*$ are admissible if they have the same risk function.*
*(ii) The Bayes solutions $\delta_\Pi^*$ are admissible if their risk functions $R(\theta, \delta_\Pi^*)$ are continuous and $\Pi(B) > 0$ for all nonempty open $B$ under a certain topology in $\Theta$.*
*(iii) The Bayes solution $\delta_\Pi^*$ is admissible if it is unique under $\mathbb{P}_\Pi$ and $\mathscr{L}(X; \mathbb{P}_\theta) \ll \mathscr{L}(X; \mathbb{P}_\Pi) \,\forall\, \theta$, where $\mathscr{L}(X; \mathbb{P})$ is the distribution of $X$ under $\mathbb{P}$.*

Proof. (i) If $\delta$ is as good as the Bayes rule $\delta_\Pi^*$, then $\delta$ is also a Bayes rule under $\Pi$.
(ii) If $\delta_1$ is better than $\delta_2$ and they both have continuous risk function and finite Bayes risks under $\Pi$, then $R(\theta, \delta_1) < R(\theta, \delta_2) - \epsilon_0$ for a certain $\epsilon_0 > 0$ and $\theta$ in a nonempty open $B$, so that $R(\Pi, \delta_2) - R(\Pi, \delta_1) \geqslant \epsilon_0 \Pi(B)$. Thus $\delta_2$ is not a Bayes rule under the given $\Pi$.
(iii) The uniqueness of the Bayes rule under $\mathbb{P}_\Pi$ implies the uniqueness under $\mathbb{P}_\theta$ for all $\theta$, and thus the risk equivalence of the Bayes solutions. □

**Example 1.18.** *Consider a density family $\{p_n(x;\theta), \theta \in \Theta\}$ with respect to a sigma-finite measure $\nu_n(dx)$. Suppose $\Theta$ is a compact Hausdorff space and $p_n(\cdot;\theta)$ is $L_1(\nu_n)$ continuous in $\Theta$. Suppose $p_n(\cdot;\theta') \ll p_n(\cdot;\theta)$ for all $\theta, \theta'$ in $\Theta$. Consider the loss $L(\theta, a) = \|\tau(\theta) - a\|_2^2$ for the estimation of a continuous function $\tau(\theta) \in \mathbb{R}^d$. Let $\mathscr{D}_0$ be the class of all randomized estimators taking value in the convex hull of $\tau(\Theta)$. Then, $\mathscr{D}_0$ is complete, and the risk set $\mathscr{S}_1 = \{R(\theta,\delta) : \delta \in \mathscr{D}_0\}$ is convex and uniformly continuous in $\Theta$. Let $\mathscr{C}_0(\Theta)$ be the space of all continuous functions on $\Theta$, and $\overline{\mathscr{S}_1}$ be the closure of $\mathscr{S}_1$ in uniform convergence. Consider a fixed estimator $\delta_0 \in \mathscr{D}_0$. Let $R(\theta,\delta_k) \downarrow f_-(\theta)$ as in condition (i) of Lemma 1.2. As $f_-(\cdot) \in \mathscr{C}_0(\Theta)$ due to the uniform continuity of $\mathscr{S}_0$, $\mathscr{Q}(f_-) \cap \overline{\mathscr{S}_1} = \{f_-\}$. As in the proof of Theorems 1.13 and 1.14 there exists a proper prior $\Pi$ with*

$$\int_\Theta f(\theta)\Pi(d\theta) \leqslant \int_\Theta f_-(\theta)\Pi(d\theta) = \lim_{k\to\infty}\int_\Theta R(\theta,\delta_k)\Pi(d\theta) \leqslant \int_\Theta R(\theta,\delta)\Pi(d\theta)$$

*for all $f \in \mathscr{Q}(f_-)$ and $\delta \in \mathscr{D}_0$. Let $\delta^*_\Pi$ be the Bayes rule under prior $\Pi$. We have*

$$\mathbb{E}_\Pi\big\{\|\delta_k(X) - \delta^*_\Pi(X)\|_2^2\big] = \textstyle\int_\Theta\big(R(\theta,\delta_k) - f_-(\theta)\big)\Pi(d\theta) \to 0.$$

*As $p_n(x;\theta)$ are absolutely continuous to each other, $\delta_k(X) \to \delta^*_\Pi(X)$ in $\mathbb{P}_\theta$, so that $R(\theta,\delta^*_\Pi) \leqslant f_-(\theta)$ by Fatou's lemma. Thus, $\delta^*_\Pi$ is as good as $\delta_0$. As $\delta^*_\Pi$ is admissible by Theorem 1.15 (iii), the class of Bayes rules is complete. Moreover, due to the strict convexity of the loss function, the Bayes rules are all non-randomized.*

1.6.3. *Admissibility in the estimation of Gaussian mean.* Let $X_i, i \leqslant n$, be iid $N(\theta, \sigma^2 I_d)$ vectors. For the estimation of the mean $\theta$ under the squared $\ell_2$ loss, the sample mean $\overline{X} = \sum_{i=1}^n X_i/n$ is ML, UMVU, MRE, minimax, and generalized Bayes. However, as mentioned in Example 1.17, $\overline{X}$ is not admissible for $d > 2$. Here we prove the admissibility of $\overline{X}$ for $d = 1, 2$. Because the admissibility of $\overline{X}$ in the estimation of $\theta$ is equivalent to the admissibility of $\overline{X}\sqrt{n}/\sigma$ in the estimation of $\theta\sqrt{n}/\sigma$ under the squared $\ell_2$ loss, we assume below without loss of generality $\sigma = 1$ and $n = 1$.

Let $X \sim N(\theta, I_d)$. Recall the Bayes risk of the MLE $\delta_{\mathrm{ml}}(x) = x$ under prior $\theta \sim N(0, \gamma I_d)$ is used to prove its minimaxity in Example 1.17. The idea of the following theorem is to prove the admissibility of $\delta_0$ by comparing the Bayes risks of $\delta_0$ and its competitors.

**Theorem 1.16** (Blyth's method)**.** *Suppose the parameter space $\Theta$ is open and decision rule $\delta_0$ has a continuous risk function. Suppose that for decision rules $\delta$ as good as $\delta_0$ the risk function is also continuous. Suppose for any nonempty open set $\Theta_0 \subseteq \Theta$ there exists a sequence of priors $\Pi_k$ such that $R(\Pi_k, \delta_0) - R(\Pi_k, \delta^*_{\Pi_k}) \ll \Pi_k(\Theta_0)$. Then, $\delta_0$ is admissible.*

Proof. Let $f(\theta)$ be a continuous function satisfying $f(\theta) \leqslant R(\theta, \delta_0)$. Suppose for some $\epsilon > 0$ the parameter set $\Theta_0 = \{\theta : R(\theta,\delta_0) - f(\theta) > \epsilon\}$ is nonempty. Then,

$$R(\Pi_k, \delta_0) - R(\Pi_k, \delta^*_{\Pi_k}) \ll \epsilon\Pi_k(B) < R(\Pi_k, \delta_0) - \textstyle\int f(\theta)\Pi_k(d\theta),$$

so that $f(\theta)$ cannot be the risk function of any decision rule $\delta$. □

In the Gaussian mean problem, $R(\theta, \delta_{\mathrm{ml}}) = d$ is continuous and $R(\theta,\delta)$ is also continuous if $\delta$ is as good as $\delta_0$. Let $\Pi_\gamma$ be the $N(0, \gamma I_d)$ prior for $\theta$. As in Example 1.17,

$$R(\Pi_k, \delta_0) - R(\Pi_k, \delta^*_{\Pi_k}) = d/(1+k),\ \Pi_k(B) = (1+o(1))(2\pi k)^{-d/2}|B| \text{ as } k \to \infty,$$

where $|B|$ is the Lebesgue measure of $B$. It follows that $\delta_{\mathrm{ml}}$ is admissible for $d = 1$.

For $d = 2$, we shall take into account the invariance of the MLE under the orthonormal group. Consider invariant decision problems with probability family $\{\mathbb{P}_\theta, \theta \in \Theta\}$ on $\mathscr{X}$, loss $L(\theta, a)$ on $\Theta \times \mathscr{A}$, and transformation groups $G$ on $\mathscr{X}$, $\overline{G}$ on $\Theta$ and $G^*$ on $\mathscr{A}$.

**Lemma 1.3.** *Consider an invariant decision problem with a finite right-Haar measure $H_r$ on the group $G$. For decision rules $\delta$ define its equivariant version $\overline{\delta}$ by*

$$\overline{\delta}(x) = \textstyle\int_{\overline{G}} (h^{-1})^* \delta(hx) H_r(dh).$$

*Suppose $L(\theta, a)$ is convex in $a$. Let $\delta$ be a decision rule better than an equivariant decision rule $\delta_0$ with $R(\theta, \delta_0) > \int R(\overline{h}\theta, \delta)\} H_r(dh)$ for some $\theta \in \Theta$. Then, $\overline{\delta}$ is better than $\delta_0$.*

Proof. By the convexity of the loss function,

$$\textstyle\int_{\overline{G}} R(\overline{h}\theta, \delta) H_r(d\overline{h}) = \int_{\overline{G}} \mathbb{E}_\theta \big[ L(\theta, (h^{-1})^* \delta(hX)) \big] H_r(d\overline{h}) \geqslant R(\theta, \overline{\delta}).$$

The invariance of $\overline{\delta}$ follows from the invariance of $H_r$. □

**Example 1.19.** *In the Gaussian mean problem, a decision rule $\delta$ is equivariant under the orthonormal transformation group if $\delta(Ux) = U\delta(x)$ for all orthonormal matrices $U$, so that $\|\delta(x)\|_2$ is a function of $\|x\|_2$. Let $\Pi_r$ be the uniform prior on the $r$-sphere $\{\theta : \|\theta\|_2 = r\}$. By symmetry, the Bayes rule $\delta_r^*(x) = \mathbb{E}_{\Pi_r}[\theta | X = x]$ is in the direction of $x$. Let $\delta'(x) = (\|\delta(x)\|_2 / \|x\|_2) x$. As the risk function of $\delta$ is invariant in the $r$-sphere, for $\|\theta\|_2 = r$,*

$$R(\theta, \delta) = \mathbb{E}_{\Pi_r} \big[ \|\delta_r^*(X) - \theta\|_2^2 + \|\delta(X) - \delta_r^*(X)\|_2^2 \big] \geqslant R(\theta, \delta').$$

*Thus, $\delta_{\rm ml}(x) = x$ is inadmissible iff some $\delta$ of the following form is better than $\delta_{\rm ml}$:*

$$\delta(x) = (1 - B(\|x\|_2^2)) x. \tag{1.64}$$

*Lemma 1.3 applies here because $R(\theta, \delta) \leqslant d$ implies the continuity of $R(\theta, \delta)$. Let*

$$b(\theta) = (b_1(\theta), \ldots, b_d(\theta))^\top = \mathbb{E}_\theta \big[ \delta(X) \big] - \theta = -\textstyle\int x B(\|x\|_2^2) (2\pi)^{-d/2} e^{-\|x-\theta\|_2^2/2} dx \tag{1.65}$$

*be the bias of $\delta$. By the Cramér-Rao inequality,*

$$R(\theta, \delta) \geqslant \|b(\theta)\|_2^2 + \textstyle\sum_{j=1}^d \big\| e_j + \nabla b_j(\theta) \big\|_2^2 \geqslant \|b(\theta)\|_2^2 + d + 2 \sum_{j=1}^d (\partial / \partial \theta_j) b_j(\theta).$$

*Taking an infinite series expansion, we write $b(\theta) = -f(\|\theta\|_2^2)\theta$ for some differentiable $f(\theta)$. As $\sum_{j=1}^d (\partial/\partial\theta_j) b_j(\theta) = -df(\|\theta\|_2^2) - 2\|\theta\|_2^2 f'(\|\theta\|_2^2)$, when $\delta$ is as good as $\delta_{\rm ml}$*

$$d \geqslant t f^2(t) + d - 2df(t) - 4tf'(t).$$

*Let $g(t) = tf(t)$. For $d = 2$, the above inequality gives $1/(4t) \leqslant g'(t)/g^2(t)$, so that*

$$\textstyle\int_{t_1}^{t_2} (4t)^{-1} dt \leqslant \int_{t_1}^{t_2} \big( g'(t)/g^2(t) \big) dt = 1/g(t_1) - 1/g(t_2).$$

*Moreover $g(t)$ is nondecreasing in $t$. If $g(t_1) > 0$, then $\int_{t_1}^{t_2} t^{-1} dt \leqslant 4/g(t_1)$ but this is impossible as $t_2 \to \infty$. If $g(t_2) < 0$, then $\int_{t_1}^{t_2} t^{-1} dt \leqslant -4/g(t_2)$ but this is also impossible as $t_1 \to 0+$. Consequently, $g(t) = f(t) = b(t) = 0$. Due to the completeness of $X$, this gives $\delta = \delta_{\rm ml}$ for all $\delta$ as good as $\delta_{\rm ml}$. Hence, $\delta_{ml}(X) = X$ is admissible for $d = 2$. This method is used in* [Ste56] *to prove the admissibility of $\delta_{ml}$ for $d = 2$. While Blyth's method aims at more general estimators, its direct application only gives the admissibility of $\delta_{\rm ml}$ for $d = 1$.*

1.7. **Problems.**

**Exercise 1.1** ([LC98], page 66, 5.3)**.**

**Exercise 1.2** ([LC98], page 72, 6.33)**.**

**Exercise 1.3** ([LC98], page 129, 1.3)**.**

**Exercise 1.4** ([LC98], page 129, 1.4)**.**

**Exercise 1.5** ([LC98], page 130, 1.7)**.**

**Exercise 1.6** ([LC98], page 130, 1.9)**.**

**Exercise 1.7** ([LC98], page 131, 1.15)**.**

**Exercise 1.8** ([LC98], page 132, 2.2)**.**

**Exercise 1.9** ([LC98], page 132, 2.12)**.**

**Exercise 1.10** ([LC98], page 133, 2.21)**.**

**Exercise 1.11** ([LC98], page 133, 2.23)**.**

**Exercise 1.12** ([LC98], page 136, 3.20)**.**

**Exercise 1.13** ([LC98], page 137, 4.2)**.**

**Exercise 1.14** ([LC98], page 138, 5.7)**.**

**Exercise 1.15** ([LC98], page 139, 5.21)**.**

**Exercise 1.16** ([LC98], page 140, 5.25)**.**

**Exercise 1.17** ([LC98], page 140, 5.25)**.**

**Exercise 1.18.** *Consider the estimation of* $\tau(\theta) = \mathbb{E}_\theta[\delta(T(X))] \in \mathbb{R}$ *in an exponential family as in Example 1.3. Prove that* $\delta(x) - \tau(\theta) = a(\theta)^\top (T(x) - \dot{\psi}(\theta))$ *for some function* $a(\theta)$ *iff* $\dot{\tau}(\theta) = \ddot{\psi}(\theta)v$ *for some vector* $v$ *not depending on* $\theta$*.*

**Exercise 1.19.** *Write the functions* $\overline{h}_k(x_{1:k})$ *in* (1.18) *explicitly as linear combinations of conditional expectations of* $h_m^s(X_{1:m})$*.*

**Exercise 1.20.** *Verify the variance formula in* (1.19)

**Exercise 1.21.** *In Theorem 1.6 let* $\overline{h}_2(x_1, x_2) = \sum_{k=1}^\infty \lambda_k \phi_k(x_1)\phi_k(x_2)$ *be the eigenvalue decomposition of* $\overline{h}_2$ *with* $L_2(\mathbb{P})$ *orthonormal* $\phi_j(X_1)$*. Prove that for some iid* $Z_j \sim N(0,1)$*,*

$$2\int_0^1 \left( \int_0^{t_2} \overline{h}_2(Q(t_1), Q(t_2)) W(dt_1) \right) W(dt_2) \sim \sum_{j=1}^\infty \lambda_j (Z_j^2 - 1).$$

**Exercise 1.22.** *Verify the limiting variance formula* (1.21) *directly by Itô calculus.*

**Exercise 1.23.** *Prove that the Itô integrals on the right-hand side of* (1.20) *are uncorrelated even through they are of different order.*

**Exercise 1.24** ([LC98], page 207, 1.5).

**Exercise 1.25** ([LC98], page 209, 2.4).

**Exercise 1.26** ([LC98], page 209, 2.6).

**Exercise 1.27** ([LC98], page 209, 2.8).

**Exercise 1.28** ([LC98], page 209, 2.11).

**Exercise 1.29** ([LC98], page 210, 2.12).

**Exercise 1.30** ([LC98], page 210, 2.18).

**Exercise 1.31** ([LC98], page 211, 3.2).

**Exercise 1.32** ([LC98], page 211, 3.3).

**Exercise 1.33** ([LC98], page 212, 3.7).

**Exercise 1.34** ([LC98], page 212, 3.19).

**Exercise 1.35** ([LC98], page 212, 3.21).

**Exercise 1.36** ([LC98], page 212, 3.23).

**Exercise 1.37** ([LC98], page 212, 3.28).

**Exercise 1.38.** *Find the explicit formulas of* $\widehat{\tau}$ *as the* $(r/2)$*-power of quadratic functions of* $X$ *in Examples 1.4 and 1.5 in the case where* $p_n(x)$ *is the standard normal density in* $\mathbb{R}^n$.

**Exercise 1.39.** *Verity the formula for* $\widehat{\mu}$ *in Example 1.6 with explicit calculation.*

**Exercise 1.40.** *Verity the two displayed formulas in Example 1.7 with explicit calculation.*

**Exercise 1.41.** *Verity the formulas for* $w_1^*$ *and* $w_2^*$ *in Example 1.8 with explicit calculation.*

**Exercise 1.42** ([LC98], page 282, 1.6).

**Exercise 1.43** ([LC98], page 282, 1.7).

**Exercise 1.44** ([LC98], page 283, 2.1).

**Exercise 1.45** ([LC98], page 283, 2.2).

**Exercise 1.46** ([LC98], page 284, 2.5).

**Exercise 1.47** ([LC98], page 284, 2.11).

**Exercise 1.48** ([LC98], page 284, 2.15).

**Exercise 1.49.** *For some fixed large d, simulate the difference between* (4.56) *and* (4.57) *and show your results in plots and tables.*

**Exercise 1.50.** *Use the formula* $\int_0^1 f(t)W(dt) \sim N(0, \|f\|^2_{L_2[0,1]})$ *to derive the covariance of* $W^{(1-\alpha)}(u)$ *and* $W^{(1-\alpha)}(v)$ *as a function of* $(u,v)^\top \in [0,1]^2$ *in Example 1.12.*

**Exercise 1.51.** *Verify the formula* (1.48) *in Example 1.13 and provide a formula for* $\mathbb{E}[\theta(u)|X_{1:n}]$ *as a function of* $u \in \mathbb{S}^2$.

**Exercise 1.52.** *In Example 1.13, derive a formula for updating the posterior from* (1.48) *when an additional observation* $(u_{n+1}, X_{n+1})$ *becomes available.*

**Exercise 1.53.** *In Example 1.15, prove that* $(Z_k, \xi_k)^\top$ *are independent vectors in* (1.49).

**Exercise 1.54.** *In the binomial case* $\mathbb{P}\{X = x|\theta\} = \binom{n}{x}\theta^x(1-\theta)^{n-x}$, $x = 0, \ldots, n$. *Prove that the Bayes estimator* $\mathbb{E}_\Pi[\theta|X = x]$ *cannot be expressed in terms of the marginal probability mass function* $p_n(x; \Pi) = \mathbb{P}_\Pi\{X = x\}$*; For any function* $f : \mathbb{R}^{n+2} \to \mathbb{R}$,

$$\mathbb{E}_\Pi[\theta|X = x] \neq f\big(x, p_n(0; \Pi), \ldots, p_n(n; \Pi)\big)$$

*for some prior* $\Pi$ *in* $[0, 1]$.

**Exercise 1.55** ([LC98], page 286, 3.8)**.**

**Exercise 1.56** ([LC98], page 286, 3.10)**.**

**Exercise 1.57** ([LC98], page 287, 4.5 (a) (b))**.**

**Exercise 1.58** ([LC98], page 287, 4.8)**.**

**Exercise 1.59** ([LC98], page 287, 4.9)**.**

**Exercise 1.60** ([LC98], page 287, 4.10)**.**

**Exercise 1.61** ([LC98], page 288, 4.13)**.**

**Exercise 1.62** ([LC98], page 288, 5.2)**.**

**Exercise 1.63** ([LC98], page 289, 5.4 (a) (b))**.**

**Exercise 1.64** ([LC98], page 290, 5.8)**.**

**Exercise 1.65** ([LC98], page 291, 5.9)**.**

**Exercise 1.66** ([LC98], page 291, 5.10)**.**

**Exercise 1.67** ([LC98], page 292, 5.11)**.**

**Exercise 1.68** ([LC98], page 390, 1.3)**.**

**Exercise 1.69** ([LC98], page 390, 1.4)**.**

**Exercise 1.70** ([LC98], page 390, 1.5)**.**

**Exercise 1.71** ([LC98], page 391, 1.6)**.**

**Exercise 1.72** ([LC98], page 391, 1.7)**.**

**Exercise 1.73** ([LC98], page 391, 1.8)**.**

**Exercise 1.74** ([LC98], page 392, 1.19)**.**

**Exercise 1.75** ([LC98], page 392, 1.20)**.**

**Exercise 1.76** ([LC98], page 392, 1.23)**.**

**Exercise 1.77** ([LC98], page 392, 1.25)**.**

**Exercise 1.78** ([LC98], page 393, 2.4).

**Exercise 1.79** ([LC98], page 394, 2.5).

**Exercise 1.80** ([LC98], page 394, 2.7).

**Exercise 1.81** ([LC98], page 394, 2.9).

**Exercise 1.82** ([LC98], page 395, 2.10).

**Exercise 1.83** ([LC98], page 395, 2.14).

**Exercise 1.84** ([LC98], page 396, 2.20).

**Exercise 1.85** ([LC98], page 397, 2.24).

**Exercise 1.86** ([LC98], page 398, 3.9).

**Exercise 1.87** ([LC98], page 398, 3.10).

**Exercise 1.88** ([LC98], page 398, 3.15).

**Exercise 1.89.** *Provide a set of sufficient conditions on more general loss functions under which the conclusions of Example 1.18 still hold.*

**Exercise 1.90.** *Prove that the minimax estimator exists and is a Bayes estimator in the setting of Example 1.18.*

**Exercise 1.91.** *In* (1.65)*, the bias is written as* $b(\theta) = -f(\|\theta\|_2^2)\theta$*. Given an infinite series formula for the function* $f(t)$*.*

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## 2. Asymptotic Theory in Point Estimation

2.1. **M- and Z-estimations.** Suppose we are interested in estimating a parameter $\theta$ in a certain parameter space $\Theta$ based on observations $X_1, \ldots, X_n$. A popular method of finding an estimator $\widehat{\theta}_n = \widehat{\theta}_n(X_1, \ldots, X_n)$, called M-estimation, is

$$\widehat{\theta}_n = \mathop{\arg\max}_{\theta \in \Theta} M_n(\theta) \quad \text{with} \quad M_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} m_\theta(X_i), \tag{2.1}$$

where $m_\theta = m_\theta(\cdot)$ are known measurable functions indexed by the parameter $\theta$. In the classical setting, $\Theta$ is fixed. However, $\Theta$ is allowed to depend on $n$ in general. Formally, (2.1) is defined as the global maximizer over $\Theta$, but one may also consider local solutions.

Often, (2.1) can be solved by setting the derivative of $M_n(\theta)$ to zero. This leads to Z-estimators (for zero) of the form

$$\Psi_n(\widehat{\theta}_n) = 0 \quad \text{with} \quad \Psi_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} \psi_\theta(X_i), \tag{2.2}$$

where $\psi_\theta = \psi_\theta(\cdot)$ are known measurable functions indexed by the parameter $\theta$. For (2.1) with differentiable $m_\theta$, $\psi_\theta = (\partial/\partial\theta)m_\theta$ gives the corresponding Z-estimator. However, $\psi_\theta$ is not required to be of the form $(\partial/\partial\theta)m_\theta$ in (2.2). When $\Theta \subseteq \mathbb{R}^d$ and $\theta = (\theta_1, \ldots, \theta_d)^\top$ is an interior point of $\Theta$, $\psi_\theta = (\psi_{\theta,1}, \ldots, \psi_{\theta,d})^\top$ with $\psi_{\theta,j} = (\partial/\partial\theta_j)m_\theta$.

**Remark 2.1.** *The notion of derivative can be extended to infinite-dimensional spaces. Among various extensions, the Fréchet derivative is most useful. Let $(\mathscr{B}_j, \|\cdot\|_{(j)})$ be Banach spaces, $\Theta \subset \mathscr{B}_1$ and $\theta_0$ an interior point of $\Theta$, the Fréchet derivative of a mapping $\theta \to f_\theta$ from $\Theta$ to $\mathscr{B}_2$ is defined as a linear operator $J_{\theta_0}$ from $\mathscr{B}_1$ to $\mathscr{B}_2$ such that*

$$\|f_{\theta_0+h} - f_{\theta_0} - J_{\theta_0}h\|_{(2)} = o\big(\|h\|_{(1)}\big).$$

*The Gâteaux differentiation is a weaker notion defined by*

$$f_{\theta_0+th} - f_{\theta_0} - t\dot{f}_{\theta_0,h} = o(t)$$

*where the derivative $\dot{f}_{\theta_0,h}$ is allowed to depend on the direction $h$ and $o(1)$ can be defined in a weaker topology in $\mathscr{B}_2$ (e.g. in terms of a dense set of linear functionals in the dual of $\mathscr{B}_2$). For example, the mapping $f_\theta = \theta/\|\theta\|$ in Banach space $(\mathscr{B}, \|\cdot\|)$ is Gâteaux differentiable but not Fréchet differentiable at $\theta_0 = 0$.*

**Example 2.1** (MLE)**.** *Suppose $X_i$ are iid observations from a population with density $p_\theta(x)$. The maximum likelihood estimator (MLE) is an M-estimator with*

$$m_\theta(x) = \log p_\theta(x).$$

*When $p_\theta$ belongs to an exponential family with $\log p_\theta(x) = \theta^\top t(x) - \psi(\theta)$, $M_n(\theta) = \theta^\top \bar{t}_n - \psi(\theta)$ is concave in $\theta \in \mathbb{R}^d$ so that the MLE is unique when $\Theta$ is convex, where $\bar{t}_n = n^{-1}\sum_{i=1}^n t(X_i)$.*

**Example 2.2** (GLM)**.** *In the generalized linear model (GLM), $X_i = \{y_i, z_i\}$ with response $y_i \in \mathbb{R}$ and covariate $z_i \in \mathbb{R}^d$, and the MLE of the regression coefficient vector $\theta$ maximizes*

$$M_n(\theta) = \frac{1}{n}\sum_{i=1}^n \Big(\theta^\top z_i y_i - \psi\big(\theta^\top z_i\big)\Big).$$

*The MLE is also given by (2.2) with $\psi_\theta(x) = z(y - \dot{\psi}(\theta^\top z))$ where $\dot{\psi}(t) = (d/dt)\psi(t)$.*

**Example 2.3** (Huber's robust estimator)**.** *Let $X_i = \{y_i, z_i\}$ with response $y_i \in \mathbb{R}$ and covariate $z_i \in \mathbb{R}^d$. Huber's robust estimator of the regression coefficient vector $\theta$ is*

$$\widehat{\theta}_n = \arg\max_\theta \sum_{i=1}^n m(y_i - z_i^\top \theta),$$

*where $m(x) = -\int_0^{|x|}\min(t,a)dt$ with a constant $a > 0$. This is also concave maximization. When $d = 1$ and $z_i = 1$ for all $i$, the above $\widehat{\theta}_n$ is Huber's robust location estimator.*

**Example 2.4** (Quantile regression)**.** *Let $X_i = \{y_i, z_i\}$ with response $y_i \in \mathbb{R}$ and covariate $z_i \in \mathbb{R}^d$. In the $q$-quantile regression, $0 < q < 1$, the coefficient vector $\theta$ is estimated by*

$$\widehat{\theta}_n = \arg\max_\theta \sum_{i=1}^n m(y_i - z_i^\top \theta),$$

*where $m(x) = -(1-q)x_- - qx_+$. This is again concave maximization. When $d = 1$ and $z_i = 1$ for all $i$, $\widehat{\theta}_n$ is the $q$-quantile of the sample $y_1, \ldots, y_n$. We note that for continuous distribution functions $F(x)$, $(d/d\theta)\int m(x-\theta)F(dx) = -(1-q)F(\theta) + q(1-F(\theta)) = 0$ if and only if (iff) $F(\theta) = q$.*

**Example 2.5** (Kernel density estimation)**.** *Let $X_i$ be iid random vectors from a population with joint density function $\theta(x)$. The kernel density estimator is a Z-estimator given by*

$$\widehat{\theta}_n(x) = \frac{1}{n}\sum_{i=1}^n K(X_i, x) \quad \forall\ x$$

*with a certain kernel $K(y,x)$. Typically, $K(y,x) = h^{-1}K((x-y)/h)$ for some function $K(\cdot)$ and bandwidth $h = h_n \to 0+$.*

2.2. **Consistency and convergence rates.** Let $\theta_0$ be the true parameter under probability $\mathbb{P}$. Let $d(\widehat{\theta}_n, \theta_0)$ be the loss of using estimator $\widehat{\theta}_n$ where $d(\cdot,\cdot)$ is a mapping from $\Theta \times \Theta$ to $\overline{\mathbb{R}}_+ = [0,\infty]$. The estimator $\widehat{\theta}_n$ is consistent if $d(\widehat{\theta}_n, \theta_0) = o_{\mathbb{P}}(1)$, $\widehat{\theta}_n$ has convergence rate $\epsilon_n > 0$ if $d(\widehat{\theta}_n, \theta_0)/\epsilon_n = O_{\mathbb{P}}(1)$, and $\widehat{\theta}_n$ is $\epsilon_n$-consistent if it has convergence rate $\epsilon_n$ with $\epsilon_n \to 0+$. We note that $\widehat{\theta}_n$ is consistent iff $\widehat{\theta}_n$ is $\epsilon_n$-consistent for some $\epsilon_n \to 0+$. Typically, $d(\cdot,\cdot)$ is a (semi-)metric on $\Theta$.

Throughout the sequel, we denote by $\overline{\mathbb{P}}_n$ the empirical measure $\overline{\mathbb{P}}_n[f] = n^{-1}\sum_{i=1}^n f(X_i)$ and by $\mathbb{P}[f]$ the expectation of $\overline{\mathbb{P}}_n[f]$ for functions $f(x)$. Here $\mathbb{P}$ is allowed to depend on $n$, e.g. when $X_i$ are not identically distributed. We assume throughout any $f$ involved with $\overline{\mathbb{P}}_n[f]$ is measurable, and any $f$ involved with $\mathbb{P}[f]$ has finite $\mathbb{P}[f_+]$ or finite $\mathbb{P}[f_-]$ so that $\mathbb{P}[f]$ is well defined. We use $\mathbb{P}\{A\}$ for the probability of event $A$ and $\mathbb{E}[Z]$ for the expectation variable $Z$ under probability $\mathbb{P}$. Thus, $\mathbb{P}[f] = \mathbb{E}\big[\mathbb{P}_n[f]\big]$. In addition, we denote by $\|\cdot\|$ a general norm, by $\|\cdot\|_p$ the $\ell_p$ norm for vectors with usual extensions to $p=\infty$ and $p \in [0,1)$, and by $\|\cdot\|_{\mathrm{S}}$, $\|\cdot\|_{\mathrm{F}}$ and $\|\cdot\|_{\mathrm{N}}$ respectively the spectrum, Frobenius and nuclear norms for matrices. A function $f(x)$ is upper-semicontinuous at $x_0$ if and only if the upper limit of $f(x)$ no greater than $f(x_0)$ as $x \to x_0$.

**Theorem 2.1.** *Let $M_n(\theta)$ be as in (2.1) and $d(\cdot,\cdot)$ a loss function. For $\epsilon > 0$ let*

$$P_n(\epsilon) = \mathbb{P}\Big\{M_n(\theta) \geqslant M_n(\theta_0) \text{ for some } \theta \in \Theta \text{ with } d(\theta,\theta_0) > \epsilon\Big\}, \quad P_n(\infty) = 0. \tag{2.3}$$

*(i) The M-estimator $\widehat{\theta}_n$ in (2.1) satisfies $\mathbb{P}\{d(\widehat{\theta}_n,\theta_0) > \epsilon\} \leqslant P_n(\epsilon)$. Thus, $\widehat{\theta}_n$ is consistent when $P_n(\epsilon) = o(1)\ \forall\ \epsilon > 0$ and $\widehat{\theta}_n$ has convergence rate $\epsilon_n$ when $P_n(\epsilon'_n) = o(1)\ \forall\ \epsilon'_n \gg \epsilon_n$.*
*(ii) (Wald consistency theorem:) Suppose $X_i$ are iid, $\Theta$ and $m_\theta$ are fixed, $d(\cdot,\cdot)$ is a metric under which $\Theta$ is compact, $\mathbb{P}[\sup_\theta m_\theta] < \infty$, and $m_\theta(X_1)$ is upper-semicontinuous in $\theta$ with probability 1. Let $M(\theta) = \mathbb{P}[m_\theta]$, $M^* = \max_{\theta\in\Theta} M(\theta)$, $\Theta_0 = \{\theta : M(\theta) = M^*\}$ and $d(\theta,\Theta_0) = \inf_{\theta_0\in\Theta_0} d(\theta,\theta_0)$. Then, the M-estimator $\widehat{\theta}_n$ is consistent in the sense of*

$$d(\widehat{\theta}_n, \Theta_0) = o_{\mathbb{P}}(1).$$

*(iii) For any deterministic $M(\theta)$ and $S_{\epsilon,j} \subset \Theta$ satisfying $\cup_{j\in J} S_{\epsilon,j} \supseteq \{\theta : d(\theta,\theta_0) > \epsilon\}$,*

$$P_n(\epsilon) \leqslant \sum_{j\in J} \mathbb{P}\Big\{\sup_{\theta\in S_{\epsilon,j}} \big(M_n(\theta) - M_n(\theta_0)\big) \geqslant 0\Big\} \leqslant \sum_{j\in J}\mathbb{P}\Big\{\sup_{\theta\in S_{\epsilon,j}} \big(\Delta_n(\theta)\big)_+ \geqslant \kappa_{\epsilon,j}\Big\}, \tag{2.4}$$

*where $\kappa_{\epsilon,j} = \inf_{\theta\in S_{\epsilon,j}}\big(M(\theta_0) - M(\theta)\big)_+$ and $\Delta_n(\theta) = M_n(\theta) - M_n(\theta_0) - M(\theta) + M(\theta_0)$. In particular, for $\epsilon_n = n^{-1/(4-2\beta)}$ with $\beta < 2$ and $S_{\epsilon_n,j} = \{2^{j-1}\epsilon_n < d(\theta,\theta_0) \leqslant 2^j\epsilon_n\}$,*

$$\mathbb{P}\big\{d(\widehat{\theta}_n,\theta_0) > 2^{j_0-1}\epsilon_n\big\} \leqslant 2^{-j_0(2-\beta)}(4C/c_*)/(1-2^{\beta-2}), \tag{2.5}$$

*when $\kappa_{\epsilon_n,j} \geqslant c_*(2^{j-1}\epsilon_n)^2$ and $\mathbb{E}\big[\sup_{\theta\in S_{\epsilon,j}}\big(\Delta_n(\theta)\big)_+\big] \leqslant C(2^j\epsilon_n)^\beta/n^{1/2}$ for all $j \geqslant j_0$.*
*(iv) (Concave case:) When $m_\theta$ are concave in $\theta$ and $d(\theta,\theta_0) = \|\theta - \theta_0\|$ for some norm, the condition on the compactness of $\Theta$ in Part (ii) can be replaced by the compactness of $\{\theta : d(\theta,\theta_0) = \epsilon\}$ for all $\epsilon \leqslant \epsilon_0$, and (2.4) becomes*

$$P_n(\epsilon) \leqslant \mathbb{P}\Big\{\sup_{\|\theta-\theta_0\|=\epsilon}\big(\Delta_n(\theta)\big)_+ \geqslant \inf_{\|\theta-\theta_0\|=\epsilon}\big(M(\theta_0) - M(\theta)\big)_+\Big\}.$$

PROOF. (i) In the event in (2.3), we have $d(\widehat{\theta}_n, \theta_0) \leqslant \epsilon$, so that the conclusions follow from the definition of $d(\widehat{\theta}_n, \theta_0) = o_{\mathbb{P}}(1)$ and the fact that $d(\widehat{\theta}_n, \theta_0)/\epsilon_n = O_{\mathbb{P}}(1)$ iff

$$\lim_{M\to\infty} \overline{\lim}_{n\to\infty} \mathbb{P}\Big\{d(\theta_n, \theta_0) \geqslant M\epsilon_n\Big\} = 0.$$

(ii) Assume $M^* > -\infty$ as $\Theta_0 = \Theta$ otherwise. Let $m_{\theta,\epsilon'}(x) = \sup_{d(\theta',\theta)<\epsilon'} m_{\theta'}(x)$. We have $\mathbb{P}\big\{\lim_{\epsilon'\to 0+} m_{\theta,\epsilon'}(X_j) = m_\theta(X_j)\big\} = 1$ by the upper-semicontinuity of $m_\theta$, so that by the monotone convergence theorem $\mathbb{P}\big[m_{\theta,\epsilon'}\big] \to \mathbb{P}\big[m_\theta\big] = M(\theta)$ as $\epsilon' \to 0+$. Thus, $\Theta_0$ is closed and nonempty. Let $\Theta_\epsilon = \{\theta \in \Theta : d(\theta, \Theta_0) < \epsilon\}$. For each $\theta \in \Theta_\epsilon^c$, $M(\theta) < M^*$, so that $\mathbb{P}\big[m_{\theta,\epsilon_\theta}\big] < M^*$ for some $\epsilon_\theta > 0$. As $\Theta_\epsilon^c$ is compact, it is covered by finitely many of the balls $B(\theta, \epsilon_\theta) = \{\theta' : d(\theta', \theta) < \epsilon_\theta\}$, say at $\theta_1, \ldots, \theta_J$. Consequently, by the LLN

$$\sup_{d(\theta,\Theta_0)>\epsilon} \big(M_n(\theta) - M_n(\theta_0)\big) \leqslant \max_{1\leqslant j\leqslant J} \overline{\mathbb{P}}_n\big[m_{\theta_j,\epsilon_{\theta_j}} - m_{\theta_0}\big] = o_{\mathbb{P}}(1) + \max_{1\leqslant j\leqslant J} \mathbb{P}\big[m_{\theta_j,\epsilon_{\theta_j}}\big] - \mathbb{P}\big[m_{\theta_0}\big]$$

where $\theta_0$ is an arbitrary member of $\Theta_0$. Thus, $P_n(\epsilon) = o(1)$ as $\mathbb{P}\big[m_{\theta_j,\epsilon_{\theta_j}}\big] < M^*$.

(iii) When $\sup_{\theta\in S_{\epsilon,j}} \big(\Delta_n(\theta)\big)_+ < \kappa_{\epsilon,j}$ and $\theta \in S_{\epsilon,j}$, we have $M(\theta_0) - M(\theta) \geqslant \kappa_{\epsilon,j} > 0$ and $M_n(\theta) - M_n(\theta_0) = \Delta_n(\theta) + M(\theta) - M(\theta_0) \leqslant \big(\Delta_n(\theta)\big)_+ - \kappa_{\epsilon,j} < 0$. Thus, (2.4) holds and under the conditions for (2.5),

$$P_n\big(2^{j_0-1}\epsilon_n\big) \leqslant \sum_{j=j_0}^{\infty} \frac{\mathbb{E}\big[\sup_{\theta\in S_{\epsilon,j}} \big(\Delta_n(\theta)\big)_+\big]}{\kappa_{\epsilon_n,j}} \leqslant \sum_{j=j_0}^{\infty} \frac{C(2^j\epsilon_n)^\beta/n^{1/2}}{c_*(2^{j-1}\epsilon_n)^2} = \sum_{j=j_0}^{\infty} \frac{4C}{c_* 2^{j(2-\beta)}}.$$

(iv) When $M_n(\theta)$ is concave in $\Theta$ and $d(\theta, \theta_0) = \|\theta - \theta_0\|$,

$$P_n(\epsilon) = \mathbb{P}\bigg\{\sup_{d(\theta,\theta_0)=\epsilon} \big(M_n(\theta) - M_n(\theta_0)\big) \geqslant 0\bigg\},$$

so that we may replace $\Theta_\epsilon^c$ by $\{\theta : d(\theta, \theta_0) = \epsilon\}$ in the above proof. $\square$

Next we consider Z-estimators.

**Theorem 2.2.** *Let $\widehat{\theta}_n$ be the Z-estimator in (2.2) and $d(\cdot,\cdot)$ a loss function. Suppose with $\mathbb{P}$ probability $1 + o(1)$, $\widehat{\theta}_n \in \Theta$ exists and $\|\Psi_n(\theta)\|$ is well defined for all $\theta \in \Theta$ and a certain norm $\|\cdot\|$. Let $\Psi(\theta)$ be deterministic with finite $\|\Psi(\theta)\|$ for all $\theta \in \Theta$. For $\epsilon > 0$, let*

$$P_n(\epsilon) = \mathbb{P}\bigg\{\big\|\Psi_n(\theta)\big\| = 0 \ \text{ for some } \theta \in \Theta \text{ with } d(\theta, \theta_0) > \epsilon\bigg\}.$$

*Then, $\widehat{\theta}_n$ is consistent when $P_n(\epsilon) = o(1)$ for all fixed $\epsilon > 0$ and $\widehat{\theta}_n$ has convergence rate $\epsilon_n$ when $P_n(\epsilon_n') = o(1)$ for all $\epsilon_n' \gg \epsilon_n$. Moreover, for $\Delta_n'(\theta) = \Psi_n(\theta) - \Psi(\theta)$ and any $S_{\epsilon,j} \subset \Theta$, $j \in J$, satisfying $\cup_{j\in J} S_{\epsilon,j} \supseteq \{\theta : d(\theta, \theta_0) > \epsilon\}$,*

$$P_n(\epsilon) \leqslant \sum_{j\in J} \mathbb{P}\bigg\{\sup_{\theta\in S_{\epsilon,j}} \big\|\Delta_n'(\theta)\big\| \geqslant \inf_{\theta\in S_{\epsilon,j}} \big\|\Psi(\theta)\big\|\bigg\}. \tag{2.6}$$

PROOF. In the complement of the event in (2.6), we have $\|\Psi_n(\theta)\| \geqslant \|\Psi(\theta)\| - \|\Delta_n'(\theta)\| > 0$ for all $\theta \in S_{\epsilon,j}$. $\square$

Both Theorems 2.1 and 2.2 apply to generic random functions $M_n(\theta)$ and $\Psi_n(\theta)$ instead of the averages in (2.1) and (2.2) respectively. When $X_i$ are independent and $M_n(\theta)$ and $\Psi_n(\theta)$ are the averages in (2.1) and (2.2) with respective expectations $M(\theta) = \mathbb{P}[m_\theta]$ and $\Psi(\theta) = \mathbb{P}[\psi_\theta]$, conditions (2.4) and (2.6) can be verified with the Glivenko-Cantelli theorem. For example, with $S_{\epsilon,1} = \{\theta : d(\theta,\theta_0) > \epsilon\}$ in general and $S_{\epsilon,1} = \{\theta : d(\theta,\theta_0) = \epsilon\}$ in the concave case, the M-estimator is consistent by (2.4) when $\inf_{\theta\in S_{\epsilon,1}} \big(M(\theta_0) - M(\theta)\big)_+ \geqslant \kappa_\epsilon$ for some $\kappa_\epsilon > 0$ depending on $\epsilon$ only and $\sup_{\theta\in S_{\epsilon,1}} \big((\overline{\mathbb{P}}_n - \mathbb{P})[m_\theta - m_{\theta_0}]\big)_+ = o_{\mathbb{P}}(1)$ for all $\epsilon > 0$. For general $\Theta$, the peeling argument in (2.4) and (2.6) can be applied to localize the analysis. Here we present an application of (2.5) under a Lipschitz condition.

**Corollary 2.1.** *Suppose $\Theta \subseteq \mathbb{R}^d$. Let $M(\theta) = \mathbb{P}[m_\theta^+]$ for some $m_\theta^+$ satisfying $m_\theta^+(x) \geqslant m_\theta(x)$ and $m_{\theta_0}^+ = m_{\theta_0}$. Suppose $M(\theta_0) - M(\theta) \geqslant c_*\|\theta-\theta_0\|_2^2$ when $\|\theta-\theta_0\|_2 \leqslant c_n$ for some constants $c_* > 0$ and $0 < c_n \leqslant \infty$. Suppose that for some $\psi = \psi(x)$ with $\mathbb{P}[\psi^2] < \infty$ and $\mu_\theta$ with $(\overline{\mathbb{P}}_n - \mathbb{P})[\mu_\theta] = 0$ (i.e. $\overline{\mathbb{P}}_n[\mu_\theta]$ deterministic under $\mathbb{P}$ as in regression models),*

$$|(m_{\theta_1}^+(x) - \mu_{\theta_1}(x)) - (m_{\theta_2}^+(x) - \mu_{\theta_2}(x))| \leqslant \psi(x)\|\theta_1-\theta_2\|_2 \tag{2.7}$$

*for all $\theta_1$ and $\theta_2$ with $\|\theta_1-\theta_0\|_2 \vee \|\theta_2-\theta_0\|_2 \leqslant c_n$. Then, the M-estimator* (2.1) *satisfies*

$$\mathbb{P}\big\{C\sqrt{d/n} < \|\widehat{\theta}_n - \theta_0\|_2 \leqslant c_n\big\} \leqslant 8\sqrt{2\pi}\big(\mathbb{P}[\psi^2]\big)^{1/2}/(c_*C), \quad \forall\ C > 0.$$

The natural choice of $m_\theta^+$ is $m_\theta$; Otherwise the choice of $m_\theta^+$ has to depend on $\theta_0$ so that $m_\theta^+$ can be only used as a vehicle to analyze the M-estimator. Here $m_\theta^+$ can be centered by functions $\mu_\theta$ of $x$ (e.g. $\mu_\theta = 0$), as long as $\mu_\theta(X_i)$ are deterministic under $\mathbb{P}$. In regression models, $\mu_\theta$ may depend on $x$ only through covariates when the covariates are treated as deterministic or $\mathbb{P}$ is treated as the conditional probability given covariates.

Proof. Let $\epsilon_n = C\sqrt{d/n}$ and $\widetilde{m}_\theta = m_\theta^+ - \mu_\theta$. As in the proof of (2.5),

$$\mathbb{P}\big\{2^{j-1}\epsilon_n < \|\widehat{\theta}_n - \theta_0\|_2 \leqslant (2^j\epsilon_n)\wedge c_n\big\} \leqslant \frac{\mathbb{E}\big[\sup_{\theta\in S_{n,j}} \big((\overline{\mathbb{P}}_n - \mathbb{P})\big[\widetilde{m}_\theta - \widetilde{m}_{\theta_0}\big]\big)_+\big]}{c_*(2^{j-1}\epsilon_n)^2}$$

where $S_{n,j} = \{\theta : 2^{j-1}\epsilon_n < \|\theta-\theta_0\|_2 \leqslant (2^j\epsilon_n)\wedge c_n\}$. Applying the Gaussian symmetrization (Lemma 2.1), we have

$$\mathbb{E}\bigg[\sup_{\theta\in S_{n,j}} \Big|(\overline{\mathbb{P}}_n - \mathbb{P})\big[\widetilde{m}_\theta - \widetilde{m}_{\theta_0}\big]\Big|\bigg] \leqslant (2\pi/n)^{1/2}\,\mathbb{E}\bigg[\big(\overline{\mathbb{P}}_n\big[\psi^2\big]\big)^{1/2} \sup_{\theta\in S_{n,j}} \big|G_\theta\big|\bigg],$$

where $G_\theta = \big(n\overline{\mathbb{P}}_n\big[\psi^2\big]\big)^{-1/2}\sum_{i=1}^n g_i\big(\widetilde{m}_\theta(X_i) - \widetilde{m}_{\theta_0}(X_i)\big)$ with a standard Gaussian sequence $\{g_i\}$ independent of $\{X_i\}$. Let $\xi \in \mathbb{R}^d$ be a standard Gaussian vector and $Y_\theta = (\theta-\theta_0)^\top\xi$. By the Lipschitz condition

$$\mathbb{E}\big[(G_{\theta_1} - G_{\theta_2})^2\big|\{X_i\}\big] = \overline{\mathbb{P}}_n\big[(\widetilde{m}_{\theta_1} - \widetilde{m}_{\theta_2})^2/\overline{\mathbb{P}}_n\big[\psi^2\big]\big] \leqslant \|\theta_1-\theta_2\|_2^2 = \mathbb{E}\big[(Y_{\theta_1} - Y_{\theta_2})^2\big],$$

so that by the Sudakov-Fernique inequality (2.24)

$$\mathbb{E}\bigg[\sup_{\theta\in S_{n,j}} \big|G_\theta\big|\bigg|\{X_i\}\bigg] \leqslant 2\mathbb{E}\bigg[\sup_{\theta\in S_{n,j}} Y_\theta\bigg] \leqslant 2\sup_{\theta\in S_{n,j}} \|\theta\|_2\Big(\mathbb{E}\big[\|\xi\|_2^2\big]\Big)^{1/2} \leqslant 2^{j+1}\epsilon_n\sqrt{d}.$$

Hence, $\mathbb{P}\big\{\epsilon_n < \|\widehat{\theta}_n - \theta_0\|_2 \leqslant c_n\big\} \leqslant \sum_{j=1}^\infty \big(c_*(2^{j-1}\epsilon_n)^2\big)^{-1}(2\pi/n)^{1/2}\big(\mathbb{P}[\psi^2]\big)^{1/2}2^{j+1}\epsilon_n\sqrt{d}.$ $\square$

**Example 2.2** (GLM, continuation). *In the GLM, $X_i = \{y_i, z_i\}$ with response $y_i \in \mathbb{R}$ and covariate $z_i \in \mathbb{R}^d$, and the MLE is given by (2.1) with $m_\theta = \theta^\top zy - \psi(\theta^\top z)$. Let $z_i$ be treated as deterministic, or $\mathbb{P}$ treated as the conditional probability given $\{z_i\}$. Let $\dot\psi(t) = (d/dt)\psi(t)$, $\ddot\psi(t) = (d/dt)\dot\psi(t)$ and $\dot m_\theta(x) = z\big(y - \dot\psi(\theta^\top z)\big)$. Let $M(\theta) = \mathbb{P}[m_\theta]$ and*

$$V_\theta = \overline{\mathbb{P}}_n\big[\ddot\psi(\theta^\top z)zz^\top\big] = \mathbb{P}\big[\ddot\psi(\theta^\top z)zz^\top\big]$$

*as its negative Hessian. As $\ddot\psi(t) \geqslant 0$, $V_\theta$ is nonnegative-definite. In Theorem 2.1 (iv),*

$$\begin{aligned}
\Delta_n(\theta) &= (\overline{\mathbb{P}}_n - \mathbb{P})[m_\theta - m_{\theta_0}] = (\theta-\theta_0)^\top\overline{\mathbb{P}}_n\big[\dot m_{\theta_0}\big],\\
M(\theta_0) - M(\theta) &= (\theta-\theta_0)^\top\bigg(\int_0^1 V_{\theta_0+t(\theta-\theta_0)}(1-t)dt\bigg)(\theta-\theta_0).
\end{aligned}$$

*Suppose for some fixed $c_* > 0$*

$$M(\theta_0) - M(\theta) \geqslant c_*\|\theta-\theta_0\|_2^2 \quad \forall\ \|\theta-\theta_0\|_2 \leqslant c_n \tag{2.8}$$

*with $c_n \gg \sqrt{d/n}$. Let $\epsilon_n = C\sqrt{d/n}$. As $\mathbb{P}[\dot m_{\theta_0}] = 0$, Theorem 2.1 (iv) yields*

$$\mathbb{P}\big\{\|\widehat\theta_n - \theta\|_2 \geqslant \epsilon_n\big\} \leqslant \frac{\mathbb{E}[\sup_{\|\theta-\theta_0\|_2=\epsilon_n}|\Delta_n(\theta)|^2]}{c_*^2\epsilon_n^4} = \frac{\mathbb{E}\big[\big\|\overline{\mathbb{P}}_n[\dot m_{\theta_0}]\big\|_2^2\big]}{c_*^2\epsilon_n^2} = \frac{\mathbb{P}[\|\dot m_{\theta_0}\|_2^2]}{c_*^2 Cd}.$$

*Thus, $\widehat\theta_n$ is $\sqrt{d/n}$-consistent when $\mathbb{P}[\|\dot m_{\theta_0}\|_2^2]/d = O(1)$. The MLE is also given as the Z-estimator* (2.2) *with $\psi_\theta(x) = \dot m_\theta$, $\Delta_n'(\theta) = (\overline{\mathbb{P}}_n - \mathbb{P})[\psi_\theta] = \overline{\mathbb{P}}_n[\dot m_{\theta_0}]$ and*

$$\Psi(\theta) = \overline{\mathbb{P}}_n\big[z\big(\dot\psi(\theta_0^\top z) - \dot\psi(\theta^\top z)\big)\big] = -\bigg(\int_0^1 V_{\theta_0+t(\theta-\theta_0)}dt\bigg)(\theta-\theta_0).$$

*Hence, Theorem 2.2 also yields the $\sqrt{d/n}$ rate under the key condition* (2.8).

2.3. **Asymptotic normality.**

**Theorem 2.3.** *Suppose* $\Theta \subseteq \mathbb{R}^d$*, the M-estimator* $\widehat{\theta}_n$ *in* (2.1) *is* $\epsilon_n$*-consistent for some* $\epsilon_n \geqslant \sqrt{d/n}$ *under the* $\ell_2$ *loss* $d(\theta,\theta_0) = \|\theta-\theta_0\|_2$*, and* $\{\theta : \|\theta-\theta_0\|_2 \leqslant \epsilon_n'\} \subset \Theta$ *for some* $\epsilon_n' \gg \epsilon_n$*. Let* $V_{\theta_0} = V_{\theta_0}^\top \in \mathbb{R}^{d\times d}$ *with* $c_*\|u\|_2^2 \leqslant u^\top V_{\theta_0} u \leqslant c^*\|u\|_2^2$ *for all* $u \in \mathbb{R}^d$ *and fixed positive constants* $c_*$ *and* $c^*$*. Let* $m_\theta^+ \geqslant m_\theta$ *with* $m_{\theta_0}^+ = m_{\theta_0}$*. Suppose for all* $C \geqslant 1$

$$\sup_{\|\theta-\theta_0\|_2 \leqslant C\epsilon_n} \left| \frac{\overline{\mathbb{P}}_n[m_\theta^+ - m_{\theta_0} - (\theta-\theta_0)^\top \dot{m}_{\theta_0}] + (\theta-\theta_0)^\top V_{\theta_0}(\theta-\theta_0)/2}{d/n + \|\theta-\theta_0\|_2^2} \right| = o_{\mathbb{P}}(1) \tag{2.9}$$

*for some* $\dot{m}_{\theta_0}(x) \in \mathbb{R}^d$ *with* $\mathbb{P}[\dot{m}_{\theta_0}] = 0$ *and* $\mathbb{P}\big[\|\dot{m}_{\theta_0}\|_2^2\big] = O(d)$*. Suppose random* $\widetilde{\theta}_n$ *exists satisfying* $\|\widetilde{\theta}_n - \theta_0 - V_{\theta_0}^{-1}\overline{\mathbb{P}}_n[\dot{m}_{\theta_0}]\|_2 = o_{\mathbb{P}}\big(\sqrt{d/n}\big)$ *and* $(\overline{\mathbb{P}}_n[m_\theta^+ - m_\theta])\big|_{\theta=\widetilde{\theta}_n} = o_{\mathbb{P}}(d/n)$*. Then,*

$$\big\|\widehat{\theta}_n - \theta_0 - V_{\theta_0}^{-1}\overline{\mathbb{P}}_n[\dot{m}_{\theta_0}]\big\|_2 = o_{\mathbb{P}}\big(\sqrt{d/n}\big). \tag{2.10}$$

*Consequently, for fixed* $d$ *and independent* $\dot{m}_{\theta_0}(X_i)$ *satisfying the Lindeberg condition,*

$$\mathscr{L}\big(\sqrt{n}(\widehat{\theta}_n - \theta_0);\mathbb{P}\big) = N\Big(0, V_{\theta_0}^{-1}\mathbb{P}\big[\dot{m}_{\theta_0}\dot{m}_{\theta_0}^\top\big]V_{\theta_0}^{-1}\Big) + o(1).$$

Here $\mathscr{L}(Z;\mathbb{P})$ is the law of $Z$ under $\mathbb{P}$, and $\mathscr{L}(Z_n;\mathbb{P}) = \mathscr{L}(Z_n';\mathbb{P}) + o(1)$ means $\mathbb{E}[f(Z_n)] = \mathbb{E}[f(Z_n')] + o(1)$ for all bounded continuous functions $f$. We also use $Z_n \rightsquigarrow_{\mathbb{P}_n} Z$, or $\mathscr{L}(Z_n;\mathbb{P}_n) \rightsquigarrow \mathscr{L}(Z;\mathbb{P})$ to denote convergence in distribution to a fixed $\mathscr{L}(Z;\mathbb{P})$, and $Z_n \rightsquigarrow Z$ to denote convergence in distribution under fixed $\mathbb{P}$.

Let $M(\theta) = \mathbb{P}[m_\theta^+]$. The numerator of the ratio in (2.9) can be written as

$$\underbrace{(\overline{\mathbb{P}}_n - \mathbb{P})[m_\theta^+ - m_{\theta_0} - (\theta-\theta_0)^\top\dot{m}_{\theta_0}]}_{\text{stochastic component}} + \underbrace{M(\theta) - M(\theta_0) + (\theta-\theta_0)^\top V_{\theta_0}(\theta-\theta_0)/2}_{\text{approximation-error component}}.$$

Because $\Delta_n(\theta) \leqslant (\overline{\mathbb{P}}_n - \mathbb{P})[m_\theta^+ - m_{\theta_0}]$ in Theorem 2.1 (iii) for $M(\theta) = \mathbb{P}[m_\theta^+]$, the stochastic component has one more term than the $\Delta_n(\theta)$ in (2.4). Similarly, compared with the condition $M(\theta_0) - M(\theta) \geqslant c_*\|\theta-\theta_0\|_2^2$ in Theorem 2.1 (iii) and Corollary 2.1, the approximation-error component is the remainder in the second-order expansion. For fixed $\{d,\Theta,m_\theta\}$, the approximation-error component of (2.9) holds when $M(\theta)$ is maximized and twice differentiable at $\theta_0$ with a negative-definite Hessian $-V_{\theta_0}$, and the stochastic component of (2.9) holds when $\mathscr{F}_n = \big\{\big(m_\theta^+ - m_{\theta_0} - (\theta-\theta_0)^\top\dot{m}_{\theta_0}\big)/\|\theta-\theta_0\|_2, 0 < \|\theta-\theta_0\|_2 \leqslant C\epsilon_n\big\}$ is a "Donsker class" of functions and $\sup_{f\in\mathscr{F}_n}\mathbb{P}[f^2] = o(1)$.

Proof. Let $M_n^+(\theta) = \overline{\mathbb{P}}_n[m_\theta^+]$. Condition (2.9) can be stated as

$$M_n^+(\theta) - M_n(\theta_0) = (\theta-\theta_0)^\top\overline{\mathbb{P}}_n[\dot{m}_{\theta_0}] - (\theta-\theta_0)^\top V_{\theta_0}(\theta-\theta_0)/2 + \mathrm{Rem}_n(\theta)$$

with $\max_{\|\theta-\theta_0\|_2\leqslant C\epsilon_n} |\mathrm{Rem}_n(\theta)|/(d/n + \|\theta-\theta_0\|_2^2) = o_{\mathbb{P}}(1)$. As $\overline{\theta}_n = \theta_0 + V_{\theta_0}^{-1}\overline{\mathbb{P}}_n[\dot{m}_{\theta_0}]$ maximizes the sum of the first two terms on the right-hand side above, we have

$$M_n^+(\theta) - M_n^+(\overline{\theta}_n) = -(\theta-\overline{\theta}_n)^\top V_{\theta_0}(\theta-\overline{\theta}_n)/2 + \mathrm{Rem}_n(\theta) - \mathrm{Rem}_n(\overline{\theta}_n).$$

As $M_n^+(\widehat{\theta}_n) \geqslant M_n(\widehat{\theta}_n) \geqslant M_n(\widetilde{\theta}_n) = M_n^+(\widetilde{\theta}_n) + o_{\mathbb{P}}(d/n)$, when $\|\widehat{\theta}_n - \theta_0\|_2 \leqslant C\epsilon_n$ we have

$$(\widehat{\theta}_n - \overline{\theta}_n)^\top V_{\theta_0}(\widehat{\theta}_n - \overline{\theta}_n)/2 \leqslant \mathrm{Rem}_n(\widehat{\theta}_n) - \mathrm{Rem}_n(\widetilde{\theta}_n) + o_{\mathbb{P}}(d/n) = o_{\mathbb{P}}(d/n + \|\widehat{\theta}_n - \overline{\theta}_n\|_2^2)$$

due to $\|\widehat{\theta}_n - \theta_0\|_2 \leqslant \|\widehat{\theta}_n - \overline{\theta}_n\|_2 + O_{\mathbb{P}}(\sqrt{d/n})$. This gives (2.10). ∎

**Corollary 2.2.** *Let $\Theta \subseteq \mathbb{R}^d$ and $m_\theta$ be fixed and $\mathbb{P}[m_\theta]$ is maximized at an interior point $\theta_0$ in $\Theta$. Suppose there exist $m_\theta^+ \geqslant m_\theta$ with $m_{\theta_0}^+ = m_{\theta_0}$ such that*

$$\mathbb{P}[m_\theta^+] = \mathbb{P}\big[m_{\theta_0}\big] - 2^{-1}(\theta-\theta_0)^\top V_{\theta_0}(\theta-\theta_0) + o(\|\theta-\theta_0\|_2^2) \tag{2.11}$$

*with a positive-definite $V_{\theta_0}$. Suppose the M-estimator (2.1) satisfies $\mathbb{P}\{\|\widehat{\theta}_n - \theta_0\|_2 \geqslant c_n\} \to 0$ for some constants $n^{-1/2} \leqslant c_n \to 0$ and that the Lipschitz condition (2.7) holds with some $\psi(x) \in \mathbb{R}$ with $\mathbb{P}[\psi^2] < \infty$. Suppose*

$$\sup_{\|\theta-\theta_0\|_2 \leqslant Cn^{-1/2}} \mathbb{P}\Big\{\Big|(\overline{\mathbb{P}}_n - \mathbb{P})[f_\theta^+]\Big| + \overline{\mathbb{P}}_n[m_\theta^+ - m_\theta] \geqslant \epsilon/n\Big\} = o(1),\ \forall C \geqslant \epsilon > 0, \tag{2.12}$$

*where $f_\theta^+ = m_\theta^+ - m_{\theta_0} - (\theta-\theta_0)^\top \dot{m}_{\theta_0}$ with $\mathbb{P}[\dot{m}_{\theta_0}] = 0$ and $\mathbb{P}\big[\|\dot{m}_{\theta_0}\|_2^2\big] = O(1)$. Then,*

$$\sqrt{n}\big(\widehat{\theta}_n - \theta_0\big) = V_{\theta_0}^{-1} n^{1/2}\overline{\mathbb{P}}_n[\dot{m}_{\theta_0}] + o_{\mathbb{P}}(1) \rightsquigarrow N\Big(0, V_{\theta_0}^{-1}\mathbb{P}\big[\dot{m}_{\theta_0}\dot{m}_{\theta_0}^\top\big]V_{\theta_0}^{-1}\Big).$$

Compared with (2.9), the supreme is taken outside the probability in (2.12).

Proof. By Corollary 2.1, $\widehat{\theta}_n$ is $n^{-1/2}$-consistent. Thus, by (2.11), it suffices to bound the stochastic component of (2.9) and verify the condition about $\widetilde{\theta}_n$. Let $\Theta_{n,C} = \{\theta : \|\theta-\theta_0\|_2 \leqslant C/n^{1/2}\}$, and $\Theta_{n,C,\delta}^2 = \{(\theta,\theta') : \|\theta-\theta'\|_2 \leqslant \delta/n^{1/2}, \theta \in \Theta_{n,C}, \theta' \in \Theta_{n,C}\}$. By (2.7) and Gaussian symmetrization,

$$\mathbb{E}\bigg[\sup_{(\theta,\theta')\in\Theta_{n,C,\delta}^2} \Big|(\overline{\mathbb{P}}_n - \mathbb{P})\big[f_\theta^+ - f_{\theta'}^+\big]\Big|\bigg] \leqslant \sqrt{2\pi/n}\,\mathbb{E}\bigg[K_n \sup_{(\theta,\theta')\in\Theta_{n,C,\delta}^2} \Big|G_\theta - G_{\theta'}\Big|\bigg],$$

where $K_n = (2\overline{\mathbb{P}}_n[\psi^2 + \|\dot{m}_{\theta_0}\|_2^2])^{1/2}$, $G_\theta = n^{-1/2}\sum_{i=1}^n g_i(f_\theta^+(X_i) - \mu_\theta)/K_n$ and $g_i$ are iid $N(0,1)$ independent of $\{X_i\}$. As $G_\theta$ are Gaussian with $\mathbb{E}[(G_\theta - G_{\theta'})^2|\{X_i\}] \leqslant \|\theta-\theta'\|_2^2$,

$$\mathbb{E}\bigg[\sup_{(\theta,\theta')\in\Theta_{n,C,\delta}^2} \Big|G_\theta - G_{\theta'}\Big|\Big|\{X_i\}\bigg] \leqslant 24\int_0^{\delta/n^{1/2}} \sqrt{\log N(\epsilon, \Theta_{n,C}, \|\cdot\|_2)}d\epsilon$$

by Dudley's inequality, where $N(\epsilon, \Theta_{n,C}, \|\cdot\|_2)$ is the $\ell_2$-covering number of $\Theta_{n,C}$. As $(\epsilon/2)^d N(\epsilon, \Theta_{n,C}, \|\cdot\|_2) \leqslant (C/n^{1/2} + \epsilon/2)^d$ via volume comparison,

$$\int_0^{\delta/n^{1/2}} \sqrt{\log N(\epsilon, \Theta_{n,C}, \|\cdot\|_2)}d\epsilon = n^{-1/2}\int_0^{\delta} \sqrt{\log N(t/n^{1/2}, \Theta_{n,C}, \|\cdot\|_2)}dt \leqslant \sqrt{d/n}J_{C,\delta}$$

with $J_{C,\delta} = \int_0^\delta \sqrt{\log(1+2C/t)}dt$. Let $C = 1/\delta_n$ and $\delta_n \to 0+$ slowly such that

$$(1+2/\delta_n^2)^d \sup_{\theta\in\Theta_{n,1/\delta_n}} \mathbb{P}\Big\{\Big|(\overline{\mathbb{P}}_n - \mathbb{P})[f_\theta^+]\Big| + \overline{\mathbb{P}}_n[m_\theta^+ - m_\theta] \geqslant \epsilon/n\Big\} = o(1)\ \forall\epsilon > 0.$$

Let $\theta_1, \ldots, \theta_N$ form a $(\delta_n/n^{1/2})$-net in $\Theta_{n,C}$ with $N \leqslant (1+2/\delta_n^2)^d$. As $J_{1/\delta_n,\delta_n} = o(1)$,

$$\sup_{\theta\in\Theta_{n,C}} \Big|(\overline{\mathbb{P}}_n - \mathbb{P})\big[f_\theta^+\big]\Big| \leqslant O_{\mathbb{P}}(J_{1/\delta_n,\delta_n}/n) + \sup_{1\leqslant j\leqslant N}\Big|(\overline{\mathbb{P}}_n - \mathbb{P})\big[f_{\theta_j}^+\big]\Big| = o_{\mathbb{P}}(1/n),$$

so that (2.9) holds. Let $\overline{\theta}_n = \theta_0 + V_{\theta_0}^{-1}\overline{\mathbb{P}}_n[\dot{m}_{\theta_0}]$ and $\widetilde{\theta}_n = \arg\min_{\theta_j, j\leqslant N}\|\theta_j - \overline{\theta}_n\|_2$. We have $\mathbb{P}\{\|\widetilde{\theta}_n - \overline{\theta}_n\|_2 \leqslant \delta_n/n^{1/2}\} \geqslant \mathbb{P}\{\|\overline{\theta}_n - \theta_0\|_2 \leqslant n^{-1/2}/\delta_n\} \to 1$ and $\overline{\mathbb{P}}_n[m_\theta^+ - m_\theta]\big|_{\theta=\widetilde{\theta}_n} \leqslant \max_{1\leqslant j\leqslant N}\overline{\mathbb{P}}_n[m_{\theta_j}^+ - m_{\theta_j}] = o_{\mathbb{P}}(1/n)$. $\square$

**Theorem 2.4.** *Suppose $\Theta \subseteq \mathbb{R}^d$, the Z-estimator $\widehat{\theta}_n$ in* (2.2) *is $\epsilon_n$-consistent for some $\epsilon_n \geqslant \sqrt{d/n}$ under the $\ell_2$ loss $d(\theta,\theta_0) = \|\theta-\theta_0\|_2$, and $\{\theta : \|\theta-\theta_0\|_2 \leqslant \epsilon'_n\} \subset \Theta$ for some $\epsilon'_n \gg \epsilon_n$. Let $V_{\theta_0} = V_{\theta_0}^\top \in \mathbb{R}^{d\times d}$ with $c_*\|u\|_2^2 \leqslant u^\top V_{\theta_0} u \leqslant c^*\|u\|_2^2$ for all $u \in \mathbb{R}^d$ and fixed positive constants $c_*$ and $c^*$. Suppose*

$$\sup_{\|\theta-\theta_0\|_2\leqslant C\epsilon_n}\left\|\frac{\overline{\mathbb{P}}_n[\psi_\theta-\psi_{\theta_0}]+V_{\theta_0}(\theta-\theta_0)}{\sqrt{d/n}+\|\theta-\theta_0\|_2}\right\|_2 = o_{\mathbb{P}}(1)\ \forall C \geqslant 1. \tag{2.13}$$

*Then,*

$$\big\|\widehat{\theta}_n-\theta_0-V_{\theta_0}^{-1}\overline{\mathbb{P}}_n[\psi_{\theta_0}]\big\|_2 = o_{\mathbb{P}}\big(\sqrt{d/n}\big). \tag{2.14}$$

*Consequently, for fixed d and independent $\psi_{\theta_0}(X_i)$ satisfying the Lindeberg condition,*

$$\mathscr{L}\big(\sqrt{n}(\widehat{\theta}_n-\theta_0);\mathbb{P}\big) = N\Big(0, V_{\theta_0}^{-1}\mathbb{P}\big[\psi_{\theta_0}\psi_{\theta_0}^\top\big]V_{\theta_0}^{-1}\Big)+o(1).$$

Again, the remainder in (2.13) is a sum of stochastic and approximation-error terms,

$$\underbrace{(\overline{\mathbb{P}}_n-\mathbb{P})[\psi_\theta-\psi_{\theta_0}]}_{\text{stochastic component}} \quad + \quad \underbrace{\Psi(\theta)+V_{\theta_0}(\theta-\theta_0)}_{\text{approximation-error component}} \quad ,$$

expressing the need of requiring one more term in the expansion than $\Delta_n(\theta) = (\overline{\mathbb{P}}_n-\mathbb{P})[\psi_\theta]$ in (2.6) and uniformity in the differentiability of $\Psi$ at $\theta_0$. When $X_i$ are iid, the stochastic component of (2.13) holds when $\{\psi_{\theta,1},\ldots,\psi_{\theta,d}, \|\theta-\theta_0\|_2 \leqslant C\epsilon_n\}$ is a "Donsker class" of functions and $\lim_{\theta\to\theta_0}\mathbb{P}\{|\psi_{\theta,j}(X_1)-\psi_{\theta_0,j}(X_1)| > \epsilon'\} = 0$ for all $\epsilon' > 0$ and $j \leqslant d$.

Proof. Under the imposed conditions, $\Psi_n(\theta) = \overline{\mathbb{P}}_n[\psi_\theta]$ has the expansion

$$\Psi_n(\theta) = \Psi_n(\theta_0) - V_{\theta_0}(\theta-\theta_0) + \mathrm{Rem}_n(\theta)$$

with $\max_{\|\theta-\theta_0\|_2\leqslant C\epsilon_n}\|\mathrm{Rem}_n(\theta)\|_2/(\sqrt{d/n}+\|\theta-\theta_0\|_2) = o_{\mathbb{P}}(1)$. Thus, as $\Psi_n(\widehat{\theta}_n) = 0$,

$$\widehat{\theta}_n-\theta_0 = V_{\theta_0}^{-1}(\Psi_n(\theta_0)+\mathrm{Rem}_n(\widehat{\theta}_n)) = V_{\theta_0}^{-1}\overline{\mathbb{P}}_n\big[\psi_{\theta_0}\big]+o_{\mathbb{P}}(\sqrt{d/n}+\|\widehat{\theta}_n-\theta_0\|_2)$$

when $\|\widehat{\theta}_n-\theta_0\|_2 \leqslant C\epsilon_n$. This gives (2.14). □

**Example 2.2** (GLM, continuation)**.** *In Theorem 2.3, set $m_\theta^+ = m_\theta$ so that the stochastic component of* (2.9) *holds automatically due to $(\overline{\mathbb{P}}_n-\mathbb{P})[m_\theta-m_{\theta_0}] = (\theta-\theta_0)^\top\overline{\mathbb{P}}_n\big[\dot{m}_{\theta_0}\big]$. Similarly, the stochastic component of* (2.13) *holds automatically due to $(\overline{\mathbb{P}}_n-\mathbb{P})[\psi_\theta-\psi_{\theta_0}] = 0$. For the approximation error components of* (2.9) *and* (2.13)*, we may add the uniform continuity of $V_\theta$ at $\theta_0$ to the key condition* (2.8)*. Because $\widehat{\theta}_n$ is the MLE, $\mathbb{P}\big[\dot{m}_{\theta_0}\dot{m}_{\theta_0}^\top\big] = \mathbb{P}\big[\psi_{\theta_0}\psi_{\theta_0}^\top\big] = V_{\theta_0}$ is the Fisher information matrix in Theorems 2.3 and 2.4.*

We apply Corollary 2.2 to study the asymptotic normality of the MLE in fixed parametric models.

**Theorem 2.5.** *Let $\Theta \subseteq \mathbb{R}^d$ and $\{p_\theta, \theta \in \Theta\}$ be a fixed density family with respect to a measure $\nu(dx)$. Suppose $X_i$ are iid observations from $p_{\theta_0}$ under $\mathbb{P}$ with an interior point $\theta_0$ in $\Theta$. Suppose the MLE $\widehat{\theta}_n$, given by (2.1) with $m_\theta = \ell_\theta = \log p_\theta$, satisfies $\mathbb{P}\{\|\widehat{\theta}_n - \theta_0\|_2 \geqslant c_n\} \to 0$ for some constants $n^{-1/2} \leqslant c_n \to 0$ and that the Lipschitz condition (2.7) holds with $m_\theta^+ = \ell_\theta^+ = \max(\ell_\theta, \ell_{\theta_0} - \log 4)$ and some $\psi(x) \in \mathbb{R}$ with $\mathbb{P}[\psi^2] < \infty$. Suppose*

$$\int\big\{\sqrt{p_\theta} - \sqrt{p_{\theta_0}} - 2^{-1}(\theta - \theta_0)^\top \dot{\ell}_{\theta_0}(x)\sqrt{p_{\theta_0}}\big\}^2 d\nu = o(\|\theta - \theta_0\|_2^2) \tag{2.15}$$

*for some measurable $\dot{\ell}_{\theta_0} \in \mathbb{R}^d$. Then, $\mathbb{P}[\dot{\ell}_{\theta_0}] = 0$ and for positive-definite $I_{\theta_0} = \mathbb{P}[\dot{\ell}_{\theta_0}\dot{\ell}_{\theta_0}^\top]$,*

$$\sqrt{n}\big(\widehat{\theta}_n - \theta_0\big) = I_{\theta_0}^{-1} n^{1/2}\overline{\mathbb{P}}_n[\dot{\ell}_{\theta_0}] + o_{\mathbb{P}}(1) \rightsquigarrow N\big(0, I_{\theta_0}^{-1}\big).$$

The $L_2(\nu)$-Fréchet differentiability of $\sqrt{p_\theta}$ in (2.15), referred to as the differentiability of the model in-quadratic-mean, also plays an important role in the local asymptotic normality and efficiency. Because $(\partial/\partial\theta)\sqrt{p_\theta} = 2^{-1}\big((\partial/\partial\theta)\log p_\theta\big)\sqrt{p_\theta}$, $\dot{\ell}_\theta(x)$ can be viewed as the score function and $I_\theta$ the Fisher information.

Proof. Let $\theta_t = \theta_0 + th$. As $\sqrt{p_{\theta_t}} \in L_2(\nu)$, $(2/t)(\sqrt{p_{\theta_t}} - \sqrt{p_{\theta_0}}) \to h^\top \dot{\ell}_{\theta_0}$ in $L_2(\nu)$. Thus, $\mathbb{P}[h^\top \dot{\ell}_{\theta_0}] = \int h^\top \dot{\ell}_{\theta_0} p_{\theta_0} d\nu = \lim_{t\to 0}\int t^{-1}(\sqrt{p_{\theta_t}} - \sqrt{p_{\theta_0}})(\sqrt{p_{\theta_t}} + \sqrt{p_{\theta_0}})d\nu = 0$.

To verify the conditions of Corollary 2.2, it suffices to prove (2.12) and

$$\mathbb{P}\big[\ell_\theta^+ - \ell_{\theta_0}\big] = -2^{-1}(\theta - \theta_0)^\top I_{\theta_0}(\theta - \theta_0) + o(\|\theta - \theta_0\|_2^2).$$

Let $\Omega_\theta = \{p_{\theta_0} > 0, |\sqrt{p_\theta}/\sqrt{p_{\theta_0}} - 1| \leqslant 1/2\}$. In $\Omega_\theta^c$, $|\ell_\theta^+ - \ell_{\theta_0}|p_{\theta_0} \leqslant p_\theta + 2p_{\theta_0} \leqslant 9\big(\sqrt{p_\theta} - \sqrt{p_{\theta_0}}\big)^2$. We also have $\mathbb{P}[\|\dot{\ell}_{\theta_0}\|_2^2 I_{\Omega_\theta^c}] = o(1)$ when $\mathbb{P}\{\Omega_\theta^c\} = o(1)$. Thus, by (2.15),

$$\mathbb{P}\Big[|\ell_\theta^+ - \ell_{\theta_0}|I_{\Omega_\theta^c}\Big] + \int_{\Omega_\theta^c}\big(p_\theta + p_{\theta_0}\big)d\nu + \mathbb{P}\Big[|(\theta - \theta_0)^\top \dot{\ell}_{\theta_0}|^2 I_{\Omega_\theta^c}\Big] = o(\|\theta - \theta_0\|_2^2). \tag{2.16}$$

In $\Omega_\theta$, $\ell_\theta^+ = \ell_\theta$ and $|2\log x - (x^2 - 1 - 2(x-1)^2)| \leqslant |x - 1|^3$ for $x = \sqrt{p_\theta}/\sqrt{p_{\theta_0}}$. Because $\mathbb{P}[|\sqrt{p_\theta}/\sqrt{p_{\theta_0}} - 1|^3 I_{\Omega_\theta}] = \mathbb{P}[|(\theta - \theta_0)^\top \dot{\ell}_{\theta_0}|^3 \wedge (1/8)] + o(\|\theta - \theta_0\|_2^2) = o(\|\theta - \theta_0\|_2^2)$ by (2.15),

$$\begin{aligned}
\mathbb{P}\Big[(\ell_\theta - \ell_{\theta_0})I_{\Omega_\theta}\Big] &= \int_{\Omega_\theta}\big\{p_\theta - p_{\theta_0} - 2\big(\sqrt{p_\theta} - \sqrt{p_{\theta_0}}\big)^2\big\}d\nu + o(\|\theta - \theta_0\|_2^2) \\
&= -2^{-1}(\theta - \theta_0)^\top I_{\theta_0}(\theta - \theta_0) + o(\|\theta - \theta_0\|_2^2).
\end{aligned}$$

It remains to prove (2.12), which just involves tail probability of sums of independent variables. For $\|\theta - \theta_0\| \leqslant C/n^{1/2}$, (2.16) yields

$$\begin{aligned}
\mathbb{P}\big\{\overline{\mathbb{P}}_n[\ell_\theta^+ - \ell_\theta] \neq 0\big\} &\leqslant n\mathbb{P}\big[I_{\{\ell_\theta^+ \neq \ell_\theta\}}\big] \leqslant n\mathbb{P}\big[I_{\{\Omega_\theta^c\}}\big] = o(1), \\
\mathbb{E}\big[\big(n(\overline{\mathbb{P}}_n - \mathbb{P})\big[(\theta - \theta_0)^\top \dot{\ell}_{\theta_0} I_{\Omega_\theta^c}\big]\big)^2\big] &\leqslant n\mathbb{E}\big[\big((\theta - \theta_0)^\top \dot{\ell}_{\theta_0}\big)^2 I_{\Omega_\theta^c}\big] = o(1),
\end{aligned}$$

and (2.15) yields

$$\mathbb{E}\big[\big(n(\overline{\mathbb{P}}_n - \mathbb{P})\big[f_\theta I_{\Omega_\theta}\big]\big)^2\big] \leqslant n\mathbb{P}\big[f_\theta^2 I_{\Omega_\theta}\big] = o(1)$$

with $f_\theta = \ell_\theta - \ell_{\theta_0} - (\theta - \theta_0)^\top \dot{\ell}_{\theta_0} = 2(\log u - v)$, $u = \sqrt{p_\theta}/\sqrt{p_{\theta_0}} \in [1/2, 3/2]$ in $\Omega_\theta$ and $v = 2^{-1}(\theta - \theta_0)^\top \dot{\ell}_{\theta_0}$, due to $|\log u - v| \leqslant |u - 1 - v| + |u - 1|^2 \leqslant 2|u - 1 - v| + (2v^2) \wedge 1$. Thus, (2.12) holds and the conclusion follows from Corollary 2.2. $\square$

2.4. **Continuous mapping theorem and delta-method.** Random elements $Z_n$ in a metric space $\mathbb{S}$ converges in distribution under $\mathbb{P}_n$ to $\mathscr{L}(Z_0;\mathbb{P}_0)$ in $\mathbb{S}$ iff $\mathbb{E}_n[f(Z_n)] \to \mathbb{E}_0[f(Z_0)]$ for all bounded continuous functions $f : \mathbb{S} \to \mathbb{R}$, or equivalently $\limsup_{n\to\infty} \mathbb{P}_n\{Z_n \in F\} \leqslant \mathbb{P}_0\{Z_0 \in F\}$ for all closed $F$. Note that we are always able to construct a probability $\mathbb{P}$ and variables $Z_n'$ such that $\mathscr{L}(Z_n;\mathbb{P}_n) = \mathscr{L}(Z_n';\mathbb{P})$ for all $n \geqslant 0$. The continuous mapping theorem concerns the convergence of functions of $Z_n$ in distribution. For simplicity, all functions, sets and random elements are assumed to be Borel measurable.

**Theorem 2.6.** *Let $Z_n$ be random elements in a metric space $\mathbb{S}$. Suppose $\mathscr{L}(Z_n;\mathbb{P}_n) \rightsquigarrow \mathscr{L}(Z_0;\mathbb{P}_0)$ and $\mathbb{P}_n\{Z_n \in \mathbb{S}_n\} = 1$ for some $\mathbb{S}_n \subset \mathbb{S}$. Let $\mathbb{S}'$ be another metric space and $g_n : \mathbb{S}_n \to \mathbb{S}'$ such that for any $n_k \to \infty$ and $x_{n_k} \in \mathbb{S}_{n_k}$, $x_{n_k} \to x \in \mathbb{S}_0$ implies $g_{n_k}(x_{n_k}) \to g_0(x)$. Then, $\mathscr{L}(g_n(Z_n);\mathbb{P}_n) \rightsquigarrow \mathscr{L}(g_0(Z_0);\mathbb{P}_0)$.*

PROOF. Let $F$ be a closed set in $\mathbb{S}'$. For any $x \in \cap_{k=1}^\infty \overline{\cup_{m=k}^\infty \{x \in \mathbb{S}_m : g_m(x) \in F\}} \cap S_0$, there exist $x_{n_k} \to x$ with $g_{n_k}(x_{n_k}) \in F$, so that $g_{n_k}(x_{n_k}) \to g_0(x) \in F$ by the continuity assumption. Thus,

$$\begin{aligned}
\limsup_n \mathbb{P}_n\big\{g_n(Z_n) \in F\big\} &\leqslant \lim_{k\to\infty} \limsup_{n\to\infty} \mathbb{P}_n\big\{Z_n \in \overline{\cup_{m=k}^\infty \{x \in \mathbb{S}_m : g_m(x) \in F\}}\big\} \\
&\leqslant \lim_{k\to\infty} \mathbb{P}_0\big\{Z_0 \in \overline{\cup_{m=k}^\infty \{x \in \mathbb{S}_m : g_m(x) \in F\}} \cup (\mathbb{S}\backslash\mathbb{S}_0)\big\} \\
&\leqslant \mathbb{P}_0\big\{g_0(Z_0) \in F\big\}.
\end{aligned}$$

The second inequality above follows from the assumption $\mathscr{L}(Z_n;\mathbb{P}_n) \rightsquigarrow \mathscr{L}(Z_0;\mathbb{P}_0)$. □

An immediate consequence of the continuous mapping theorem is Slutsky's lemma:

$$\mathscr{L}\big((1 + o_{\mathbb{P}_n})Z_n + o_{\mathbb{P}_n}(1);\mathbb{P}_n\big) \rightsquigarrow \mathscr{L}(Z;\mathbb{P}) \text{ when } \mathscr{L}(Z_n;\mathbb{P}_n) \rightsquigarrow \mathscr{L}(Z;\mathbb{P}). \tag{2.17}$$

In statistics, a main application of the continuous mapping theorem is the delta method.

**Theorem 2.7.** *Suppose $\Theta \subseteq \mathbb{R}^d$ and $\mathscr{L}((\widehat{\theta}_n - \theta_{0,n})/\epsilon_n;\mathbb{P}_n) \rightsquigarrow \mathscr{L}(Z;\mathbb{P}\}$ for some random variable $Z \in \mathbb{R}^d$ and deterministic $\epsilon_n \to 0+$ and $\theta_{0,n} \in \mathbb{R}^d$. Let $\phi_n(\cdot)$ be mappings from $\mathbb{R}^d$ to another Euclidean space such that $\sup\{\|\big(\phi_n(\theta_{0,n} + \epsilon_n h) - \phi_n(\theta_{0,n})\big)/\epsilon_n - A_n h\|_2 : \|h\|_2 \leqslant C\} = o(1)$ for every $C > 0$ and some matrices $A_n$ with $\|A_n\|_{\mathrm{S}} = O(1)$. Then,*

$$\mathscr{L}\big(\big(\phi_n(\widehat{\theta}_n) - \phi_n(\theta_{0,n})\big)/\epsilon_n;\mathbb{P}_n\big) = \mathscr{L}(A_n Z;\mathbb{P}) + o(1).$$

PROOF. Let $Z_n = (\widehat{\theta}_n - \theta_{0,n})/\epsilon_n$ and $R_n(h) = \big(\phi_n(\theta_{0,n} + \epsilon_n h) - \phi_n(\theta_{0,n})\big)/\epsilon_n - A_n h$. As $R_n(h) = o(1)$ uniformly on bounded sets and $\|Z_n\|_2 = O_{\mathbb{P}}(1)$, $R_n(Z_n) = o_{\mathbb{P}_n}(1)$. Consequently, $\big(\phi_n(\widehat{\theta}_n) - \phi_n(\theta_{0,n})\big)/\epsilon_n = A_n Z_n + R_n(Z_n)$. The conclusion follows. □

We illustrate the delta method in the following example.

**Example 2.6** (Estimation of covariance and precision matrices)**.** *Let $X_1, \ldots, X_n$ be iid observations from a fixed distribution in $\mathbb{R}^d$ with mean $\mu$ and covariance matrix $\Sigma$. Suppose $\Sigma$ is positive-definite and $\mathbb{E}[\|X_1\|_2^4] < \infty$. The precision matrix, also called concentration matrix, is inverse $\Omega = \Sigma^{-1}$. We note that both $\Sigma$ and $\Omega$ live in $\mathbb{R}^{d\times d}_{\rm sym} = \{A \in \mathbb{R}^{d\times d} : A = A^\top\}$, which should be viewed as a $d(d+1)/2$ dimensional linear space with inner-product*

$$\langle A, B\rangle = \text{trace}(A^\top B).$$

*By the method of moments, we estimate $\mu$ by $\widehat{\mu} = \overline{\mathbb{P}}_n[x]$ and estimate $\Sigma$ by*

$$\widehat{\Sigma} = \overline{\mathbb{P}}_n\big[(x-\widehat{\mu})(x-\widehat{\mu})^\top\big].$$

*As $\widehat{\Sigma} = \overline{\mathbb{P}}_n\big[(x-\mu)(x-\mu)^\top\big] - \overline{\mathbb{P}}_n\big[(x-\mu)\big]\overline{\mathbb{P}}_n\big[(x-\mu)^\top\big]$,*

$$\sqrt{n}\big(\widehat{\Sigma} - \Sigma\big) \rightsquigarrow N\big(0, V_\Sigma\big),$$

*where $V_\Sigma$ is the nonnegative-definite operator in $\mathbb{R}^{d\times d}_{\rm sym}$ defined by*

$$\langle A, V_\Sigma B\rangle = \mathbb{E}\big[(X_1-\mu)^\top A(X_1-\mu)(X_1-\mu)^\top B(X_1-\mu)\big] - \langle A, \Sigma\rangle\langle B, \Sigma\rangle$$

*as the covariance of $\langle A, (X_1-\mu)(X_1-\mu)^\top\rangle$ and $\langle B, (X_1-\mu)(X_1-\mu)^\top\rangle$, $\{A, B\} \subset \mathbb{R}^{d\times d}_{\rm sym}$.*

*Because $\widehat{\Sigma}$ is consistent, it is positive-definite with high probability, so that we may use*

$$\widehat{\Omega} = \widehat{\Sigma}^{-1}$$

*to estimate the precision matrix $\Omega$. As $\widehat{\Omega} - \Omega = \Omega(\Sigma - \widehat{\Sigma})\widehat{\Omega}$,*

$$\sqrt{n}\big(\widehat{\Omega} - \Omega\big) = \Omega\big(\sqrt{n}\big(\Sigma - \widehat{\Sigma}\big)\big)\Omega + o_{\mathbb{P}}(1) \rightsquigarrow N\big(0, V_\Omega\big),$$

*where $V_\Omega$ is the nonnegative-definite operator in $\mathbb{R}^{d\times d}_{\rm sym}$ defined by*

$$\langle A, V_\Omega B\rangle = \mathbb{E}\big[(X_1-\mu)^\top\Omega A\Omega(X_1-\mu)(X_1-\mu)^\top\Omega B\Omega(X_1-\mu)\big] - \langle A, \Omega\rangle\langle B, \Omega\rangle$$

*due to $\langle A, \Omega(X_1-\mu)(X_1-\mu)^\top\Omega\rangle = \langle\Omega A\Omega, (X_1-\mu)(X_1-\mu)^\top\rangle$.*

*The operators can be identified with $A$ and $B$ of the form $A_{j,k} = (u_j u_k^\top + u_k u_j^\top)/2$ in any basis. Let $\Sigma = \sum_{j=1}^d \lambda_j u_j u_j^\top$ be the eigenvalue decomposition of $\Sigma$. In the Gaussian case,*

$$\langle A_{j,k}, V_\Sigma A_{\ell,m}\rangle = \lambda_j\lambda_k\big(1 + I_{\{j=k\}}\big)I_{\{A_{j,k}=A_{\ell,m}\}},$$

*so that*

$$\langle A, V_\Sigma B\rangle = 2\,\text{trace}\big(\Sigma A\Sigma B\big), \quad \langle A, V_\Omega B\rangle = 2\,\text{trace}\big(\Omega A\Omega B\big).$$

*In particular, for the estimation of the elements $\Sigma_{j,k} = \langle A_{j,k}, \Sigma\rangle$ and $\Omega_{j,k} = \langle A_{j,k}, \Omega\rangle$ with $A_{j,k} = (e_j e_k^\top + e_k e_j^\top)/2$,*

$$\sqrt{n}\big(\widehat{\Sigma}_{j,k} - \Sigma_{j,k}\big) \rightsquigarrow N\big(0, \Sigma_{j,j}\Sigma_{k,k} + \Sigma_{j,k}^2\big),$$

*and*

$$\sqrt{n}\big(\widehat{\Omega}_{j,k} - \Omega_{j,k}\big) \rightsquigarrow N\big(0, \Omega_{j,j}\Omega_{k,k} + \Omega_{j,k}^2\big).$$

The delta method can be used to construct confidence regions of the unknown $\theta$ through variance stabilizing transformations. Suppose

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) \rightsquigarrow_{\mathbb{P}} N(0, \sigma^2(\theta))$$

in $\mathbb{R}$ with unknown $\sigma^2(\theta)$. Let

$$\phi(\theta) = \int_a^\theta \frac{1}{\sigma(x)} dx.$$

By the delta method

$$\sqrt{n}\big(\phi(\widehat{\theta}_n) - \phi(\theta_0)\big) \rightsquigarrow_{\mathbb{P}} (\sigma(\theta))^{-1} N(0, \sigma^2(\theta)) = N(0,1).$$

**Example 2.7** (Fisher's transformation)**.** *Consider the sample correlation*

$$\widehat{\rho}_n = \frac{\overline{\mathbb{P}}_n[xy] - \overline{\mathbb{P}}_n[x]\overline{\mathbb{P}}_n[y]}{\big(\overline{\mathbb{P}}_n[x^2] - \overline{\mathbb{P}}_n^{\,2}[x]\big)^{1/2}\big(\overline{\mathbb{P}}_n[y^2] - \overline{\mathbb{P}}_n^{\,2}[y]\big)^{1/2}}$$

*based on bivariate data $(X_1, Y_1), \ldots, (X_n, Y_n)$. Suppose $(X_i, Y_i)$ are iid with mean $(\mu_X, \mu_Y)$, variances $\sigma_X^2$ and $\sigma_Y^2$, correlation $\rho \in (-1, 1)$ and finite fourth moment. We have*

$$\sqrt{n} \begin{pmatrix} \overline{\mathbb{P}}_n[xy] - \overline{\mathbb{P}}_n[x]\overline{\mathbb{P}}_n[y] - \mathbb{E}\big[(X_1 - \mu_X)(Y_1 - \mu_Y)\big] \\ \overline{\mathbb{P}}_n[x^2] - \overline{\mathbb{P}}_n^{\,2}[x] - \mathbb{E}\big[(X_1 - \mu_X)^2\big] \\ \overline{\mathbb{P}}_n[y^2] - \overline{\mathbb{P}}_n^{\,2}[y] - \mathbb{E}\big[(Y_1 - \mu_Y)^2\big] \end{pmatrix} \rightsquigarrow N(0, V),$$

*where $V$ is the covariance matrix of $((X_1 - \mu_X)(Y_1 - \mu_Y), (X_1 - \mu_X)^2, (Y_1 - \mu_Y)^2)^\top$. Let $\phi(t, x, y) = t/\sqrt{xy}$. Because $\rho = \phi(\mathrm{Cov}(Y, Y), \mathrm{Var}(X), \mathrm{Var}(Y))$, the delta method yields*

$$\sqrt{n}\big(\widehat{\rho}_n - \rho\big) \rightsquigarrow N\big(0, a^\top V a\big),$$

*where $a = \dot{\phi}(\rho\sigma_X\sigma_Y, \sigma_X^2, \sigma_Y^2)$ with $\dot{\phi}$ being the gradient of $\phi$. In the Gaussian case, it suffices to consider $\sigma_X = \sigma_Y = 1$ due to scale invariance, so that*

$$a^\top V a = \begin{pmatrix} 1 \\ -\rho/2 \\ -\rho/2 \end{pmatrix}^\top \begin{pmatrix} 1+\rho^2 & 2\rho & 2\rho \\ 2\rho & 2 & 2\rho^2 \\ 2\rho & 2\rho^2 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -\rho/2 \\ -\rho/2 \end{pmatrix} = (1-\rho^2)^2.$$

*The variance stabilizing transformation is*

$$\phi(\rho) = \int_0^\rho (1-x^2)^{-1} dx = \frac{1}{2}\log\left(\frac{1+\rho}{1-\rho}\right).$$

*It follows that in the Gaussian case and for $|\rho| < 1$*

$$n^{1/2}\big(\phi(\rho) - \phi\big(\widehat{\rho}_n\big)\big) \rightsquigarrow N(0,1).$$

2.5. **Problems.**

**Exercise 2.1.** *Consider the nonparametric regression model* $y_i = f(z_i) + \varepsilon_i, 1 \leqslant i \leqslant n$, *where the noise variables* $\varepsilon_i$ *are iid* $N(0, \sigma^2)$ *and independent of* $\{\varepsilon_i\}$ *the design points* $z_i$ *are iid uniform variables in* $[0, 1]$. *Suppose* $|f(y) - f(z)| \leqslant |y - z|^\alpha$ *for some* $\alpha \in (0, 1)$. *Let*

$$\widehat{\sigma}^2 = \frac{\sum_{1 \leqslant i < j \leqslant n} (y_i - y_j)^2 I\{|z_i - z_j| \leqslant n^{-\beta}\}}{2\#\{(i,j) : 1 \leqslant i < j \leqslant n, |z_i - z_j| \leqslant n^{-\beta}\}}.$$

*Find the largest possible (by your method) a constant* $\gamma = \gamma_{\alpha,\beta}$ *satisfying* $\widehat{\sigma}^2 - \sigma^2 = O_{\mathbb{P}}(n^{-\gamma})$.

**Exercise 2.2.** *Let* $X_i = \theta + \varepsilon_i$ *with unknown* $\theta \in \mathbb{R}$ *and iid* $\varepsilon_i \sim N(0, 1)$. *However, instead of observing* $\{X_1, \ldots, X_n\}$, *only the* $X_i$ *with* $|\varepsilon_i| \geqslant 1$, $\{Y_1, \ldots, Y_N\} = \{X_i, i \in I\}$, *are observed, where* $I = \{i \leqslant n : |\varepsilon_i| \geqslant 1\}$. *Let* $\widehat{\theta}_n$ *be the MLE of* $\theta_n$ *based on* $\{Y_1, \ldots, Y_N\}$. *What is the convergence rate of* $\widehat{\theta}_n$*? What is the asymptotic distribution of* $\widehat{\theta}_n$*?*

**Exercise 2.3.** *Consider the quantile regression* $\widehat{\theta}_n$ *in Example 2.4 with* $y_i = z_i^\top \theta + \varepsilon_i, i = 1, \ldots, n$. *Suppose* $\{z_i\}$ *are iid random vectors with joint density function* $g_0$ *in* $\mathbb{R}^d$, *and* $\{\varepsilon_i\}$ *are iid variables independent of* $\{z_i\}$ *with density function* $p_0$ *in* $\mathbb{R}$. *Suppose* $\mathbb{P}[zz^\top]$ *is fixed and positive-definite. For fixed d write condition(s) under which* $\sqrt{n}(\widehat{\theta} - \theta_0)$ *is asymptotically normal. Prove the asymptotic normality and find the asymptotic covariance matrix. For* $d = 1$ *what would happen if your condition is violated? Is it possible to have faster rate than* $n^{-1/2}$*?*

**Exercise 2.4.** *Consider the kernel density estimator* $\widehat{\theta}_n$ *in Example 2.5.*
*(i) For the Gaussian kernel* $K(y, x) = h^{-1} e^{-(x-y)^2/(2h^2)}$, *derive the rate of convergence under proper conditions on* $h = h_n$ *and the density function* $\theta$.
*(ii) Is it possible to achieve unbiasedness in kernel density estimation,* $\mathbb{P}[\widehat{\theta}_n] = \theta$ *possibly with a non-Gaussian kernel?*

**Exercise 2.5.** *In the Gaussian sequence model, independent* $X_i \sim N(\theta_i, 1)$ *are observed and* $\theta = (\theta_1, \ldots, \theta_n) \in \mathbb{R}^n$ *is the unknown parameter. Suppose* $\Theta = \Theta_n = \{\theta : \|\theta\|_1 \leqslant n^{1-\alpha}\}$ *is the parameter space with a fixed* $\alpha \in (0, 1)$. *Find the convergence rate for the maximum average MSE,* $\max_{\boldsymbol{\theta} \in \Theta_n} \mathbb{E}\big[\|\widehat{\theta}_n - \theta\|_2^2/n\big]$, *for the MLE based on* $\{X_1, \ldots, X_n\}$. *Note that only one* $X_i$ *is observed for each* $\theta_i$.

**Exercise 2.6** (Elliptical distribution)**.** *A random vector X has an elliptical distribution in* $\mathbb{R}^d$ *if it can be written as* $X = \mu + \xi A U$, *where* $\mu \in \mathbb{R}^d$ *and* $A \in \mathbb{R}^{d \times d}$ *are deterministic fixed, U has the uniform distribution on the unit sphere in* $\mathbb{R}^d$, *and* $\xi$ *is a random variable independent of U with* $\mathbb{E}[\xi^2] = d$. *It can be seen that X has mean* $\mu$ *and covariance matrix* $\Sigma = \mathbb{P}[xx^\top] = AA^\top$. *Suppose* $\mathbb{E}\big[\xi^4\big] < \infty$ *and* $\Sigma$ *is fixed and positive-definite. Find the asymptotic covariance matrices* $V_\Sigma$ *and* $V_\Omega$ *in Example 2.6 based on iid observations from the elliptical distribution.*

**Exercise 2.7** (Correlation matrix)**.** *In Example 2.6 the sample correlation matrix can be written as* $\widehat{\rho} = \widehat{D}^{-1/2} \widehat{\Sigma} \widehat{D}^{-1/2} \in \mathbb{R}^{d \times d}$ *with* $\widehat{D} = \mathrm{diag}(\widehat{\Sigma})$. *Use the delta method and the calculation in Example 2.6 to find the limiting distribution of* $\widehat{\rho}$ *in the Gaussian case. Identify the asymptotic covariance operator, and check your formula against Example 2.7. What is the asymptotic covariance between* $\widehat{\rho}_{j,k}$ *and* $\widehat{\rho}_{\ell,m}$*?*

2.6. **Maximal inequalities.** Let $X_1, \ldots, X_n$ be independent random elements in a certain measurable space under $\mathbb{P}$. Here we discuss upper bounds for $\sup_{f\in\mathscr{F}} \big|(\overline{\mathbb{P}}_n - \mathbb{P})[f]\big|$ for countable function classes $\mathscr{F}$ and the maximum of Gaussian processes.

Recall that $\overline{\mathbb{P}}_n[f] = n^{-1}\sum_{i=1}^n f(X_i)$ and $\mathbb{P}[f]$ the expectation of $\overline{\mathbb{P}}_n[f]$ under $\mathbb{P}$. Let $\{r_i\}$ be a Rademacher sequence, iid with $\mathbb{P}\{r_i = \pm 1\} = 1/2$, independent of $\{X_i\}$. The symmetrized empirical measure is defined as $\overline{\mathbb{P}}_n^r[f] = n^{-1}\sum_{i=1}^n r_i f(X_i)$. Let $\{g_i\}$ be a standard Gaussian sequence, iid $N(0,1)$, independent of $\{r_i\}$ and $\{X_i\}$. For Gaussian symmetrization, we define $\overline{\mathbb{P}}_n^g[f] = n^{-1}\sum_{i=1}^n g_i f(X_i)$.

**Lemma 2.1.** *Let $\mathscr{F}$ be a countable collection of functions $f(x)$ with $\mathbb{P}[|f|] < \infty$. Let $F$ be a nonnegative non-decreasing convex function in $[0,\infty)$. Then,*

$$\mathbb{E}\bigg[F\bigg(\sup_{f\in\mathscr{F}}\Big|(\overline{\mathbb{P}}_n - \mathbb{P})[f]\Big|\bigg)\bigg] \leqslant \mathbb{E}\bigg[F\bigg(2\sup_{f\in\mathscr{F}}\Big|\overline{\mathbb{P}}_n^r[f]\Big|\bigg)\bigg] \leqslant \mathbb{E}\bigg[F\bigg(\sqrt{2\pi}\sup_{f\in\mathscr{F}}\Big|\overline{\mathbb{P}}_n^g[f]\Big|\bigg)\bigg]. \tag{2.18}$$

It follows from the proof below that (2.18) is also valid when $\overline{\mathbb{P}}_n^r[f]$ and $\overline{\mathbb{P}}_n^g[f]$ are replaced respectively by $n^{-1}\sum_{i=1}^n r_i\big(f(X_i) - \mathbb{E}[f(X_i)]\big)$ and $n^{-1}\sum_{i=1}^n g_i\big(f(X_i) - \mathbb{E}[f(X_i)]\big)$.

Proof. Let $\{X_i'\}$ be an independent copy of $\{X_i\}$ such that $\{X_i, X_i'\}$ is independent of $\{r_i\}$. By the Jensen inequality

$$\begin{aligned}
\mathbb{E}\bigg[F\bigg(\sup_{f\in\mathscr{F}}\Big|(\overline{\mathbb{P}}_n - \mathbb{P})[f]\Big|\bigg)\bigg] &= \mathbb{E}\bigg[F\bigg(\sup_{f\in\mathscr{F}}\bigg|\sum_{i=1}^n \frac{f(X_i)}{n} - \mathbb{E}\bigg[\sum_{i=1}^n \frac{f(X_i')}{n}\bigg]\bigg|\bigg)\bigg] \\
&\leqslant \mathbb{E}\bigg[F\bigg(\sup_{f\in\mathscr{F}}\bigg|\sum_{i=1}^n \frac{f(X_i) - f(X_i')}{n}\bigg|\bigg)\bigg].
\end{aligned}$$

Because $\{f(X_i) - f(X_i'), f \in \mathscr{F}\}$ has the same distributions as $\{r_i(f(X_i) - f(X_i')), f \in \mathscr{F}\}$ and $F(a+b) \leqslant (F(2a) + F(2b))/2$ for nonnegative $a$ and $b$,

$$\begin{aligned}
\mathbb{E}\bigg[F\bigg(\sup_{f\in\mathscr{F}}\Big|(\overline{\mathbb{P}}_n - \mathbb{P})[f]\Big|\bigg)\bigg] &\leqslant \mathbb{E}\bigg[F\bigg(\sup_{f\in\mathscr{F}}\bigg|\sum_{i=1}^n \frac{r_i(f(X_i) - f(X_i'))}{n}\bigg|\bigg)\bigg] \\
&\leqslant \mathbb{E}\bigg[F\bigg(2\sup_{f\in\mathscr{F}}\Big|\overline{\mathbb{P}}_n^r[f]\Big|\bigg)\bigg].
\end{aligned}$$

Similarly, as $\mathbb{E}[|g_i|] = \sqrt{2/\pi}$, the Jensen inequality also yields

$$\begin{aligned}
\mathbb{E}\bigg[F\bigg(2\sup_{f\in\mathscr{F}}\Big|\overline{\mathbb{P}}_n^r[f]\Big|\bigg)\bigg] &= \mathbb{E}\bigg[F\bigg(\sqrt{2\pi}\sup_{f\in\mathscr{F}}\bigg|\sum_{i=1}^n \Big(\mathbb{E}\Big[|g_i|\Big|\{X_i, r_i\}\Big]\Big)\frac{r_i f(X_i)}{n}\bigg|\bigg)\bigg] \\
&\leqslant \mathbb{E}\bigg[F\bigg(\sqrt{2\pi}\sup_{f\in\mathscr{F}}\bigg|\sum_{i=1}^n \frac{|g_i| r_i f(X_i)}{n}\bigg|\bigg)\bigg].
\end{aligned}$$

The second inequality in (2.18) follows as $|g_i| r_i \sim N(0,1)$. $\square$

The Rademacher and Gaussian symmetrizations in Lemma (2.18) allow applications of sub-Gaussian and Gaussian concentration inequalities respectively to the processes $\{\overline{\mathbb{P}}_n^r[f], f \in \mathscr{F}\}$ and $\{\overline{\mathbb{P}}_n^g[f], f \in \mathscr{F}\}$ by first conditioning on $\{X_i\}$. We first present Dudley's inequality [Dud67, Sud69, LT91] which applies to general sub-Gaussian processes. A stochastic process $\{X(t), t \in T\}$ is sub-Gaussian with respect to a semi-metric $d(\cdot,\cdot) : T \times T \to [0,\infty]$ if

$$\mathbb{P}\Big\{|X(s) - X(t)| > x\Big\} \leqslant 2\exp\bigg(-\frac{x^2}{2d^2(s,t)}\bigg), \quad \forall x. \tag{2.19}$$

Note that $\mathbb{E}\big[|X(s)-X(t)|^2\big] \leqslant 4d^2(s,t)$ by (2.19). Conditionally on $\{X_i\}$, $\overline{\mathbb{P}}_n^r[f]$ is sub-Gaussian with $d^2(f_1,f_2) = \overline{\mathbb{P}}_n[(f_1-f_2)^2]$. For $\epsilon > 0$ and the semi-metric $d = d(\cdot,\cdot)$ in (2.19), let $N(\epsilon, T, d)$ be the covering number of $T$ by $d$-balls of radius $\epsilon$.

**Theorem 2.8** (Dudley's inequality)**.** *Let $\{X(t), t \in T\}$ be a sub-Gaussian process with a countable $T$ and a semi-metric $d(\cdot,\cdot)$ in (2.19). Then, for all $\delta > 0$*

$$\mathbb{E}\bigg[\sup_{s\in T, t\in T, d(s,t)\leqslant\delta}\big|X(s)-X(t)\big|\bigg] \leqslant 24\int_0^\delta \sqrt{\log\big(N(\epsilon,T,d)\big)}\,d\epsilon. \tag{2.20}$$

Proof. By the monotone convergence theorem, we assume $2 \leqslant |T| < \infty$ and $d(s,t) > 0$ for $s \neq t$ without loss of generality. For $N \geqslant 2$ and $\{s_i, t_i\} \subseteq T$ satisfying $d(s_i,t_i) \leqslant \sigma$,

$$\mathbb{E}\bigg[\max_{1\leqslant i\leqslant N}\big|X(s_i)-X(t_i)\big|\bigg] \leqslant L_N = \int_0^\infty \min\Big(1, 2Ne^{-x^2/(2\sigma^2)}\Big)dx = C_N\sigma\sqrt{\log N},$$

where $C_N = \big\{\sqrt{2\log(2N)} + \int_0^\infty e^{-x^2/2 - x(2\log(2N))^{1/2}}dx\big\}/(\log N)^{1/2} \leqslant 2.58$. Let $T_k \subseteq T$ satisfying $|T_k| = N(\delta/2^k, T, d)$ and $\max_{s\in T}\min\{d(s,t) : t \in T_k\} \leqslant \delta/2^k$. For $s \in T_k$ let $s_j = s$ for $j \geqslant k$ and $s_j \in T_j$ satisfying $d(s_j, s_{j+1}) \leqslant \delta/2^j$ for $j < k$. Let $k_1 = \arg\min_k\{|T_k| \geqslant 2 \text{ or } k = 1\}$ and $k_2 = \arg\min_k |T_k| = T$. For $d(s,t) \leqslant \delta$, $d(s_0,t_0) \leqslant 5\delta I_{\{|T_0|>1\}}$ and

$$\begin{aligned}
&\mathbb{E}\bigg[\sup_{s\in T, t\in T, d(s,t)\leqslant\delta}\big|X(s)-X(t)\big|\bigg] \\
\leqslant\ & \sum_{j=k_1}^{k_2} 2\mathbb{E}\bigg[\sup_{s_j\in T_j}\big|X(s_j)-X(s_{j-1})\big|\bigg] + I_{\{|T_0|>1\}}\mathbb{E}\bigg[\sup_{s\in T, t\in T, d(s,t)\leqslant\delta}\big|X(s_0)-X(t_0)\big|\bigg] \\
\leqslant\ & \sum_{j=k_1}^{k_2} 2(\delta/2^{j-1})L_{|T_j|} + 5\delta I_{\{|T_0|>1\}}L_{|T_0|^2/2} \\
\leqslant\ & 8C_2\int_0^{\delta/2}\sqrt{\log\big(N(\epsilon,T,d)\big)}\,d\epsilon + 5\delta\max_{n\geqslant 2}\Big(C_{n^2/2}\sqrt{\log(n^2/2)}/\sqrt{\log n}\Big)\Big)\sqrt{\log N(\delta,T,d)}
\end{aligned}$$

as $N(\epsilon,T,d) \downarrow \epsilon$. The conclusion follows as the maximum on the right-hand side above is attained at $n = 2$ and $9C_2 \leqslant 23.22$. ☐

For Gaussian processes, contraction theorems may provide sharper bounds. Slepian's [Sle62] inequality below provides sufficient conditions for the stochastic ordering of the maxima of two Gaussian vectors.

**Theorem 2.9** (Slepian's inequality)**.** *Let $X = (X_1, \ldots, X_n)^\top$ and $Y = (Y_1, \ldots, Y_n)^\top$ be two Gaussian vectors. Suppose $X_i \sim Y_i$ and $\mathrm{Var}(X_i - X_j) \leqslant \mathrm{Var}(Y_i - Y_j)$ $\forall i \neq j$. Then,*

$$\mathbb{P}\Big\{ \max_{1\leqslant i\leqslant n} X_i > t\Big\} \leqslant \mathbb{P}\Big\{ \max_{1\leqslant i\leqslant n} Y_i > t\Big\}, \quad \forall t \in \mathbb{R}. \tag{2.21}$$

*Consequently, $\mathbb{P}\{\|X\|_\infty > t\} \leqslant \mathbb{P}\{\|Y\|_\infty > t\}$ for all $t \in \mathbb{R}$.*

Proof. Let $\mu = \mathbb{E}[X]$, $\Sigma^X = (\Sigma^X_{i,j})_{n\times n} = \mathbb{E}[(X-\mu)(X-\mu)^\top]$, $\Sigma^Y = (\Sigma^Y_{i,j})_{n\times n} = \mathbb{E}[(Y-\mu)(Y-\mu)^\top]$. Assume without loss of generality $X$ and $Y$ are independent and $\mathrm{rank}(\Sigma^X) = \mathrm{rank}(\Sigma^Y) = n$. Let $X(s) = \sqrt{1-s}(X-\mu) + \sqrt{s}(Y-\mu) + \mu$, $s \in [0,1]$. As

$$\varphi_s(x) = (2\pi)^{-n}\int_{\mathbb{R}^n} \exp\Big[\sqrt{-1}(\mu - x)^\top u - u^\top((1-s)\Sigma^X + s\Sigma^Y)u/2\Big]du$$

is the joint density of $X(s)$ by the Fourier inversion formula,

$$\frac{\partial \varphi_s(x)}{\partial s} = \frac{1}{2}\sum_{i=1}^n\sum_{j=1}^n \Big(\Sigma^Y_{i,j} - \Sigma^X_{i,j}\Big)\frac{\partial^2\varphi_s(x)}{\partial x_i \partial x_j}.$$

Thus, for functions $f(x)$ with uniformly integrable second derivative with respect to $\varphi_s$,

$$\begin{aligned} \frac{d}{ds}\mathbb{E}\big[f(X(s))\big] &= \frac{1}{2}\sum_{i=1}^n\sum_{j=1}^n \big(\Sigma^Y_{i,j} - \Sigma^X_{i,j}\big)\int f(x)\frac{\partial^2\varphi_s(x)}{\partial x_i\partial x_j}dx \\ &= \frac{1}{2}\sum_{i=1}^n\sum_{j=1}^n \big(\Sigma^Y_{i,j} - \Sigma^X_{i,j}\big)\int \Big(\frac{\partial^2 f(x)}{\partial x_i\partial x_j}\Big)\varphi_s(x)dx. \end{aligned} \tag{2.22}$$

Let $f(x) = 1 - \prod_{i=1}^n f_i(x_i)$ with non-increasing twice continuously differentiable functions $f_i$. For $i \neq j$, $\partial^2 f/(\partial x_i \partial x_j) \leqslant 0$ and $\Sigma^Y_{i,j} - \Sigma^X_{i,j} = \mathrm{Var}(X_i - X_j)/2 - \mathrm{Var}(Y_i - Y_j)/2 \leqslant 0$ by assumption. Thus, $(d/ds)\mathbb{E}\big[f(X(s))\big] \geqslant 0$, which implies $\mathbb{E}\big[f(Y)\big] \geqslant \mathbb{E}\big[f(X)\big]$. The proof is complete by letting $f(x)$ converge to $1 - \prod_{i=1}^n I\{x_i \leqslant t\} = I\{\max_i x_i > t\}$. $\square$

Compared with Slepian's inequality, the Sudakov-Fernique inequality [Sud71, Sud76, Fer74], which compares only the means of the maximum of Gaussian vectors, is easier to apply as it does not require $X_i$ and $Y_i$ to have the same variance. Here we present the Sudakov-Fernique inequality in (2.24) below along with Chatterjee's [Cha05] error bound in (2.23) below for comparison of the means of Gaussian maxima.

**Theorem 2.10.** *Let $X \in \mathbb{R}^n$ and $Y \in \mathbb{R}^n$ be two Gaussian vectors with $\mathbb{E}[X] = \mathbb{E}[Y]$. Let $\gamma = \max_{1\leqslant i\leqslant j\leqslant n} \big|\mathrm{Var}(Y_i - Y_j) - \mathrm{Var}(X_i - X_j)\big|$. Then,*

$$\Big|\mathbb{E}\Big[\max_{i\leqslant n} Y_i\Big] - \mathbb{E}\Big[\max_{i\leqslant n} X_i\Big]\Big| \leqslant \sqrt{\gamma \log n}. \tag{2.23}$$

*If in addition $\mathrm{Var}(X_i - X_j) \leqslant \mathrm{Var}[(Y_i - Y_j)$ for all $1 \leqslant i < j \leqslant n$, then*

$$\mathbb{E}\Big[\max_{i\leqslant n} X_i\Big] \leqslant \mathbb{E}\Big[\max_{i\leqslant n} Y_i\Big]. \tag{2.24}$$

Proof. For $\beta > 0$ let $g = g(x) = \beta^{-1}\log\sum_{i=1}^n e^{\beta x_i}$, $x = (x_1, \ldots, x_n)^\top$, be the "soft-max" function. We have $g(x) - \beta^{-1}\log n \leqslant \max_{1\leqslant i_2\leqslant n} x_i \leqslant g(x)$. Let $p_i = p_i(x) = e^{\beta x_i}/\sum_{j=1}^n e^{\beta x_j}$. As $(\partial g(x))/\partial x_i = p_i$ and $(\partial^2 g(x))/(\partial x_i \partial x_j) = \beta p_i I_{\{i=j\}} - \beta p_i p_j$,

$$\begin{aligned}\sum_{i,j}\left(\frac{\partial^2 g(x)}{\partial x_i \partial x_j}\right)\Sigma^X_{i,j} &= \beta\sum_i p_i \Sigma^X_{i,i} - \beta\sum_{i,j} p_i p_j \Sigma^X_{i,j} \\ &= -\beta\sum_{i,j} p_i p_j \big(\Sigma^X_{i,j} - \Sigma^X_{i,i}/2 - \Sigma^X_{j,j}/2\big).\end{aligned}$$

Let $\gamma^X_{i,j} = \Sigma^X_{i,i} + \Sigma^X_{j,j} - 2\Sigma^X_{i,j}$ and $\gamma^Y_{i,j} = \Sigma^Y_{i,i} + \Sigma^Y_{j,j} - 2\Sigma^Y_{i,j}$. We have

$$\sum_{i,j}\left(\frac{\partial^2 g(x)}{\partial x_i \partial x_j}\right)\Sigma^X_{i,j} = \frac{\beta}{2}\sum_{i,j} p_i p_j \gamma^X_{i,j}.$$

It follows from the above identity, its $Y$-version and (2.22) that

$$\begin{aligned}\mathbb{E}\big[g(Y) - g(X)\big] &= \frac{1}{2}\int_0^1 \mathbb{E}\Big[\sum_{i,j}\Big(\Big(\frac{\partial^2 g}{\partial x_i \partial x_j}\Big)(X(t))\Big)\big(\Sigma^Y_{i,j} - \Sigma^X_{i,j}\big)\Big]dt \\ &= \frac{\beta}{4}\int_0^1 \mathbb{E}\Big[\sum_{i,j} p_i(X(t))p_j(X(t))\big(\gamma^Y_{i,j} - \gamma^X_{i,j}\big)\Big]dt\end{aligned}$$

which is nonnegative under the condition for (2.24). This proves (2.24) by letting $\beta \to \infty$. Moreover, $\big|\mathbb{E}\big[g(Y) - g(X)\big]\big| \leqslant \beta\gamma/4$ as $\gamma = \max_{i,j}\big|\gamma^Y_{i,j} - \gamma^X_{i,j}\big|$, and

$$\Big|\big\{g(Y) - g(X)\big\} - \Big\{\max_{1\leqslant i\leqslant n} Y_i - \max_{1\leqslant i\leqslant n} X_i\Big\}\Big| \leqslant \beta^{-1}\log n.$$

These give (2.23) as $\min_{\beta>0}(\gamma\beta/4 + \beta^{-1}\log n) = \sqrt{\gamma \log n}$. $\square$

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## 3. Hypothesis Testing

3.1. **Uniformly most powerful tests.** Suppose data $X_{1:n}$ are observed under probability $\mathbb{P}_\theta$ with an unknown parameter $\theta \in \Theta$. In hypothesis testing we are compelled to choose between a null hypothesis $H_0 : \theta \in \Theta_0$ and an alternative hypothesis $H_1 : \theta \in \Theta_1$ with two disjoint subsets $\Theta_0$ and $\Theta_1$ of the parameter space $\Theta$. A randomized test is a function

$$\phi = \phi(X_{1:n}) \in [0, 1]$$

representing the probability of rejecting $H_0$ (and thus accepting $H_1$) when $X_{1:n}$ are observed. The test is deterministic when $\phi \in \{0, 1\}$ almost surely under $\mathbb{P}_\theta$ for all $\theta \in \Theta_0 \cup \Theta_1$. Let

$$\text{Type-I error} = \sup_{\theta\in\Theta_0} \mathbb{E}_\theta[\phi], \quad \text{Type-II error} = \sup_{\theta\in\Theta_1} \mathbb{E}_\theta[1-\phi]. \tag{3.1}$$

To control both types of errors, we may minimize Type-II error subject to a certain constraint on Type-I error. This is solved by linear programming:

$$\phi^* = \arg\min_\phi \Big\{\beta : \sup_{\theta\in\Theta_0} \mathbb{E}_\theta[\phi] \leqslant \alpha, \sup_{\theta\in\Theta_1} \mathbb{E}_\theta[1-\phi] \leqslant \beta, 0 \leqslant \phi \leqslant 1\Big\}.$$

For testing a simple null $H_0 : \theta = \theta_0$ against a simple alternative $H_1 : \theta = \theta_1$, the solution of the linear programming, which maximizes the power $\mathbb{E}_{\theta_1}[\phi]$ subject to $\mathbb{E}_{\theta_0}[\phi] \leqslant \alpha$, is given by the well known Neyman-Pearson lemma.

**Theorem 3.1** (Neyman-Pearson lemma)**.** *For testing $H_0 : \theta = \theta_0$ against $H_1 : \theta = \theta_1$, $\phi^*$ is a most powerful (MP) level-$\alpha$ test if for a certain constant $K$*

$$\phi^* = \begin{cases} 1, & d\mathbb{P}_{n,\theta_1}/d\mathbb{P}_{n,\theta_0} > K, \\ 0, & d\mathbb{P}_{n,\theta_1}/d\mathbb{P}_{n,\theta_0} < K, \end{cases} \text{ a.s. under both } \mathbb{P}_{n,\theta_0} \text{ and } \mathbb{P}_{n,\theta_1}, \text{ and } \mathbb{E}_{\theta_0}[\phi^*] = \alpha,$$

*where $\mathbb{P}_{n,\theta}$ is the restriction of $\mathbb{P}_\theta$ in $\sigma(X_{1:n})$. Such $\phi^*$ always exists with a proper choice of $K$. Moreover, $\phi^*$ is the unique MP test when its power $\mathbb{E}_{\theta_1}[\phi^*]$ is less than 1.*

Proof. Let $d\nu = d\mathbb{P}_{n,\theta_1} + d\mathbb{P}_{n,\theta_0}$ and $p_\theta = d\mathbb{P}_{n,\theta}/d\nu$. We have

$$\mathbb{E}_{\theta_1}[\phi^* - \phi] = \int (\phi^* - \phi)(p_{\theta_1} - Kp_{\theta_0})d\nu + K\big(\mathbb{E}_{\theta_0}[\phi^*] - \mathbb{E}_{\theta_0}[\phi]\big) \geqslant 0. \qquad \square$$

For composite alternative hypotheses, a problem with criterion (3.1) is that when the null and alternative hypotheses have a common limiting point, the trivial solution $\phi^* = \alpha$ is optimal. A main aim of the classical testing theory is to construct the uniformly most powerful (UMP) test,

$$\phi^* = \arg\max_\phi \Big\{\mathbb{E}_\theta\big[\phi\big] : \sup_{\theta\in\Theta_0} \mathbb{E}_\theta\big[\phi\big] \leqslant \alpha\Big\} \quad \forall\ \theta \in \Theta_1,$$

which maximizes the power $\mathbb{E}_\theta[\phi]$ simultaneously for *all* $\theta \in \Theta_1$ subject to Type-I error $\leqslant \alpha$. For each $\theta \in \Theta_1$, linear programming provides a set of solutions for possible UMP $\phi^*$, and the UMP test exists iff the intersection of these solution sets is nonempty. When the UMP test does not exist, one may consider the UMP test among all tests satisfying certain constraints.

Explicit solutions of the UMP test can be obtained for one-sided tests when the likelihood ratio is monotone in a certain statistic $T(X_{1:n}) \in \mathbb{R}^k$ under a relationship, say $\preceq$:

$$\theta \preceq \theta' \ \Rightarrow \ \frac{d\mathbb{P}_{n,\theta'}}{d\mathbb{P}_{n,\theta}} \uparrow T(X_{1:n}), \quad \theta \preceq \theta_0 \preceq \theta_1 \ \forall \theta \in \Theta_0 \ \& \ \theta_1 \in \Theta_1, \ \exists\, \theta_0 \in \Theta_0. \tag{3.2}$$

**Theorem 3.2.** *Under* (3.2), *$\phi^*$ is the UMP level-$\alpha$ test if for a certain constant $K_1$*

$$\phi^* = \begin{cases} 1, & T(X_{1:n}) > K_1, \\ 0, & T(X_{1:n}) < K_1, \end{cases} \quad \textit{and} \quad \mathbb{E}_{\theta_0}[\phi^*] = \alpha.$$

Proof. By Theorem 3.1, $\phi^*$ is MP for testing $\theta = \theta_0$ against $\theta = \theta_1$, simultaneously for all $\theta_1 \in \Theta_1$, as $K_1$ and $\phi|(T = K_1)$ are uniquely determined by $\mathbb{E}_{\theta_0}[\phi^*] = \alpha$. For $\theta_0' \in \Theta_0$, $\phi^*$ is MP for testing $\theta = \theta_0'$ against $\theta = \theta_0$ at the level $\mathbb{E}_{\theta_0'}[\phi^*]$, so that $\mathbb{E}_{\theta_0'}[\phi^*] \leqslant \alpha$. □

The above MP and UMP tests provides approximate MP and UMP tests under the LAN condition in view of Theorem 2.2 (ii).

For certain hypothesis testing problems involving two-sided alternatives and nuisance parameters, the UMP test can be obtained under the constraint that the test be unbiased:

$$\mathbb{E}_\theta[\phi] \geqslant \alpha \ \ \forall\, \theta \in \Theta_1.$$

A UMP test under the above constraint is called a UMP unbiased (UMPU) test. We say that a density family $p_\theta$ with respect to a measure $d\nu$ is $L_1(\nu)$-continuous at $\theta_0$ if $\int |p_\theta - p_{\theta_0}| d\nu = o(1)$ as $\theta \to \theta_0$, and $L_1(\nu)$-differentiable at $\theta_0$ if $\int |p_\theta - p_{\theta_0} - (\theta - \theta_0)^\top \dot{p}_{\theta_0}| d\nu = o(\|\theta - \theta_0\|_2)$ for some function $\dot{p}_{\theta_0}$.

**Theorem 3.3.** *Consider testing $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$ for $\theta \in \Theta \subseteq \mathbb{R}$, where $\theta_0$ is an interior point of $\Theta$. Suppose $X_{1:n} \sim p_\theta d\nu$ and $p_\theta$ is $L_1(\nu)$-continuous and differentiable in $\theta$ at $\theta = \theta_0$ with derivative $\dot{p}_{\theta_0}$. Then, $\phi^*$ is UMPU if*

$$\phi^* = \begin{cases} 1, & p_\theta - K_\theta p_{\theta_0} + c_\theta \dot{p}_{\theta_0} > 0, \\ 0, & p_\theta - K_\theta p_{\theta_0} - c_\theta \dot{p}_{\theta_0} < 0, \end{cases} \forall\, \theta \neq \theta_0, \quad \int \phi^* p_{\theta_0} d\nu = \alpha, \quad \int \phi^* \dot{p}_{\theta_0} d\nu = 0, \tag{3.3}$$

*for some constants $K_\theta$ and $c_\theta$ depending on $\theta$ only.*

Proof. Due to the smoothness $p_\theta$, any unbiased test $\phi$ must satisfy $\int \phi p_{\theta_0} d\nu = \alpha$ and $\int \phi \dot{p}_{\theta_0} d\nu = 0$. Thus, $\mathbb{E}_\theta[\phi^* - \phi] = \int (\phi^* - \phi)(p_\theta - K_\theta p_{\theta_0} - c_\theta \dot{p}_{\theta_0}) d\nu \geqslant 0$ for all $\theta \in \Theta_1$. □

**Example 3.1.** *Under the conditions of Theorem 3.3, consider an exponential family*

$$p_\theta(X_{1:n}) \propto \exp\big[\theta T - \psi(\theta)\big].$$

*with certain $T = T(X_{1:n})$. Then, $\phi^*$ is UMPU for testing $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$ if*

$$\phi^* = \begin{cases} 1, & T \notin [a, b], \\ 0, & T \in (a, b), \end{cases} \forall\, \theta \neq \theta_0, \quad \mathbb{E}_{\theta_0}[\phi^*] = \alpha, \quad \mathbb{E}_{\theta_0}[(\phi^* - \alpha)T] = 0,$$

*for certain constants $a < b$. We note that*

$$p_\theta(t) - K_\theta p_{\theta_0}(t) - c_\theta \dot{p}_{\theta_0}(t) = p_{\theta_0}(t)\Big(p_\theta(t)/p_{\theta_0}(t) - K_\theta - c_\theta(t - \dot{\psi}(\theta_0))\Big),$$

*so that $(K_\theta, c_\theta)$ is the solution of $p_\theta(t)/p_{\theta_0}(t) - K_\theta - c_\theta(t - \dot{\psi}(\theta_0)) = 0$, $t \in \{a, b\}$. The solution satisfied* (3.3) *due to the convexity of $p_\theta(t)/p_{\theta_0}(t)$ in $t$.*

For composite hypotheses involving nuisance parameters, the UMPU tests can be constructed by conditioning.

**Lemma 3.1.** *Suppose $X_{1:n} \sim p_{\tau,\eta} d\nu$ with $\theta = (\tau, \eta^\top)^\top \in \Theta$, where $\tau$ is real and $\Theta$ is an open set in a Euclidean space. Suppose $p_{\tau,\eta}$ is $L_1(\nu)$-continuous in $\tau$ at $\tau_0$ and there exists a sufficient and complete statistic $U = U(X_{1:n})$ with the family $\{p_\theta : \theta \in \Theta, \tau = \tau_0\}$. Then, any level-$\alpha$ unbiased one-sided test for $H_0 : \tau \leqslant \tau_0$ against $H_1 : \tau > \tau_0$ must satisfy $\mathbb{E}_{\tau_0,\eta}[\phi|U] = \alpha$. If in addition $p_{\tau,\eta}$ is $L_1(\nu)$-differentiable in $\tau$ at $\tau_0$ with a score function $\dot{\ell}_{\tau_0|\eta} = (\partial/\partial\tau) \log p_{\tau,\eta}\big|_{\tau=\tau_0}$ of the form $\dot{\ell}_{\tau_0|\eta} = T - \mathbb{E}_{\tau_0,\eta}[T]$ with a random variable $T$ not dependent on $\eta$, then any level-$\alpha$ unbiased two-sided test for $H_0 : \tau = \tau_0$ against $H_1 : \tau \neq \tau_0$ must satisfy both equations $\mathbb{E}_{\tau_0,\eta}[\phi|U] = \alpha$ and $\mathbb{E}_{\tau_0,\eta}[(\phi - \alpha)T|U] = 0$.*

PROOF. Due to the $L_1$ continuity, any unbiased test $\phi$ must satisfy $\mathbb{E}_{\tau_0,\eta}\big[\mathbb{E}_{\tau_0,\eta}[\phi|U]\big] = \mathbb{E}_{\tau_0,\eta}[\phi] = \alpha$. Due to the sufficiency of $U$ on the boundary, $\mathbb{E}_{\tau_0,\eta}[\phi|U]$ is a statistic. Thus, due to the completeness of $U$ on the boundary, $\mathbb{E}_{\tau_0,\eta}[\phi|U] = \alpha$ almost surely. Similarly, for two-sided test, $\mathbb{E}_{\tau_0,\eta}[\phi\, \dot{\ell}_{\tau_0|\eta}] = 0$ as in the proof of Theorem 3.3, so that $\mathbb{E}_{\tau_0,\eta}[(\phi - \alpha)T] = 0$ for all $\eta$. Hence, the sufficiency and completeness of $U$ also yield $\mathbb{E}_{\tau_0,\eta}[(\phi - \alpha)T|U] = 0$. □

**Theorem 3.4.** *Consider an exponential family*

$$p_{\tau,\eta}(X_{1:n}) = \exp\big[\tau T(X_{1:n}) + \eta^\top U(X_{1:n}) - \psi(\tau, \eta)\big], \theta = (\tau, \eta^\top) \in \Theta \subset \mathbb{R}^{1+k},$$

*with real $\tau$ and open $\Theta$. Then, $\phi^*$ is UMPU for testing $H_0 : \tau \leqslant \tau_0$ vs $H_1 : \tau > \tau_0$ if*

$$\phi^*(T, U) = \begin{cases} 1, & T > K(U), \\ 0, & T < K(U), \end{cases} \quad \mathbb{E}_{\tau_0,\eta}\big[\phi^*\big|U\big] = \alpha,$$

*and $\phi^*$ is UMPU for testing $H_0 : \tau = \tau_0$ vs $H_1 : \tau \neq \tau_0$ if*

$$\phi^*(T, U) = \begin{cases} 1, & T \notin [a(U), b(U)], \\ 0, & T \in (a(U), b(U)), \end{cases} \quad \mathbb{E}_{\tau_0,\eta}\big[\phi^*\big|U\big] = \alpha,\ \mathbb{E}_{\tau_0,\eta}\big[(\phi^* - \alpha)T\big|U\big] = 0.$$

PROOF. Given $\tau = \tau_0$, $U$ is sufficient and complete for the subfamily $p_{\tau_0,\eta}(X_{1:n}) \propto \exp\big[\eta^\top U(X_{1:n}) - \psi(\tau_0, \eta)\big]$, so that $\mathbb{E}_{\tau_0,\eta}\big[\phi\big|U\big] = \alpha$ for unbiased tests $\phi$ by Lemma 3.1. Consequently, because the conditional density given $U$ has monotone likelihood ratio, the one-sided $\phi^*$ as specified is UMP conditionally on $U$ by Theorem 3.2. This implies the UMPU property of $\phi^*$ without conditioning. For the two-sided tests, Lemma 3.1 provided the additional constraint $\mathbb{E}_{\tau_0,\eta}\big[(\phi - \alpha)T\big|U\big] = 0$ for unbiased $\phi$, and Theorem 3.3 can be used instead of Theorem 3.2 in the same proof. □

**Example 3.2** (t-tests)**.** *Suppose iid $X_1, \ldots, X_n$ from $N(\mu, \sigma)$ are observed. The one-sided t-test is UMPU for testing $H_0 : \mu \leqslant \mu_0$ against $H_1 : \mu > \mu_0$, and the two-sided t-test is UMPU for testing $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$. Similarly, the t-tests are UMPU for testing about an estimable linear combination $a_0^\top \beta$ of the regression coefficients in the linear model $Y_{1:n}|X_{1:n} \sim N(X_{1:n}\beta, \sigma I_n)$ with unknown $\beta \in \mathbb{R}^d$ and $\sigma > 0$.*

The above explicit UMPU tests provides approximate UMPU tests under the LAN condition in view of Theorem 2.2 (ii).

UMP tests can be also obtained in certain problems under invariance constraints. A statistical model $X_{1:n} \sim p_\theta, \theta \in \Theta$, is invariant if there exist groups of invertible transformations $\mathscr{G}$ and $\overline{\mathscr{G}}$ such that for every $g \in \mathscr{G}$ there exists $\overline{g} \in \overline{\mathscr{G}}$ satisfying

$$X_{1:n} \sim p_\theta \;\Rightarrow\; g(X_{1:n}) \sim p_{\overline{g}\theta}.$$

In this case a testing problem is invariant if

$$\overline{g}\Theta_0 = \Theta_0,\ \overline{g}\Theta_1 = \Theta_1,\ \forall \overline{g} \in \overline{\mathscr{G}}.$$

Given an invariant testing problem as the above, a decision rule $\phi$ is invariant if

$$\phi(g(X_{1:n})) = \phi(X_{1:n}),\ \forall g \in \mathscr{G}.$$

A statistic $T = T(X_{1:n})$ is invariant if it depends on $X_{1:n}$ only through its orbit $\{g(X_{1:n}) : g \in \mathscr{G}\}$, i.e. $T(g(X_{1:n})) = T(X_{1:n})$, $\forall g \in \mathscr{G}$, and $T = T(X_{1:n})$ is maximum invariant if it is invariant and takes different values in different orbits of $T$, i.e. $T(X'_{1:n}) \neq T(X_{1:n})$ when $X'_{1:n} \neq g(X_{1:n})$ for all $g \in \mathscr{G}$. If $T$ is maximum invariant, then all invariant tests are functions of $T$, say a function of the form $\phi(T) = \phi(T(X_{1:n}))$.

**Example 3.3** (F-test). *Suppose $X_{1:n} \sim N(\mu, \sigma I_n)$ are observed, and we wish to test $H_0 : \mu \in V_0$ against $H_1 : \mu \in V_1 \backslash V_0$ for some nested subspaces $V_0 \subset V_1 \subset \mathbb{R}^n$. Let $P_j$, $j = 0, 1, 2$ be respectively the orthogonal projections from $\mathbb{R}^n$ to $V_0$, $V_0^\perp \cap V_1$ and $V_1^\perp$. Let $d_j = \mathrm{rank}(P_j)$. Assume $d_1 \wedge d_2 > 0$. Let $\mathscr{G}$ be the collection of all mappings $g \in \mathbb{R}^n \to \mathbb{R}^n$ of the form*

$$g(X_{1:n}) = \big(P_0 X_{1:n} - u_0 + U_1 P_1 X_{1:n} + U_2 P_2 X_{1:n}\big)/c$$

*with $c > 0$, $u_0 \in V_0$, orthonormal $U_j$ satisfying $U_j P_j = P_j U_j$, $j = 1, 2$, and the corresponding*

$$\overline{g}(\mu, \sigma) = \big(\big(P_0\mu - u_0 + U_1 P_1 \mu + U_2 P_2 \mu\big)/c, \sigma/c\big).$$

*Let $T = (\|P_1 X_{1:n}\|_2^2/d_1)/(\|P_2 X_{1:n}\|_2^2/d_2)$ be the F-statistic, and for $j = 1, 2$, $u_j$ fixed unit vectors satisfying $u_j = P_j u_j$. Given $X_{1:n}$, $g(X_{1:n}) = \sqrt{T d_1/d_2}\, u_1 + u_2$ is in its orbit with $u_0 = P_0 X_{1:n}$, $U_j P_j X_{1:n} = \|P_j X_{1:n}\|_2 u_j$, $j = 1, 2$ and $c = \|P_2 X_{1:n}\|_2$. Because $T$ is a one-to-one mapping of this $g(X_{1:n})$, $T$ is maximum invariant. Under $H_0$, $T$ has the $F_{d_2,d_1}$-density*

$$f_{d_2,d_1}(t) = \big((d_2/d_1)^{d_2/2}/B(d_2/2, d_1/2)\big)t^{d_2/2-1}/(1 + t d_2/d_1)^{(d_2+d_1)/2}$$

*with $B(\alpha, \beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha + \beta)$. Under $H_1$, $T$ has the non-central $F_{d_2,d_1,\kappa}$-density*

$$f_{d_2,d_1,\kappa}(t) = \sum_{k=0}^{\infty} \frac{e^{-\kappa/2}(\kappa/2)^k (d_2/d_1)^{d_2/2+k} t^{d_2/2+k-1}}{k! B(d_2/2 + k, d_1/2)(1 + t d_2/d_1)^{(d_2+d_1)/2+k}}.$$

*Because the likelihood ratio*

$$\frac{f_{d_2,d_1,\kappa}(t)}{f_{d_2,d_1}(t)} = \sum_{k=0}^{\infty} \frac{C_{d_2,d_1,\kappa,k} t^k}{(1 + t d_2/d_1)^k}, \quad C_{d_2,d_1,\kappa,k} > 0,$$

*is increasing in $t$, the F-test is UMP invariant (UMPI) by Theorem 3.2.*

Again the F-test provides approximate UMPI tests under the LAN condition in view of Theorem 2.2 (ii).

**Bayes tests:** Consider a prior $G$ under which

$$\mathbb{P}_G\{H_j\} = \pi_j,\ \theta\big|H_j \sim G_j,\ j = 0, 1,$$

with $\pi_0 + \pi_1 = 1$ and distributions $G_j$ satisfying $G_j(\Theta_j) = 1$. Consider the loss function

$$L_\lambda(\phi, \theta) = \lambda\phi I\{\theta \in \Theta_0\} + (1 - \phi)I\{\theta \in \Theta_1\}.$$

Given the data, the posterior risk is

$$\mathbb{E}_G\big[L_\lambda(a, \theta)\big|X_{1:n}\big] = \begin{cases} \lambda\mathbb{P}_G\big\{\theta \in \Theta_0\big|X_{1:n}\big\}, & a = 1, \\ \mathbb{P}_G\big\{\theta \in \Theta_1\big|X_{1:n}\big\}, & a = 0. \end{cases}$$

Thus, the Bayes rule is

$$\phi = \begin{cases} 1, & \mathbb{P}_G\big\{\theta \in \Theta_1\big|X_{1:n}\big\}/\mathbb{P}_G\big\{\theta \in \Theta_0\big|X_{1:n}\big\} \geqslant \lambda, \\ 0, & \mathbb{P}_G\big\{\theta \in \Theta_1\big|X_{1:n}\big\}/\mathbb{P}_G\big\{\theta \in \Theta_0\big|X_{1:n}\big\} < \lambda. \end{cases}$$

When $X_{1:n} \sim p_\theta d\nu$ given $\theta$, $\mathbb{P}_G\big\{\theta \in \Theta_1\big|X_{1:n}\big\}/\mathbb{P}_G\big\{\theta \in \Theta_0\big|X_{1:n}\big\} = (\pi_1/\pi_0)\Lambda$, where

$$\Lambda = \frac{\int p_\theta(X_{1:n})G_1(d\theta)}{\int p_\theta(X_{1:n})G_0(d\theta)} = \frac{p_\theta(X_{1:n}|H_1)}{p_\theta(X_{1:n}|H_0)},$$

called the Bayes factor, is the ratio of the marginal densities of the data given the hypotheses. This relationship can be written as the multiplicative rule

$$\text{posterior odds} = \text{Bayes factor} \times \text{prior odds}.$$

As $\Lambda$ is the marginal likelihood ratio, the Bayes test is the Neyman-Pearson test using the marginal densities $\int p_\theta(X_{1:n})G_1(d\theta)$ and $\int p_\theta(X_{1:n})G_0(d\theta)$.

3.2. **Generalized likelihood ratio tests.** For testing $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1 = \Theta \backslash \Theta_0$, the generalized likelihood ratio test (GLRT) can be written as

$$\Lambda_n = 2 \log \left( \frac{\sup_{\theta \in \Theta} L_n(\theta)}{\sup_{\theta \in \Theta_0} L_n(\theta)} \right), \quad \phi = \begin{cases} 1, & \Lambda_n > K, \\ 0, & \Lambda_n < K, \end{cases} \tag{3.4}$$

where $L_n(\theta)$ is the likelihood function and $K$ is a constant. The idea is to use the MLEs

$$\widehat{\theta}_n = \arg\max_{\theta \in \Theta} L_n(\theta), \quad \widehat{\theta}_{j,n} = \arg\max_{\theta \in \Theta_j} L_n(\theta), \tag{3.5}$$

to approximate the ideal members $\Theta$ and $\Theta_j$, $j = 0, 1$ under respective hypotheses in the Neyman-Pearson lemma,

$$\Lambda_n = 2 \log \left( L_n(\widehat{\theta}_n) / L_n(\widehat{\theta}_{0,n}) \right) = \max \left( 0, 2 \log \left( L_n(\widehat{\theta}_{1,n}) / L_n(\widehat{\theta}_{0,n}) \right) \right).$$

The GLRT can be approximated by its limiting Gaussian version under conditions comparable to those for the asymptotic normality of the MLE.

**Theorem 3.5.** *Let $\Theta$ be an open set in $\mathbb{R}^d$ and $\{p_\theta, \theta \in \Theta\}$ be a fixed density family with respect to a measure $\nu$. Consider testing $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta \backslash \Theta_0$ based on $X_{1:n}$ with iid elements from $\mathbb{P}_\theta$. Let $\theta_0 \in \Theta_0$. Suppose for some $\psi = \psi(x)$ with $\mathbb{E}_{\theta_0}[\psi^2] < \infty$,*

$$\Big| \log \left( (p_{\theta_1}/p_{\theta_0}) \vee (1/4) \right) - \log \left( (p_{\theta_2}/p_{\theta_0}) \vee (1/4) \right) \Big| \leqslant \psi \|\theta_1 - \theta_2\|_2 \tag{3.6}$$

*when $\|\theta_1 - \theta_0\|_2 \vee \|\theta_2 - \theta_0\|_2 \leqslant \epsilon_0$, for some constant $\epsilon_0 > 0$, and that*

$$\int \left( \sqrt{p_\theta} - \sqrt{p_{\theta_0}} - 2^{-1} (\theta - \theta_0)^\top \dot{\ell}_{\theta_0}(x) \sqrt{p_{\theta_0}} \right)^2 d\nu = o(\|\theta - \theta_0\|_2^2) \tag{3.7}$$

*for some $\dot{\ell}_{\theta_0} \in \mathbb{R}^d$ with a invertible $I_{\theta_0} = \mathbb{P}[\dot{\ell}_{\theta_0} \dot{\ell}_{\theta_0}^\top]$. Suppose $\|\widehat{\theta}_n - \theta_0\|_2 = o_{\mathbb{P}_{\theta_0}}(1)$ and $\|\widehat{\theta}_{0,n} - \theta_0\|_2 = o_{\mathbb{P}_{\theta_0}}(1)$ for the MLEs in (3.5). Suppose $\{h : \theta_0 + n^{-1/2} h \in \Theta_0\} \to \mathscr{T}_0$. Then,*

$$\mathscr{L}\left( \Lambda_n ; \mathbb{P}_{\theta_0 + n^{-1/2} h^*} \right) \rightsquigarrow \mathscr{L}\left( \Lambda_\infty = \inf_{h \in \mathscr{T}_0} \left\| I_{\theta_0}^{-1/2} Z - I_{\theta_0}^{1/2} h \right\|_2^2 ; \mathbb{P}_{h^*} \right) \tag{3.8}$$

*with $\mathscr{L}(Z; \mathbb{P}_{h^*}) \sim N(I_{\theta_0} h^*, I_{\theta_0})$. Consequently, when $\mathscr{T}_0$ is a $d_0$-dimensional subspace of $\mathbb{R}^d$, $\mathscr{L}\left( \Lambda_\infty ; \mathbb{P}_{h^*} \right) \sim \chi^2_{d_1, \kappa_{h^*}}$ with $d_1 = d - d_0$ and $\kappa_{h^*} = \inf_{h \in \mathscr{T}_0} \left\| I_{\theta_0}^{1/2} h^* - I_{\theta_0}^{1/2} h \right\|_2^2$.*

When $\theta_0$ is an interior point of $\Theta$ and $\Theta_0 = \{\theta : \tau(\theta) = \tau(\theta_0)\}$ for a differentiable $\tau : \mathbb{R}^d \to \mathbb{R}^{d_0}$, $\mathscr{T}_0 = \dot{\tau}_{\theta_0} \mathbb{R}^d$ is a $d_0$-dimensional subspace of $\mathbb{R}^d$ when $\dot{\tau}_{\theta_0}$ is of rank $d_0$.

Proof. By (3.6), (3.7) and Theorem 1.5, the MLEs $\widehat{\theta}_n$ and $\widehat{\theta}_{0,n}$ are $n^{-1/2}$-consistent under $\mathbb{P}_{\theta_0}$. Moreover, as in the proofs of Corollary 1.2 and Theorem 1.5,

$$\sup_{\|h\|_2 \leqslant C} \left| n \overline{\mathbb{P}}_n \left[ \log \left( p_{\theta_0 + n^{-1/2} h} / p_{\theta_0} \right) \right] - \left( h^\top Z_n - h^\top I_{\theta_0} h/2 \right) \right| = o_{\mathbb{P}_{\theta_0}}(1), \ \forall C > 0,$$

with $Z_n = n^{1/2} \overline{\mathbb{P}}_n \left[ \dot{\ell}_{\theta_0} \right] \rightsquigarrow Z$ under $\mathbb{P}_{\theta_0}$. This gives (3.8) with

$$\begin{aligned} \Lambda_\infty &= 2 \sup_{h \in \mathbb{R}^d} \left( h^\top Z - h^\top I_{\theta_0} h/2 \right) - 2 \sup_{h \in \mathscr{T}_0} \left( h^\top Z - h^\top I_{\theta_0} h/2 \right) \\ &= -\inf_{h \in \mathbb{R}^d} \left\| I_{\theta_0}^{-1/2} Z - I_{\theta_0}^{1/2} h \right\|_2^2 + \inf_{h \in \mathscr{T}_0} \left\| I_{\theta_0}^{-1/2} Z - I_{\theta_0}^{1/2} h \right\|_2^2 \end{aligned}$$

for $h^* = 0$ and then by Theorem 2.2 for general $h^* \in \mathbb{R}^d$. $\square$

**Definition 3.1** (Uniform LAN). *Let $\{(\mathbb{P}_{n,\theta}, \mathscr{X}_n, \mathscr{A}_n) : \theta \in \Theta_n\}$ be a sequence of statistical models. Let $\mathscr{H}_0$ be a closed separable subspace of a Hilbert space $\mathscr{H}$ equipped with inner product $\langle \cdot, \cdot \rangle_{\mathscr{H}}$, and $\mathbb{P}_0$ be a probability under which $\{Z(h), h \in \mathscr{H}_0\}$ is a centered Gaussian process with $\mathrm{Cov}_0(Z(h), Z(h')) = \langle h, h' \rangle_{\mathscr{H}}$ for all paires $\{h, h'\}$ in $\mathscr{H}_0$. Let $K_n$ be subsets of $\Theta_n$ and $h_n = h_n(\theta)$ be mappings from $K_n$ into $\mathscr{H}_0$ with $h_n(\theta_{0,n}) = 0$ for certain $\theta_{0,n} \in K_n$. Let $K_{n,C} = K_n \cap \{\theta : \|h_n(\theta)\|_{\mathscr{H}} \leqslant C\}$. The model sequence is uniformly LAN in the neighborhoods $K_{n,C}$ of $\theta_{0,n}$ if*

$$\mathbb{E}^*_{n,\theta_{0,n}}\left[f\left(\log\frac{d\,\mathbb{P}_{n,\theta}}{d\,\mathbb{P}_{n,\theta_{0,n}}}, \theta \in K_{n,C}\right)\right] = \mathbb{E}_0\left[f\left(Z(h_n(\theta)) - \frac{\|h_n(\theta)\|^2_{\mathscr{H}}}{2}, \theta \in K_{n,C}\right)\right] + o(1)$$

*for all uniformly bounded and continuous mappings $f$ from $L_\infty(K_{n,C}) \to \mathbb{R}$. The local experiments $\{(\mathbb{P}_{n,\theta}, \mathscr{X}_n, \mathscr{A}_n) : \theta \in K_{n,C}\}$ converges to a limit if*

$$\mathscr{T}_C = \cap_{\epsilon>0}\overline{\left\{\liminf_n \big(h_n(K_{n,C})\big)_\epsilon\right\}} = \cap_{\epsilon>0}\overline{\left\{\limsup_n \big(h_n(K_{n,C})\big)_\epsilon\right\}},$$

*where $(A)_\epsilon = \cup_{h' \in A}\{h : \|h - h'\|_{\mathscr{H}} \leqslant \epsilon\}$. We call $\overline{\mathscr{T}} = \overline{\sup_{C>0}\mathscr{T}_C}$ the tangent set.*

**Theorem 3.6.** *Suppose $\{(\mathbb{P}_{n,\theta}, \mathscr{X}_n, \mathscr{A}_n) : \theta \in \Theta_n\}$ is uniformly LAN in the neighborhoods $K_{n,C}$ of $\theta_{0,n}$ with $h_n(\theta_{0,n}) = 0$ and a tangent set $\overline{\mathscr{T}}$ in the limit. Consider testing $H_0 : \theta \in \Theta_{0,n}$ against $H_1 : \theta \in \Theta_{1,n} = \Theta_n \backslash \Theta_{0,n}$. Suppose*

$$\mathscr{T}_{0,C} = \cap_{\epsilon>0}\overline{\left\{\liminf_n \big(h_n(K_{n,C} \cap \Theta_{0,n})\big)_\epsilon\right\}} = \cap_{\epsilon>0}\overline{\left\{\limsup_n \big(h_n(K_{n,C} \cap \Theta_{0,n})\big)_\epsilon\right\}}.$$

*Let $\overline{\mathscr{T}}_0 = \overline{\cup_{C>0}\mathscr{T}_{0,C}}$ be the tangent set for $H_0$. Suppose the MLEs in (3.5) satisfy*

$$\lim_{C\to\infty}\liminf_{n\to\infty}\mathbb{P}_{\theta_{0,n}}\left\{\widehat{\theta}_n \in K_{n,C},\ \widehat{\theta}_{0,n} \in K_{n,C} \cap \Theta_{0,n}\right\} = 1.$$

*Then, for any $h^* \in \mathscr{T}$ and $\theta_n \in K_n$ satisfying $\|h_n(\theta_n) - h^*\|_{\mathscr{H}} = o(1)$,*

$$\mathscr{L}\big(\Lambda_n; \mathbb{P}_{\theta_n}\big) \rightsquigarrow \lim_{C\to\infty}\mathscr{L}\big(\Lambda_{\infty,C}; \mathbb{P}_{h^*}\big) \tag{3.9}$$

*with a $\mathbb{P}_{h^*}$ satisfying $\mathscr{L}(Z(h), h \in \mathscr{H}_0; \mathbb{P}_{h^*}) \sim \mathscr{L}(Z(h) + \langle h^*, h \rangle_{\mathscr{H}}, h \in \mathscr{H}_0; \mathbb{P}_0)$ and*

$$\Lambda_{\infty,C} = 2\sup_{h\in\overline{\mathscr{T}}_C}\big(Z(h) - \|h\|^2_{\mathscr{H}}/2\big) - 2\sup_{h\in\overline{\mathscr{T}}_{0,C}}\big(Z(h) - \|h\|^2_{\mathscr{H}}/2\big).$$

*If in addition $\overline{\mathscr{T}}$ and $\overline{\mathscr{T}}_0$ are cones, then $\mathscr{L}\big(\Lambda_{\infty,C}; \mathbb{P}_{h^*}\big) \rightsquigarrow \mathscr{L}\big(\Lambda_\infty; \mathbb{P}_{h^*}\big)$ where*

$$\Lambda_\infty = \sup_{u\in\overline{\mathscr{T}}, \|u\|_{\mathscr{H}}=1}\big(Z(u)\big)^2_+ - \sup_{u\in\overline{\mathscr{T}}_0, \|u\|_{\mathscr{H}}=1}\big(Z(u)\big)^2_+.$$

We note that Theorem 3.6 is an extension of Theorem 3.5 with $\mathscr{H} = L_2(\mathbb{P}_{\theta_0})$, $h_n(\theta) = n^{1/2}(\theta - \theta_0)^\top \dot{\ell}_{\theta_0}$, $K_n = \Theta$ and $\mathscr{T}_C = \left\{h : \mathbb{E}_{\theta_0}[h^2] \leqslant C^2\right\}$. When $d\mathbb{P}_{h^*}/d\mathbb{P}_0 = e^{Z(h^*) - \|h^*\|^2_{\mathscr{H}}/2}$,

$$\mathbb{E}_{h^*}\big[e^{tZ(h)}\big] = e^{\|th+h^*\|^2_{\mathscr{H}}/2 - \|h^*\|^2_{\mathscr{H}}/2} = \mathbb{E}_0\big[e^{tZ(h) + t\langle h^*, h\rangle_{\mathscr{H}}}\big].$$

Proof. Let $\Lambda_{n,C} = 2\sup_{\theta\in K_{n,C}}\log(L_n(\theta)) - 2\sup_{\theta\in K_{n,C}\cap\Theta_{0,n}}\log(L_n(\theta))$. It follows from the uniform LAN condition and the convergence of $h_n(K_{n,C})$ to $\mathscr{T}_C$ and $h_n(K_{n,C} \cap \Theta_{0,n})$ to $\mathscr{T}_{0,C}$ that $\mathscr{L}\big(\Lambda_{n,C}; \mathbb{P}_{\theta_n}\big) \rightsquigarrow \mathscr{L}\big(\Lambda_{\infty,C}; \mathbb{P}_{h^*}\big)$. Thus, (3.9) holds because $\widehat{\theta}_n \in K_{n,C}$ and $\widehat{\theta}_{0,n} \in K_{n,C} \cap \Theta_{0,n}$ with high probability. When $\overline{\mathscr{T}}$ and $\overline{\mathscr{T}}_0$ are cones, $2\max_{t>0}\big(Z(tu) - \|tu\|^2_{\mathscr{H}}/2\big) = (Z(u))^2_+$ for any $u \in \mathscr{T}$ with $\|u\|_{\mathscr{H}} = 1$, so that the limit is given by $\Lambda_\infty$. $\square$

The following theorem provides sufficient conditions for the convergence of the MLE of the underlying density in the Hellinger distance and the uniform LAN condition.

**Theorem 3.7.** *Let $\{p_\theta, \theta \in \Theta_n\}$ be a sequence of density families and $X_1, \ldots, X_n$ be iid observations from $p_{\theta_{0,n}}$ under $\mathbb{P}_{0,n}$. Let $\mathscr{H}$ be a fixed Hilbert space and $g_n$ be distance preserving mappings from $L_2(\mathbb{P}_{0,n})$ to $\mathscr{H}$. For $\theta \in \Theta_n$, let*

$$H_\theta^2 = \mathbb{E}_{0,n}\big[\big(\sqrt{p_\theta}/\sqrt{p_{\theta_{0,n}}} - 1\big)^2\big] = 2\,\mathbb{E}_{0,n}\big[1 - \sqrt{p_\theta}/\sqrt{p_{\theta_{0,n}}}\big]$$

*be the squared Hellinger distance between $p_\theta$ and $p_{\theta_{0,n}}$, $\sigma_\theta = (4H_\theta^2 - H_\theta^4)^{1/2}$,*

$$u_\theta = \big(2\sqrt{p_\theta}/\sqrt{p_{\theta_{0,n}}} - 2 + H_\theta^2\big)/\sigma_\theta \quad \text{and} \quad h_n(\theta) = n^{1/2} g_n\big(\sigma_\theta u_\theta\big).$$

*Let $\theta_{0,n} \in K_n \subset \Theta_n$ with $\sup_{\theta \in K_n} H_\theta^2 \leqslant c_0 < 4$ and $\widehat{\theta}_n = \arg\max_{\theta \in K_n} \overline{\mathbb{P}}_n[\log p_\theta]$. Suppose*

$$\mathbb{E}_{0,n}^*\big[f\big(n^{1/2}\overline{\mathbb{P}}_n\big[u_\theta\big], \theta \in K_{n,C}\big)\big] = \mathbb{E}_0\big[f\big(Z(u_\theta), \theta \in K_{n,C}\big)\big] + o(1) \tag{3.10}$$

*for all $f$ uniformly bounded and continuous mappings from $L_\infty(K_{n,C}) \to \mathbb{R}$, and*

$$\sup_{\theta \in K_n} \Big\{\big|\overline{\mathbb{P}}_n\big[u_\theta^2\big] - 1\big| + \overline{\mathbb{P}}_n\big[(u_\theta^2 - M_n)_+\big]\Big\} + M_n/n = o_{\mathbb{P}_{0,n}}(1) \tag{3.11}$$

*for some constants $M_n > 0$. Then, $H_{\widehat{\theta}_n} = O_{\mathbb{P}_{0,n}}(n^{-1/2})$ and family $\{p_\theta, \theta \in K_n\}$ is uniformly LAN.*

Suppose $\Theta \subset \mathbb{R}^d$ is fixed and $p_\theta$ is differentiable in quadratic mean as in (3.7). Let $K_n = \Theta$. Then, $h^* \in \mathscr{T}$ iff there exist $\theta_n \in \Theta$ such that $2H_{\theta_n} = (1 + o(1))n^{-1/2}\|h^*\|_{\mathscr{H}}$,

$$n^{1/2}(\theta_n - \theta_0) \to I_{\theta_0}^{-1}\mathbb{E}_{\theta_0}\big[\dot{\ell}_{\theta_0} g_n^{-1}(h^*)\big], \quad \text{and} \quad \big\|g_n\big(n^{1/2}(\theta_n - \theta_0)^\top \dot{\ell}_{\theta_0}\big) - h^*\big\|_{\mathscr{H}} \to 0.$$

Proof. By assumption, $\sqrt{4 - c_0}H_\theta \leqslant \sigma_\theta \leqslant 2H_\theta$ for $\theta \in K_n$. Due to $\log(1+x) \leqslant x$,

$$\overline{\mathbb{P}}_n\big[\log\big(p_\theta/p_{\theta_{0,n}}\big)\big] \leqslant \overline{\mathbb{P}}_n\big[2\big(\sqrt{p_\theta}/\sqrt{p_{\theta_{0,n}}} - 1\big)\big] = \overline{\mathbb{P}}_n\big[\sigma_\theta u_\theta\big] - H_\theta^2.$$

Thus, because $0 \leqslant \overline{\mathbb{P}}_n\big[\log\big(p_{\widehat{\theta}_n}/p_{\theta_{0,n}}\big)\big]$, (3.10) provides

$$H_{\widehat{\theta}_n}^2 \leqslant \sigma_{\widehat{\theta}_n} \sup_{\theta \in K_n} \overline{\mathbb{P}}_n\big[u_\theta\big] = O_{\mathbb{P}_{0,n}}(n^{-1/2}) H_{\widehat{\theta}_n}.$$

It follows that $H_{\widehat{\theta}_n} = O_{\mathbb{P}_{0,n}}(n^{-1/2})$. For $\theta \in K_{n,C}$, $n\sigma_\theta^2 = \|h_n(\theta)\|_{\mathscr{H}}^2 \leqslant C^2$, so that by (3.11)

$$\sup_{\theta \in K_{n,C}} \max_{i \leqslant n} \sigma_\theta^2 u_\theta^2(X_i) \leqslant (C^2/n)M_n + C^2\overline{\mathbb{P}}_n\big[(u_\theta^2 - M_n)_+\big] = o_{\mathbb{P}_{0,n}}(1).$$

Thus, the Taylor expansion of $\log(\sqrt{p_\theta}/\sqrt{p_{\theta_{0,n}}})$ is uniformly valid in $\theta \in K_{n,C}$,

$$\sup_{\theta \in K_{n,C}} \Big| n\overline{\mathbb{P}}_n\Big[2\log\big(\sqrt{p_\theta}/\sqrt{p_{\theta_{0,n}}}\big) - \big(\sigma_\theta u_\theta - H_\theta^2 - \sigma_\theta^2 u_\theta^2/4\big)\Big]\Big| = o_{\mathbb{P}_{0,n}}(1) \sup_{\theta \in K_{n,C}} \overline{\mathbb{P}}_n\big[u_\theta^2\big].$$

The conclusion then follows from (3.10), (3.11) and $\sup_{\theta \in K_{n,C}} |2H_\theta/\sigma_\theta - 1| = o(1)$. $\square$

The following theorem is a variation of Theorem 3.7 for square integrable likelihood ratios.

**Theorem 3.8.** *Let $\{p_\theta, \theta \in \Theta_n\}$ be a sequence of density families and $X_1, \ldots, X_n$ be iid observations from $p_{\theta_{0,n}}$ under $\mathbb{P}_{0,n}$. Let $\mathscr{H}$ be a fixed Hilbert space and $g_n$ be distance preserving mappings from $L_2(\mathbb{P}_{0,n})$ to $\mathscr{H}$. For $\theta \in \Theta_n$, let*

$$s_\theta = p_\theta/p_{\theta_{0,n}} - 1,\ D_\theta = (\mathbb{E}_{0,n}[s_\theta^2])^{1/2},\ v_\theta = s_\theta/D_\theta \ \text{ and } \ h_n(\theta) = n^{1/2} g_n\big(s_\theta\big).$$

*Let $\theta_{0,n} \in K_n \subset \Theta_n$ with $\sup_{\theta\in K_n} D_\theta^2 \leqslant 1/M_n$. Let $\widehat{\theta}_n = \arg\max_{\theta\in K_n} \overline{\mathbb{P}}_n[\log p_\theta]$. Suppose*

$$\mathbb{E}^*_{0,n}\big[f\big(n^{1/2}\overline{\mathbb{P}}_n\big[v_\theta\big], \theta \in K_{n,C}\big)\big] = \mathbb{E}_0\big[f\big(Z(v_\theta), \theta \in K_{n,C}\big)\big] + o(1) \tag{3.12}$$

*for all $f$ uniformly bounded and continuous mappings from $L_\infty(K_{n,C}) \to \mathbb{R}$, and*

$$\sup_{\theta\in K_n} \Big\{\big|\overline{\mathbb{P}}_n\big[v_\theta^2\big] - 1\big| + \overline{\mathbb{P}}_n\big[(v_\theta^2 - M_n)_+\big]\Big\} = o_{\mathbb{P}_{0,n}}(1), \quad M_n/n + 1/M_n = o(1). \tag{3.13}$$

*Then, family $\{p_\theta, \theta \in K_n\}$ is uniformly LAN and $D_{\widehat{\theta}_n} = O_{\mathbb{P}_{0,n}}(n^{-1/2})$.*

Proof. Because $\sqrt{p_\theta}/\sqrt{p_{\theta_{0,n}}} = \sqrt{s_\theta + 1}$.

$$\log\big(p_\theta/p_{\theta_{0,n}}\big) \leqslant 2\big(\sqrt{p_\theta}/\sqrt{p_{\theta_{0,n}}} - 1\big) = s_\theta - (\sqrt{s_\theta+1} - 1)^2 = s_\theta - s_\theta^2/(1 + \sqrt{s_\theta + 1})^2.$$

Due to the conditions $\sup_{\theta\in K_n} D_\theta^2 \leqslant 1/M_n$ and (3.13),

$$\overline{\mathbb{P}}_n\bigg[\frac{s_\theta^2}{(1+\sqrt{s_\theta+1})^2}\bigg] \geqslant \overline{\mathbb{P}}_n\bigg[\frac{s_\theta^2 \wedge 1}{(1+\sqrt{2})^2}\bigg] \geqslant \frac{D_\theta^2\overline{\mathbb{P}}_n\big[v_\theta^2 \wedge M_n\big]}{(1+\sqrt{2})^2} \geqslant \frac{D_\theta^2(1 + o_{\mathbb{P}_{0,n}}(1))}{(1+\sqrt{2})^2}$$

uniformly in $\theta \in K_n$. Thus, because $0 \leqslant \overline{\mathbb{P}}_n\big[\log\big(p_{\widehat{\theta}_n}/p_{\theta_{0,n}}\big)\big]$, (3.12) provides

$$\frac{D_{\widehat{\theta}}^2(1 + o_{\mathbb{P}_{0,n}}(1))}{(1+\sqrt{2})^2} \leqslant \sup_{\theta\in K_n} \overline{\mathbb{P}}_n\big[s_\theta\big] \leqslant D_{\widehat{\theta}_n} \sup_{\theta\in K_n} \overline{\mathbb{P}}_n\big[v_\theta\big] = O_{\mathbb{P}_{0,n}}(n^{-1/2}) D_{\widehat{\theta}_n}.$$

It follows that $D_{\widehat{\theta}_n} = O_{\mathbb{P}_{0,n}}(n^{-1/2})$. For $\theta \in K_{n,C}$, $nD_\theta^2 = \|h_n(\theta)\|_{\mathscr{H}}^2 \leqslant C^2$, so that by (3.13)

$$\sup_{\theta\in K_{n,C}} \max_{i\leqslant n} s_\theta^2(X_i) \leqslant (C^2/n)M_n + C^2\overline{\mathbb{P}}_n\big[(v_\theta^2 - M_n)_+\big] = o_{\mathbb{P}_{0,n}}(1).$$

Thus, the Taylor expansion of $\log(p_\theta/p_{\theta_{0,n}})$ is uniformly valid in $\theta \in K_{n,C}$,

$$\sup_{\theta\in K_{n,C}} \Big|n\overline{\mathbb{P}}_n\Big[\log\big(p_\theta/p_{\theta_{0,n}}\big) - \big(s_\theta - s_\theta^2/2\big)\Big]\Big| = o_{\mathbb{P}_{0,n}}(1) \sup_{\theta\in K_{n,C}} n\overline{\mathbb{P}}_n\big[s_\theta^2/2\big)\big].$$

The conclusion the follows from (3.12) and (3.13) because $\sup_{\theta\in K_{n,C}} nD_\theta^2 \leqslant C$. □

The Hellinger distance between densities $p$ and $q$ is bounded by

$$H^2(p,q) = \int \big(q^{1/2} - p^{1/2}\big)^2 d\nu = \int \frac{(q-p)^2}{\big(q^{1/2} + p^{1/2}\big)^2} d\nu \leqslant \int (q-p)^2(1/p) d\nu.$$

**Lemma 3.2.** *Let $p = p(x)$ and $q = q(x)$ be densities with respect to a measure $\nu$. Suppose $\int_{q>Mp}(q-p)^2(1/p)d\nu \leqslant \int(q-p)^2(1/p)d\nu/2$. Then, $\int(q-p)^2(1/p)d\nu \leqslant 2(1+M^{1/2})^2 H^2(p,q)$.*

Proof. $\int(q/p-1)^2 p d\nu \leqslant (1+M^{1/2})^2 \int(\sqrt{q/p}-1)^2 p d\nu + \int(q/p-1)^2 p d\nu/2$. □

**Example 3.4** (Gaussian mixtures)**.** *Suppose iid $X_1, \ldots, X_n$ are observed with density $p_G(x)$ in the Gaussian mixture model $\mathscr{P}_m = \{p_G = \sum_{j=1}^m w_j p_{\theta_j} : |\theta_j| \leqslant M, G(\{\theta_j\}) = w_j\}$ with $p_\theta(x) = \varphi(x-\theta) \sim N(\theta, 1)$ and a finite $M > 0$. Consider testing $H_0 : p_G = p_0 = \varphi$ against $H_1 : p_G \in \mathscr{P}_m \backslash \{p_0\}$. The GLRT can be written as $\phi = I\{\Lambda_n \geqslant K\}$ with*

$$\Lambda_n = n\overline{\mathbb{P}}_n\big[\log(p_{\widehat{G}_n}/\varphi)\big], \quad \widehat{G}_n = \arg\max_G \big\{n\overline{\mathbb{P}}_n\big[\log(p_G)\big] : p_G \in \mathscr{P}_m\big\}.$$

*Here the mixing distribution $G$ is treated as the parameter and $m = \infty$ is allowed. Let*

$$D_G = \big(\mathbb{E}_0\big[(p_G/p_0 - 1)^2\big]\big)^{1/2}, \quad K_n = \{G : D_G^2 \leqslant (n^{-1}\log\log n)^{1/2}\}.$$

*Let $\mathbb{P}_0$ be the measure corresponding to the density $p_0 = \varphi$. We shall prove the $L_2$ consistency of the MLE $\mathbb{P}_0\big\{\widehat{G}_n \in K_n\big\} \to 1$ and verify the conditions of Theorem 3.8 under $\mathbb{P}_0$. We begin with the Hermite polynomial expansion of $p_G/\varphi$. The Hermite polynomials are*

$$H_k = H_k(x) = (-1)^k e^{x^2/2}(d/dx)^k e^{-x^2/2} = (d/d\theta)^k e^{x\theta - \theta^2/2}\big|_{\theta=0}, \tag{3.14}$$

*with $H_0 = 1$, $H_1 = x$, $H_2 = x^2 - 1$, etc. They form an orthonormal basis in $L_2(\mathbb{P}_0)$,*

$$\mathbb{E}_0\big[H_j H_k\big] = \frac{\partial^j}{\partial y^j}\frac{\partial^k}{\partial z^k}\int e^{xy - y^2/2 + xz - z^2/2}\varphi(x)dx\Big|_{y=z=0} = \frac{\partial^j}{\partial y^j}\frac{\partial^k}{\partial z^k}e^{yz}\Big|_{y=z=0} = k! I_{\{j=k\}}.$$

*Let $\mu_k(G) = \int u^k dG(u)$. By (3.14) the Gaussian mixture has the expansion*

$$\frac{p_G}{\varphi} = \int e^{x\theta - \theta^2/2}G(d\theta) = \sum_{k=0}^\infty \frac{\mu_k(G)}{k!}H_k(x),\ \ v_G(x) = \sum_{k=1}^\infty c_k(G)\frac{H_k(x)}{\sqrt{k!}},\ \ c_k(G) = \frac{\mu_k(G)}{D_G\sqrt{k!}}.$$

*By the orthonormality of $H_k/\sqrt{k!}$, $\sum_{k=1}^\infty c_k^2(G) = 1$. Moreover, as $|\mu_k(G)| \leqslant M^{k-2}\mu_2(G)$ for $k \geqslant 2$, $c_k^2(G) \leqslant 2M^{2(k-2)_+}/k!$. Let $v_{G,k^*} = \sum_{k=1}^{k^*} c_k(G)H_k/\sqrt{k!}$. We have*

$$\mathbb{E}\bigg[\sup_{G \in \mathscr{P}_\infty}\Big|(\overline{\mathbb{P}}_n - \mathbb{P}_0)\big[v_G - v_{G,k^*}\big]\Big|\bigg] \leqslant \sum_{k=k^*+1}^\infty \big(2M^{2(k-2)_+}/k!\big)^{1/2}\mathbb{E}\Big[\Big|\overline{\mathbb{P}}_n\big[H_k/\sqrt{k!}\big]\Big|\Big] \leqslant \frac{\epsilon_{k^*}}{n^{1/2}}$$

*with $\epsilon_{k^*} = \sum_{k=k^*+1}^\infty \big(2M^{2(k-2)_+}/k!\big)^{1/2} \to 0$ as $k^* \to \infty$. Thus, as $v_{G,k^*}$ is a linear combination of $k^*$ orthonormal $H_k/\sqrt{k!}$ with uniformly bounded coefficients, the class $\{v_G : G([-M, M]) = 1\}$ is $\mathbb{P}_0$-Donsker. Consequently, (3.12) holds with $C = \infty$. Similarly, (3.13) holds and $\sup_{G \in \mathscr{P}_\infty} D_G^2/H^2(p_G, \varphi) < \infty$ by Lemme 3.2 due to $\mathbb{E}_0[H_k^4] < \infty$ and*

$$\big(\mathbb{E}_0\big[\sup_{G \in \mathscr{P}_\infty}\overline{\mathbb{P}}_n\big[(v_G - v_{G,k^*})^2\big]\big]\big)^{1/2} \leqslant \textstyle\sum_{k=k^*+1}^\infty \sup_{G \in \mathscr{P}_\infty}|c_k(G)|\big(\mathbb{E}_G\big[H_k^2/k!\big]\big)^{1/2} \leqslant \epsilon_{k^*}.$$

*It remains to prove $\mathbb{P}_0\{\widehat{G}_n \in K_n\} \to 1$. Let $\Phi_n$ be the ECDF of the data and $\Phi(x) = \mathbb{E}_0[\Phi_n]$ be the $N(0,1)$ CDF. Bacause $|x + p_G'(x)/p_G(x)| = \mathbb{E}_G[\theta|X_1 = x] \leqslant M$,*

$$D_{\mathrm{KL}}(\varphi\|p_{\widehat{G}_n}) \leqslant \sup_{p_G \in \mathscr{P}_m}(\overline{\mathbb{P}}_n - \mathbb{P}_0)\big[\log(p_G/\varphi)\big] \leqslant M\textstyle\int\big|\Phi_n(x) - \Phi(x)\big|dx$$

*via integration by parts, so that the Hellinger distance $H(p_{\widehat{G}_n}, \varphi)$ is bounded by*

$$\mathbb{E}_0\big[H^2(p_{\widehat{G}_n}, \varphi)\big] \leqslant \mathbb{E}_0\big[D_{\mathrm{KL}}(\varphi\|p_{\widehat{G}_n})\big] \leqslant Mn^{-1/2}\textstyle\int\sqrt{\Phi(x)(1 - \Phi(x))}dx.$$

*This gives $D_{\widehat{G}_n} = O_{\mathbb{P}_0}(n^{-1/2})$ due to $\sup_{G \in \mathscr{P}_\infty} D_G^2/H^2(p_G, \varphi) < \infty$, Thus, Theorem 3.8 is directly applicable in this example. This analysis accommodates the case of $m = \infty$ where $\widehat{G}_n$ is the nonparametric MLE and the tangent cone $\mathscr{T}$ is infinite-dimensional.*

3.3. **Multiple-testing.** Multiple testing concerns a collection of statistical hypothesis $H_j, j = 1, \ldots, m$. For $m = 1$, the size of a test is measured by the probability of type I error (false rejection). There are three popular ways to extend this concept to the case of $m > 1$: The per comparison error rate (PCER) is the average of the probabilities of type one errors; The family wise error rate (FWER) is the probability of rejecting at least one true hypothesis; The false discovery rate (FDR) is the expected ratio of the number of false rejections and the total number of rejections, with the convention $0/0 \equiv 0$. Let

$$\theta_j = I\{H_j \text{ is not true}\},\ \delta_j = I\{\text{ reject } H_j\}.$$

The three error measures can be expressed as PCER $= \mathbb{E}\big[\sum_{j=1}^m \delta_j(1-\theta_j)/m\big]$, FWER $= \mathbb{E}\big[\max_{j\leqslant m}\delta_j(1-\theta_j)\big]$, and FDR $= \mathbb{E}\big[\sum_{j=1}^m \delta_j(1-\theta_j)/\sum_{j=1}^m \delta_j\big]$. Since $\sum_{j=1}^m \delta_j \leqslant m$,

$$\text{PCER} \ \leqslant\ \text{FDR} \leqslant\ \text{FWER}.$$

Let $m_0 = \sum_{j=1}^m (1-\theta_j)$ be the number of true hypotheses and $m_1 = m - m_0$ the number of false ones. For $m_0\alpha \asymp m_1$, controlling the PCER at level $\alpha$ may still allow too many false rejections. Among the other two error measures, the constraint FDR $\leqslant \alpha$ may provide significantly higher power than FWER $\leqslant \alpha$ does. The FDR is closely related to the positive predictive value (PPV) of a medical test $\mathbb{P}[\theta = 1|\delta = 1]$, where $\delta = 1$ iff the test is positive test and $\theta = 1$ iff the subject has the disease. In fact, the $1 -$ FDR can be viewed as the expectation of the realized PPV. Since controlling the PCER can be treated by the theory for testing a single hypothesis, we discuss here the FWER and FDR.

**Controlling the family wise error rate.** For notational simplicity, we discuss multiple testing procedures based on p-values $p_j$ for $H_j$ satisfying

$$\theta_j = 0 \ \Rightarrow\ \mathbb{P}\{p_j \leqslant c\} \leqslant c\ \forall\, 0 \leqslant c \leqslant 1.$$

Let $p_{(1)} \leqslant \cdots \leqslant p_{(m)}$ be the ordered values of $\{p_j, j \leqslant m\}$. The FWER can be controlled by single step procedures comparing $p_j$ with certain critical levels $c_j$, and by step-down and step-up procedures comparing $p_{(j)}$ with $c_j$ sequentially. In a more general setting, closure methods can be used to control the FWER.

*Single step tests:* Reject $H_j$ iff $p_j \leqslant c_j$. Bonferroni's adjustment $c_j = \alpha/m$ gives

$$\text{FWER} = \mathbb{P}\{p_j \leqslant c_j \text{ for some } \theta_j = 0\} \leqslant \sum_{j=1}^m c_j(1-\theta_j) \leqslant \alpha.$$

If $\{p_j : j \leqslant m, \theta_j = 0\}$ are independent p-values, the larger $c_j = 1 - (1-\alpha)^{1/m}$ suffices.

*Step-down tests:* Reject $H_{(j)}$ iff $p_{(k)} \leqslant c_k$ for all $k \leqslant j$, where $H_{(j)}$ is the hypothesis corresponding to $p_{(j)}$. It begins with the test for $H_{(1)}$ with the smallest/most significant p-value $p_{(1)}$ and continues until a certain $H_{(\widehat{k}+1)}$ is accepted with $p_{\widehat{k}+1} > c_{\widehat{k}+1}$, along with all $H_{(j)}$, $j > \widehat{k}+1$, with larger p-values, where $\widehat{k} = \min\{k : p_{(k)} > c_k\} - 1$.

**Theorem 3.9.** *Let $p_{1,0} = \min_{\theta_j=0} p_j$ be the smallest p-value for the true nulls and $m_1 = \sum_{j=1}^m \theta_j$. Then, for step-down tests with nondecreasing critical values $c_1 \leqslant \cdots \leqslant c_m$,*

$$\mathbb{P}\{p_{1,0} \leqslant c_1\} \leqslant \mathrm{FWER} \leqslant \mathbb{P}\{p_{1,0} \leqslant c_{m_1+1}\}, \quad m_1 = \textstyle\sum_{j=1}^m \theta_j.$$

*Thus,* FWER $\leqslant \alpha$ *for the Holm* [Hol79] *test with $c_j = \alpha/(m-j+1)$ . If the p-values $\{p_j, \theta_j = 0\}$ for the true nulls are independent, the larger $c_j = 1-(1-\alpha)^{1/(m-j+1)}$ suffice.*

Proof. Type-I error occurs iff the true null with p-value $p_{1,0}$ is rejected, $p_{1,0} \leqslant c_k$ for some $1 \leqslant k \leqslant m_1+1$. Formally because $p_{(\widehat{k}+1)} > c_{\widehat{k}+1} \geqslant c_{\widehat{k}} \geqslant p_{(\widehat{k})}$, $H_j$ is accepted iff $p_j > c_{\widehat{k}}$. If $p_{1,0} > c_{m_1+1}$, then $p_{(m_1+1)} \geqslant p_{1,0} > c_{m_1+1} \geqslant c_{\widehat{k}}$, so that all true $H_j$ are accepted due to $p_j \geqslant p_{1,0} > c_{\widehat{k}}$. □

*Step-up tests:* Reject $H_{(j)}$ iff $p_{(k)} \leqslant c_k$ for some $k \geqslant j$, where $H_{(j)}$ is the hypothesis corresponding to $p_{(j)}$. It begins with the test for $H_{(m)}$ with the least significant $p_{(m)}$ and continues until a certain $H_{(\widehat{k})}$ is rejected, along with all $H_{(j)}$ with $j < \widehat{k}$, where $\widehat{k} = \max\{k : p_{(k)} \leqslant c_k\}$.

**Theorem 3.10.** *Let $p_{(1,0)} \leqslant \cdots \leqslant p_{(m_0,0)}$ be the ordered values of $\{p_j, \theta_j = 0\}$ corresponding to the true nulls. Then, for step-up tests with nondecreasing critical values $c_1 \leqslant \cdots \leqslant c_m$,*

$$\mathbb{P}\{p_{(j,0)} \leqslant c_j \text{ for some } j \leqslant m_0\} \leqslant \mathrm{FWER} \leqslant \mathbb{P}\{p_{(j,0)} \leqslant c_{m_1+j} \text{ for some } j \leqslant m_0\}$$

Proof. The ordered $p_{(1,0)} \leqslant \cdots \leqslant p_{(m_0,0)}$ are compared with a subsequence of $\{c_j\}$ depending on the $p$-values of the false nulls, with $c_1 \leqslant \cdots \leqslant c_{m_0}$ being the most favorable subsequence and $c_{m_1+1} \leqslant \cdots \leqslant c_m$ being the least favorable. Formally, because $p_{(\widehat{k}+1)} > c_{\widehat{k}+1} \geqslant c_{\widehat{k}} \geqslant p_{(\widehat{k})}$, $H_j$ is accepted iff $p_j > c_{\widehat{k}+1}$ and $H_j$ is rejected iff $p_j \leqslant c_{\widehat{k}}$. If $p_{(j,0)} > c_{m_1+j}$ for all $j \leqslant m_0$, then $p_{(m_1+j)} \geqslant p_{(j,0)} > c_{m_1+j}$ for all $1 \leqslant j \leqslant m_0$, so that $c_{\widehat{k}+1} \leqslant c_{m_1+1}$. Thus, all true $H_j$ are accepted with $p_j > c_{m_1+1} \geqslant c_{\widehat{k}+1}$. If $p_{(j,0)} \leqslant c_j$, then $p_{(j)} \leqslant p_{(j,0)} \leqslant c_j$, so that $H_{(j)}$ is rejected. Consequently, $c_{\widehat{k}} \geqslant c_j \geqslant p_{(j,0)}$ and the true hypothesis corresponding to $p_{(j,0)}$ is rejected. □

**Example 3.5.** *Suppose $\mathbb{P}\{p_{(j,0)} \leqslant \alpha/(m_0-j+1)$ for some $j \leqslant m_0\} \leqslant \alpha$ when $\sum_{j=1}^m (1-\theta_j) = m_0$ for all $1 \leqslant m_0 \leqslant m$. Then,* FWER $\leqslant \alpha$ *for the step-up test with $c_j = \alpha/(m-j+1)$. Compared with the Holm step-down test* [Hol79] *with the same $c_j$, the step-up test is more powerful but requires a stronger condition for controlling the* FWER.

*Closure methods:* Let $X \sim \mathbb{P}_\theta$ with $\theta \in \Theta$. Consider null hypotheses $H_w : \theta \in \Theta_w$ with $\Theta_w \subseteq \Theta$. Suppose the collection $\{\Theta_w, w \in W\}$ is closed under intersection: For any pair $\{w_1, w_2\} \subset W$, there exists a $w \in W$ with $\Theta_{w_1} \cap \Theta_{w_2} = \Theta_w$. Let $\phi_w = \phi_w(X)$ be tests for the hypotheses $\theta \in \Theta_w$ with level $\sup_{\theta\in\Theta_w} \mathbb{E}_\theta \phi_w \leqslant \alpha$. Let $U$ be a uniform random variable on $[0,1]$ independent of $X$. The closure of the tests $\{\phi_w, w \in W\}$ is defined as

$$\text{Reject } \Theta_w \text{ when } U \leqslant \min\left\{\phi_{w'} : \Theta_{w'} \subseteq \Theta_w, w' \in W\right\},\ \forall\, w \in W.$$

**Theorem 3.11.** [MEG76] *The closure test has size $\alpha$ simultaneously for all $\Theta_w, w \in W$:* FWER $\leqslant \mathbb{P}_\theta\{$*Reject* $\Theta_w$ *for some* $\Theta_w \ni \theta\} \leqslant \alpha$.

PROOF. Let $\theta$ be fixed and $\Theta_{w_\theta} = \cap_{\theta\in\Theta_w} \Theta_w$. For true $\Theta_w \ni \theta$, we have $\theta \in \Theta_{w_\theta} \subseteq \Theta_w$, so that $\cup_{\theta\in\Theta_w}\{\text{Reject } \Theta_w\} \subset \{U < \phi_{w_\theta}\}$. □

Now consider controlling the FWER for hypotheses $H_j, j \leqslant m$. Let $J = \{1, \ldots, m\}$ and $H_A = \cap_{j\in A} H_j$. The collection $\{H_A : \varnothing \neq A \subseteq J\}$ is the smallest one containing all $H_i$ and closed under intersection. Let $\phi_A$ be level-$\alpha$ tests for $H_A$. The closure test rejects $H_j$ when $U \leqslant \min_{j\in A} \phi_A$.

*Adjusted p-values.* It will be convenient if p-values for individual hypotheses can be adjusted for multiplicity such that for controlling a type I error measure in multiple testing, the hypotheses are rejected whenever their adjusted p-values are no greater than $\alpha$.

A p-value $p$ is valid for a hypothesis $H$ if $\mathbb{P}\{p \leqslant \alpha\} \leqslant \alpha$ for all $\alpha$ whenever $H$ is true. If p-values $p_j$ are valid for $H_j$ individually, $j = 1, \ldots, m$, the adjusted p-values $p_j^* = mp_j$ yields the Bonferroni test for controlling the FWER. If p-values $p_w$ are valid for $\Theta_w$ individually, the adjusted p-vaues $p_w^* = \max_{\Theta_{w'}\subseteq\Theta_w} p_{w'}$ yield the closure test for controlling the FWER.

**Example 3.6.** *Suppose p-values $p_j$ are valid for $H_j$ individually, $j = 1, \ldots, m$. For each $\varnothing \neq A \subseteq \{1, \ldots, m\}$, $p_A = |A| \min_{j\in A} p_j$ is a valid p-value for $H_A = \cap_{j\in A} H_j$ by the Bonferroni adjustment. Thus, an application of the closure method yields the adjusted p-values $p_j^* = \max_{j\in A} |A| \min_{k\in A} p_k$, or equivalently, according to the order $p_{(1)} \leqslant \ldots \leqslant p_{(m)}$,*

$$p^*_{(j)} = \max_{k\leqslant j} \max\left\{|A| p_{(k)} : p_{(k)} = \min_{i\in A} p_i, (j) \in A\right\} = \max_{k\leqslant j}(m-k+1)p_{(k)}$$

*for $H_{(j)}$. This is Holm's step-down test.*

**Remark 3.1.** *For independent p-values $p_j$ valid for $H_j$, $j \leqslant m$,* [See68, Sim86] *proved that the step-up test with $c_j = \alpha j/m$ has size $\alpha$ for testing $H_J = \cap_{j=1}^m H_j$. This implies $\mathbb{P}\{p_{(j,0)} \leqslant j\alpha/m_0$ for some $j \leqslant m_0\} \leqslant \alpha$ whenever $\sum_{j=1}^m (1-\theta_j) = m_0$ for all $1 \leqslant m_0 \leqslant m$, which then implies the condition of Example 3.5 due to $j/m_0 \geqslant 1/(m_0 - j + 1), 1 \leqslant j \leqslant m_0$.* [Hoc88] *proved that the closure test based on the Seeger-Simes tests for $H_A$ is the step-up test with $c_j = \alpha/(m-j+1)$.*

**Controlling the false discovery rate:** A random vector $Y_{1:m} = (Y_1, \ldots, Y_m)'$ is positive regression dependent on each one from a subset $J_0$ (PRDS on $J_0$) at a level $t_0$ iff $\mathbb{E}[h(Y_i \wedge t_0, i \neq j)|Y_j = t]$ is nondecreasing in $t$ for all $t \leqslant t_0$, $j \in J_0$ and nondecreasing functions $h : \mathbb{R}^{m-1} \to \mathbb{R}$. Actually, what we need is the weaker condition that $\mathbb{E}[h(Y_i \wedge t_0, i \neq j)|Y_j \leqslant t]$ is nondecreasing in $t$ for all $t \leqslant t_0$, $j \in J_0$ and nondecreasing function $h$.

**Theorem 3.12.** *Let $0 \leqslant q \leqslant q_0$ and $c_j = jq/m$. Suppose that the vector of p-values $p_{1:m} = (p_1, \ldots, p_m)'$ is* PRDS *on the set $J_0 = \{j : \theta_j = 0\}$ at the level $q_0$ and that $\mathbb{P}\{p_j \leqslant c_k\} \leqslant c_k$ for all $k \leqslant m$ and $j \in J_0$. Then,* FDR $\leqslant qm_0/m$ *for the step-up test with critical levels $c_j$,*

$$\text{Reject } H_{(j)} \text{ for all } j \leqslant \widehat{k} = \max\big\{j \leqslant m : p_{(j)} \leqslant qj/m\big\}.$$

Proof. Assume $J_0 = \{1, \ldots, m_0\}$ without loss of generality. Let $\widehat{k} = \max\{k : p_{(k)} \leqslant c_k\}$ be the total number of rejections. Since $H_j$ is rejected iff $p_j \leqslant c_k$ in the event $\widehat{k} = k$ and $\mathbb{P}\{p_j \leqslant c_k\}/k \leqslant c_k/k = q/m$ for $j \leqslant m_0$,

$$\text{FDR} = \mathbb{E}\sum_{k=1}^{m}\sum_{j=1}^{m_0}\frac{I\{p_j \leqslant c_k, \widehat{k} = k\}}{k} \leqslant \sum_{j=1}^{m_0}\sum_{k=1}^{m}\frac{q}{m}\mathbb{P}\big\{\widehat{k} = k\big|p_j \leqslant c_k\big\}. \tag{3.15}$$

Let $N_k = \#\{i : p_i \leqslant c_k\}$ and $N_k^{(j)} = \#\{i \neq j : p_i \wedge q_0 \leqslant c_k\}$. Since $p_{(k)} \leqslant c_k$ iff $N_k \geqslant k$, $\widehat{k} \leqslant k$ iff $N_j < j$ for all $j > k$. For $p_j \leqslant c_{k+1}$, $N_\ell = N_\ell^{(j)} + 1$ for all $\ell > k$. Thus, in the event $p_j \leqslant c_{k+1}$, $I\{\widehat{k} \leqslant k\} = I\{1 + N_\ell^{(j)} < \ell, \forall \ell > k\}$ is a nondecreasing function of $(p_i \wedge q_0, i \neq j)'$, so that $P\big\{\widehat{k} \leqslant k\big|p_j \leqslant c_k\big\} \leqslant P\big\{\widehat{k} \leqslant k\big|p_j \leqslant c_{k+1}\big\}$ by the PRDS assumption. It follows that

$$\begin{aligned}\sum_{k=1}^{m} P\big\{\widehat{k} = k\big|p_j \leqslant c_k\big\} &= \sum_{k=1}^{m} P\big\{\widehat{k} \leqslant k\big|p_j \leqslant c_k\big\} - \sum_{k=1}^{m} P\big\{\widehat{k} < k\big|p_j \leqslant c_k\big\} \\ &\leqslant \sum_{k=1}^{m} P\big\{\widehat{k} < k+1\big|p_j \leqslant c_{k+1}\big\} - \sum_{k=2}^{m} P\big\{\widehat{k} < k\big|p_j \leqslant c_k\big\} = 1.\end{aligned}$$

This and (3.15) yields FDR $\leqslant qm_0/m$. □

**Remark 3.2.** *Benjamini and Hochberg* [BH95] *proposed the* FDR *as an error measure for multiple testing and proved* FDR $\leqslant qm_0/m$ *for the step-up test with $c_j = jq/m$ and independence $\{p_1, \ldots, p_m\}$. The extension to the* PRDS *case was established by Benjamini and Yekutieli* [BY01]. *Since the* FDR *is the probability of type I error for testing $H_J = \cap_{j=1}^m H_j$, their results can be viewed as extensions of the Seeger-Simes bound in Remark 3.1 from testing $H_J$ to controlling the* FDR *in multiple testing.*

**Remark 3.3.** *The* PRDS *is a consequence of the positive regression dependence* (PRD) *property* [Leh66, Sar69]*: $\mathbb{E}[h(Y_{1:m})|Y_{j_i} = t_i, i = 1, \ldots, k]$ is nondecreasing in $t_1, \ldots, t_k$ for all $j_1 < \cdots < j_k$ and nondecreasing $h$. The joint distribution or density of $Y_{1:m}$ is* PRD *if $Y_{1:m}$ is* PRD. *Of course, $Y_{1:m}$ is* PRD *if the components of $Y_{1:m}$ are independent.*

**Example 3.7** (One-sided z-test). *Let $\widehat{\mu}_{1:m} \sim N(\mu_{1:m}, \Sigma)$ with a covariance matrix $\Sigma = (\sigma_{jk})_{m\times m}$. Let $p_j = 1 - \Phi(\widehat{\mu}_j/\sigma_{jj})$ be the p-value for testing $H_j : \mu_j = 0$ against $K_j : \mu_j > 0$ based on $\widehat{\mu}_j$ and known $\sigma_{jj}$. Suppose $\sigma_{ij} \geqslant 0$ for all $i \neq j$. Then, $p_{1:n}$ is* PRDS *on $J_0 = \{j : \mu_j = 0\}$. This can be seen as follows. Since $p_j$ are all decreasing in $\widehat{\mu}_j$, $p_{1:n}$ is* PRDS *on $J_0$ iff $\widehat{\mu}_{1:m}$ is* PRDS *on $J_0$. Conditionally on $\widehat{\mu}_j$, $\widehat{\mu}^{(j)}_{1:m} = (\widehat{\mu}_i, i \neq j) = \mathbb{E}[\widehat{\mu}^{(j)}_{1:m}|\widehat{\mu}_j] + \boldsymbol{\varepsilon}^{(j)}$, where $\boldsymbol{\varepsilon}^{(j)}$ is independent of $\widehat{\mu}_j$. Since $\sigma_{ij} \geqslant 0$, $\mathbb{E}[\widehat{\mu}_i|\widehat{\mu}_j = t] = a_{ij} + t\sigma_{ij}/\sigma_{jj}$ for certain constants $a_{ij}$ and thus are nondecreasing in $t$. It follows that $\mathbb{E}[h(\widehat{\mu}^{(j)}_{1:m})|\widehat{\mu}_j = t] = \mathbb{E}[h(a_{ij} + t\sigma_{ij}/\sigma_{jj} + \epsilon^{(j)}_i, i \neq j)]$ is nondecreasing in $t$ for all nondecreasing $h$.*

**Example 3.8** (One-sided t-tests). *Let $\widehat{\mu}_{1:m} \sim N(\mu_{1:m}, \Sigma)$ and $\widehat{\sigma}^2 \sim \sigma^2\chi^2_\nu/\nu$ be independent statistics, where $\mu_{1:m}$ and $\sigma$ are unknown, and $\Sigma = (\sigma_{jk})_{m\times m}$ with known $v_j = \sigma_{jj}/\sigma^2$. Let $H_j$ be as in Example 2.2.1 and $p_j = 1 - \Phi_\nu(\widehat{\mu}_j/(\widehat{\sigma}\sqrt{v_j}))$ be the p-values for univariate one-sided t-tests, where $\Phi_\nu$ is the t-distribution function with $\nu$ degrees of freedom. Suppose $\sigma_{ij} \geqslant 0$ for all $i \neq j$. Then, $p_{1:n}$ is* PRDS *on $J_0 = \{j : \mu_j = 0\}$ at the level $1/2$. This can be seen as follows. Let $j \in J_0$ and $h$ be nondecreasing. Since $p_j \wedge (1/2) = 1 - \Phi_\nu(\widehat{\mu}^+_j/(\widehat{\sigma}\sqrt{v_j}))$, the question is if $\mathbb{E}[h(\mu^+_i/\widehat{\sigma}, i \neq j)|\widehat{\mu}_j/\widehat{\sigma} = t]$ is nondecreasing in $t \geqslant 0$. Let $h_1(u, 1/v) = \mathbb{E}[h(\mu^+_i/\widehat{\sigma}, i \neq j)|\widehat{\mu}^+_j = \sqrt{u}, \widehat{\sigma}^2 = v]$. Since $h$ is nondecreasing, $h_1(u, 1/v)$ is nondecreasing in $1/v$. By Example 2.2.1, $h_1(u, 1/v)$ is nondecreasing in $u$. It remains to prove that $\mathbb{E}[h_1(\widehat{\mu}^2_j, 1/\widehat{\sigma}^2)|\widehat{\mu}^+_j/\widehat{\sigma} = t]$ is nondecreasing in $t \geqslant 0$. Let $\chi^2_{\nu+1} = \widehat{\mu}^2_j/\sigma_{jj} + \nu\widehat{\sigma}^2/\sigma^2$. For $\mu_j = 0$, $\chi^2_{\nu+1}$ is independent of $\widehat{\mu}_j/\widehat{\sigma}$. Given $\widehat{\mu}^+_j/\widehat{\sigma} = t$, $\chi^2_{\nu+1} = (t^2/\sigma_{jj} + \nu/\sigma^2)\widehat{\sigma}^2 = (1/\sigma_{jj} + \nu t^{-2}/\sigma^2)\widehat{\mu}^2_j$. Thus,*

$$\mathbb{E}\Big[h_1(\widehat{\mu}^2_j, 1/\widehat{\sigma}^2)\Big|\widehat{\mu}^+_j/\widehat{\sigma} = t\Big] = \mathbb{E}\Big[h_1\Big(\frac{\chi^2_{\nu+1}}{1/\sigma_{jj} + \nu t^{-2}/\sigma^2}, \frac{t^2/\sigma_{jj} + \nu/\sigma^2}{\chi^2_{\nu+1}}\Big)\Big]$$

*is nondecreasing in $t$.*

**FDR control with knockoffs.** Consider the linear model

$$Y = X\beta + \varepsilon$$

with a design matrix $X = (X_1, \ldots, X_p) \in \mathbb{R}^{n\times p}$ and a coefficient vector $\beta = (\beta_1, \ldots, \beta_p)^\top$.

Let $\Sigma = X^\top X$. For $2p \leqslant n$, $\widetilde{X} = (\widetilde{X}_1, \ldots, \widetilde{X}_p) \in \mathbb{R}^{n\times p}$ is a knockoff matrix if

$$\widetilde{X}^\top\widetilde{X} = \Sigma, \quad X^\top\widetilde{X} = \Sigma - \mathrm{diag}(s),$$

for a $p$-dimensional nonnegative vector $s$. Such $\widetilde{X}$ can be constructed as

$$\widetilde{X} = X(I_p - \Sigma^{-1}\mathrm{diag}(s)) + C$$

with $C \in \mathbb{R}^{n\times p}$ satisfying $X^\top C = 0$ and $C^\top C = 2\mathrm{diag}(s) - \mathrm{diag}(s)\Sigma^{-1}\mathrm{diag}(s)$. Let $S_0 = \{j : \beta_j = 0\}$. Let $[X, \widetilde{X}]_{\mathrm{swap}(S)}$ denote the $n \times (2p)$ matrix with columns $X_j$ and $\widetilde{X}_j$ swapped for $j \in S$. As $X^\top_j\widetilde{X}_k = X^\top_j X_k$ for $j \neq k$,

$$[X, \widetilde{X}]^\top_{\mathrm{swap}(S)}[X, \widetilde{X}]_{\mathrm{swap}(S)} = [X, \widetilde{X}]^\top[X, \widetilde{X}] \ \ \forall\ S \subseteq [p],$$

so that

$$[X, \widetilde{X}]^\top_{\mathrm{swap}(S)}Y \sim [X, \widetilde{X}]^\top Y \ \ \forall\ S \subseteq S_0.$$

Suppose $\mathrm{rank}(\Sigma) = p$. Given a knockoff $\widetilde{X}$, the LSE and its knockoff version are

$$\widehat{\beta} = \Sigma^{-1}X^\top Y, \quad \widetilde{\beta} = \Sigma^{-1}\widetilde{X}^\top Y.$$

For $W_j = \widehat{\beta}_j - \widetilde{\beta}_j$ and other statistics $W_j = w_j(X, \widetilde{X}, Y)$ satisfying

$$w_j([X, \widetilde{X}]_{\mathrm{swap}(S)}, Y) = (-1)^{I\{j\in S\}} w_j(X, \widetilde{X}, Y), \quad j \in [p], \tag{3.16}$$

$W \sim (\epsilon_1 W_1, \ldots, \epsilon_p W_p)^\top$ for any deterministic $\epsilon_{1:p}$ satisfying $\epsilon_j = 1$ for $\beta_j \neq 0$ and $\epsilon_j = \pm 1, j \in S_0$. Thus, $(\epsilon_1 W_1, \ldots, \epsilon_p W_p)^\top \big| \epsilon_{1:p} \sim W$ for any random $\epsilon_{1:p}$ independent of $W$ satisfying $\epsilon_j = 1$ for $\beta_j \neq 0$ and $\epsilon_j = \pm 1, j \in S_0$. It follows that

$$\mathbb{P}\Big\{\mathrm{sgn}(W_j) = \pm 1, j \in S_0 \Big| W_j, j \notin S_0, |W_j|, j \in S_0\Big\} = 2^{-|S_0|}. \tag{3.17}$$

The knockoff+ rule is to reject all $H_j : \beta_j = 0$ with $W_j \geqslant T$, where

$$T = \inf\left\{t \geqslant 0 : \frac{1 + \#\{j : W_j \leqslant -t\}}{\#\{j : W_j \geqslant t\} \vee 1} \leqslant q\right\}.$$

The ideal is $\mathbb{E}[\#\{j : \beta_j = 0, W_j \geqslant T\}] = \mathbb{E}[\#\{j : \beta_j = 0, W_j \leqslant -T\}]$ by symmetry.

**Theorem 3.13.** $\mathrm{FDR} \leqslant q$ *for the knockoff+ rule.*

Proof. The false discovery proportion (FDP) is bounded by

$$\mathrm{FDP} = \frac{\#\{j : \beta_j = 0, W_j \geqslant T\}}{\#\{j : W_j \geqslant T\} \vee 1} \leqslant q \times \frac{\#\{j : \beta_j = 0, W_j \geqslant T\}}{1 + \#\{j : \beta_j = 0, W_j \leqslant -T\}}.$$

Because $\#\{j : \beta_j = 0, W_j \geqslant t\}/(1 + \#\{j : \beta_j = 0, W_j \leqslant -t\})$ is a super-martingale,

$$\mathbb{E}\left[\frac{\#\{j : \beta_j = 0, W_j \geqslant T\}}{1 + \#\{j : \beta_j = 0, W_j \leqslant -T\}}\right] \leqslant \sum_{j=1}^{m_0'} \binom{m_0'}{j} \frac{j/2^{m_0'}}{1 + m_0' - j} = \sum_{j=0}^{m_0'-1} \binom{m_0'}{j} 2^{m_0'} \leqslant 1,$$

where $m_0' = \#\{j : \beta_j = 0, W_j \neq 0\}$. Here is the martingale argument. It suffices to consider $0 < |W_j| \neq |W_k|$ for all $j \neq k$. Let $V_\pm(t) = \#\{j : \beta_j = 0, \pm W_j \geqslant t\}$ and $\mathscr{F}_t = \sigma(V_\pm(s), s \leqslant t; W_j, j \notin S_0)$. Let $t_1 < t_2 < \ldots < t_{m_0}$ be the ordered values of $|W_j| I\{\beta_j = 0\}, j \in S_0$. Given $t = t_k$ and $V_-(t-) > 0$,

$$\mathbb{E}\left[\frac{V_+(t)}{1 + V_-(t)}\middle|\mathscr{F}_{t-}\right] = \frac{(V_+(t-) - 1)V_+(t-)}{(1 + V_-(t-))(V_-(t-) + V_+(t-))} + \frac{V_+(t-)}{(V_-(t-) + V_+(t-))} = \frac{V_+(t-)}{1 + V_-(t-)}.$$

Given $t = t_k$ and $V_-(t-) = 0$, $\mathbb{E}\big[V_+(t)/(1 + V_-(t))\big|\mathscr{F}_{t-}\big] = (V_+(t-) - 1)/(1 + V_-(t-))$. □

Alternatively, for the more aggressive

$$T = \inf\left\{t \geqslant 0 : \frac{\#\{j : W_j \leqslant -t\}}{\#\{j : W_j \geqslant t\} \vee 1} \leqslant q\right\}$$

with $q \in (0, 1)$. A similar analysis leads to the control of a modified FDR as follows,

$$\mathbb{E}\left[\frac{\#\{j : \beta_j = 0, W_j \geqslant T\}}{\#\{j : W_j \geqslant T\} + 1/q}\right] \leqslant q.$$

The use of knockoffs to control FDR was proposed in [BC15]. [CFJL18] proposed "model-X knockoffs" to weaken the knockoff condition to

$$[X, \widetilde{X}]_{\mathrm{swap}(S)} \sim [X, \widetilde{X}] \;\; \forall\, S \subseteq [p], \quad \widetilde{X} \perp\!\!\!\perp Y | X.$$

3.4. **Problems.**

**Exercise 3.1.** *Let* $X \sim N(\mu, 1)$*. Consider* $H_0 : \mu = 0$ *against* $H_1 : \mu \neq 0$*. Prove that the test* $\phi = I\{|X| > K\}$ *is not a UMP test at level* $\alpha = 0.05$ *for any* $K$*.*

**Exercise 3.2.** *Show that the F-test in Example 3.3 is a GLRT.*

**Exercise 3.3.** *Let* $X_{1:n}$ *be an iid sample from* $p_G$*. Consider testing the simple* $H_0 : p_G = \varphi \sim N(0, 1)$ *against simple* $H_1 : p_G = (1-\pi_n)\varphi + \pi_n p_{\theta_n}$ *with* $\pi_n = n^{-\beta}$ *and* $\theta_n = \sqrt{2r \log n}$*. For what* $(\beta, r)$ *the alternative is detectable in the sense of*

$$\text{Type-I error} + \text{Type-II error} = o(1)?$$

**Exercise 3.4.** *Derive the GLRT for testing* $H_0 : (\mu, \sigma) = (0, 1)$ *against* $H_1 : \sigma \geqslant 1$ *based on iid* $X_{1:n}$ *from* $N(\mu, \sigma)$*. Find the limiting distribution of the GLRT statistic* $\Lambda_n$*.*

**Exercise 3.5.** *Find the tangent set* $\mathscr{T}$ *for testing* $H_0 : p_G = \varphi \sim N(0, 1)$ *against the two-component alternative* $H_1 : p_G \in \mathscr{P}_2 \backslash \{\varphi\}$ *for Gaussian mixtures with means in* $[-M, M]$ *and unit variance as in Example 3.4 with fixed* $M$*. Let* $\Lambda_{\infty,M}$ *be the limit of the GLRT statistic. Does* $\Lambda_{\infty,M}$ *converge to a finite limit as* $M \to \infty$*?*

**Exercise 3.6.** *Let* $\widehat{\mu}_j \sim N(\mu_j, 1)$*,* $j \leqslant m$*. The Bonferroni adjustment yields simultaneous confidence intervals (SCI)* $\mu_j \in \widehat{\mu}_j \pm \widehat{\sigma} z_{\alpha/(2m)}$*, where* $z_\alpha$ *is* $(1-\alpha)$ *quantile of the standard normal distribution. Such SCI can be obtained by converting the Bonferroni single step test into a confidence region. What is the shape of the confidence region converted from Holm's step-down test?*

**Exercise 3.7.** *Suppose* $\widehat{\mu}_{1:m} \sim N(\mu_{1:m}, I_m)$ *is available with unknown* $\mu_{1:m} = (\mu_1, \ldots, \mu_m)'$*. What are the rejection/acceptance regions of the closure test for* $H_j : \mu_j = 0, j = 1, \ldots, m$*, based on the two-sided* $z$*- and* $\chi^2$*-tests for* $H_A = \{\mu_{1:m} : \mu_j = 0, j \in A\}$*? Is this closure test (always) more powerful than Bonferroni's single step test?*

**Exercise 3.8.** *Suppose* $\widehat{\mu}_{1:m} \sim N(\mu_{1:m}, \Sigma)$ *and* $\widehat{\sigma}^2 \sim \sigma^2\chi^2_\nu/\nu$ *are independent statistics as in linear regression, where* $\Sigma \in \mathbb{R}^{m\times m}$ *with known* $\Sigma/\sigma^2$*,* $\mu_{1:m}$ *and* $\sigma$ *are unknown, and* $\chi^2_\nu$ *denotes a chi-square variable with* $\nu$ *degrees of freedom. For a small* $m$*, say* $m = 2$ *or* $3$*, what are the rejection/acceptance regions of the closure test for* $H_j : \mu_j = 0, j \leqslant m$*, based on the F-tests for* $H_A$*?*

**Exercise 3.9.** *For univariate two-sided* $z$*-tests based on* $\widehat{\mu}_{1:m} \sim N(\mu_{1:m}, \Sigma)$ *with known* $\Sigma$*, the p-values are* $p_j = 2\{1 - \Phi(|\widehat{\mu}_j|/\sigma_{jj})\}$*. Is* $p_{1:n}$ PRDS *when* $\mu_{1:m} = 0$*?*

**Exercise 3.10.** *In linear regression, we observe* $(Y_{1:m}, X) \in \mathbb{R}^{n\times(1+d)}$ *satisfying* $Y_{1:m} \sim N(X\beta, \sigma^2 I_n)$*. Assume* $\operatorname{rank}(X) = d$*. Find some examples* $X$ *for which the LSE* $\widehat{\beta}$ *satisfies (or does not satisfy) the* PRDS *condition. How about ANOVA?*

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## 4. Nonparametric estimation

4.1. **Function spaces and bases.** In nonparametric estimation, smoothness conditions are typically imposed to motivate developments of sensible methods and provide theoretical guarantee. Smoothness can be expressed in terms of differentiability or speed of the decay of coefficients in certain bases.

Consider functions defined on an open domain $\Omega$ in $\mathbb{R}^d$. Typically, $\Omega$ is convex. For more explicit discussion, we consider $\Omega = [0,1]^d$. Let $\mathbb{N}_+$ be the set of nonnegative integers. For $k = (k_1, \ldots, k_d) \in \mathbb{N}_+^d$, let $f^{(k)} = f^{(k)}(x) = (\partial/\partial x_1)^{k_1} \cdots (\partial/\partial x_d)^{k_d} f(x)$. A function $f$ on $\Omega$ satisfies the Hölder smoothness condition of index $\alpha$ at a point $a_0 \in \Omega$ if

$$\max_{\alpha-1 \leqslant \|k\|_1 < \alpha} \sup_{b:\|b-a_0\|_2 \leqslant h} \frac{|f^{(k)}(b) - f^{(k)}(a_0)|}{\|b-a_0\|_2^{\alpha-\|k\|_1}} \leqslant C_0 \tag{4.1}$$

for some fixed $C_0 < \infty$ and $h > 0$ with $\{b : \|b - a_0\|_2 \leqslant h\} \subseteq \Omega$. The Hölder condition holds uniformly if $\|f\|_{L_\infty^\alpha(\Omega)} \leqslant C_0$, where

$$\|f\|_{L_\infty^\alpha(\Omega)} = \max_{\alpha-1 \leqslant \|k\|_1 < \alpha} \sup_{x\in\Omega, y\in\Omega, x\neq y} \frac{|f^{(k)}(x) - f^{(k)}(y)|}{\|x-y\|_2^{\alpha-\|k\|_1}}. \tag{4.2}$$

For $\alpha = 1$, the Hölder condition becomes the Lipschitz condition with

$$\|f\|_{L_\infty^1(\Omega)} = \sup_{x\in\Omega, y\in\Omega, x\neq y} \frac{|f(x) - f(y)|}{\|x-y\|_2}.$$

The Sobolev condition relaxes the supreme norm in (4.2) with the $L_p$ norm. Let $\alpha > 0$ and $1 \leqslant p \leqslant \infty$. When $\alpha$ is an integer, the Sobolev norm of an $\alpha$-times differentiable function $f$ on $\Omega$ can be defined as

$$\|f\|_{W_p^\alpha(\Omega)} = \big(\textstyle\sum_{m=0}^{\alpha} \|f\|_{L_p^m(\Omega)}^p\big)^{1/p} \quad \text{with} \quad \|f\|_{L_p^m(\Omega)} = \big(\textstyle\sum_{\|k\|_1 = m} \|f^{(k)}\|_{L_p(\Omega)}^p\big)^{1/p}, \tag{4.3}$$

where $\|f\|_{L_p(\Omega)} = \big(\int_\Omega |f|^p dx\big)^{1/p}$. For non-integer $\alpha$, the Sobolev norm can be defined as

$$\|f\|_{W_p^\alpha(\Omega)} = \big(\textstyle\sum_{m=0}^{\lfloor\alpha\rfloor} \|f\|_{L_p^m(\Omega)}^p + \|f\|_{L_p^\alpha(\Omega)}^p\big)^{1/p}, \tag{4.4}$$

where $\|f\|_{L_p^\alpha(\Omega)}$ is the Hölder norm for $p = \infty$ and for $1 \leqslant p < \infty$

$$\|f\|_{L_p^\alpha(\Omega)} = \sum_{\|k\|_1 = \lfloor\alpha\rfloor} \left( \int_\Omega \int_\Omega \frac{|f^{(k)}(x) - f^{(k)}(y)|^p}{\|x-y\|_2^{p(\alpha-\lfloor\alpha\rfloor)+d}} dxdy \right)^{1/p}.$$

The Sobolev space $W_p^\alpha(\Omega)$ is defined as the completion of the set of all $\lceil\alpha\rceil$-times continuously differentiable functions with compact support under the norm $\|f\|_{W_p^\alpha(\Omega)}$. The Sobolev space can be also defined with the $L_p(\Omega)$ norm $\big(\int_\Omega |f|^p dx\big)^{1/p}$ replaced by $\big\{\int_\Omega |f|^p \gamma(dx)\big\}^{1/p}$ for a general measure $\gamma$, e.g. the standard Gaussian measure for $\Omega = \mathbb{R}^d$.

In the unit cube $\Omega = [0,1]^d$, it is often more convenient to treat the Sobolev space in terms of the speed of decay of coefficients in commonly used bases. For $d = 1$ let

$$f^{(k)}(x) = \theta_0^{(k)} + \textstyle\sum_{j=1}^{\infty}\theta_j^{(k)}\sqrt{2}\cos^{(k)}(\pi jx), \quad x \in [0,1],$$

be the Fourier series expansion of $f^{(k)} \in L_2([0,1])$. For $f \in W_2^\alpha([0,1])$,

$$f(x) = \theta_0 + \textstyle\sum_{j=1}^{\infty}\theta_j\sqrt{2}\cos(\pi jx) = Q_f(x) + \sum_{j=1}^{\infty}\theta_j^{(\lfloor\alpha\rfloor)}(\pi j)^{-\lfloor\alpha\rfloor}\sqrt{2}\cos(\pi jx). \tag{4.5}$$

where $\theta_j = \theta_j^{(0)}$ and $Q_f$ is a polynomial of degree $\lfloor\alpha\rfloor - 1$.

**Lemma 4.1.** *Let $J_\alpha(x) = \int_0^x t^{-2\alpha-1}\{(\cos(\pi t)-1)^2+\sin^2(\pi t)\}dt$ and $\|\theta\|_{2,\alpha} = (\sum_{j=0}^{\infty} j^{2\alpha}\theta_j^2)^{1/2}$ for $\theta = (\theta_1, \theta_2, \ldots)^\top$. Then, for functions $f$ on $[0,1]$ with the Fourier series expansion (4.5),*

$$(2/3)J_\alpha(1)\|\theta\|_{2,\alpha}^2 \leqslant \|f\|_{L_2^\alpha([0,1])}^2 \leqslant 2J_\alpha(\infty)\|\theta\|_{2,\alpha}^2, \ \forall\ 0 < \alpha < 1. \tag{4.6}$$

*For $\alpha > 0$, $Q_f$ and $\theta_j^{(\lfloor\alpha\rfloor)}$ in (4.5) and $g_m(x) = \theta_0 + \sum_{j=1}^{m-1}\theta_j^{(\lfloor\alpha\rfloor)}(\pi j)^{-\lfloor\alpha\rfloor}\sqrt{2}\cos(\pi jx)$,*

$$\big\|f^{(\beta)} - Q_f^{(\beta)} - g_m^{(\beta)}\big\|_{L_q([0,1])} \leqslant C_{\alpha,\beta,q}m^{\beta-\alpha+1/2-1/q}\|f\|_{L_2^\alpha([0,1])} \tag{4.7}$$

*for $0 \leqslant 1/2 - 1/q < \alpha - \beta$, $\beta \in \mathbb{N}_+$ and certain $C_{\alpha,\beta,q}$ depending on $(\alpha,\beta,q)$ only. Moreover,*

$$\|\theta\|_{2,0}^2 + c_\alpha^2\|\theta\|_{2,\alpha}^2 \leqslant \|f\|_{W_2^\alpha([0,1])}^2 \leqslant \|\theta\|_{2,0}^2 + C_\alpha^2\|\theta\|_{2,\alpha}^2, \ \forall\ \alpha > 0, \tag{4.8}$$

*when $Q_f = \theta_0$, where $c_\alpha$ and $C_\alpha$ are positive constants depending on $\alpha$ only.*

Lemma 4.1 provides the equivalence between the Sobolev norm $\|f\|_{W_2^\alpha([0,1])}$ on $f$ and the sequence norm $(\|\theta\|_{2,0}^2 + \|\theta\|_{2,\alpha}^2)^{1/2}$ on the Fourier coefficients when $Q_f = \theta_0$. Lemma 4.1 also provides error bounds for the approximation $Q_f + g_m$ as consequences of this equivalence. These features of $W_2^\alpha([0,1])$ can be extended to Sobolev spaces in more general $\Omega \subseteq \mathbb{R}^d$ with proper adjustments. In particular, an extension of Lemma 4.1 can be directly carried out to $W_2^\alpha([0,1]^d)$ with the multivariate Fourier basis.

For any orthonormal basis $\{\phi_j, j \geqslant 1\}$ in $L_2([0,1])$, the tensor products $\phi_{j_1,\ldots,j_d}(x) = \prod_{a=1}^d \phi_{j_a}(x_a)$, $(j_1,\ldots,j_d)^\top \in \mathbb{N}_+^d$, form an orthonormal basis in $L_2([0,1]^d)$ and

$$f(x) = \sum_{(j_1,\ldots,j_d)^\top \in \mathbb{N}_+^d} \theta_{j_1,\ldots,j_d}\phi_{j_1,\ldots,j_d}(x), \quad \theta_{j_1,\ldots,j_d} = \int_{[0,1]^d} f(x)\phi_{j_1,\ldots,j_d}(x)dx, \tag{4.9}$$

as an infinite series expansion of $f \in L_2([0,1]^d)$. Let $\theta = (\theta_{j_1,\ldots,j_d}, j_a \in \mathbb{N}_+, 1 \leqslant a \leqslant d)^\top$ be the coefficient sequence in this product basis. For the Fourier basis $\phi_j(t) = \sqrt{2}\cos(\pi jt), t \in [0,1]$, Lemma 4.1 implies that the Sobolev norm $\|f\|_{W_2^\alpha([0,1]^d)}$ on functions $f$ is equivalent to the sequence norm $(\|\theta\|_{2,0,d}^2 + \|\theta\|_{2,\alpha,d}^2)^{1/2}$ when the sequence norm is finite, where

$$\|\theta\|_{2,\alpha,d} = \big(\textstyle\sum_{(j_1,\ldots,j_d)^\top \in \mathbb{N}_+^d}(j_1^{2\alpha} + \cdots + j_d^{2\alpha})\theta_{j_1,\ldots,j_d}^2\big)^{1/2}. \tag{4.10}$$

**Lemma 4.2.** *For certain positive constants $c_{\alpha,d}$ and $C_{\alpha,d}$ depending on $(\alpha, d)$ only,*

$$\|\theta\|_{2,0,d}^2 + c_{\alpha,d}^2\|\theta\|_{2,\alpha,d}^2 \leqslant \|f\|_{W_2^\alpha([0,1]^d)}^2 \leqslant \|\theta\|_{2,0,d}^2 + C_{\alpha,d}^2\|\theta\|_{2,\alpha,d}^2, \ \forall\ \alpha > 0$$

*for the Sobolev norm in (4.3) and (4.4) and the sequence norm $\|\theta\|_{2,\alpha,d}$ in (4.10), provided that $\|\theta\|_{2,\alpha,d} < \infty$. For such a function $f$ and its Fourier series approximation $g_m(x)$ with $m$-terms with the smallest $j_1^2 + \ldots + j_d^2$,*

$$\|f^{(\beta)} - Q_f^{(\beta)} - g_m^{(\beta)}\|_{L_q([0,1]^d)} \leqslant C_{\alpha,\beta,d,q} m^{\beta/d-\alpha/d+1/2-1/q}\|f\|_{L_2^\alpha([0,1]^d)} \tag{4.11}$$

*for $0 \leqslant 1/2 - 1/q < \alpha/d - \beta/d$, $\beta \in \mathbb{N}_+$, a certain polynomial $Q_f$ of degree $\lfloor\alpha\rfloor$, and certain constants $C_{\alpha,\beta,d,q}$ depending on $(\alpha, \beta, d, q)$ only.*

Lemma 4.2 gives $W_2^\alpha([0,1]^d) \subset W_q^\beta([0,1]^d)$ for $\alpha/d - 1/2 > \beta/d - 1/q$ with $\beta \in N_+$ and $p \geqslant 2$ as a special case of the Sobolev and Kondrachov embedding theorems for more general $\Omega$ and $\beta$. For convex $\Omega \in \mathbb{R}^d$, such relationships can be studied via the Riemann–Liouville formula in Taylor's expansion: For $x = (x_1, \ldots, x_d)^\top \in \mathbb{R}^d$ and $k = (k_1, \ldots, k_d)^\top \in \mathbb{N}_+^d$,

$$\begin{aligned} f(x) &= \sum_{0 \leqslant \|k\|_1 < \lfloor\alpha\rfloor} f^{(k)}(a_0)\frac{(x-a_0)^k}{k!} \\ &\quad + \sum_{\|k\|_1 = \lfloor\alpha\rfloor} \int_0^1 \frac{f^{(k)}(a_0 + t(x-a_0))(x-a_0)^k}{k!/\lfloor\alpha\rfloor}(1-t)^{\lfloor\alpha\rfloor-1}dt, \end{aligned} \tag{4.12}$$

where $x^k = \prod_{a=1}^d x_a^{k_a}$ and $k! = \prod_{a=1}^d k_a!$. For $\beta \in \mathbb{N}_+$, let $C^\beta(\Omega)$ be the space of all functions $f$ with continuous and uniformly bound $f^{(k)}$ in $\Omega$ for all $0 \leqslant \|k\|_1 \leqslant \beta$. Lemma 4.2 also gives $W_2^\alpha([0,1]^d) \subset C^\beta([0,1]^d)$ for $\alpha/d - 1/2 > \beta/d$. By (4.11) with $q = \infty$ and the Riesz representation theorem, $W_2^\alpha([0,1]^d)$ is a reproducing kernel Hilbert space (RKHS) for $\alpha > d/2$: $f(x) = \langle\delta_x^\alpha, f\rangle_{W_2^\alpha(\Omega)}$ with a unique $\delta_x^\alpha \in W_2^\alpha([0,1]^d)$. In fact, for $2 > d/\alpha$, $W_2^\alpha(\Omega)$ is an RKHS for general open $\Omega \subset \mathbb{R}^d$.

For $\Omega = \mathbb{R}^d$, the following lemma is parallel to Lemma 4.2.

**Lemma 4.3.** *Let $\breve{f}(\omega) = (2\pi)^{-d/2}\int\exp(-\sqrt{-1}\omega^\top x)f(x)dx$ be the Fourier transformation of $f$. For certain positive constants $c_{\alpha,d}$ and $C_{\alpha,d}$ depending on $(\alpha, d)$ only,*

$$\|\breve{f}\|_{L_2}^2 + c_{\alpha,d}^2\big\|\|\omega\|_2^\alpha\breve{f}\big\|_{L_2}^2 \leqslant \|f\|_{W_2^\alpha([0,1]^d)}^2 \leqslant \|\breve{f}\|_{L_2}^2 + C_{\alpha,d}^2\big\|\|\omega\|_2^\alpha\breve{f}\big\|_{L_2}^2, \ \forall\ \alpha > 0.$$

Proof. For $0 < \alpha < 1$, Plancherel's identity yields

$$\begin{aligned} \int\int\frac{|f(x)-f(y)|^2}{\|x-y\|_2^{2\alpha+d}}dxdy &= \int\left(\int|f(u+v)-f(u-v)|^2du\right)\frac{2^d}{\|2v\|_2^{2\alpha+d}}dv \\ &= \int\left(\int\big|2\sin(\omega^\top v)\breve{f}(w)\big|^2d\omega\right)\frac{2^{-2\alpha}}{\|v\|_2^{2\alpha+d}}dv \\ &= \int\frac{|2\sin(z_1)|^2}{2^{2\alpha}\|z\|_2^{2\alpha+d}}dz\int\big|\breve{f}(w)\big|^2\|\omega\|_2^{2\alpha}d\omega \end{aligned}$$

with $\|z\|_2 = \|v\|_2\|\omega\|_2$. The conclusion follows as the Fourier transformation of $(\partial/\partial x_j)f(x)$ is $\sqrt{-1}\omega_j\breve{f}(\omega)$ by the Fourier inversion formula $f(x) = (2\pi)^{-d/2}\int\exp(\sqrt{-1}\omega^\top x)\breve{f}(\omega)d\omega$. $\square$

In addition to the Fourier basis, spline bases are also commonly used. Consider $\Omega = [0,1]$ and positive integer $\alpha$. A function $g(x)$ is a spline of degree $\alpha$ (order $\alpha+1$) if it is a piecewise polynomial of degree $\alpha$ with continuous $g^{(k)}(x)$ for all $k < \alpha$. For $\alpha = 0$, $g(x)$ is piecewise constant. For $\alpha > 0$, $g^{(\alpha)}$ is piecewise constant. The discontinuities of $g^{(\alpha)}(x)$ are called knots for splines $g(x)$ of degree $\alpha$. It follows from the univariate version of (4.12),

$$g(x) = \sum_{k=0}^{\alpha-1} g^{(k)}(a_0)\frac{(x-a_0)^k}{k!} + \int_{a_0}^{x} g^{(\alpha)}(y)\frac{(x-y)^{\alpha-1}}{(\alpha-1)!}dy, \tag{4.13}$$

that given knots $0 = t_1 < \cdots < t_m = 1$, the collection of all splines of degree $\alpha$ form a linear space of dimension $\alpha + m - 1$. The following lemma gives a spline approximation error bound for functions in the Sobolev space.

**Lemma 4.4.** *Let $\alpha$ be a positive integer and $f \in W_p^\alpha([0,1])$. Let $S^\alpha(t_{1:m})$ be the space of all splines of degree $\alpha$ with knots $0 = t_1 < \cdots < t_m = 1$. Let $g_m \in S^\alpha(t_{1:m})$ be given by*

$$g_m^{(\beta)}(t_{(\alpha j+1)\wedge m}) = f^{(\beta)}(t_{(\alpha j+1)\wedge m}), \quad 0 \leqslant \alpha j + \beta + 1 < \alpha + m,\ 0 \leqslant \beta < \alpha. \tag{4.14}$$

*Suppose* $\min_{1\leqslant j<m}(t_{j+1} - t_j) \geqslant \Delta$ *and* $\max_{1\leqslant j<m}(t_{j+1} - t_j) \leqslant 3\Delta$. *Then,*

$$\|f^{(\beta)} - g_m^{(\beta)}\|_{L_q([0,1])} \leqslant C_{\alpha,\beta,p,q}\|f\|_{L_p^\alpha([0,1])}\Delta^{\alpha-\beta-1/p+1/q} \tag{4.15}$$

*for $0 \leqslant 1/p - 1/q \leqslant \alpha - \beta$, $\beta \in \mathbb{N}_+$ and certain $C_{\alpha,\beta,p,q}$ depending on $(\alpha, \beta, p, q)$ only.*

We note that the number of boundary conditions in (4.14), $\alpha + m - 1$, matches the dimension of $S^\alpha(t_{1:m})$, and the spline $g_m$ in the interval $[t_{\alpha j+1}, t_{(\alpha j+\alpha+1)\wedge m}]$ is completely determined by the boundary conditions at two endpoints of the interval. For $\alpha = 1$, $g_m$ is the linear interpolation of $(t_j, f(t_j)), 1 \leqslant j \leqslant m$.

When the knots are equally spaced and $p = 2$, the error bound in (4.15) for the spline approximation is of the same order as the error bound in (4.7) for the Fourier series approximation in subspaces of comparable dimension. However, compared with the Fourier basis, the spline basis requires a specification of the degree $\alpha$. In Lemma 4.4, the knots are required to be roughly equally spaced, but this condition is not very restrictive. For any knot sequence $0 = t_1 < \cdots < t_m = 1$ with $\Delta = \max_{1\leqslant j<m}(t_{j+1} - t_j)$, there exists a subsequence of knots $0 = t'_1 < \cdots < t'_{m'} = 1$ such that $\min_{1\leqslant j<m'}(t'_{j+1} - t'_j) \geqslant \Delta$ and $\max_{1\leqslant j<m'}(t'_{j+1} - t'_j) < 3\Delta$.

Splines can be expressed on terms of local basis functions supported on small intervals. Specifically, given knots $t_1 < \cdots < t_m$, the B-spline basis is defined, with external knots $\cdots < t_{-1} < t_0 < t_1$ and $t_m < t_{m+1} < \cdots$ and the initialization $B_{j,1}(x) = I\{t_j \leqslant x < t_{j+1}\}$, recursively by

$$B_{j,k+1}(x) = w_{j,k}(x)B_{j,k}(x) + \big(1 - w_{j+1,k}(x)\big)B_{j+1,k}(x), \tag{4.16}$$

with weights $w_{j,k}(x) = I\{t_{j+k} \neq t_j\}(x - t_j)/(t_{j+k} - t_j)$. We note that by induction

$$\mathrm{supp}(B_{j,k}) = [t_j, t_{j+k}), \quad \sum_{j=-\infty}^{\infty} B_{j,k}(x) = 1. \tag{4.17}$$

It follows that the value of $B_{j,k}(x)$ for $t_1 \leqslant x < t_m$, determined by (4.17), does not depend on the choice of the external knots.

The B-splines $B_{j,\alpha+1}(x)I\{x \geqslant t_1\}, j = -\alpha + 1, \ldots, m - 1$, form an algebraic basis in the space $S^\alpha(t_{1:m})$ of all degree-$\alpha$ splines. Due to $B_{j-1,2}(t_j) = 1$ in (4.17), for $\alpha = 1$ the spline approximation in (4.14) is the linear interpolation conveniently given by

$$g_m(x) = \sum_{j=1}^{m} f(t_j)B_{j-1,2}(x) \ \text{ with } \ g_m(t_j) = f(t_j), 1 \leqslant j \leqslant m.$$

We may also introduce tied knots to allow violation of the spline boundary conditions. The recursive construction of B-splines in (4.16) still works with tied knots. For example, when $t_0 \to t_1-$, $B_{0,2}(x) \to I\{t_1 \leqslant x < t_2\}(t_2 - x)/(t_2 - t_1)$ with a discontinuity at $t_1$. It turns out that the multiplicity of the tie at $t_j$ is the number of the constraints removed from the highest order. From this point of view a spline in $S^\alpha(t_{1:m})$ is identified with a spline in $S^\alpha(t_1 1_{1:\alpha}, t_{1:m}, t_m 1_{1:\alpha})$ in $\mathbb{R}$ with the tied knots to indicate the removal of all the continuity constraints for $g^{(\beta)}(x), 0 \leqslant \beta < \alpha$, at $t_1$ and $t_m$. Here $t1_{1:k}$ represents a string of $t's$ of length $k$. Internal tied knots are also allowed as in the following example.

**Example 4.1.** *Given an $\alpha - 1$ times differentiable function $f$ in $[0, 1]$ and knots $0 = t_1 < \cdots < t_m = 1$, there exists uniquely an $\alpha - 1$ times continuously differentiable piecewise polynomial $g_m$ of degree $2\alpha - 1$ such that*

$$g_m^{(\beta)}(t_j) = f^{(\beta)}(t_j), \ 0 \leqslant \beta < \alpha, 1 \leqslant j \leqslant m, \tag{4.18}$$

*and the solution is a spline in the $(\alpha m)$-dimensional space*

$$S^{2\alpha-1}\big(t_1 1_{1:(2\alpha)}, t_2 1_{1:\alpha}, \ldots, t_{m-1} 1_{1:\alpha}, t_m 1_{1:(2\alpha)}\big)$$

*with the convention $g_m^{(\beta)}(t_m) = g_m^{(\beta)}(t_m-), 0 \leqslant \beta < \alpha$. Moreover, (4.15) holds for $g_m$ with $\Delta = \max_{1 \leqslant j < m}(t_j - t_{j-1})$.*

4.2. **Density estimation.**

4.2.1. *Kernel methods.* Let $X_1, \ldots, X_n$ be iid data points from a density $p(\cdot)$ with an open support $\Omega \subseteq \mathbb{R}^d$. For the estimation of the density $p(a_0)$ at a given point $a_0 \in \Omega$, a kernel density estimator [Ros56, Par62] can be written as

$$\widehat{p}(a_0) = \overline{\mathbb{P}}_n\big[K(a_0, \cdot)\big] = n^{-1}\textstyle\sum_{i=1}^n K\big(a_0, X_i\big) \tag{4.19}$$

for some kernel function $K(a_0, y)$. We note that

$$\mathbb{P}\big\{\widehat{p}(x) \text{ is a density function in } \Omega\big\} = 1$$

when $K(x, y) \geqslant 0$ and $\int_\Omega K(x, y)dx = 1$.

We typically choose a kernel satisfying $K(x, y) = K_1((x - y)/h)/h^d$ for some density function $K_1(x)$ in $\mathbb{R}^d$ and bandwidth $h = h_n \to 0+$. Examples of $K_1(\cdot)$ include the uniform density in the unit ball, and the standard Gaussian density in $\mathbb{R}^d$. However, the form of the kernel is not crucial, and we may actually want to use different kernels at different location $a_0$, especially near the boundary of $\Omega$.

The kernel density estimator is relatively easy to analyze as it is an average of iid random variables. For simplicity assume that for a suitable bandwidth $h > 0$

$$\operatorname{supp}(K(a_0, \cdot)) \subset \operatorname{Ball}(a_0, h) \cap \Omega, \quad \textstyle\int_\Omega \big|K(a_0, y)\big|^2 dy \leqslant C_1^2/h^d, \tag{4.20}$$

to guarantee a finite variance of $\widehat{p}(a_0)$ and the usage of only data points near $a_0$. The simplest kernel satisfying the above condition is the density of the uniform distribution in the ball $\operatorname{Ball}(a_0, h)$ with center $a_0$ and radius $h$. However, to remove the bias of higher order, we use higher order kernels to estimate densities of higher order of smoothness. Specifically, to utilize smoothness conditions of order $\alpha > 1$ on the underlying density $p(x)$, we select a kernel satisfying the following conditions:

$$\textstyle\int_\Omega K(a_0, y)dy = 1, \quad \int_\Omega K(a_0, y)(y - a_0)^k dy = 0,\ 0 < \|k\|_1 < \alpha, \tag{4.21}$$

where $x^k = \prod_{a=1}^d x_a^{k_a}$ for $x = (x_1, \ldots, x_d)^\top \in \mathbb{R}^d$ and $k = (k_1, \ldots, k_d)^\top \in \mathbb{N}_+^d$.

**Theorem 4.1.** *Let $a_0 \in \Omega$ and $p(x)$ be a density in $\Omega$ satisfying Hölder's condition of order $\alpha$ at $a_0$ as in* (4.1). *Let $\widehat{p}(a_0)$ be the kernel density estimator* (4.19) *based on iid observations from $p(\cdot)$. Suppose that the kernel $K(a_0, y)$ satisfies conditions* (4.20) *and* (4.21) *with $h = h_n \asymp n^{-1/(d+2\alpha)}$. Then, the mean squared risk of $\widehat{p}(a_0)$ is bounded by*

$$\mathbb{E}\Big[\big(\widehat{p}(a_0) - p(a_0)\big)^2\Big] \leqslant \left(\frac{C_1 \pi^{d/4} C_0 d^{\alpha/2} h^\alpha}{\Gamma^{1/2}(d/2+1)\alpha!}\right)^2 + \frac{C_1^2}{h^d n} \lesssim n^{-2\alpha/(d+2\alpha)}.$$

*If in addition $\alpha$ is an integer and $p(x)$ is $\alpha$-times continuously differentiable at $a_0$, then*

$$n^{-\alpha/(d+2\alpha)}\big(\widehat{p}(a_0) - p(a_0)\big) \rightsquigarrow N\big(\mu(a_0), \sigma^2(a_0)\big) \tag{4.22}$$

*for $K(x, y) = K_1((x - y)/h)/h^d$ and $h = (1 + o(1))C_3 n^{-1/(d+2\alpha)}$, where*

$$\mu(a_0) = C_3^\alpha \sum_{\|k\|_1 = \alpha} \frac{p^{(k)}(a_0)}{k!} \int K_1(y) y^k dy, \quad \sigma^2(a_0) = \frac{p(a_0)}{C_3^d} \int K_1^2(y)dy.$$

We leave the proof of Theorem 4.1 as an exercise. When the conditions Theorem 4.1 hold uniformly for all $a_0$ in $\Omega$ and the volume $|\Omega|$ is finite, Theorem 4.1 automatically provides an $L_2(\Omega)$ error bound for $\widehat{p}(\cdot)$. For $\Omega = \mathbb{R}^d$, $L_2(\Omega)$ error bounds can be obtained under a moment condition on the density $p(x)$ with proper adjustments of the analysis. For the $L_2(\mathbb{R}^d)$ risk of projections in the Fourier domain, see Theorem 4.4.

For $K(x,y) = K_1((x-y)/h)/h^d$, (4.20) and (4.21) hold when

$$\begin{aligned}&\mathrm{supp}(K_1) \subset \mathrm{Ball}(0,1), \quad \|K_1\|_{L_2} \leqslant C_1, \quad \mathrm{Ball}(a_0,h) \subset \Omega,\\ &\textstyle\int_{\mathbb{R}^d} K_1(x)dx = 1, \quad \int_{\mathbb{R}^d} K_1(x)x^k dx = 0 \ \forall 0 < \|k\|_1 < \alpha.\end{aligned} \tag{4.23}$$

Condition (4.23) holds for $\alpha \leqslant 2$ and $\mathrm{Ball}(a_0,h) \subset \Omega$ when $K_1(x)$ is a density function in $\mathrm{Ball}(0,1)$ with zero first moment and finite second moment. However, for $\alpha > 2$, (4.23) implies that $K_1(x) < 0$ with positive Lebesgue measure.

For $a_0$ near the boundary of $\Omega$, we need to take special care in the construction of the kernel. For example, we may take $K(a_0,y) \propto \{1 - Q((a_0-y)/h)\}K_1((a_0-y)/h)$ on $\mathrm{Ball}(a_0,h) \cap \Omega$ with a suitable polynomial $Q(\cdot)$ of degree $\lceil \alpha \rceil - 1$ to fulfill (4.21). See Example 4.2 below in our discussion of kernel regression. Again, for $\alpha > 1$ and $a_0$ close to the boundary of $\Omega$, $K(a_0,y) < 0$ typically has positive Lebesgue measure. Thus, to achieve a squared $L_2(\Omega)$ convergence rate of faster order than $n^{-2/(d+2)}$ with $\alpha > 1$, such a construction does not guarantee $\mathbb{P}\{\widehat{p}(x) \geqslant 0 \ \forall x\} = 1$.

The rate in Theorem 4.1 is optimal. Let

$$p_t(x) = p_0(x) + t^\alpha g_0((x-a_0)/t) \tag{4.24}$$

with an $\alpha$-times continuously differentiable $g_0(x)$ in $\mathbb{R}^d$ satisfying

$$g_0(0) \neq 0, \quad \mathrm{supp}(g_0(x)) = \mathrm{Ball}(0,1), \quad \textstyle\int_{\|x\|_2 \leqslant 1} g_0(x)x^k dx = 0 \ \forall 0 \leqslant \|k\|_1 \leqslant \alpha.$$

**Theorem 4.2.** *Let $p_0(x)$ be a density in an open $\Omega$ satisfying Hölder's condition of order $\alpha$ at an $a_0 \in \Omega$ with a fixed $h$ in* (4.1). *Suppose $p_0(a_0) > 0$. Let $p_t(x)$ be as in* (4.24). *Let $\mathbb{P}_t$ be the probability under which the observations $X_1, \ldots, X_n$ are iid with density $p_t$. Then, Hölder's condition* (4.1) *also holds for $p_t$ for $t \in [0,h]$ and for a certain fixed $c_3 > 0$*

$$\min_{\widehat{p}} \max_{t=0,c_3 n^{-1/(d+2\alpha)}} \mathbb{P}_t\Big\{|\widehat{p}(a_0) - p_t(a_0)| > |g_0(0)/2| c_3^\alpha n^{-\alpha/(d+2\alpha)}\Big\} > 1/4.$$

Proof. The Hellinger distance between $p_t$ and $p_0$ is bounded by

$$\begin{aligned} d_H^2(p_t,p_0) &= \int_\Omega \big(1 - \sqrt{p_t(x)/p_0(x)}\big)^2 p_0(x)dx \\ &\leqslant \int_{\|x-a_0\|_2 \leqslant t} \frac{\big(t^\alpha g_0((x-a_0)/t)\big)^2}{p_0(x)} dx \\ &\leqslant \big(\pi^{d/2}/\Gamma(d/2+1)\big) t^{2\alpha+d} \|g_0\|_{L_\infty}^2 \|1/p_0\|_{L_\infty(\mathrm{Ball}(a_0,h))}. \end{aligned}$$

Thus, for $t = t_n = c_3 n^{-1/(d+2\alpha)}$, $\max_{n \geqslant 1} n d_H^2(p_{t_n}, p_0)$ is small when $c_3$ is small. The conclusion follows from Le Cam's lower bound or the risk of the Bayes rule for testing the simple null $p_0$ against the simple alternative $p_t$ with equal prior probability. □

4.2.2. *Orthogonal projections.* Consider the estimation of a density $p(x)$ in $\Omega$ under the $L_2(\Omega)$ loss. Suppose $p(x)$ is believed to be close to its $L_2(\Omega)$ projection to the linear span of linearly independent functions $\phi_j, 1 \leqslant j \leqslant m$, in $L_2(\Omega)$. Write the projection as

$$p_\theta = \sum_{j=1}^{m} \theta_j \phi_j, \quad \int_\Omega \phi_j(x)\Big(\sum_{j=1}^{m} \theta_j \phi_j(x) - p(x)\Big)dx = 0. \tag{4.25}$$

Replacing $\int \phi_j(x)p(x)dx$ with $\overline{\mathbb{P}}_n[\phi_j]$ in the above estimating equation, we obtain

$$\widehat{p}(x) = \begin{pmatrix} \phi_1(x) \\ \vdots \\ \phi_m(x) \end{pmatrix} \Phi^{-1}_{1:m,1:m} \begin{pmatrix} \overline{\mathbb{P}}_n[\phi_j] \\ \vdots \\ \overline{\mathbb{P}}_n[\phi_j] \end{pmatrix} = \frac{1}{n}\sum_{i=1}^{n} K(x, X_i), \tag{4.26}$$

where $\Phi_{1:m,1:m}$ is the $m \times m$ matrix with elements $\Phi_{j,k} = \int_\Omega \phi_j(x)\phi_k(x)dx$, and

$$K(x,y) = \sum_{j=1}^{m}\sum_{k=1}^{m} \phi_j(x)(\Phi^{-1})_{j,k}\phi_k(y). \tag{4.27}$$

Although the motivation is different, (4.26) is again a kernel estimator. By (4.27)

$$\int_\Omega K(x,y)\phi_j(y)dy = \phi_j(x), \quad j = 1, \dots, m,$$

so that the kernel gives the $L_2(\Omega)$ orthogonal projection to the subspace generated by $\{\phi_1, \dots, \phi_m\}$ and the function $K(x, y)$ does not depend on the specific choice of the basis in the subspace. Moreover, since the trace of an orthogonal projection is its rank,

$$\int_\Omega K(x,x)dx = \int_\Omega\int_\Omega K^2(x,y)dxdy = m.$$

**Theorem 4.3.** *Let $p(\cdot)$ be a density in $\Omega$ and $p_\theta(\cdot)$ be its $L_2(\Omega)$ projection to an $m$-dimensional subspace with basis $\{\phi_1, \dots, \phi_m\}$ as in* (4.25). *Let $\widehat{p}(\cdot)$ be the estimator in* (4.26) *with the kernel in* (4.27) *based on iid observations $X_1, \dots, X_n$ from $p(\cdot)$. Then,*

$$\mathbb{E}\Big[\big\|\widehat{p}(\cdot) - p(\cdot)\big\|^2_{L_2(\Omega)}\Big] = \frac{m\sigma^2_\theta}{n} + \big\|p(\cdot) - p_\theta(\cdot)\big\|^2_{L_2(\Omega)} \tag{4.28}$$

*with $\sigma^2_\theta = \big(\int K(x,x)p(x)dx - \|p_\theta(\cdot)\|^2_{L_2(\Omega)}\big)/m \leqslant \|p(\cdot)\|_{L_\infty(\Omega)} - \|p_\theta(\cdot)\|^2_{L_2(\Omega)}/m$.*

The methodological and theoretical development described so far are dimension free due to their validity for any space $L_2(\Omega)$ with potentially infinite-dimensional $\Omega$. For example, the observations $X_i$ can be iid Gaussian processes.

Proof. As $K(x, y)$ is the projection kernel, by (4.25) and (4.26) $\widehat{p}(x)$ is an unbiased estimator of $p_\theta(x)$, so that the $L_2(\Omega)$ risk is given by (4.28) with $m\sigma^2_\theta = \int_\Omega \big(\int_\Omega K^2(x,y)p(y)dy\big)dx - \|p_\theta\|^2_{L_2(\Omega)}$. The conclusion follows as $\int_\Omega K^2(x,y)dx = K(y,y)$ by (4.27). □

Next we consider a density $p(\cdot)$ in an open set $\Omega$ in $\mathbb{R}^d$ with finite a certain Sobolev-type norm $\|\cdot\|_{\alpha,d}$ satisfying

$$\|f\|_{L_\infty(\Omega)} \leqslant C_1\|f\|_{\alpha,d}. \tag{4.29}$$

Moreover, we consider an $L_2(\Omega)$ orthogonal projection $K_{\phi_1,\ldots,\phi_m}$ to certain basis functions $\phi_1, \ldots, \phi_m$ such that

$$\|f - K_{\phi_1,\ldots,\phi_m} f\|_{L_2(\Omega)} \leqslant C_2 m^{-\alpha/d}\|f\|_{\alpha,d}. \tag{4.30}$$

**Corollary 4.1.** *Suppose $\|p(\cdot)\|_{\alpha,d} \leqslant C_0$ in Theorem 4.3 for a norm satisfying* (4.29) *and basis functions satisfying* (4.30). *Then, for the estimator $\widehat{p}$ in* (4.26) *and $m \asymp n^{1/(1+2\alpha/d)}$,*

$$\mathbb{E}\Big[\big\|\widehat{p}(\cdot) - p(\cdot)\big\|^2_{L_2(\Omega)}\Big] \leqslant C_1\|p(\cdot)\|_{\alpha,d}(m/n) + C_2^2 m^{-2\alpha/d}\|p(\cdot)\|^2_{\alpha,d} \lesssim n^{-2\alpha/(d+2\alpha)}. \tag{4.31}$$

*Furthermore, when $\phi_1, \ldots, \phi_m$ are orthonormal in $L_2(\Omega)$, the factor $C_1\|p(\cdot)\|_{\alpha,d}$ in* (4.31) *can be replaced by $\|\sum_{j=1}^m \phi_j^2(x)/m\|_{L_\infty(\Omega)}$.*

Condition (4.29) holds for the Sobolev norm $\|\cdot\|_{\alpha,d} = \|\cdot\|_{W_2^\alpha([0,1]^d)}$ with $\alpha > d/2$ by Lemma 4.2 with $q = \infty$ and also for the Sobolev norm $\|\cdot\|_{W_p^\alpha(\Omega)}$ with $\alpha > d/p$ for general $\Omega$ by the general Sobolev embedding theorem. For $\Omega = [0,1]^d$, condition (4.30) holds for the product Fourier and spline bases by Lemmas 4.2 and 4.4 respectively with $q = 2$. When $\phi_1, \ldots, \phi_m$ are orthonormal in $L_2(\Omega)$, $K(x,y) = \sum_{j=1}^m \phi_j(x)\phi_j(y)$ by (4.27), so that

$$\int K(x,x)p(x)dx \leqslant \big\|\textstyle\sum_{j=1}^m \phi_j^2\big\|_{L_\infty(\Omega)}.$$

For the Fourier basis $\phi_{2j-1}(x) = \sqrt{2}\sin(2\pi jx)$ and $\phi_{2j}(x) = \sqrt{2}\cos(2\pi jx)$, $j \geqslant 1$, in $[0,1]$,

$$\textstyle\sum_{j=0}^{2m}\phi_j(x)\phi_j(y) = 1 + \sum_{j=1}^m 2\cos(2\pi j(x-y)) = \dfrac{\sin(\pi(2m+1)(x-y))}{\sin(\pi(x-y))}$$

and

$$1 + K(x,x) = \textstyle\sum_{j=0}^{2m}\phi_j^2(x) = 1 + 2m.$$

The convergence rate $n^{-\alpha/(d+2\alpha)}$ is again minimax optimal in Sobolev balls. Given a smoothness index $\alpha$, the optimal rate $n^{-\alpha/(d+2\alpha)} = n^{-(\alpha/d)/(1+2(\alpha/d))}$ deteriorates quickly as the dimension $d$ grows. A heuristic explanation of this phenomenon, commonly referred to as "the curse of dimensionality", is that the smoothness condition is useful only in $\mathrm{Ball}(a_0, h)$ for a small $h > 0$ but the number of observations in the ball is of the order $nh^d$.

For $\Omega = \mathbb{R}^d$, it is convenient to analyze projections in the Fourier domain. The idea is to estimate the Fourier transformation $\breve{p}(\omega) = (2\pi)^{-d/2}\langle\phi_\omega, p\rangle_{L_2} = (2\pi)^{-d/2}\int\overline{\phi}_\omega p(x)dx$ of the density $p(x)$ by its sample version $(2\pi)^{-d/2}\overline{\mathbb{P}}_n\big[\overline{\phi}_\omega\big]$, where $\phi_\omega = \phi_\omega(x) = e^{\sqrt{-1}\omega^\top x}$. The variance of $\overline{\mathbb{P}}_n\big[\overline{\phi}_\omega\big]$ is of order $O(1/n)$, and the signal to noise ratio deteriorates as the frequency grows. Thus, it makes sense to trim the high-frequency portion of the estimate. This leads to the following projection in the Fourier domain,

$$\widehat{p} = \widehat{p}(x) = (2\pi)^{-d}\int_{\|\omega\|_2\leqslant r_n}\phi_\omega(x)\overline{\mathbb{P}}_n\big[\overline{\phi}_\omega\big]d\omega = \overline{\mathbb{P}}_n\big[K_n(x,\cdot)\big] \tag{4.32}$$

with $K_n(x,y) = (2\pi)^{-d}\int_{\|\omega\|_2\leqslant r_n}\phi_\omega(x)\overline{\phi}_\omega(y)d\omega = (2\pi)^{-d}\int_{\|\omega\|_2\leqslant r_n}\phi_\omega(x-y)d\omega$.

**Theorem 4.4.** *Let $p(\cdot)$ be a density in $\mathbb{R}^d$ and $\widehat{p}(\cdot)$ be the estimator in* (4.32) *based on iid observations $X_1,\ldots,X_n$ from $p(\cdot)$. Let $k\in\mathbb{N}_+^d$ with $\|k\|_1=\beta$. Then,*

$$\mathbb{E}\Big[\big\|\widehat{p}^{(k)} - p^{(k)}\big\|_{L_2}^2\Big] \leqslant \frac{r_n^{d+2\beta}/n}{(\beta+d/2)\Gamma(d/2)2^d\pi^{d/2}} + \frac{\|p(\cdot)\|^2_{L_2^\alpha(\mathbb{R}^d)}}{c^2_{\alpha,d}r_n^{2(\alpha-\beta)}}.$$

*for the constant $c_{\alpha,d}$ in Lemma 4.3. Consequently, for $r_n \asymp \big(n\|p(\cdot)\|^2_{L_2^\alpha(\mathbb{R}^d)}\big)^{1/(d+2\alpha)}$,*

$$\mathbb{E}\Big[\big\|\widehat{p}^{(k)} - p^{(k)}\big\|_{L_2}^2\Big] \lesssim n^{-2(\alpha-\beta)/(d+2\alpha)}\|p(\cdot)\|^{2(d+2\beta)/(d+2\alpha)}_{L_2^\alpha(\mathbb{R}^d)}$$

Proof. By Plancherel's identity and Lemma 4.3, $\mathbb{E}\big[\|\widehat{p}^{(k)} - p^{(k)}\|^2_{L_2}\big]$ is no greater than

$$\begin{aligned}
&\mathbb{E}\bigg[(2\pi)^{-d}\int_{\|\omega\|_2\leqslant r_n}\|\omega\|_2^{2\beta}\big|(\overline{\mathbb{P}}_n-\mathbb{P})[\overline{\phi}_\omega]\big|^2d\omega\bigg] + \int_{\|\omega\|_2>r_n}\|\omega\|_2^{2\beta}\big|\breve{p}(\omega)\big|^2d\omega\\
\leqslant\;& \frac{r_n^{d+2\beta}/n}{(\beta+d/2)\Gamma(d/2)2^d\pi^{d/2}} + \frac{\|p(\cdot)\|^2_{L_2^\alpha(\mathbb{R}^d)}}{c^2_{\alpha,d}r_n^{2(\alpha-\beta)}}.
\end{aligned}$$

The second conclusion follows immediately. ☐

The projection in the Fourier domain provides near parametric rate of convergence for Gaussian mixtures.

**Theorem 4.5.** *Suppose $p(x) = \int\varphi(x-y)G(dy)$, where $\varphi(\cdot)$ denotes the standard normal density in $\mathbb{R}^d$ and $G$ is an arbitrary distribution in $\mathbb{R}^d$. Let $\widehat{p}(\cdot)$ be the estimator in* (4.32) *based on iid observations $X_1,\ldots,X_n$ from $p(\cdot)$. Then,*

$$\mathbb{E}\Big[\big\|\widehat{p}(\cdot) - p(\cdot)\big\|_{L_2}^2\Big] \leqslant \frac{r_n^d/n}{\Gamma(d/2+1)2^d\pi^{d/2}} + \frac{\mathbb{P}\big\{\chi^2_d > 2r_n^2\big\}}{2^d\pi^{d/2}}. \tag{4.33}$$

*Consequently, for $r_n = \sqrt{d/2}+\sqrt{\log n}$,*

$$\mathbb{E}\Big[\big\|\widehat{p}(\cdot) - p(\cdot)\big\|_{L_2}^2\Big] \leqslant \frac{1}{n}\bigg(\frac{(\sqrt{d}+\sqrt{2\log n})^d}{\Gamma(d/2+1)2^d\pi^{d/2}} + \frac{1}{2^d\pi^{d/2}}\bigg). \tag{4.34}$$

We note that (4.34) follows from (4.33) and the Gaussian concentration inequality for $\chi_d$ as a function with unit Lipschitz norm.

4.2.3. *Log-density projections.* A shortcoming of the kernel estimator (4.19) with $\alpha > 2$ and the projection estimator (4.26) is that $\widehat{p}$ is not guaranteed to be a density, i.e. the condition $\widehat{p}(x) \geqslant 0$ may fail to hold. As an alternative, we may consider the approximation of the log-density by a function in a certain finite dimensional space. We write such a candidate for the approximation of $p(x)$ as a member of the exponential family

$$p_\theta(x) = \exp\big(\phi_0(x) + \theta^\top \phi_{1:m}(x) - \psi(\theta)\big), \tag{4.35}$$

with a baseline measure $e^{\phi_0(x)}dx$, basis functions as elements of $\phi_{1:m} = (\phi_1, \ldots, \phi_m)^\top$, $\theta = (\theta_1, \ldots, \theta_m)^\top$ and $\psi(\theta) = \log \int_\Omega \exp\big(\phi_0(x) + \theta^\top \phi_{1:m}(x)\big)dx$. The idea is to pretend that $p(x)$ is actually of the form (4.35) and proceed to estimate it by the MLE [KS91],

$$\widehat{p}(x) = p_{\widehat{\theta}}(x), \quad \widehat{\theta} = \arg\max_\theta \overline{\mathbb{P}}_n\Big[\theta^\top \phi_{1:m} - \psi(\theta)\Big]. \tag{4.36}$$

As the true $p(x)$ is typically not of the form (4.35), (4.36) is a misspecified MLE.

Let $S_m$ be the linear span of $\{\phi_1, \ldots, \phi_m\}$. As the MLE (4.36) can be written as

$$\widehat{p}(x) = \frac{e^{\phi_0(x)+\widehat{f}(x)}}{\int_\Omega e^{\phi_0(y)+\widehat{f}(y)}dy}, \quad \widehat{f} = \arg\max_{f \in S_m} \prod_{i=1}^n \frac{e^{\phi_0(X_i)+f(X_i)}}{\int_\Omega e^{\phi_0(y)+f(y)}dy}, \tag{4.37}$$

it depends on the basis functions $\phi_j$ only through the space $S_m$ and centered $f \in S_m$. Thus, when $\int \phi_j^2(x)p(x)dx < \infty$, we may study (4.37) with orthonormal basis functions satisfying

$$\int_\Omega \phi_j(x)p(x)dx = 0, \quad \int_\Omega \phi_j(x)\phi_k(x)p(x)dx = I\{j = k\}, \quad 1 \leqslant j \leqslant k \leqslant m, \tag{4.38}$$

even though (4.37) is computed with a different set of basis of $S_m$ in (4.36). Under a system satisfying (4.38), the mapping $\theta \to p_\theta$ in (4.35) is invertible.

For $\theta \in \mathbb{R}^m$, the Kullback-Leibler divergence between $p(x)$ and $p_\theta(x)$ is

$$\mathrm{KL}(p\|p_\theta) = \int \log\big(p(x)/p_\theta(x)\big)p(x)dx = \mu_0 + \psi(\theta) - \int_\Omega \theta^\top \phi_{1:m}(x)p(x)dx. \tag{4.39}$$

with $\mu_0 = \int_\Omega \big(\log p(x) - \phi_0(x)\big)p(x)dx$. Because $\mu_0$ does not depend on $\theta$, the population version of (4.36), say $p^*$, must satisfy

$$p^* = p_{\theta^*}, \quad \theta^* = \arg\min_\theta K(p\|p_\theta), \quad \nabla\psi(\theta^*) = \mathbb{P}\big[\phi_{1:m}\big], \tag{4.40}$$

where $\nabla\psi(\theta) = (\partial/\partial\theta)\psi(\theta)$ is the gradient of $\psi(\theta)$. The solution $\theta^*$ is unique under (4.38), due to the strict convexity of $\psi(\theta)$, with $\mathbb{P}\big[\phi_{1:m}\big] = 0$. As in (4.37) we may also write

$$p^* = \frac{e^{\phi_0+f^*}}{\int_\Omega e^{\phi_0(x)+f^*(x)}dx}, \quad f^* = \arg\min_{f \in S_m}\bigg(\mu_0 + \log\int_\Omega e^{\phi_0(x)+f(x)}dx - \mathbb{P}[f]\bigg), \tag{4.41}$$

to emphasize that the population MLE $p^*$ depends on the basis only through the space $S_m$ and centered $f \in S_m$. In view of the condition $\nabla\psi(\theta^*) = \mathbb{P}\big[\phi_{1:m}\big]$ in (4.40), the MLE (4.36) is expected to have small $\mathrm{KL}(p\|\widehat{p})$ when $\mathrm{KL}(p\|p^*)$ is small because (4.36) must satisfy

$$\overline{\mathbb{P}}_n\big[\phi_{1:m}\big] = \nabla\psi(\widehat{\theta}) = \int_\Omega \phi_{1:m}(x)p_{\widehat{\theta}}(x)dx.$$

We first provide a bound on $\mathrm{KL}(p\|p^*)$ for the population (misspecified) MLE.

**Proposition 4.1.** *Let $p(\cdot)$ be a density in $\Omega$ and $p^*(\cdot)$ be the population MLE as in (4.41) with an $m$-dimensional space $S_m$. Let $\{\phi_1,\ldots,\phi_m\}$ be an orthonormal basis of $\{f-\mathbb{P}[f]: f\in S_m\}$ in the sense of (4.38). Let $K(x,y)=\sum_{j=1}^m \phi_j(x)\phi_j(y)$, $\|K\|_{\max}=\sup_x K(x,x)$, $\eta_\theta(x)=\big(\log p(x)-\phi_0(x)-\mu_0\big)-\theta^\top\phi_{1:m}(x)$ and $\sigma_\theta=\big\{\int_\Omega \eta_\theta^2(x)p(x)dx\big\}^{1/2}$. Suppose*

$$\sigma_{\overline{\theta}}\leqslant \epsilon_0 \text{ and } \big\|\eta_{\overline{\theta}}\big\|_{L_\infty(\Omega)}\leqslant c_0 \text{ for some } \overline{\theta}\in\mathbb{R}^m \tag{4.42}$$

*with $\|K\|_{\max}^{1/2}\big(e^{3c_0/2}+e^{c_0}\big)\epsilon_0<c_0$. Then, for the $\theta^*$ in (4.40),*

$$\mathrm{KL}(p\|p^*)\leqslant \log\big(1+e^{c_0}\epsilon_0^2/2\big),\quad \sigma_{\theta^*}\leqslant e^{3c_0/2}\epsilon_0,\quad \big\|\eta_{\theta^*}\big\|_{L_\infty(\Omega)}\leqslant 2c_0. \tag{4.43}$$

In $L_2(p(x)dx)$, $p\to\int K(\cdot,y)p(x)dx$ is the orthogonal projection to $\{f-\mathbb{P}[f]: f\in S_m\}$.

Proof. Due to $\int\phi_{1:m}(x)p(x)dx=0$, (4.39) and (4.35),

$$\mathrm{KL}(p\|p_\theta)=\mu_0+\psi(\theta)=\log\bigg(\int_\Omega e^{\mu_0+\phi_0(x)+\theta^\top\phi_{1:m}(x)}dx\bigg)=\log\bigg(\int_\Omega e^{-\eta_\theta(x)}p(x)dx\bigg).$$

Because $e^{-|x|}x^2/2\leqslant e^x-1-x\leqslant e^{|x|}x^2/2$ and $\mu_0=\int_\Omega\big(\log p(x)-\phi_0(x)\big)p(x)dx$,

$$\log\big(1+e^{-c}\sigma_\theta^2/2\big)\leqslant \mathrm{KL}(p\|p_\theta)\leqslant\log\big(1+e^{c}\sigma_\theta^2/2\big) \tag{4.44}$$

when $\|\eta_\theta\|_{L_\infty(\Omega)}\leqslant c$. Let $\theta^*(t)=t\theta^*+(1-t)\overline{\theta}$ for $t\geqslant 0$. By (4.40) and (4.43),

$$\mathrm{KL}(p\|p_{\theta^*})\leqslant\mathrm{KL}(p\|p_{\theta^*(t)})\leqslant\mathrm{KL}(p\|p_{\overline{\theta}})\leqslant\log\big(1+e^{c_0}\epsilon_0^2/2\big)\quad\forall\, t\in[0,1]$$

due to the convexity of $\mathrm{KL}(p\|p_\theta)$ in $\theta$ and the optimality of $\theta^*$. For $t\in[0,1]$ with $\|\eta_{\theta^*(t)}\|_{L_\infty(\Omega)}\leqslant 2c_0$, we must have

$$\log\big(1+e^{-2c_0}\sigma^2_{\theta^*(t)}/2\big)\leqslant\mathrm{KL}(p\|p_{\theta^*(t)})\leqslant\log\big(1+e^{c_0}\epsilon_0^2/2\big), \tag{4.45}$$

so that by (4.38) $\|\theta^*(t)-\overline{\theta}\|_2=\|\eta_{\theta^*(t)}-\eta_{\overline{\theta}}\|_{L_2(\mathbb{P})}\leqslant\sigma_{\theta^*(t)}+\sigma_{\overline{\theta}}\leqslant\big(e^{3c_0/2}+e^{c_0}\big)\epsilon_0$ and

$$\begin{aligned}\big\|\eta_{\theta^*(t)}\big\|_{L_\infty(\Omega)} &\leqslant \big\|\eta_{\overline{\theta}}\big\|_{L_\infty(\Omega)}+\|K\|_{\max}^{1/2}\|\theta^*(t)-\overline{\theta}\|_2\\ &\leqslant \big\|\eta_{\overline{\theta}}\big\|_{L_\infty(\Omega)}+\|K\|_{\max}^{1/2}\big(e^{3c_0/2}+e^{c_0}\big)\epsilon_0.\end{aligned}$$

As $\|K\|_{\max}^{1/2}\big(e^{3c_0/2}+e^{c_0}\big)\epsilon_0<c_0$, we have $\max_{0\leqslant t\leqslant 1}\big\|\eta_{\theta^*(t)}\big\|_{L_\infty(\Omega)}<2c_0$. Therefore, $\max_{0\leqslant t\leqslant 1}\big\|\eta_{\theta^*(t)}\big\|_{L_\infty(\Omega)}<2c_0$. The conclusions follow from (4.45). □

The following lemma gives a useful relationship among three densities $p$, $p_{\theta^*}$ and $p_\theta$.

**Lemma 4.5.** *For any $\theta$ and $\theta^*$ in $\mathbb{R}^m$ with finite $\psi(\theta)$ and $\psi(\theta^*)$,*

$$\mathrm{KL}(p_\theta\|p_{\theta^*})+\mathrm{KL}(p\|p_\theta)=\mathrm{KL}(p\|p_{\theta^*})+\langle\mathbb{P}[\phi_{1:m}]-\nabla\psi(\theta),\theta^*-\theta\rangle.$$

Such three-point identity holds for the Bregman divergence

$$D_\psi(a,b)=\psi(a)-\psi(b)-\langle\nabla\psi(b),a-b\rangle$$

for general convex function $\psi$ with the gradient $\nabla\psi(b)$. While the Kullback-Leibler divergence in exponential family is a special case of the Bregman divergence, the density $p(\cdot)$ may not belong to the exponential family (4.35).

**Theorem 4.6.** *Let $p = p(\cdot)$ be a density in $\Omega$ and $\widehat{p} = \widehat{p}(\cdot)$ be the estimator in* (4.37) *with an $m$-dimensional space $S_m$. Let $\{\phi_1, \ldots, \phi_m\}$ be an orthonormal basis of $\{f - \mathbb{P}[f] : f \in S_m\}$ satisfying* (4.38). *Let $\widehat{\theta}$, $\theta^*$ and $p^*$ be as in* (4.36), (4.39) *and* (4.40) *respectively with $\widehat{p} = p_{\widehat{\theta}}$ and $p^* = p_{\theta^*}$. Let $\|K\|_{\max}$, $\eta_\theta(x)$ and $\sigma_\theta$ be as in Proposition 4.1. Suppose* (4.42) *holds with $\|K\|_{\max}^{1/2}\big(e^{3c_0/2} + e^{c_0}\big)\epsilon_0 < c_0$. Then,* (4.43) *holds and*

$$\begin{aligned} & 1 - \mathbb{P}\left\{\begin{matrix} \mathrm{KL}(\widehat{p}\|p^*) + \mathrm{KL}(p\|\widehat{p}) \leqslant \mathrm{KL}(p\|p^*) + C\sqrt{m/n}\epsilon_1 \\ \sigma_{\widehat{\theta}} \leqslant \epsilon_1/2, \big\|\eta_{\widehat{\theta}}\big\|_{L_\infty(\Omega)} < 3c_0 \end{matrix}\right\} \\ \leqslant\ & \mathbb{P}\Big\{\big\|(\overline{\mathbb{P}}_n - \mathbb{P})[\phi_{1:m}]\big\|_2 > C\sqrt{m/n}\Big\} \end{aligned}$$

*when $2e^{4c_0}\epsilon_0^2 + 4e^{3c_0}C\sqrt{m/n}\epsilon_1 \leqslant (\epsilon_1/2)^2 \leqslant 1$ and $\|K\|_{\max}^{1/2}\epsilon_1 < c_0$. In particular, if $\|K\|_{\max} \lesssim m \asymp n^{1/(1+2\alpha/d)}$ and* (4.42) *holds with $\epsilon_0 \asymp m^{-\alpha/d} \ll m^{-1/2}$ and $c_0 = O(1)$, then*

$$\mathrm{KL}(p_{\theta^*}\|\widehat{p}) + \mathrm{KL}(\widehat{p}\|p_{\theta^*}) + \mathrm{KL}(p\|\widehat{p}) + \mathrm{KL}(p\|p^*) + \sigma_{\widehat{\theta}}^2 + \sigma_{\theta^*}^2 = O_{\mathbb{P}}\big(n^{-2\alpha/(d+2\alpha)}\big) \tag{4.46}$$

*and $\big\|\eta_{\widehat{\theta}}\big\|_{L_\infty(\Omega)} + \big\|\eta_{\theta^*}\big\|_{L_\infty(\Omega)} = O_{\mathbb{P}}(1)$.*

Proof. By (4.35) and (4.38), $\int_\Omega \big(u^\top\phi_{1:m}(x)\big)^2 p_\theta(x)dx > 0$ for all $u \neq 0$ and $\theta$ in $\mathbb{R}^m$, as $u^\top\phi_{1:m}(x) = 0$ almost everywhere would imply $\int_\Omega \big(u^\top\phi_{1:m}(x)\big)^2 p(x)dx = 0$. Thus, the Hessian of $\psi(\theta)$ is invertible and by the inverse function theorem we can construct a continuous curve $\theta(t)$ in $\mathbb{R}^m$ satisfying

$$\nabla\psi(\theta(t)) = t\nabla\psi(\widehat{\theta}) + (1-t)\nabla\psi(\theta^*) = \mathbb{P}[\phi_{1:m}] + t(\overline{\mathbb{P}}_n - \mathbb{P})[\phi_{1:m}]$$

with $\theta(0) = \theta^*$ and $\theta(1) = \widehat{\theta}$. For $t \in [0,1]$ with

$$\big\|\eta_{\theta(t)}\big\|_{L_\infty(\Omega)} \leqslant 3c_0, \quad \|\theta^* - \theta(t)\|_2 \leqslant \epsilon_1, \tag{4.47}$$

(4.44) and Lemma 4.5 yield

$$\log\big(1 + e^{-3c_0}\sigma_{\theta(t)}^2/2\big) \leqslant \mathrm{KL}(p\|p_{\theta(t)}) \leqslant \mathrm{KL}(p\|p_{\theta^*}) + \langle \mathbb{P}[\phi_{1:m}] - \nabla\psi(\theta(t)), \theta^* - \theta(t)\rangle.$$

In the event $\|(\overline{\mathbb{P}}_n - \mathbb{P})[\phi_{1:m}]\|_2 \leqslant C\sqrt{m/n}$, we have

$$\log\big(1 + e^{-3c_0}\sigma_{\theta(t)}^2/2\big) \leqslant \log\big(1 + e^{c_0}\epsilon_0^2/2\big) + C\sqrt{m/n}\epsilon_1.$$

Thus, as $x \leqslant -\log(1-x)$ and $C\sqrt{m/n}\epsilon_1 \leqslant 1/4$,

$$1 - C\sqrt{m/n}\epsilon_1 + e^{-3c_0}\sigma_{\theta(t)}^2/4 \leqslant \big(1 - C\sqrt{m/n}\epsilon_1\big)\big(1 + e^{-3c_0}\sigma_{\theta(t)}^2/2\big) \leqslant 1 + e^{c_0}\epsilon_0^2/2,$$

which implies $\sigma_{\theta(t)}^2 \leqslant 2e^{4c_0}\epsilon_0^2 + 4e^{3c_0}C\sqrt{m/n}\epsilon_1 \leqslant (\epsilon_1/2)^2$ by the imposed condition. It follows that $\|\theta(t) - \theta^*\|_2 \leqslant \sigma_{\theta(t)} + \sigma_{\theta^*} \leqslant \epsilon_1/2 + e^{3c_0/2}\epsilon_0 < \epsilon_1$, so that

$$\big\|\eta_{\theta(t)}\big\|_{L_\infty(\Omega)} \leqslant \big\|\eta_{\theta^*}\big\|_{L_\infty(\Omega)} + \|K\|_{\max}^{1/2}\|\theta^*(t) - \overline{\theta}\|_2 \leqslant 2c_0 + \|K\|_{\max}^{1/2}\epsilon_1 < 3c_0.$$

Consequently, (4.47) holds for all $t \in [0,1]$. The conclusions follow from another application of Lemma 4.5 and the fact that $\mathbb{E}\big[\big\|(\overline{\mathbb{P}}_n - \mathbb{P})[\phi_{1:m}]\big\|_2^2\big] \leqslant \mathbb{P}\big[\|\phi_{1:m}\|_2^2/n\big] = m/n$ by (4.38). $\square$

4.2.4. *Remarks.* The projection estimators can be viewed as empirical risk minimizers,

$$\widehat{p} = \arg\min_{f\in\mathscr{F}} \overline{\mathbb{P}}_n\big[L(p,f)\big].$$

In (4.26), $L(p,f)(x) = \{\|f\|^2_{L_2(\Omega)} - 2f(x)\} + \|p\|^2_{L_2(\Omega)}$ and $\mathbb{E}[L(p,f)] = \|f-p\|^2_{L_2(\Omega)}$ with $\mathscr{F} = S_m$, where $S_m$ is the linear span of the basis functions $\phi_1,\ldots,\phi_m$. In (4.36), $L(p,f)(x) = -\log f(x) + \int \log(p(x))p(x)dx$ and $\mathbb{E}[L(p,f)] = \mathrm{KL}(p\|f)$ with

$$\mathscr{F} = \Big\{ f : \log f(x) - \phi_0(x) + c \in S_m, c \in \mathbb{R}, \int_\Omega f(x)dx = 1\Big\}.$$

In both cases, $L(p,f)$ is convex in $f$, and the optimization over $f \in \mathscr{F}$ can be carried out because $L(p,f)(x)$ is a sum of two components respectively depending on $f$ and $p$ only. The choice of $\mathscr{F}$ is quite flexible, including infinite-dimensional smooth density classes and neural networks.

When $m$ is very large and the coefficient vector $\theta$ is believed to be sparse, it would be sensible to add a penalty to the empirical risk,

$$\widehat{p} = p_{\widehat{\theta}}, \quad \widehat{\theta} = \arg\min_{\theta} \Big\{ \overline{\mathbb{P}}_n\big[L(p,f_\theta)\big] + \mathrm{pen}(\theta)\Big\},$$

e.g. $\mathrm{pen}(\theta) = \|\theta\|_1$ when $\mathscr{F} = \{f_\theta : \theta \in \mathbb{R}^m\}$.

In a game theoretical approach, generative adversarial networks (GANs) [GPAM$^+$14] pit a generator $f$ against a discriminator $d$ in a minimax problem

$$\min_{f\in\mathscr{F}} \max_{d\in\mathscr{D}} \Big\{ \int_\Omega \big(\log d(x)\big)p(x)dx + \int_\Omega \big(\log(1-d(x))\big)f(x)dx\Big\},$$

where $\mathscr{F}$ is a density family on $\Omega$ and $\mathscr{D}$ is a family of discriminators, both in terms of neural networks. For example, $f$ can be the density of $g(Z)$ with a high-dimensional Gaussian $Z$ and a neural network $g(\cdot)$. If $\mathscr{D}$ is sufficiently rich, the optimal discriminator is approximately $d^*(x) = p(x)/(p(x)+f(x))$, so that the distribution of the optimal generator is approximately

$$f^* = \arg\min_{f\in\mathscr{F}} \mathrm{JS}(p\|f)$$

where $\mathrm{JS}(p\|f) = \mathrm{KL}\big(p\|(p+f)/2\big) + \mathrm{KL}\big(f\|(p+f)/2\big)$ is the Jensen-Shannon divergence between densities $p$ and $g$. From this density estimation point of view, the GANs approach can be taken with other distance functions expressible in minimax forms.

4.3. **Nonparametric regression.** In nonparametric regression, $X_i = (y_i, z_i)$ with

$$y_i = f(z_i) + \varepsilon_i, \tag{4.48}$$

where $y_i \in \mathbb{R}$ and $z_i \in \Omega \subseteq \mathbb{R}^d$ are respectively the response variable and the design point with the $i$-th observation, $f : \Omega \to \mathbb{R}$ is an unknown function of interest and $\varepsilon_i$ are unobservable noise variables. For simplicity we assume throughout this section that $\Omega$ is open and that for deterministic design points $\{z_i\}$ or conditionally on random $\{z_i\}$ the noise variables $\varepsilon_i$ are independent random variables with $\mathbb{E}[\varepsilon_i] = 0$ and $\mathrm{Var}[\varepsilon_i] \leqslant \sigma^2$.

4.3.1. *Kernel regression.* Suppose $f(z)$ is smooth in a neighborhood of a given point $a_0 \in \Omega$. As $\lim_{z\to a_0} f(z) = f(a_0)$, it makes sense to estimate $f(a_0)$ by the average those $y_i$ with small $\|z_i - a_0\|_2$. The kernel regression method [Nad64, Wat64] estimates $f(a_0)$ by

$$\widehat{f}(a_0) = \frac{\overline{\mathbb{P}}_n[K(a_0, z)y]}{\overline{\mathbb{P}}_n[K(a_0, z)]} = \frac{\sum_{i=1}^n K(a_0, z_i)y_i}{\sum_{i=1}^n K(a_0, z_i)} \tag{4.49}$$

with $\widehat{f}(a_0) = 0$ for $\sum_{i=1}^n K(a_0, z_i) = 0$. Similar to density estimation, we consider conditions

$$\mathrm{supp}(K(a_0, \cdot)) \subset \mathrm{Ball}(a_0, h) \cap \Omega, \quad \int_\Omega \big|K(a_0, z)\big|^2 dz \leqslant C_1^2/h^d, \tag{4.50}$$

for some $h > 0$ and

$$\int_\Omega K(a_0, z)dz = 1, \quad \int_\Omega K(a_0, z)(z - a_0)^k dz = 0,\ 0 < \|k\|_1 < \alpha, \tag{4.51}$$

**Theorem 4.7.** *Suppose $f(\cdot)$ satisfies Hölder's condition of order $\alpha$ in* (4.1)*. Suppose the design points $z_i$ are iid from a density $p(\cdot)$ in $\Omega$ with $\pi_* \leqslant p(z) \leqslant \pi^*$ in* $\mathrm{Ball}(a_0, h) \cap \Omega$*. For $\alpha > 1$, assume further that* $\max_{0\leqslant\|k\|_1<\alpha} |f^{(k)}(a_0)| \leqslant C_0$, $\max_{0\leqslant\|k\|_1<\alpha-1} |p^{(k)}(a_0)| \leqslant C_0$, *and $p(\cdot)$ satisfies* (4.1) *with $\alpha$ replaced by $\alpha-1$. Let $\widehat{f}(a_0)$ be the kernel regression estimator* (4.49) *with a kernel $K(a_0, \cdot)$ satisfying conditions* (4.50) *and* (4.51)*, with $K(a_0, z) \geqslant 0$ for $\alpha \leqslant 2$. Then, for some constant $C$ depending on $(\alpha, C_0, C_1)$ only,*

$$\mathbb{E}\Big[\big(\widehat{f}(a_0) - f(a_0)\big)^2\Big] \leqslant C\big(h^{2\alpha} + 1/(nh^d)\big),\ \alpha \leqslant 2,$$
$$\big(\widehat{f}(a_0) - f(a_0)\big)^2 \leqslant O_{\mathbb{P}}\big(h^{2\alpha} + 1/(nh^d)\big),\ \alpha > 2.$$

*In particular, $h^{2\alpha} + 1/(nh^d) \asymp n^{-2\alpha/(d+2\alpha)}$ when $h = h_n \asymp n^{-1/(d+2\alpha)}$.*

Proof. Consider $\alpha \leqslant 2$. As $\mathbb{P}[K_n(a_0, z)] = \int K(a_0, z)p(z)dz \geqslant \pi_*$,

$$\mathbb{P}\Big\{\overline{\mathbb{P}}_n[K_n(a_0, z)] \leqslant \pi_*/2\Big\} \leqslant \frac{\mathbb{E}\big[\big|(\overline{\mathbb{P}}_n - \mathbb{P})[K(a_0, z)]\big|^2\big]}{(\pi_*/2)^2} \leqslant \frac{\pi^* C_1^2}{nh^d(\pi_*/2)^2}$$

by Chebyshev's inequality. By conditioning on the design points, we find that

$$\begin{aligned}
&\mathbb{E}\Big[\big(\widehat{f}(a_0) - f(a_0)\big)^2\Big] \\
\leqslant\ & \mathbb{E}\bigg[\bigg(\frac{\overline{\mathbb{P}}_n[K(a_0, z)(f(z) - f(a_0))]}{\overline{\mathbb{P}}_n[K(a_0, z)]}\bigg)^2\bigg] + \frac{\sigma^2}{n}\mathbb{E}\bigg[\frac{\overline{\mathbb{P}}_n[K^2(a_0, z)]}{(\overline{\mathbb{P}}_n[K(a_0, z)])^2}\bigg] \\
\leqslant\ & \mathbb{E}\Big[\big(\overline{\mathbb{P}}_n[K(a_0, z)(f(z) - f(a_0))]\big)^2(2/\pi_*)^2\Big] + (\sigma^2/n)\mathbb{E}\big[\overline{\mathbb{P}}_n[K^2(a_0, z)](2/\pi_*)^2\big] \\
&+ \big(\sup_{\|z-a_0\|_2\leqslant h}\big|f(z) - f(a_0)\big| + \sigma^2\big)(2/\pi_*)^2\pi^* C_1^2/(nh^d).
\end{aligned}$$

The rest of the proof, involving the second moment of sums of iid variables, is omitted. $\square$

The rate in Theorem 4.7 is optimal. Let

$$f_t(x) = f_0(x) + t^\alpha g_0((x-a_0)/t) \tag{4.52}$$

with the same function $g_0(x)$ as in (4.24).

**Theorem 4.8.** *Let $f_0(x)$ be a regression function in $\Omega$ satisfying Hölder's condition of order $\alpha$ at an $a_0 \in \Omega$ with a fixed $h$ in (4.1). Suppose $z_i$ are iid with a density $p(\cdot)$ satisfying $\pi_* \leqslant p(z) \leqslant \pi^*$ for some positive constants $\pi_*$ and $\pi^*$. Let $f_t(x)$ be as in (4.52). Let $\mathbb{P}_t$ be the probability under which $f = f_t$ and $\varepsilon_i$ are iid $N(0,1)$ in (4.48). Then, Hölder's condition (4.1) also holds for $f_t$ for $t \in [0,h]$ and for a certain fixed $c_3 > 0$*

$$\min_{\widehat{f}} \max_{t=0, c_3 n^{-1/(d+2\alpha)}} \mathbb{P}_t\Big\{|\widehat{f}(a_0) - f_t(a_0)| > |g_0(0)/2| c_3^\alpha n^{-\alpha/(d+2\alpha)}\Big\} > 1/2.$$

As discussed below Theorem 4.1, Theorem 4.7 automatically provides an $L_2(\Omega)$ error bound for kernel regression when the conditions of Theorem 4.7 hold uniformly for all $a_0$ in the domain $\Omega$ and the volume $|\Omega|$ is finite. The following example demonstrates the feasibility of fulfilling conditions (4.50) and (4.51) on the boundary of $\Omega$.

**Example 4.2.** *Let $K_1(z)$ be a probability density function with support $\{z : \|z\|_2 < 1\}$ and $\mathscr{Q}_{\alpha,0}$ be the collection of all polynomials $Q$ of degree $\lceil\alpha\rceil - 1$ with $Q(0) = 0$. Let*

$$Q_{\alpha,h,a_0} = \operatorname*{arg\,min}_{Q \in \mathscr{Q}_{\alpha,0}} \int_\Omega \big(1 - Q((a_0-z)/h)\big)^2 K_1((a_0-z)/h)dz, 0 < h \leqslant 1,$$

*so that $\int_\Omega Q((a_0-z)/h)\big(1 - Q_{\alpha,h,a_0}((a_0-z)/h)\big)K_1((a_0-z)/h)dz = 0, \forall\, Q \in \mathscr{Q}_{\alpha,0}$. Let*

$$K(a_0,z) = \frac{\big(1 - Q_{\alpha,h,a_0}((a_0-z)/h)\big)K_1((a_0-z)/h)I\{z \in \Omega\}}{\int_\Omega \big(1 - Q_{\alpha,h,a_0}((a_0-z)/h)\big)^2 K_1((a_0-z)/h)dz}.$$

*Suppose $\int_\Omega h^{-d}K_1((a_0-z)/h)dz \geqslant c_0 > 0$. Then, (4.50) and (4.51) hold with $C_1$ depending on $c_0$ and $K_1(\cdot)$ only. Condition (4.51) and the first part of (4.50) hold automatically. To prove the uniform boundedness of the $C_1$ in the second part of (4.50), we recognize that $\int_{\Omega_{a,h}} Q^2(z)K_1(z)dz$ is a uniformly bounded positive-definite quadratic form of the coefficients of the polynomial $Q(z)$ and that $\int_{\Omega_{a,h}} (1-Q(z))^2 K_1(z)dz$ is uniformly bounded away from zero in $\mathscr{Q}_{\alpha,0}$ under the condition $\int_{\Omega_{a,h}} K_1(z)dz \geqslant c_0$, where $\Omega_{a,h} = \{(a_0-z)/h : z \in \Omega\}$. The problem is location-scale invariant.*

Kernel density estimation and kernel regression are developed based on the idea that local smoothing would prevent overfitting by averaging the noise. However, in machine learning, there is a considerable interest in interpolation estimators satisfying $\widehat{f}(z_i) = y_i$, i.e. completely fitting the data. Theorem 4.7 provides a theoretical justification of the kernel regression as an interpolation estimator in the following example [BRT19].

**Example 4.3.** *Let $K_1(z) = \|z\|_2^{-\gamma}(1 - \|z\|_2^2)/c_{\gamma,d}$ with $c_{\gamma,d} = \pi^{d/2}\int_0^1 t^{d/2-\gamma-1}(1-t)dt/\Gamma(d/2)$ and $0 < \gamma < d/2$. Construct $K(a_0,z)$ as in Example 4.2. Then, for bounded open $\Omega$, $\inf_{a\in\Omega} \overline{\mathbb{P}}_n[K(a,\cdot)] > 0$ with probability $1 + o(1)$ and in this event the kernel regression yields an interpolation estimator satisfying $\lim_{z\to z_i} \widehat{f}(z) = y_i$, $1 \leqslant i \leqslant n$. Still, by Theorem 4.7, $\widehat{f}$ achieves the minimax rate $n^{-\alpha/(d+2\alpha)}$ when $z_i$ are iid uniform on $\Omega$.*

4.3.2. *Local least squares.* The local LSE [Cle79] can be written as

$$\widehat{f}(a_0) = \widehat{Q}(0), \quad \widehat{Q} = \mathop{\arg\min}_{Q\in\mathscr{Q}_\alpha} \overline{\mathbb{P}}_n\Big[K(a_0,z)\big(y - Q((a_0-z)/h)\big)^2\Big], \tag{4.53}$$

where $\mathscr{Q}$ is the collection of all polynomials $Q$ of degree $\lceil\alpha\rceil - 1$ and $K(a_0, z)$ is a kernel satisfying conditions (4.50) and

$$\textstyle\int_\Omega K(a_0,y)dy = 1, \quad K(a_0,z) \geqslant 0. \tag{4.54}$$

The idea is that $f(z_i) = Q_{f,a_0}((a_0 - z_i)/h) + \Delta_i$ with a polynomial $Q_{f,a_0} \in \mathscr{Q}_\alpha$ and small remainder $\Delta_i$ when $f$ is smooth and $z_i \in \mathrm{Ball}(a_0, h)$, so that

$$\widehat{f}(a_0) - f(a_0) = \widetilde{Q}(0), \quad \widetilde{Q} = \mathop{\arg\min}_{Q\in\mathscr{Q}_\alpha} \overline{\mathbb{P}}_n\Big[K(a_0,z)\big(\Delta + \varepsilon - Q((a_0-z)/h)\big)^2\Big] \tag{4.55}$$

by the equi-variance of the local LSE under the subtraction of a polynomial from $f$. In particular, under Hölder's condition of order $\alpha$ in (4.1), $|\Delta_i| \leqslant C_0 h^\alpha/\Gamma(\lfloor\alpha\rfloor)$, so that $|\widetilde{Q}(a_0)| \lesssim h^\alpha$ is expected when $h \asymp n^{-1/(d+2\alpha)}$ as in Theorems 4.1 and 4.7.

**Theorem 4.9.** *Suppose $f(\cdot)$ satisfies Hölder's condition of order $\alpha$ in* (4.1)*. Suppose the design points $z_i$ are iid from a density $p(\cdot)$ in $\Omega$ with $\pi_* \leqslant p(z) \leqslant \pi^*$ in* $\mathrm{Ball}(a_0,h) \cap \Omega$*. Let $\widehat{f}(a_0)$ be the local LSE in* (4.53) *with a kernel $K(a_0,\cdot)$ satisfying conditions* (4.50) *and* (4.54)*. Then, for $h = h_n \asymp n^{-1/(d+2\alpha)}$,*

$$\mathbb{E}\Big[\big(\widehat{f}(a_0) - f(a_0)\big)^2\Big| z_{1:n}\Big] = O_{\mathbb{P}}\big(n^{-2\alpha/(d+2\alpha)}\big).$$

Proof. Let $I_0 = \{i \leqslant n : \|z_i - a_0\|_2 \leqslant h\}$. By (4.50) and the condition $\pi_* \leqslant p(z) \leqslant \pi^*$, the local sample size is of the order $|I_0| \asymp nh^d$ with high probability. By (4.55), the conclusion follows from the well known properties of the least squares estimator of the intercept in polynomial regression with the sample size of the order $nh^d$ and biases $\Delta_i$ bounded by $\sum_{i\in I_0}\Delta_i^2 \lesssim nh^{d+2\alpha} = O(1)$. The eigenvalues of the Gram matrix for this polynomial regression are bounded away from 0 and $\infty$ with large probability in view of the discussion in Example 4.2. We omit details. $\square$

**Example 4.4.** *Let $\Omega = [-1,1]$, $\varepsilon_i \sim N(0,\sigma^2)$, $p(x) = 1/2$, $\alpha = 2$, $a_0 = 0$ and $h \in (0,1)$. The local LSE of $f(0)$ is the intercept of the least squares estimator in simple linear regression based on data with $|z_i| \leqslant h$. Let $I_0 = \{i : |z_i| \leqslant h\}$. Then,*

$$\mathrm{Var}\big(\widehat{f}(0)\big|z_{1:n}\big) = \frac{\sigma^2}{|I_0|} + \frac{\sigma^2\big(\sum_{i\in I_0} z_i/|I_0|\big)^2}{\sum_{i\in I_0}\big(z_i - \sum_{j\in I_0} z_j/|I_0|\big)^2}.$$

*However, $\mathbb{E}\big[\mathrm{Var}\big(\widehat{f}(0)\big|z_{1:n}\big)I\{|I_0| = 2\}\big] = \infty$ when $z_i$ are iid uniform in* $[-1,1]$.

4.3.3. *Smoothing spline and RKHS.* In nonparametric regression (4.48) on $[0,1]$, smoothing spline [Wah90] can be written as

$$\widehat{f} = \arg\min_f \Big\{ \overline{\mathbb{P}}_n\big[(y-f(z))^2\big] + \lambda \|f\|^2_{L_2^\alpha([0,1])} \Big\} \tag{4.56}$$

with an integer smoothness index $\alpha > 0$ and a penalty level $\lambda \geqslant 0$. Let $\mathscr{B}_\alpha$ be the space of all functions $g$ in $W_2^\alpha([0,1])$ with $g^{(k)}(0) = 0$ for $k = 0, \ldots, \alpha - 1$. By (4.13)

$$f(z) = \sum_{k=0}^{\alpha-1} f^{(k)}(0)\frac{z^k}{k!} + \int_0^1 G_\alpha(z,t) f^{(\alpha)}(t)dt \tag{4.57}$$

where $G_\alpha(z,t) = (z-t)_+^{\alpha-1}/(\alpha-1)!$ is Green's function for the mapping $g^{(\alpha)} \to g \in \mathscr{B}_\alpha$. It follows that $g(z) = \langle \delta_z, g\rangle_{L_2^\alpha([0,1])}$ for all $g \in \mathscr{B}_\alpha$ with

$$\delta_z(t) = (-1)^\alpha (z-t)_+^{2\alpha-1}/(2\alpha-1)! - \textstyle\sum_{k=0}^{\alpha-1}(-1)^{\alpha-k}\{z^{2\alpha-k-1}/(2\alpha-k-1)!\}(t^k/k!), \tag{4.58}$$

which is a member of $\mathscr{B}_\alpha$ satisfying $\delta_z^{(\alpha)}(u) = G_\alpha(z,u)$. Define $K(\cdot,\cdot) : [0,1]^2 \to \mathbb{R}$ by

$$K(s,t) = \langle \delta_s, \delta_t\rangle_{L_2^\alpha([0,1])} = \textstyle\int_0^1 G_\alpha(s,u)G_\alpha(t,u)du = \delta_t(s). \tag{4.59}$$

As $\delta_z \in \mathscr{B}_\alpha$ and $\delta_z^{(\alpha)}(u) = G_\alpha(z,u)$, (4.56) can be written as

$$\widehat{f} = \arg\min_f \bigg\{ \frac{1}{n}\sum_{i=1}^n \bigg( y_i - \sum_{k=0}^{\alpha-1} f^{(k)}(0)\frac{z_i^k}{k!} - \langle \delta_{z_i}, f\rangle_{L_2^\alpha([0,1])}\bigg)^2 + \lambda\|f\|^2_{L_2^\alpha([0,1])}\bigg\}.$$

Given the values of $\langle \delta_{z_i}, f\rangle_{L_2^\alpha([0,1])}$, $\|f\|^2_{L_2^\alpha([0,1])}$ is minimized at an $f$ as a linear combination of $\delta_{z_i}$, say $\sum_{i=1}^n \beta_i \delta_{z_i}$, so that in matrix notation (4.56) becomes

$$\begin{aligned} \widehat{f}(z) &= \sum_{k=0}^{\alpha-1}\widehat{\gamma}_k z^k + \sum_{i=1}^n \widehat{\beta}_i \delta_{z_i}(z), \\ \{\widehat{\beta}, \widehat{\gamma}\} &= \arg\min_{\beta,\gamma}\Big\{ \big\| y_{1:n} - T_n\gamma - K_n\beta\big\|_2^2 + n\lambda \beta^\top K_n\beta\Big\}, \end{aligned} \tag{4.60}$$

where $T_n = (z_i^k)_{n\times\alpha}$, and $K_n = (K(z_i,z_j))_{n\times n}$ with the kernel in (4.59). By (4.58) $\delta_{z_i}$ are splines of degree $2\alpha-1$, so that $\widehat{f}(z)$ is a spline of the same degree.

**Lemma 4.6.** *Let $(U_1, U_2)$ be an orthonormal matrix with $T_n = U_1 R$ being the QR decomposition of $T_n$. Suppose $z_{1:n}$ are distinct. Then, (4.60) is explicitly solved by*

$$\widehat{\beta} = M_n^{-1}(y_{1:n} - T_n\widehat{\gamma}), \quad \widehat{\gamma} = \big(T_n^\top M_n^{-1} T_n\big)^{-1} T_n^\top M_n^{-1} y_{1:n}, \tag{4.61}$$

*with $M_n = K_n + n\lambda I_n$. Consequently, with $T_n^\dagger$ being the generalized inverse*

$$\widehat{f}(z_{1:n}) = y_{1:n} - n\lambda\widehat{\beta},\ T_n^\top\widehat{\beta} = 0,\ \widehat{\beta} = U_2(U_2^\top M_n U_2)^{-1}U_2^\top y_{1:n},\ \widehat{\gamma} = T_n^\dagger(y_{1:n} - M_n\widehat{\beta}). \tag{4.62}$$

Proof. As $z_{1:n}$ are distinct, $\text{rank}(K_n) = n$. The first equation of (4.61) can be seen by differentiation of the penalized loss with respect to $\beta$. The second equation of (4.61) follows by inserting the first in the penalized loss. The solution (4.61) can be equivalently written as $M_n\widehat{\beta} + T_n\widehat{\gamma} = \widehat{f}(z_{1:n}) + n\lambda\widehat{\beta} = y_{1:n}$ and $T_n^\top\widehat{\beta} = 0$. Due to $U_2U_2^\top\widehat{\beta} = \widehat{\beta}$ and $U_2^\top T_n = 0$, $U_2^\top M_n\widehat{\beta} = U_2^\top y_{1:n}$, so that the solution is also given by the last two equation in (4.62). $\square$

The above program can be extended to penalized least squares in a semi-linear model

$$f(z) = \gamma_1\phi_1(z) + \cdots + \gamma_m\phi_m(z) + g(z) \tag{4.63}$$

with a penalty proportional to a squared RKHS norm. A Hilbert space $\mathscr{H}$ with norm $\|g\|^2_{\mathscr{H}} = \langle g, g\rangle_{\mathscr{H}}$ for functions $g$ on $\Omega$ is an RKHS iff for each $z \in \Omega$, $g(z) = \langle \delta_z, g\rangle_{\mathscr{H}}$ for a certain $\delta_z \in \mathscr{H}$. Assume $\|\phi_j\|_{\mathscr{H}} = 0$ for $1 \leqslant j \leqslant m$, so that $\|f\|_{\mathscr{H}} = \|g\|_{\mathscr{H}}$ is a seminorm for $f$ in (4.63). The penalized LSE can be written as

$$\widehat{f} = \arg\min_f \Big\{\overline{\mathbb{P}}_n\big[(y - f(z))^2\big] + \lambda\|f\|^2_{\mathscr{H}} : f = \textstyle\sum_{k=1}^m \gamma_k\phi_k + g\Big\}. \tag{4.64}$$

Similar to (4.60) for $\mathscr{H} = \mathscr{B}_\alpha$, the solution is given by $\widehat{f}(z) = \widehat{\gamma}^\top\phi_{1:m}(z) + \widehat{\beta}^\top\delta_{z_{1:n}}(z)$ with

$$\{\widehat{\beta}, \widehat{\gamma}\} \quad = \quad \arg\min_{\beta,\gamma} \Big\{\big\|y_{1:n} - T_n\gamma - K_n\beta\big\|_2^2 + n\lambda\beta^\top K_n\beta\Big\}, \tag{4.65}$$

where $T_n = (\phi_j(z_i))_{n\times m}$, $M_n = K_n + n\lambda I_n$ and $K_n = (K(z_i, z_j))_{n\times n}$ with the kernel

$$K(s,t) = \langle \delta_s, \delta_t\rangle_{\mathscr{H}},\ \ s \in \Omega, t \in \Omega.$$

As the decomposition (4.63) is not unique, the problem is unchanged if we replace $K(s,t)$ by a different kernel $\widetilde{K}(s,t)$ satisfying

$$\widetilde{K}(s,t) = \langle \widetilde{\delta}_s, \widetilde{\delta}_t\rangle_{\mathscr{H}}, \quad \langle \widetilde{\delta}_z - \delta_z, f\rangle_{\mathscr{H}} \in \overline{\{\phi_1, \ldots, \phi_m\}}. \tag{4.66}$$

As $T_n^\top\widehat{\beta} = 0$ by (4.62), $\widehat{f}(z)$ and $\widehat{\beta}$ are unchanged if instead of (4.65) the following are used,

$$\begin{aligned} &\widehat{f}(z) = \widehat{\gamma}^\top\phi_{1:m}(z) + \widehat{\beta}^\top\widetilde{\delta}_{z_{1:n}}(z), \\ &\widehat{\beta} = \widetilde{M}_n^{-1}(y_{1:n} - T_n\widehat{\gamma}), \quad \widehat{\gamma} = \big(T_n^\top\widetilde{M}_n^{-1}T_n\big)^{-1}T_n^\top\widetilde{M}_n^{-1}y_{1:n}, \end{aligned}$$

where $\widetilde{M}_n = \widetilde{K}_n + n\lambda I_n$ with $\widetilde{K}_n = (\widetilde{K}(z_i, z_j))_{n\times n}$, and $T_n = (\phi_j(z_i))_{n\times m}$.

**Theorem 4.10.** *Let $f$ be as in (4.63) and $\widehat{f}$ be as in (4.64) with the data in (4.48). Then,*

$$\mathbb{E}\Big[\overline{\mathbb{P}}_n\big[(\widehat{f} - f)^2\big] + (\lambda/2)\|\widehat{f}\|^2_{\mathscr{H}}\Big] \leqslant \mathbb{E}\big[(\sigma^2/n)\mathrm{trace}\big(\widetilde{K}_n\widetilde{M}_n^{-1}\big)\big] + \sigma^2 m/n + (\lambda/2)\|f\|^2_{\mathscr{H}}. \tag{4.67}$$

*If $\widetilde{K}(s,t) = \sum_{j=1}^\infty \tau_j\varphi_j(s)\varphi_j(t)$ in (4.66) with some $\tau_j \geqslant 0$ and basis functions $\varphi_j(t)$, then*

$$\mathbb{E}\Big[\overline{\mathbb{P}}_n\big[(\widehat{f} - f)^2\big] + (\lambda/2)\|\widehat{f}\|^2_{\mathscr{H}}\Big] \leqslant 2\sigma^2 k_*/n + \sigma^2 m/n + (\lambda/2)\|f\|^2_{\mathscr{H}} \tag{4.68}$$

*for the smallest $k_*$ satisfying $\sum_{j=k_*+1}^\infty \tau_j\mathbb{P}[\varphi_j^2] \leqslant k_*\lambda$.*

Proof. We have $\overline{\mathbb{P}}_n\big[-h(z)(y - \widehat{f}(z))\big] + \lambda\langle h, \widehat{f}\rangle_{\mathscr{H}} = 0$ by the optimality of $\widehat{f}$ along the path $\widehat{f} + th$ at $t = 0$. Taking $h = \widehat{f} - f$, we find that

$$\overline{\mathbb{P}}_n\big[(\widehat{f} - f)^2\big] + \lambda\|\widehat{f}\|^2_{\mathscr{H}}/2 \leqslant \overline{\mathbb{P}}_n\big[(\widehat{f} - f)\varepsilon\big] + \lambda\|f\|^2_{\mathscr{H}}/2.$$

Because $\widehat{f}(z_{1:n}) = T_n\widehat{\gamma} + \widetilde{K}_n\widehat{\beta} = \widetilde{K}_n\widetilde{M}_n^{-1}y_{1:n} + n\lambda\widetilde{M}_n^{-1}T_n\big(T_n^\top\widetilde{M}_n^{-1}T_n\big)^{-1}T_n^\top\widetilde{M}_n^{-1}y_{1:n}$ and $\mathrm{trace}(n\lambda\widetilde{M}_n^{-1}T_n\big(T_n^\top\widetilde{M}_n^{-1}T_n\big)^{-1}T_n^\top\widetilde{M}_n^{-1}) \leqslant m\|n\lambda\widetilde{M}_n^{-1}\|_{\mathrm{S}} \leqslant m$, (4.67) holds. For (4.68),

$$\mathbb{E}\big[\mathrm{trace}\big(\widetilde{K}_n\widetilde{M}_n^{-1}\big)\big] = \textstyle\sum_{j=1}^\infty\mathbb{E}\big[\tau_j\varphi_j^\top(z_{1:n})\widetilde{M}_n^{-1}\varphi_j(z_{1:n})\big] \leqslant k_* + \sum_{j=k_*+1}^\infty \dfrac{\tau_j}{n\lambda}\mathbb{E}\big[\|\varphi_j(z_{1:n})\|_2^2\big]$$

due to $\widetilde{M}_n = \widetilde{K}_n + n\lambda I_n \geqslant \sum_{j=1}^{k_*}\tau_j\varphi_j(z_{1:n})\varphi_j^\top(z_{1:n}) + n\lambda I_n$. □

For the smoothing spline (4.56), we may choose $\{\varphi_j, j \geqslant 1\}$ in Theorem 4.10 such that $\{\tau_j^{1/2}\varphi_j^{(\alpha)}, j \geqslant 1\}$ forms an orthonormal basis in $L_2([0,1])$, and write Green's function as

$$G_\alpha(z,t) = \sum_{j=1}^\infty \tau_j \varphi_j^{(\alpha)}(t) \int_0^1 G_\alpha(z,u)\varphi_j^{(\alpha)}(u)du = \sum_{j=1}^\infty \tau_j\varphi_j^{(\alpha)}(t)\Big(\varphi_j(z) - \sum_{k=0}^{\alpha-1}\frac{z^k}{k!}\varphi_j^{(k)}(0)\Big)$$

with an application of (4.57) to $f(z) = \varphi_j(z)$. After removing the monomials of order up to $\alpha - 1$, we replace Green's function by $\widetilde{G}_\alpha(z,t) = \sum_{j=1}^\infty \tau_j\varphi_j^{(\alpha)}(t)\varphi_j(z)$ so that

$$f(z) = \textstyle\sum_{k=0}^{\alpha-1}(z^k/k!)\big\{f^{(k)}(0) - \sum_{j=1}^\infty \tau_j\varphi_j^{(k)}(0)\int_0^1 \varphi_j^{(\alpha)}(t)f^{(\alpha)}(t)dt\big\} + \int_0^1 \widetilde{G}(z,t)f^{(\alpha)}(t)dt$$

as in (4.63) with an $(\alpha-1)$-degree polynomial and $g(z) = \int_0^1 \widetilde{G}(z,t)f^{(\alpha)}(t)dt$. Due to the orthonormality of $\{\tau_j^{1/2}\varphi_j^{(\alpha)}\}$, the corresponding kernel is given by

$$\widetilde{K}(s,t) = \textstyle\int_0^1 \widetilde{G}_\alpha(s,u)\widetilde{G}_\alpha(t,u)du = \sum_{j=1}^\infty \tau_j\varphi_j(s)\varphi_j(t). \tag{4.69}$$

Specifically, for the orthonormal basis $\tau_j^{1/2}\varphi_j^{(\alpha)}(t) = \sqrt{2}\sin(\pi jt)$, $j \geqslant 1$, we take

$$\varphi_j = \sin(\pi jt - \alpha\pi/2), \quad \tau_j^{1/2} = 1/(\pi j)^\alpha,$$

due to $(d/dt)\sin(t-\pi/2) = \cos(t-\pi/2) = \sin(t)$, so that $\sum_{j=k_*+1}^\infty 2\tau_j \leqslant 2\int_{k_*}^\infty (\pi x)^{-2\alpha}dx = (2/\pi)(\pi k_*)^{1-2\alpha}/(2\alpha-1)$. Because $\varphi_j$ are also orthonormal in $L_2([0,1])$, (4.69) gives the eigenvalue decomposition of the kernel.

**Corollary 4.2.** *Let $f \in \mathscr{H} = L_2^\alpha([0,1])$ with a positive integer $\alpha$ and $\widehat{f}$ be the smoothing spline estimator in (4.56) with $\lambda \asymp n^{-2\alpha/(1+2\alpha)}$. Then,*

$$\mathbb{E}\Big[\overline{\mathbb{P}}_n\big[(\widehat{f}-f)^2\big] + (\lambda/2)\|\widehat{f}\|_{\mathscr{H}}^2\Big] \leqslant 2\sigma^2 k_*/n + \sigma^2\alpha/n + (\lambda/2)\|f\|_{\mathscr{H}}^2 \asymp n^{-2\alpha/(1+2\alpha)} \tag{4.70}$$

*with $k_* = \lceil\{(\alpha-1/2)\pi^{2\alpha}\lambda\}^{-1/(2\alpha)}\rceil \asymp n^{1/(1+2\alpha)}$.*

An interesting feature of the error bound (4.70) is its validity for any design $z_{1:n}$.

*Bayesian interpretation.* The smoothing spline estimator has the following Bayesian interpretation. Let the prior distribution of $f$ be given by

$$f(z) = \gamma_0 + \gamma_1 z + \cdots + \gamma_{\alpha-1}z^{\alpha-1} + g(z), \quad g(z) = b\int_0^1 G_\alpha(z,t)dW(t),$$

with independent $\gamma_k \sim N(0,a^2)$ and a Brownian motion process $W(t)$. As $\mathbb{E}[g(s)g(t)] = b^2K(s,t)$ for the kernel in (4.59), $b^2K_n$ is the covariance matrix of $g(z_{1:n})$, it follows that the negative log joint density of $\{y_{1:n}, g(z_{1:n}), \gamma\}$ is

$$n\overline{\mathbb{P}}_n\big[\big(y - \textstyle\sum_{k=0}^{\alpha-1}\gamma_k z^k - g(z)\big)^2/(2\sigma^2)\big] + g(z_{1:n})^\top K_n^{-1}g(z_{1:n})/(2b^2) + \|\gamma\|_2^2/(2a^2) + C$$

for some constant $C$. Let $\lambda = \sigma^2/(nb^2)$ and $g(z_{i:n}) = K_n\beta$. The Bayes estimator, the posterior mean and also the minimizer of

$$\big\|y_{1:n} - \textstyle\sum_{k=0}^{\alpha-1}\gamma_k z_{1:n}^k - K_n\beta\big\|_2^2 + n\lambda\beta^\top K_n\beta + \|\gamma\|_2^2\sigma^2/a^2, \tag{4.71}$$

converges to the smoothing spline estimator (4.60) as $a \to \infty$.

4.4. **Adaptive estimation.**

4.4.1. *Elliptical classes.* In the white noise model we observe a stochastic process

$$y(t) = \int_0^t f(u)du + \sigma_n W(t), \tag{4.72}$$

where $W(t)$ is a standard Brownian motion process and $\sigma_n \to 0+$. Suppose

$$f(t) = \sum_{k=1}^{\infty} \theta_k \phi_k(t), \quad 0 \leqslant t \leqslant 1, \tag{4.73}$$

with the Fourier basis $\phi_k(z) = \sqrt{2}\sin(\pi k z)$ in $[0,1]$, and unknown Fourier coefficients $\theta_k$. Assume that $f(z)$ is $\alpha$-times differentiable under the above sum and

$$\sum_{k=1}^{\infty} (\pi k)^{2\alpha} \theta_k^2 = \int_0^1 \big|f^{(\alpha)}(t)\big|^2 dt = \|f\|^2_{L_2^\alpha([0,1])} \leqslant C^2. \tag{4.74}$$

Let $\mathscr{F}_{\alpha,C}$ be the collection of all $f$ satisfying (4.73) and (4.74). [Pin80] proved that

$$\inf_{\widehat{f}} \sup_{f\in\mathscr{F}_{\alpha,C}} \mathbb{E}\Big[\|\widehat{f} - f\|^2_{L_2([0,1])}\Big] = (1+o(1))\sigma_n^{4\alpha/(1+2\alpha)}(C/\pi^\alpha)^{2/(1+2\alpha)} C_\alpha^*, \tag{4.75}$$

where $C_\alpha^* = (\alpha/(\alpha+1))^{2\alpha/(1+2\alpha)}(2\alpha+1)^{1/(1+2\alpha)}$ is known as Pinsker's constant.

Let $\{\phi_k, k \geqslant 1\}$ be a general orthonormal bases in $L_2([0,1])$. In the white noise model (4.72), the following statistics are mutually independent and jointly sufficient,

$$\xi_k = \int_0^1 \phi_k(t)dy(t) \sim N(\theta_k, \sigma_n^2), \;\; k \geqslant 1. \tag{4.76}$$

By Parseval's identity the estimation of $f$ under the $L_2([0,1])$ loss in the white noise model (4.72) is equivalent to the estimation of $\theta = (\theta_k, k \geqslant 1)^\top$ under the $\ell_2$ loss in the Gaussian sequence model (4.76). In particular, they have the same minimax risk,

$$\inf_{\widehat{f}} \sup_{f\in\mathscr{F}} \mathbb{E}\Big[\|\widehat{f} - f\|^2_{L_2([0,1])}\Big] = \inf_{\widehat{\theta}} \sup_{\theta\in\Theta} \mathbb{E}\big[\|\widehat{\theta} - \theta\|_2^2\big],$$

when $\Theta = \{\theta : \sum_{k=1}^\infty \theta_k\phi_k = f \in \mathscr{F}\}$. The class $\Theta$, or equivalently $\mathscr{F}$, is elliptical if

$$\Theta = \left\{\theta : \sum_{k=1}^{\infty} (\theta_k/r_k)^2 \leqslant 1\right\} \tag{4.77}$$

for a proper choice of basis and some constants $0 < r_k \leqslant \infty$. In (4.74), $r_k = C/(\pi k)^\alpha$. The minimax linear estimator provides an upper bound for the minimax risk.

**Lemma 4.7.** *In the Gaussian sequence model* (4.76) *with the Sobolev class* (4.74)*, the right-hand side of* (4.75) *is attained by linear estimators of the form* $\widehat{f}(t) = \sum_{k=1}^\infty \widehat{\theta}_k \phi_k(t)$*,* $\widehat{\theta}_k = b_k\xi_k, k \geqslant 1$*, with certain constant shrinkage factors* $b_k \in [0,1]$*,*

$$\inf_{0\leqslant b_k\leqslant 1, k\geqslant 1} \sup_{\theta\in\Theta} \sum_{k=1}^{\infty} \mathbb{E}\big[\big(b_k\xi_k - \theta_k\big)^2\big] = (1+o(1))\sigma_n^{4\alpha/(1+2\alpha)}(C^2/\pi^{2\alpha})^{1/(1+2\alpha)} C_\alpha^*.$$

*Hence, the right-hand side of* (4.75) *is no smaller than the left-hand side.*

The minimax risk is bounded from below by the Bayes risk of the Bayes rule. Here we consider such a lower bound in a block structure. Let $k_0 < k_1 < k_2 < \cdots$ be a sequence of nonnegative integers, $[j] = \{k \in N_+ : k_{j-1} < k \leqslant k_j\}$ with $k_{-1} = 0$ and

$$\Theta(\eta, p, q) = \left\{\theta : \sum_{j=0}^{\infty} \left(|[j]|^{-1} \sum_{k\in[j]} |\theta_k/\eta_j|^p\right)^{q/p} \leqslant 1\right\}. \tag{4.78}$$

where $\eta = (\eta_0, \eta_1, \ldots)^\top$ with $\eta_j > 0$, $0 \leqslant p \leqslant \infty$ and $1 \leqslant q \leqslant \infty$. For the ellipse $\Theta$ in (4.77),

$$\Theta(\eta_-, 2, 2) \subset \Theta \subset \Theta(\eta_+, 2, 2) \tag{4.79}$$

when $\eta_\pm = (\eta_{\pm,0}, \eta_{\pm,1}, \ldots)^\top$ with $\eta_{-,j} \leqslant r_k \leqslant \eta_{+,j}$ for all $k \in [j]$ and $j \geqslant 0$.

**Lemma 4.8.** *Let $\Theta_j(\eta_j, p, C_j) = \{\theta_{[j]} : |[j]|^{-1/p}\|\theta_{[j]}/\eta_j\|_p \leqslant C_j\}$ with the $\{[j], \eta_j\}$ in (4.78), and $\mathbb{E}_{C_j}$ denote an expectation with a prior satisfying $\mathbb{P}_{C_j}\{\theta_{[j]} \in \Theta_j(\eta_j, p, C_j)\} = 1$. Then, in the Gaussian sequence model (4.76),*

$$\inf_{\widehat\theta} \sup_{\theta\in\Theta(\eta,p,q)} \mathbb{E}\big[\|\widehat\theta - \theta\|_2^2\big] \geqslant \sup\left\{\sum_{j=0}^{\infty} \inf_{\widehat\theta_{[j]}} \mathbb{E}_{C_j}\big[\|\widehat\theta_{[j]} - \theta_{[j]}\|_2^2\big] : \sum_j C_j^q \leqslant 1\right\}.$$

**Lemma 4.9.** *Let $\Theta_{p,r_0} = \{\mu \in \mathbb{R}^d : \|\mu\|_p/d^{1/p} \leqslant r_0\}$, $X|\mu \sim N(\mu, \sigma^2 I_d)$ and $\mathbb{E}_G$ be the expectation with the prior $G$ under which $\mu \sim N(0, r_1^2 I_d)$. Let $b_j = r_j^2/(\sigma^2 + r_j^2)$. Then,*

$$\min\nolimits_b \max\nolimits_{\mu\in\Theta_{2,r_0}} \mathbb{E}_\mu\big[\|bX - \mu\|_2^2\big] = d\sigma^2 b_0,$$
$$d\sigma^2 b_0 - \inf\nolimits_{\delta(\cdot)} \mathbb{E}_G\big[\|\delta(X) - \mu\|_2^2 \big| \mu \in \Theta_{2,r_0}\big] \leqslant d\sigma^2(b_0 - b_1) + \mathbb{E}_G\big[\big(d^{1/2}r_0 + \|\mu\|_2\big)^2; \mu \notin \Theta_{2,r_0}\big].$$

*In particular, the right-hand side of the above inequality is of the order $o(d\sigma^2)$ when $\sigma \asymp r_0$, $r_0/r_1 = 1 + o(1)$ and $d^{1/2}(r_0/r_1 - 1) \to \infty$.*

Proof. For $\mu \in \Theta_{2,r_0}$, the minimax shrinkage estimator is given by

$$\min_b \max_{\mu\in\Theta_{2,r_0}} \mathbb{E}_\mu\big[\|bX - \mu\|_2^2\big] = \min_b \big\{b^2 d\sigma^2 + (1-b)^2 dr_0^2\big\} = d\sigma^2 b_0.$$

Similarly, $\mathbb{E}_G\big[\|b_1 X - \mu\|_2^2\big] = d\sigma^2 b_1$. As $b_1 X$ is the Bayes rule under prior $G$,

$$\mathbb{E}_G[\|b_1 X - \mu\|_2^2] \quad\leqslant\quad \mathbb{E}_G[\|\delta(X) - \mu\|_2^2; \mu \in \Theta_{2,r_0}] + \mathbb{E}_G[\|\delta(X) - \mu\|_2^2; \mu \notin \Theta_{2,r_0}]$$

for any estimator $\delta$. For the Bayes rule under the conditional prior $G(d\mu|\Theta_{2,r_0})$, $\|\delta(X)\|_2^2 \leqslant dr_0^2$, so that $\mathbb{E}_G[\|\delta(X) - \mu\|_2^2; \mu \notin \Theta_{2,r_0}] \leqslant \mathbb{E}_G\big[\big(d^{1/2}r_0 + \|\mu\|_2\big)^2; \mu \notin \Theta_{2,r_0}\big]$. The asymptotic result follows from the Gaussian concentration inequality $\mathbb{P}\{\|\mu\|_2/r_1 \geqslant \sqrt{d} + t\} \leqslant e^{-t^2/2}$. $\square$

Lemma 4.9 implies that the minimax linear estimator is an approximate Bayes rule in $\ell_2$ balls. Thus, if the ellipse (4.77) can be approximated by the block ellipse as in (4.79), the minimax linear estimator in the ellipse is approximately minimax. This is applicable to the Sobolev balls $\mathscr{F}_{\alpha,C}$ in (4.75), giving Pinsker's theorem.

**Theorem 4.11.** *Equation (4.75) holds for all fixed $(\alpha, C)$ as $\sigma_n \to 0+$. Moreover, as $\sigma_n \to 0+$, the general minimax risk on the left-hand side is attained within an infinitesimal fraction by a linear estimator of the form $\widehat\theta_k = b_k \xi_k$ with proper $b_k = b_k(\sigma_n, \alpha, C)$.*

In nonparametric regression, careful analyses of the kernel regression, local LSE or smoothing spline would show that these methods achieve

$$\sup_{f\in\mathscr{F}_{\alpha,C}} \mathbb{E}\Big[\|\widehat{f}-f\|_{L_2([0,1])}^2\Big] = (1+o(1))C_1 n^{-2\alpha/(1+2\alpha)},$$

where $C_1$ is a constant depending only on $(\alpha, C)$ and the method used. However, to achieve the rate, the choice of the tuning parameters, e.g. the bandwidth, kernel or penalty level, may depend on $(\alpha, C)$ and the optimality of the resulting $C_1$ is unclear.

The analysis leading to Theorem 4.11 is valid for much more general elliptical classes. When an ellipse (4.77) can be approximated by block ellipses of the form (4.78) with proper $\eta_{+,j}^2/(\sigma_n^2+\eta_{+,j}^2) \approx \eta_{-,j}^2/(\sigma_n^2+\eta_{-,j}^2)$ and $1/|[j]| + \eta_{+,j}^2 = o(1)$ in (4.79), the minimax risk is approximately attained by a certain block-wise shrinkage estimator by Lemma 4.9.

Theorem 4.11 and its extensions assert that when the Gaussian mean belongs to an elliptical class, the general minimax MSE is approximately attained by linear estimators, provided proper mild conditions on $\{r_k\}$ for the ellipse in (4.77). Moreover, Theorem 4.11 gives explicitly the asymptotic minimax constant in Sobolev balls. Still, the approximately minimax linear estimator depends on $\{r_k\}$ in (4.77) through the $\eta_j$ in (4.79).

The goal of adaptive estimation is to achieve optimality within an infinitesimal fraction simultaneously in multiple classes of parameters. For the Sobolev classes given by (4.73) and (4.74), this problem, solved by Efroimovich and Pinsker [EP84], means to achieve (4.75) in both the rate and the optimal constant factor without the knowledge of $\alpha$ and $C$. Their result has since been extended to more general settings and beyond the Sobolev balls.

Consider the Gaussian sequence model (4.76) with the mean $\theta$ in an ellipse $\Theta$ in (4.77) with $0 < r_k \to 0$. Given $\sigma_n$ let $1 \leqslant k_j < k_{j+1}$ and $[j] = \{k \in N_+ : k_{j-1} < k \leqslant k_j\}$ be as in (4.78) with $k_j - k_{j-1} \uparrow \infty$. It follows from Lemmas 4.8 and 4.9 that under mild conditions, the minimax risk is approximately achieved by a linear estimator of the form

$$\widehat{\theta} = \big(\widehat{\theta}_{[0]}^\top, \widehat{\theta}_{[1]}^\top, \ldots\big)^\top, \quad \widehat{\theta}_{[j]} = b_j \xi_{[j]}.$$

The idea of Efroimovich and Pinsker is to choose the shrinkage factors $b_j$ adaptively with a certain version of the James-Stein estimator. Such estimators can be written as

$$\widehat{\theta} = \big(\widehat{\theta}_{[0]}^\top, \widehat{\theta}_{[1]}^\top, \ldots\big)^\top, \quad \widehat{\theta}_{[j]} = \big(1 - \sigma_n^2 c_j/\|\xi_{[j]}\|_2^2\big)_+ \xi_{[j]} I\{j \leqslant j_n^*\} \tag{4.80}$$

with $|[j]| - 2 \leqslant c_j < 2(|[j]| - 2)$.

**Theorem 4.12.** *Let $\widehat{\theta}$ be the block James-Stein estimator* (4.80) *with $\xi$ in* (4.76), *$[j] = \{k \in N_+ : k_{j-1} < k \leqslant k_j\}$ depending on $\sigma_n$ only, and $c_j = |[j]| - 2$. Let $\eta_j^+ = \max_{k\in[j]} r_k$, $\eta_j^- = \min_{k\in[j]} r_k$ and $b_{\pm,j} = \eta_{\pm,j}^2/(\sigma_n^2 + \eta_{\pm,j}^2)$ with the ellipse $\Theta$ in* (4.77). *Suppose*

$$\lim_{C\to\infty} \liminf_{\sigma_n\to 0+} \frac{\sum_{j\in J_{n,C}} |[j]|\sigma_n^2 b_{-,j}}{j_n^*\sigma_n^2 + \sum_{j=0}^{j_n^*} |[j]|\sigma_n^2 b_{+,j} + \sum_{j=j_n^*+1}^{\infty} \|\theta_{[j]}\|_2^2} = 1,$$

*where $J_{n,C} = \{j \leqslant j_n^* : b_{-,j} \geqslant 1/C\}$. Then,*

$$\lim_{\sigma_n\to 0+} \frac{\sup_{\theta\in\Theta} \mathbb{E}\big[\|\widehat{\theta} - \theta\|_2^2\big]}{\inf_{\delta(\cdot)} \sup_{\theta\in\Theta} \mathbb{E}\big[\|\delta(\xi) - \theta\|_2^2\big]} = 1. \tag{4.81}$$

PROOF. It follows from Lemma 4.9 that $\sum_{j\in J_{n,C}}|[j]|b_{-,j}$ is within $(1+o(1))$ of the Bayes risk of a Bayes estimator, so that

$$\lim_{C\to\infty}\limsup_{\sigma_n\to 0+}\frac{\sum_{j\in J_{n,C}}|[j]|b_{-,j}}{\inf_{\delta(\cdot)}\sup_{\theta\in\Theta}\mathbb{E}\big[\|\delta(\xi)-\theta\|_2^2\big]}\leqslant 1.$$

For the James Stein estimator, $\mathbb{E}[\|\widehat{\theta}_{[j]}-\theta_{[j]}\|_2^2]\leqslant|[j]|\sigma_n^2 b_{+,j}+4\sigma_n^2$ for $j\leqslant j_n^*$, so that

$$\limsup_{\sigma_n\to 0+}\frac{\mathbb{E}[\|\widehat{\theta}-\theta\|_2^2]}{j_n^*\sigma_n^2+\sum_{j=0}^{j_n^*}|[j]|\sigma_n^2 b_{+,j}+\sum_{j=j_n^*+1}^{\infty}\|\theta_{[j]}\|_2^2}\leqslant 1.$$

The conclusion follows by applying the imposed condition to the product of the above two inequalities. ☐

**Corollary 4.3.** *Let $k_0=c_n-1$ and $k_{j+1}=k_j+\max(c_n,k_j/c_n)$ with $c_n=2\vee\lceil\log(1/\sigma_n^2)\rceil$ and $j_n^*=c_n^3$ in the construction of $\widehat{\theta}$ in* (4.80)*. Then,* (4.81) *holds for the Sobolev classes given by* (4.73), (4.74) *and* (4.76)*. Consequently, for $\widehat{f}(t)=\sum_{j=0}^{j_n^*}\sum_{k\in[j]}\widehat{\theta}_k\sqrt{2}\cos(\pi kt)$,*

$$\begin{aligned}\sup_{f\in\mathscr{F}_{\alpha,C}}\mathbb{E}\Big[\|\widehat{f}-f\|_{L_2([0,1])}^2\Big] &= (1+o(1))\inf_{\delta(\cdot)}\sup_{f\in\mathscr{F}_{\alpha,C}}\mathbb{E}\Big[\|\delta-f\|_{L_2([0,1])}^2\Big]\\ &= (1+o(1))\sigma_n^{4\alpha/(1+2\alpha)}(C/\pi^{\alpha})^{2/(1+2\alpha)}C_\alpha^*,\end{aligned}$$

*with Pinsker's constant $C_\alpha^*$ in* (4.75)*, where the infimum is taken over all estimators of the function $f$ based on the white noise process $\{y(t),0\leqslant t\leqslant 1\}$ with noise level $\sigma_n$ in* (4.72)*.*

Now consider nonparametric regression with design points $z_i=(i-1/2)/n, i=1,\ldots,n$, and iid noise $\varepsilon_i\sim N(0,\sigma^2)$ in (4.48). For this discrete uniform design, the Fourier basis in (4.73) is also orthogonal in the discrete case

$$\overline{\mathbb{P}}_n\big[\phi_j(z)\phi_k(z)\big]=I\{j=k\},\ 0\leqslant j\leqslant k<n.$$

It follows from this dual orthonormality in both the continuous and discrete settings that

$$\overline{\mathbb{P}}_n\big[y\phi_k(z)\big]\sim N(\mu_k,\sigma_n^2),\ 1\leqslant j\leqslant k\leqslant n, \tag{4.82}$$

are independent and jointly sufficient for the estimation of the unknown $f$. Moreover, for $2\alpha>1$, the means $\mu_k$ are very close to the Fourier coefficients $\theta_k$ of $f$,

$$\sum_{k=1}^{n}(\mu_k-\theta_k)^2\leqslant C'_\alpha/n^{2\alpha},\ 1\leqslant k\leqslant n,\ \sum_{k=n+1}^{\infty}\theta_k^2\leqslant C''_\alpha/n^{2\alpha},\ \ \sigma_n^2=\sigma^2/n.$$

Thus, by the Parseval identity and fact that $n^{-2\alpha}\ll\sigma_n^{4\alpha/(1+2\alpha)}$, for $2\alpha>1$,

$$\begin{aligned}\sup_{f\in\mathscr{F}_{\alpha,C}}\mathbb{E}\Big[\|\widehat{f}-f\|_{L_2([0,1])}^2\Big] &= \inf_{\delta(\cdot)}\sup_{f\in\mathscr{F}_{\alpha,C}}\mathbb{E}\Big[\|\delta-f\|_{L_2([0,1])}^2\Big]\\ &= (1+o(1))\inf_{\widehat{f}}\sup_{\mu\in B_{\alpha,C,n}}\mathbb{E}\Big[\|\widehat{\mu}-\mu\|_2^2\Big]\end{aligned}\tag{4.83}$$

with $\widehat{f}=\sum_{k=1}^{n}\widehat{\theta}_k\phi_k$ as in Corollary 4.3 and $B_{\alpha,C,n}=\{\mu\in\mathbb{R}^n:\sum_{k=1}^{n}(\pi k)^{2\alpha}\mu_k^2\leqslant C^2\}$.

4.4.2. *Sparsity and discontinuity.* We have discussed linear estimators including kernel regression, local LSE and smoothing spline. The block-wise James-Stein estimator is not linear but it adaptively mimics the optimal linear estimator under proper conditions. However, linear estimators are not rate minimax when the regression function $f$ is piecewise smooth with unknown discontinuities.

**Proposition 4.2.** *Let $y$ be as in nonparametric regression* (4.48) *or white noise* (4.72)*. Let $\mathscr{F}_0$ be a class of functions and $\mathscr{F}$ be the convex hull of $\mathscr{F}_0$. Then, for any convex loss function $L(\cdot)$ in $\mathbb{R}^d$ and linear estimator $Ay$ for a linear parameter $\tau(f) \in \mathbb{R}^d$,*

$$\sup_{f \in \mathscr{F}_0} \mathbb{E}\big[L\big(Ay - \tau(f)\big)\big] = \sup_{f \in \mathscr{F}} \mathbb{E}\big[L\big(Ay - \tau(f)\big)\big].$$

Proposition 4.2 asserts that for linear estimators the minimax risk for any class $\mathscr{F}_0$ of regression functions $f$ is identical to that of the convex hull of $\mathscr{F}_0$. Thus, when $\mathscr{F}_0$ is much smaller than its convex hull, linear estimators may not be optimal under the assumption $f \in \mathscr{F}_0$, e.g. when the regression function $f$ is piecewise smooth. An alternative to the linear estimator is the soft-threshold estimator

$$s_\lambda(x) = \mathrm{sgn}(x)(|x| - \lambda)_+ = \arg\min_t \big\{(x-t)^2/2 + \lambda|t|\big\},\ x \in \mathbb{R}, \lambda \geqslant 0.$$

**Lemma 4.10.** *For $X \sim N(\mu, \sigma^2)$ and $\lambda \geqslant 0$*

$$\mathbb{E}\big[\big(s_\lambda(X) - \mu\big)^2\big] \leqslant 2\sigma^2\Phi(-\lambda/\sigma)/\{1 + (\lambda/\sigma)^2/2\} + |\mu|^2 \wedge (\lambda^2 + \sigma^2)$$

**Example 4.5.** *In the white noise model* (4.72) *let $\mathscr{F}_0$ be the class of all functions of the form $f(z) = \beta_1 I\{z \leqslant \beta_0\} + \beta_2 I\{z > \beta_0\}$ with $|\beta_1 - \beta_2| \leqslant 1$ and $\beta_0 \in [0,1]$. The convex hull of $\mathscr{F}_0$ is $\mathscr{F} = \{f : \|f\|_{TV} \leqslant 1\}$ where $\|f\|_{TV}$ is the total variation of $f$. For differentiable $f$, $\|f\|_{TV} = \int_0^1 |f'(t)|dt$, so that $\mathscr{F} \supset \{f : \|f'\|_2 \leqslant 1\}$. By Theorem 4.11 and Proposition 4.2,*

$$\inf_{\widehat{f} \in \mathscr{L}} \sup_{f \in \mathscr{F}_0} \mathbb{E}\Big[\|\widehat{f} - f\|^2_{L_2([0,1])}\Big] \geqslant (1 + o(1))\sigma_n^{4/3}\pi^{-2/3}C_1^*,$$

*where $\mathscr{L}$ is the set of all linear estimators. Let $\phi_{-1,1}(t) = 1$ and $\phi_{j,k}(t) = 2^{j/2}\phi(2^j t - k + 1)$, $1 \leqslant k \leqslant 2^j$, $j \geqslant 0$, be the Haar basis with $\phi(t) = I\{0 < t \leqslant 1/2\} - I\{1/2 < t \leqslant 1\}$. The white noise is then turned into a Gaussian sequence model with*

$$\xi_{j,k} = \int_0^1 \phi_{j,k}(t)dy(t) \sim N(\theta_{j,k}, \sigma_n^2),\ 1 \leqslant k \leqslant 2^{j\vee 0}, j = -1, 0, 1, 2, \ldots \tag{4.84}$$

*As $f$ has a single jump, $\#\{k : \theta_{j,k} \neq 0\} = 1$ and $\max_{1 \leqslant k \leqslant 2^{j\vee 1}} |\theta_{j,k}| \leqslant 2^{-j/2}$ for all $j \geqslant 1$. Thus, it makes sense to threshold the noisy Haar coefficients $\xi_{j,k}$ and estimate $f$ by*

$$\widehat{f} = \xi_{-1,1} + \textstyle\sum_{j=0}^{j_n^*}\sum_{k=1}^{2^j} s_{\lambda_j}(\xi_{j,k})\phi_{j,k}$$

*with $j_n^*$ satisfying $2^{-j_n^*} \leqslant \sigma_n^2 \leqslant 2^{1-j_n^*}$ and the threshold levels $\lambda_j = \sigma_n\sqrt{2\log(2^j)}$. By Lemma 4.10 and the Parseval identity*

$$\mathbb{E}\Big[\|\widehat{f} - f\|^2_{L_2([0,1])}\Big] \leqslant \sigma_n^2\big(1 + 2j_n^* + 1 + 2j_n^*(j_n^* + 1)\log(2)\big) + 2\sigma_n^2 \asymp \sigma_n^2 \log^2(1/\sigma_n^2).$$

*This nearly parametric rate $\sigma_n^2$ is faster than what linear estimators can achieve.*

Example 4.5 demonstrates the rate sub-optimality of linear estimators for the estimation of a piecewise smooth function with unknown locations of the discontinuities. In a more general setting, the smoothness and the inhomogeneity of the smoothness of a signal function can be conveniently expressed in a wavelet expansion of the signal.

The simplest wavelet basis in $L_2[0,1]$ is the Haar basis $\phi_{j,k}$ in (4.84). In general, a wavelet basis can be written as an orthonormal system $\phi_{-1,1}$, $\phi_{j,k}$, $1 \leqslant k \leqslant 2^j$, $j \geqslant 0$, of the form $\phi_{j,k}(t) = 2^{j/2}\phi(2^j t - k + 1)$, with boundary adjustment if necessary, where $\phi_{-1,1}$ is called the father wavelet and $\phi$ the mother wavelet. Compared with the special case of the Haar system, wavelets can be differentiable. Similar to the B-spline, the support of a differentiable mother wavelet can be compact but must be larger than $[0,1]$. The wavelet expansion of a function $f$ is then

$$f(t) = \sum_{j=-1}^{\infty} \sum_{k=1}^{1\vee 2^j} \theta_{j,k}\phi_{j,k}(t) \tag{4.85}$$

with the coefficient sequence $\theta = (\theta_{[j]}, j \geqslant -1)$, where $\theta_{[j]} = (\theta_{j,k}, k \leqslant 1 \vee 2^j)$.

Compared with the Fourier series expansion $f(t) = \sum_j \theta_j^{(F)} e^{\sqrt{-1}2\pi t}$ in which the coefficient $\theta_j^{(F)}$ represents the projection of the signal $f$ to frequency $j$, the wavelet coefficient $\theta_{j,k}$ represents the projection of the signal to frequency $2^j$ in a neighborhood of $k/2^j$ as the wavelet $\phi_{j,k}(t) = 2^{j/2}\phi(2^j t - k + 1)$ localizes in both the frequency and time domains. In a multi-resolution (scale, frequency) analysis, this feature of wavelets allows us to express the general smoothness of the signal in terms of the overall speed of decay of the coefficient vector $\theta_{[j]}$ as the resolution level gets finer and also to express the locations of the less smooth parts of the signal in terms of the spikes of the coefficients within individual resolution levels. When the discontinuities are sparse, the spikes are also sparse. Thus, the $\ell_p$ norm with small $p$, possibly $p < 1$ to reduce sensitivity to the spikes caused by uneven smoothness, can be used to measure the overall magnitude of $\theta_{[j]}$ in individual resolution levels. Similarly, an $\ell_q$ norm, with small $q$ to reduce sensitivity to spikes in $\|\theta_{[j]}\|_p$, can be used to combine the signal strength in different resolution levels. This motivates the use of the Besov norm

$$\begin{aligned}
\|f\|_{\alpha,p,q} = \|\theta\|_{\alpha,p,q} &= \left\{|\theta_{-1,1}|^q + \sum_{j=0}^{\infty}\left(2^{j/2}\left(\frac{1}{2^j}\sum_{k=1}^{2^j}|2^{\alpha j}\theta_{j,k}|^p\right)^{1/p}\right)^q\right\}^{1/q} \\
&= \left\{|\theta_{-1,1}|^q + \sum_{j=0}^{\infty}\left(2^{j(\alpha+1/2-1/p)}\|\theta_{[j]}\|_p\right)^q\right\}^{1/q}
\end{aligned}$$

with $\alpha + 1/2 > 1/p$ for functions with the wavelet expansion (4.85). For $p \leqslant 2$, $\|\theta\|_{\alpha,p,q} \leqslant \|\theta\|_{\alpha,2,q}$ so that $\sigma_n^{4\alpha/(1+2\alpha)}C^{2/(1+2\alpha)}$ is a lower bound for the minimax risk in the Basov ball $\{f : \|f\|_{\alpha,p,q} \leqslant C\}$ by the proof of Pinsker's theorem (4.75).

The Sobolev norm is equivalent to $\|f\|_{\alpha,p,p}$. In Example 4.5 the Besov norm of $f \in \mathscr{F}_0$ is bounded in the Haar system by $\|f\|_{\alpha,p,\infty} \leqslant |\theta_{-1,1}| \vee 1$ with $1/p = \alpha$ to reflect the smoothness index $\alpha = 1$ for the convex hull $\mathscr{F}$ with $p = 1$ and the overall level of smoothness in most neighborhoods with index $\alpha \to \infty$ when the sparse spikes in $|\theta_{j,k}|$ are suppressed with $p = 1/\alpha \to 0+$.

In the white noise model (4.72), the wavelet transformation of the data yields the Gaussian sequence of independent observations

$$\xi_{j,k} = \int \phi_{j,k}(t)dy(t) \sim N\big(\theta_{j,k}, \sigma_n^2\big),\ 1 \leqslant k \leqslant 1 \vee 2^j,\ j \geqslant -1. \tag{4.86}$$

In nonparametric regression with lattice design $z_i = i/n$ with $n = 2^J$, the discrete wavelet transformation of the data yields

$$\big(\xi_{j,k}, 1 \leqslant k \leqslant 1 \vee 2^j, -1 \leqslant j \leqslant J\big)^\top = n^{-1/2} W_n (y_1, \ldots, y_n)^\top$$

where $W_n \in \mathbb{R}^{n\times n}$ is an orthonormal "finite wavelet transformation matrix" satisfying $n^{1/2} W_{(j,k),i} \approx \phi_{j,k}(z_i)$ with rows indexed by $(j,k)$. As in (4.82), this gives independent

$$\xi_{j,k} \sim N\big(\mu_{j,k}, \sigma_n^2\big), \quad \mu_{j,k} \approx \theta_{j,k},\ \sigma_n^2 = \sigma^2/n.$$

The problem is then to estimate $f$ or equivalently $\theta$ based on available $\xi = \{\xi_{j,k}\}$ with

$$\widehat{f} = \sum_{j,k} \widehat{\theta}_{j,k}\phi_{j,k}, \tag{4.87}$$

and the goal of rate adaptive estimation is to achieve

$$\sup_{\|f\|_{\alpha,p,q}\leqslant C} \mathbb{E}\big[\|\widehat{f} - f\|_{L_2([0,1])}^2\big] = \sup_{\|\theta\|_{\alpha,p,q}\leqslant C} \mathbb{E}\big[\|\widehat{\theta} - \theta\|_2^2\big] \lesssim \sigma_n^{4\alpha/(1+2\alpha)} C^{2/(1+2\alpha)} \tag{4.88}$$

without the knowledge of $(\alpha, p, q, C)$.

Donoho and Johnstone [DJ94, DJ98] proved that the rate in (4.88) is optimal. Moreover, they proved that the minimax rate can be achieved by threshold estimators but for $p < 2$ not by linear estimators. Consider the soft-threshold estimator as in Example 4.5,

$$\widehat{\theta}_{j,k} = s_{\lambda_j}(\xi_{j,k}),\ \lambda_j = \sigma_n\sqrt{2(j\vee 0)\log 2}. \tag{4.89}$$

**Theorem 4.13.** *Let the estimator* (4.87) *be given by* (4.89) *with the* $\xi_{j,k}$ *in* (4.86)*. Then, as* $\sigma_n \to 0+$ *and for* $\alpha + 1/2 > 1/p \geqslant 1/2$,

$$\sup_{\|f\|_{\alpha,p,q}\leqslant C} \mathbb{E}\big[\|\widehat{f} - f\|_{L_2([0,1])}^2\big] = \sup_{\|\theta\|_{\alpha,p,q}\leqslant C} \mathbb{E}\big[\|\widehat{\theta} - \theta\|_2^2\big] \lesssim \big\{\sigma_n^2 \log(1/\sigma_n^2)\big\}^{2\alpha/(1+2\alpha)} C^{2/(1+2\alpha)}.$$

Proof. It follows from Lemma 4.10 that

$$\begin{aligned}
\sup_{\|\theta\|_{\alpha,p,q}\leqslant C} \mathbb{E}\big[\|\widehat{\theta} - \theta\|_2^2\big] &\leqslant 2\sigma_n^2 + \sum_{j=1}^{j_n^*}\big\{\sigma_n^2 + 2^j(\lambda_j^2 + \sigma_n^2)\big\} \\
&\quad + \sum_{j=j_n^*+1}^{\infty}\bigg\{\frac{\sigma_n^2}{\sqrt{j\log 4}(1+j\log 2)} + \sum_{k=1}^{2^j}|\theta_{j,k}|^p(\lambda_j^2+\sigma_n^2)^{1-p/2}\bigg\} \\
&\lesssim 2^{j_n^*}\lambda_{j_n^*}^2
\end{aligned}$$

with $2^{j_n^*}\lambda_{j_n^*}^2 \leqslant 2^{-j_n^* p(\alpha+1/2-1/p)} C^p \lambda_{j_n^*}^{2-p} \leqslant 2^{j_n^*+1}\lambda_{j_n^*+1}^2$, or $\sigma_n^2 j_n^* \asymp \lambda_{j_n^*} \asymp 2^{-j_n^*(\alpha+1/2)}C$. □

Theorem 4.13 asserts that with the "universal threshold level", without the knowledge of $(\alpha, p, q, C)$, the threshold estimator (4.89) adaptively but nearly achieves the minimax rate (4.88) with an extra logarithmic factor.

4.4.3. *Empirical Bayes.* For $p = q = 2$, the Besov ball $\{\theta : \|\theta\|_{\alpha,p,q} \leqslant C\}$ is elliptical. In fact, it is of the block form with $p = 2$, $|[j]| = 2^{j-1}$ and $\eta_j = 2^{-j(\alpha+1/2)}$ in (4.78), so that the block-wise James-Stein estimator is adaptive to the minimax constant as in Theorem 4.12. In the analysis leading to the theorem, the crucial point is that in each block, the maximum risk of the James-Stein estimator in the $\ell_2$ ball $\Theta_j(\eta_j, 2, C_j)$ is approximately the Bayes risk of the Bayes estimator with the truncated Gaussian prior, so that the sums of these upper and lower bounds over $j$ accurately approximate the minimax risk in Besov balls through Lemma 4.8. To extend this analysis to Besov balls with general $p$, we need to construct expectations $\mathbb{E}_{C_j}$ with priors supported in $\Theta_j(\eta_j, p, C_j) = \{\theta_{[j]} : |[j]|^{-1/p}\|\theta/\eta_j\|_p \leqslant C_j\}$ as in Lemma 4.8 and approximately minimax estimators $\widehat{\theta}_{[j]}$ satisfying

$$\frac{\sup\left\{\mathbb{E}\left[\|\widehat{\theta}_{[j]} - \theta_{[j]}\|_2^2\right] : \theta_{[j]} \in \Theta_j(\eta_j, p, C_j)\right\}}{\inf_{\delta_j} \mathbb{E}_{C_j}\left[\|\delta_j(\xi_{[j]}) - \theta_{[j]}\|_2^2\right]} = 1 + o(1) \tag{4.90}$$

in an extension of Lemma 4.9.

Because (4.90) does not involve the coefficients and observations outside the $j$-th resolution level, the problem is actually the estimation of a Gaussian mean vector with the $\ell_2$ loss in an $\ell_p$ ball. For $|[j]|^{-1/p} \ll c_j = \eta_j C_j/\sigma_n \to 0$, [DJ94] proved (4.90) with

$$\inf_{\delta_j} \mathbb{E}_{C_j}\left[\|\delta_j(\xi_{[j]}) - \theta_{[j]}\|_2^2\right]/|[j]| = (1 + o(1))\sigma_n^2 c_j^p (\log(1/c_j^p))^{1-p/2}, p < 2,$$

for a prior of the form $\theta_{[j]} \sim \mu_j(\delta_k, k \in [j])$ conditionally on $\mu_j(\delta_k, k \in [j]) \in \Theta_j(\eta_j, p, C_j)$ with $\mu_j = (1 + o(1))(\log(1/c_j^p))^{1/2}$ and iid $\delta_k \sim$Bernoulli$((1 + o(1))c_j^p)$. However, for $p < 2$ and at the crucial resolution level with $c_j \asymp 1$, the feasibility of a parametric solution of (4.90) is unclear.

Robbins [Rob51, Rob56] formulated the compound decision problem as statistical inferences about many parameters $\theta_i$ based on independent observations $X_i \sim p(x|\theta_i), i = 1, \ldots, n$, under the average of losses $n^{-1}\sum_{i=1}^n L(\widehat{\theta}_i, \theta_i)$, and proposed the empirical Bayes approach as follows. For any decision rule $\delta(x)$, the compound risk satisfies the identity

$$\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n L(\delta(X_i), \theta_i)\right] = \int L(\delta(x), \theta) p(x|\theta) G_n(d\theta) \tag{4.91}$$

where $G_n(A) = n^{-1}\sum_{i=1}^n I\{\theta_i \in A\}$ is the empirical distribution of the unknown parameters. The optimal solution of (4.91) is the Bayes rule $\delta^*_{G_n}$, where

$$\delta^*_G = \arg\min_{\delta(\cdot)} \int L(\delta(x), \theta) p(x|\theta) G(d\theta). \tag{4.92}$$

If the Bayes rule $\delta^*_{G_n}$ is unavailable, an estimated version $\widehat{\delta}$ can be used and such an empirical Bayes rule is asymptotically optimal if

$$\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n L(\widehat{\delta}(X_i), \theta_i)\right] \approx \int L(\delta^*_{G_n}(x), \theta) p(x|\theta) G_n(d\theta). \tag{4.93}$$

This problem includes (4.90) as a special case with $[j] = \{1, \ldots, n\}$, the squared estimation loss $L(\widehat{\theta}_i, \theta_i) = (\widehat{\theta}_i - \theta_i)^2$ and distribution family $p(x|\theta_i) \sim N(\theta_i, \sigma_n^2)$.

In the Gaussian sequence model with independent $X_i \sim N(\theta_i, \sigma_n^2)$ for deterministic $\theta_{1:n}$ or $X_i|\theta_{1:n} \sim N(\theta_i, \sigma_n^2)$, the Bayes estimator (4.92) under the squared loss is

$$\delta_G^*(x) = \frac{\int \theta p(x|\theta)G(d\theta)}{\int p(x|\theta)G(d\theta)} = x + \frac{\sigma_n^2 p_G'(x)}{p_G(x)} \tag{4.94}$$

with the mixture density $p_G(x) = \int p(x|\theta)G(d\theta)$. This formula appeared in a more general form in [Rob56] who gave credits to M.C.K. Tweedie. [Zha97, Zha05] proposed to use the Fourier method to estimate $p_G$ and $p_G'$ in (4.94) as in Theorem 4.5 and proved that a block-wise application of such an estimator in the white noise and nonparametric regression models achieves adaptive minimaxity in Besov balls

$$\sup_{\|\theta\|_{\alpha,p,q}\leqslant C} \mathbb{E}\Big[\sum_{j=-1}^{\infty} \big\|\widehat{\delta}_j\big(\xi_{[j]}\big) - \theta_{[j]}\big\|_2^2\Big] = (1+o(1)) \inf_{\widehat{\theta}} \sup_{\|\theta\|_{\alpha,p,q}\leqslant C} \mathbb{E}\Big[\big\|\widehat{\theta} - \theta\big\|_2^2\Big] \tag{4.95}$$

as $\sigma_n \to 0+$ simultaneously in all $(\alpha, p, q, C)$ satisfying $\alpha + 1/2 > 1/p \geqslant 1/2$ and $C \geqslant 0$. A block-wise application of the threshold estimator, even with the knowledge of $(\alpha, p, q, C)$, achieves (4.95) only when $C = C_n \ll \sigma_n$ because threshold functions do not approximate well nonlinear analytic functions such as $\delta_G^*$ (4.94) at the crucial resolution levels $j$ satisfying $(\|\theta_{[j]}\|_p/2^{j/p})/C \leqslant 2^{-j(\alpha+1/2)} \asymp \sigma_n$.

Another method of estimating the empirical distribution $G_n$ of the unknown $\theta_{1:n}$ is the generalized maximum likelihood estimator [Rob50, KW56]

$$\widehat{G}_n = \arg\max_G \overline{\mathbb{P}}_n\big[\log p_G(x)\big] \tag{4.96}$$

with $p_G(x) = \int p(x|\theta)G(d\theta)$, where the maximization is taken over all distributions. [JZ09] proposed to directly plug-in this estimator $\widehat{G}_n$ in (4.94) and proved the adaptive optimality of the method in the estimation of $\theta_{1:n} = \mathbb{E}[X_{1:n}]$ in $\ell_p$ balls

$$\sup_{n^{-1/p}\|\theta_{1:n}/\sigma_n\|_p\leqslant C} \frac{\mathbb{E}\big[\big\|\delta_{\widehat{G}_n}\big(X_{1:n}\big) - \theta_{1:n}\big\|_2^2\big]}{n\epsilon_n^2(\log n)^3\sigma_n^2 + \mathbb{E}\big[\big\|\delta_{G_n}\big(X_{1:n}\big) - \theta_{1:n}\big\|_2^2\big]} = 1 + o(1) \tag{4.97}$$

when $\log n \geqslant 2/p$ and $\epsilon_n \gg n^{-p/(2+2p)}(\log n)^{(2+3p)/(4+4p)}$ without the knowledge of $(p, C)$. This gives (4.93) when the normalized $\ell_2$ risk is greater than $\epsilon_n^2(\log n)^3\sigma_n^2$. Consequently, (4.90) holds with $|[j]| = n$ when $\epsilon_n^2(\log n)^3\sigma_n^2 \leqslant c_j^p(\log(1/c_j^p))^{1-p/2}$. By Lemma 4.8, (4.90) and (4.97), a block-wise application of this general maximum likelihood empirical Bayes estimator also achieves adaptive minimaxity in Besov balls as stated in (4.95).

4.5. **Problems.**

**Exercise 4.1.** *Prove* (4.6) *in Lemma 4.1. Note that*

$$\|f\|^2_{L_2^\alpha([0,1])} \leqslant \int_{-1}^1\int_0^1 \frac{|f(x)-f(x+t)|^2}{t^{2\alpha+1}}dtdx \leqslant 3\|f\|^2_{L_2^\alpha([0,1])}$$

*if we extend $f$ to $[-1,2]$ by setting $f(-x)=f(x)$ for $x<0$ and $f(x)=f(2-x)$ for $x>1$ according to its Fourier series expansion, and that the double integral has an explicit Fourier series expansion.*

**Exercise 4.2.** *Use* (4.6) *to prove* (4.8) *and* (4.7) *in Lemma 4.1.*

**Exercise 4.3.** *Use* (4.8) *in Lemma 4.1 to prove Lemma 4.2*

**Exercise 4.4.** *Use the Riemann–Liouville formula* (4.13) *to prove Lemma 4.4.*

**Exercise 4.5.** *Use* (4.13) *to prove* (4.15) *for the $g_m$ in* (4.18) *and $\Delta = \max_{1\leqslant j<m}(t_j - t_{j-1})$.*

**Exercise 4.6.** *Let $\alpha \in \mathbb{N}_+$ and $H_\alpha = \{h \in L_2([0,1]) : \int_0^1 x^j h(x)dx = 0, j = 0,\ldots,\alpha-1\}$ as a subspace of $L_2([0,1])$. For $\alpha > \beta \in \mathbb{N}_+$, let*

$$K_{\alpha,\beta}(s,t) = (-1)^{\alpha-\beta}\frac{|s-t|^{2(\alpha-\beta)-1}}{2(2\alpha-2\beta-1)!}.$$

*Prove that for the $g_m$ in* (4.18) *and $\Delta = \max_{1\leqslant j<m}(t_j - t_{j-1})$,*

$$\|f^{(\beta)} - g_m^{(\beta)}\|_{L_2([0,1])} \leqslant C^{1/2}_{\alpha,\beta}\|f\|_{L_2^\alpha([0,1])}\Delta^{\alpha-\beta}, \tag{4.98}$$

*where $C_{\alpha,\beta}$ is the spectral norm of the kernel $K_{\alpha,\beta}(s,t)$ in $H_\alpha$,*

$$C_{\alpha,\beta} = \sup_{h\in H_\alpha, \|h\|_{L_2([0,1])}=1}\int_0^1\int_0^1 K_{\alpha,\beta}(s,t)h(s)h(t)dsdt. \tag{4.99}$$

*The above $C_{\alpha,\beta}$ is sharp as the equality is attained in* (4.98) *for some $f \in W_2^\alpha([0,1])$.*

**Exercise 4.7.** *Let $C_{\alpha,\beta}$ be as in* (4.99)*. Prove $C_{1,0} = 1/\pi^2$, $C_{2,1} = 1/(2\pi)^2$ and $C_{2,0} \leqslant 1/\pi^4$.*

**Exercise 4.8.** *Prove Theorem 4.1.*

**Exercise 4.9.** *Provide an alternative proof of Theorem 4.2 using the Chapman-Robbins inequality. The Chapman-Robbins bound asserts that*

$$\mathrm{Var}_{\theta_1}(\delta(X)) \geqslant \frac{\left|\mathbb{E}_{\theta_2}[\delta(X)] - \mathbb{E}_{\theta_1}[\delta(X)]\right|^2}{\mathbb{E}_{\theta_1}[(p_{\theta_2}(X)/p_{\theta_1}(X)-1)^2]}$$

*for any decision rule $\delta(X)$, when $X$ has density $p_\theta$ under $\mathbb{E}_\theta$.*

**Exercise 4.10.** *Prove that for $d=1$ and $\alpha=2$ and with $K_1(x)\geqslant 0$, the asymptotic mean integrated squared error (MISE) of the kernel density estimator, defined as $\int(\mu^2(x)+\sigma^2(x))dx$ with functions $\mu(a_0)$ and $\sigma^2(a_0)$ in Theorem 4.1, is minimized when*

$$C_3 = \left(\frac{\int K_1^2(y)dy}{\left(\int K_1(y)y^2dy\right)^2\int(p^{(2)}(y))^2dy}\right)^{1/5}$$

*and $K_1$ is chosen as the Epanechnikov kernel $K_1(t) = (3/4)(1-t^2)_+$.*

**Exercise 4.11.** *Prove Theorem 4.5 and provide a corresponding risk bound for the estimation of the derivative* $p^{(k)}$ *of the Gaussian mixture density* $p(x) = \int \varphi(x-y)G(dy)$ *for* $k \in \mathbb{N}_+^d$ *with* $\|k\|_1 = \beta$*, any finite* $\beta$.

**Exercise 4.12.** *Prove Lemma 4.5.*

**Exercise 4.13.** *Embed the density* $p(\cdot)$ *and the exponential family* (4.35) *in an* $(m+1)$*-dimensional exponential family of densities and express the identity in Lemma 4.5 in terms of the Bregman divergence* $D_\psi$.

**Exercise 4.14.** *Let* $p(x)$ *be a density in* $[0,1]$ *with* $\|\log p\|_{W_2^\alpha([0,1])} \leqslant 1$*. Prove that the estimator* (4.36) *with the Fourier basis* $\phi_j(x) = \sqrt{2}\cos(\pi j x), 1 \leqslant j \leqslant m$ *and* $m \asymp n^{1/(1+2\alpha)}$ *converges to the true* $p(\cdot)$ *with rate* $n^{-\alpha/(1+2\alpha)}$ *in the sense of* (4.46).

**Exercise 4.15.** *Provide the omitted details in the proof of Theorem 4.7 and specify proper conditions for a central limit theorem for* $\widehat{f}(a_0)$ *analogues to* (4.22).

**Exercise 4.16.** *Provide the omitted details in Example 4.2 to show that under the condition* $\int_\Omega h^{-d}K_1((a_0-z)/h)dz \geqslant c_0 > 0$*, the* $C_1$ *in the second part of* (4.50) *is indeed uniformly bounded for the specific* $K(a_0,z)$ *in the example.*

**Exercise 4.17.** *Prove that the condition* $\int_\Omega h^{-d}K_1((a_0-z)/h)dz \geqslant c_h$ *holds for the* $K_1$ *in Example 4.3 and the unit ball* $\Omega$ *in* $\mathbb{R}^d$ *with* $\lim_{h\to 0+} c_h = 1/2$*. Prove* $\lim_{z\to z_i} \widehat{f}(z) = y_i$*,* $1 \leqslant i \leqslant n$ *in this example.*

**Exercise 4.18.** *Provide the omitted details in the proof of Theorem 4.9 and specify proper conditions for a central limit theorem for* $\widehat{f}(a_0)$ *analogues to* (4.22).

**Exercise 4.19.** *Verify the claims in Example 4.4.*

**Exercise 4.20.** *Prove that* $\widehat{f}(z)$ *and* $\widehat{\beta}$ *are unchanged when* $K(s,t)$ *in* (4.59) *is replaced by* $\widetilde{K}(s,t)$ *in* (4.69)*. What is the difference between the two* $\widehat{\gamma}$ *with the two kernels?*

**Exercise 4.21.** *Prove that the posterior mean and also the minimizer of* (4.71) *with the prior specified above* (4.71).

**Exercise 4.22.** *Prove that the minimizer of* (4.71) *converges to the smoothing spline estimator given in* (4.60).

**Exercise 4.23.** *Prove Lemma 4.7.*

**Exercise 4.24.** *Prove Lemma 4.8.*

**Exercise 4.25.** *Prove Theorem 4.11.*

**Exercise 4.26.** *Prove Corollary 4.3 based on Theorem 4.11.*

**Exercise 4.27.** *Prove* (4.83) *based on Theorem 4.11 and Corollary 4.3.*

**Exercise 4.28.** *Prove Lemma 4.10. Let* $R_\lambda(\mu)$ *be the risk function in the case of* $\sigma = 1$*. You may verify by differentiation that* $R_\lambda(\mu) \leqslant R_\lambda(0) + \mu^2 \wedge R_\lambda(\infty)$.

4.6. **Sobolev embedding.** Let $\alpha > 0$, $1 \leqslant p \leqslant \infty$ and $\Omega$ be an open set in $\mathbb{R}^d$. When $\alpha$ is an integer, the Sobolev norm of an $\alpha$-times differentiable function $f$ on $\Omega$ is

$$\|f\|_{W_p^\alpha(\Omega)} = \textstyle\sum_{m=0}^{\alpha}\|f\|_{L_p^m(\Omega)} \quad \text{with} \quad \|f\|_{L_p^m(\Omega)} = \textstyle\sum_{\|k\|_1=m}\|f^{(k)}\|_{L_p(\Omega)},$$

where $k = (k_1, \dots, k_d)^\top$, $f^{(k)}(x) = \prod_{j=1}^d (\partial/\partial x_j)^{k_j} f(x)$ and $\|f\|_{L_p(\Omega)} = \big(\int_\Omega |f|^p dx\big)^{1/p}$. For non-integer $\alpha$, the Sobolev norm is $\|f\|_{W_p^\alpha(\Omega)} = \sum_{m=0}^{\lfloor\alpha\rfloor} \|f\|_{L_p^m(\Omega)} + \|f\|_{L_p^\alpha(\Omega)}$ with

$$\|f\|_{L_p^\alpha(\Omega)} = \sum_{\|k\|_1=\lfloor\alpha\rfloor} \bigg(\int_\Omega\int_\Omega \frac{|f^{(k)}(x) - f^{(k)}(y)|^p}{\|x-y\|_2^{p(\alpha-\lfloor\alpha\rfloor)+d}} dxdy\bigg)^{1/p}.$$

The Sobolev space $W_p^\alpha(\Omega)$ is defined as the completion of the set of all $\lceil\alpha\rceil$-times continuously differentiable functions with compact support under the norm $\|f\|_{W_p^\alpha(\Omega)}$.

Consider bounded convex $\Omega \subset \mathbb{R}^d$ with volume $|\Omega|$ and diameter $|\Omega|_\infty$. By Taylor expansion, functions $f \in W_p^\alpha(\Omega)$ can be approximated by the polynomial

$$(\mathrm{Poly}^\alpha f)(x) = \mathrm{Poly}_x^\alpha(f) = \sum_{0\leqslant\|k\|_1<\alpha} \int_\Omega \frac{f^{(k)}(y)(x-y)^k}{|\Omega| k!} dy, \tag{4.100}$$

where $v^k = \prod_{i=1}^d v_i^{k_i}$ $\forall\, v \in \mathbb{R}^d$, $k! = k_1! \cdots k_d!$, $\mathrm{Poly}_x^\alpha$ are viewed as linear functionals and $\mathrm{Poly}^\alpha$ linear mappings. We write the remainder of this average Taylor expansion, $(\mathrm{Rem}^\alpha f)(x) = \mathrm{Rem}_x^\alpha(f) = f(x) - (\mathrm{Poly}^\alpha f)(x)$, as

$$\mathrm{Rem}_x^\alpha(f) = \begin{cases} f(x) - |\Omega|^{-1}\int_\Omega f(y)dy, & \alpha < 1, \\ \displaystyle\sum_{\|k\|_1=\lceil\alpha\rceil-1} \int_\Omega\int_0^1 \frac{\{f^{(k)}(y+t(x-y)) - f^{(k)}(y)\}(x-y)^k}{|\Omega| k!(1-t)^{2-\lceil\alpha\rceil}/(\lceil\alpha\rceil-1)} dtdy, & \alpha > 1. \end{cases} \tag{4.101}$$

It is clear by the Hölder inequality that $\mathrm{Poly}_x^\alpha$ are uniformly bounded linear functionals on $W_p^\alpha(\Omega)$. Sobolev's embedding theorem, given below, implies that when $p > d/\alpha$, $\mathrm{Rem}_x^\alpha$ are also uniformly bounded linear functionals on $W_p^\alpha(\Omega)$. Let $\delta_x^\alpha = \mathrm{Poly}_x^\alpha + \mathrm{Rem}_x^\alpha$. This gives the representation

$$f(x) = \delta_x^\alpha(f) = \mathrm{Poly}_x^\alpha(f) + \mathrm{Rem}_x^\alpha(f), \quad \|f\|_\infty \leqslant C\|f\|_{W_p^\alpha(\Omega)},$$

with $C = \sup_{x\in\Omega} \|\delta_x^\alpha\|_{W_p^{\alpha *}(\Omega)}$, where $\|\cdot\|_{W_p^{\alpha *}(\Omega)}$ is the dual norm of $\|\cdot\|_{W_p^\alpha(\Omega)}$. In this case $f$ is also uniformly continuous as it is the $L_\infty$ limit of uniformly continuous functions in the completion process.

In what follows, $|B|_p = \sup\{\|x-y\|_p : x \in \Omega, y \in B\}$ is the $\ell_p$ diameter of measurable sets $B \subset \mathbb{R}^d$, $|\{x : f(x) > t\}|$ is the Lebesgue measure of the upper set of $f$, $\|f\|_{L_{p,w}} = t|\{x : |f(x)| > t\}|^{1/p}$ is the weak $L_p$ norm, and $\mathbb{N}_+$ is the set of all nonnegative integers. Sobolev's embedding theorem, stated below, describes the tradeoff between the orders of the moments and smoothness of functions in terms of the Sobolev norm.

**Theorem 4.14.** *Let $0 < \alpha < \infty$, $1 \leqslant p \leqslant \infty$, $\beta \wedge \gamma \geqslant 0$ and $\Omega$ be a bounded open convex set in $\mathbb{R}^d$. Let $C_{d,\alpha,p,\beta,q}$ denote finite constants depending on $(d,\alpha,p,\beta,q)$ only. Then,*

$$\big\|f - \mathrm{Poly}^\alpha f\big\|_{L^\beta_\infty(\Omega)} \leqslant C_{d,\alpha,p,\beta,\infty}(|\Omega|^d_\infty/|\Omega|)^\delta |\Omega|^{\gamma d}_\infty \|f\|_{L^\alpha_p(\Omega)} \tag{4.102}$$

*when $\alpha > d/p$, $\beta = \alpha - d/p - \gamma d \geqslant 0$ and $\gamma + p > 1$, where $\delta = I\{\alpha > 1\}$. Moreover,*

$$\big\|f - \mathrm{Poly}^\alpha f\big\|_{L^\beta_q(\Omega)} \leqslant C_{d,\alpha,p,\beta,q}(|\Omega|^d_\infty/|\Omega|)^\delta |\Omega|^{\gamma d}_\infty \|f\|_{L^\alpha_p(\Omega)} \tag{4.103}$$

*when $\alpha > \beta \in \mathbb{N}_+$, $1/q = \gamma + 1/p - (\alpha-\beta)/d \in (0, 1/p)$ and $\gamma + p > 1$, where $\delta = I\{\alpha \geqslant 1\}$.*

The above version of Sobolev's embedding theorem immediately implies

$$1/p - \alpha/d < 1/q - \beta/d \quad \Rightarrow \quad W^\alpha_p(\Omega) \subset W^\beta_q(\Omega) \tag{4.104}$$

for $\alpha > \beta$, provided $\beta \in \mathbb{N}_+$ for $q < \infty$. In fact, (4.104) holds without the side constraint $\beta \in \mathbb{N}_+$ for $q < \infty$. The embedding theorem translates higher moments in higher order of smoothness to higher order of the guaranteed uniform smoothness.

We provide a proof of the theorem based on two inequalities. The first one is the Hardy-Littlewood maximal inequality stated in the following lemma.

**Lemma 4.11.** *Let $\|\cdot\|$ be a norm in $\mathbb{R}^d$. For $f \in L_1(\mathbb{R}^d)$ let*

$$\overline{f}_t(x) = |B_{x,t}|^{-1}\textstyle\int_{B_{x,t}}|f(y)|dy, \quad f_{\max}(x) = \sup_{t>0}\overline{f}_t(x), \tag{4.105}$$

*with $B_{x,t} = \{y : \|x-y\| < t/2\}$. Then, for $\|f\|_{L_1} > 0$,*

$$\frac{\|f_{\max}\|_{L_{1,w}}}{\|f\|_{L_1}} \leqslant 2^d, \quad \frac{\|f_{\max}\|_{L_{p,w}}}{\|f\|_{L_{p,w}}} \vee \frac{\|f_{\max}\|_{L_p}}{\|f\|_{L_p}} \leqslant \left(\frac{2^{d+p}}{1-1/p}\right)^{1/p}, \quad p > 1.$$

The proof also use the Hardy–Littlewood–Sobolev inequality which bounds the $L_q$ norm of the convolution

$$\big(I_\alpha f\big)(x) = \int_{\mathbb{R}^d} \frac{f(y)}{\|x-y\|^{d-\alpha}}dy. \tag{4.106}$$

When $\|\cdot\|^{d-\alpha} = \{\pi^{d/2}2^\alpha\Gamma(\alpha/2)/\Gamma(d/2-\alpha/2)\}\|\cdot\|_2^{d-\alpha}$, $I_\alpha$ is the Riesz potential as fractional inverse of the Laplacian, but the choice of the norm does not matter in the proof of Sobolev's embedding.

**Lemma 4.12.** *Let $\alpha > 0$, $1 \leqslant p \leqslant d/\alpha$, $\epsilon \geqslant 0$ and $q \in (1,\infty)$ with $1/q = \epsilon + 1/p - \alpha/d$ and $\epsilon + p > 1$. Let $I_\alpha$ be as in (4.106) with a norm standardized to $|\{x : \|x\| \leqslant t/2\}| = 1$. Suppose* $\mathrm{supp}(f) = \Omega$. *Then, for some constant $C_{d,\alpha,p,\epsilon}$ depending on $(d,\alpha,p,\epsilon)$ only,*

$$\big\|I_\alpha f\big\|_{L_q} \leqslant C_{d,\alpha,p,\epsilon}|\Omega|^\epsilon \|f\|_{L_p}. \tag{4.107}$$

PROOF OF THEOREM 4.14. Throughout the proof we use $C_0, C_1, C_2, \ldots$ to denote constants depending on $(d, \alpha, p, \beta, q, \epsilon)$ only. We divide the proof into 6 cases.

*Case 1:* $1 > \alpha > d/p$. Let $B_{x,t}$ and $\overline{f}_t(x)$ be as in (4.105) with the $\ell_\infty$ norm. By the Hölder inequality

$$\begin{aligned} (4.108) \quad \big|\overline{f}_s(x) - \overline{f}_t(y)\big| &\leqslant \frac{(s/2 + t/2 + \|x-y\|_\infty)^{\alpha+d/p}}{|B_{x,s}| \times |B_{y,t}|} \int_{B_{y,t}} \int_{B_{x,s}} \frac{|f(u) - f(v)|}{\|u-v\|_\infty^{\alpha+d/p}} du\, dv \\ &\leqslant d^{(\alpha+d/p)/2} \|f\|_{L_p^\alpha(\Omega)} (s/2 + t/2 + \|x-y\|_\infty)^{\alpha+d/p}/(st)^{d/p}. \end{aligned}$$

Thus, with $t_k = \|x-y\|_\infty/2^k$, (4.102) follows from

$$\begin{aligned} &\big|f(x) - f(y)\big| \\ \leqslant\ & \big|\overline{f}_{t_0}(x) - \overline{f}_{t_0}(y)\big| + \textstyle\sum_{k=0}^\infty \big|\overline{f}_{t_{k+1}}(x) - \overline{f}_{t_k}(x)\big| + \sum_{k=0}^\infty \big|\overline{f}_{t_{k+1}}(y) - \overline{f}_{t_k}(y)\big| \\ \leqslant\ & C_0 \|x-y\|_\infty^{\alpha-d/p} \|f\|_{L_p^\alpha(\Omega)}. \end{aligned}$$

*Case 2:* $\alpha < 1$, $\alpha \leqslant d/p$ *and* $\beta = 0$. Let $\overline{g}_t(x)$ and $g_{\max}(x)$ be as in (4.105) with

$$g(x) = \Big(\textstyle\int_\Omega |f(x) - f(y)|^p \|x-y\|_\infty^{-\alpha p - d} dy\Big)^{1/p}.$$

By applying the Hölder inequality to the inside integral in (4.108), we have $\big|\overline{f}_{t/2}(x) - \overline{f}_t(x)\big| \leqslant (t/2)^\alpha (3/2)^{\alpha+d/p} \overline{g}_t(x)$, so that

$$\big|f(x) - \overline{f}_t(x)\big| \leqslant t^\alpha C_1 g_{\max}(x).$$

Moreover, because $\gamma + 1/p > (\alpha - \beta)/d = \alpha/d$ and $\overline{f}_t(x) = \overline{f}$ for $t \geqslant 2|\Omega|_\infty$, (4.108) yields

$$\big|\overline{f}_t(x) - \overline{f}\big| \leqslant \textstyle\sum_{k=0}^\infty \big|\overline{f}_{2^k t}(x) - \overline{f}_{2^{k+1}t}(x)\big| \leqslant C_2 t^{\alpha - d/p_\gamma} |\Omega|_\infty^{\gamma d} \|f\|_{L_p^\alpha(\Omega)}$$

with $p_\gamma = 1/(\gamma + 1/p)$. Thus, with $t = \big\{C_2(2|\Omega|_\infty)^{\gamma d}\|f\|_{L_p^\alpha(\Omega)}/(C_1 g_{\max}(x))\big\}^{p_\gamma/d}$

$$\begin{aligned} \big|f(x) - \overline{f}\big| &\leqslant t^\alpha C_1 g_{\max}(x) + t^{\alpha - d/p_\gamma} C_2 (2|\Omega|_\infty)^{\gamma d} \|f\|_{L_p^\alpha(\Omega)} \\ &\leqslant 2(C_1 g_{\max}(x))^{p_\gamma/q} \big\{C_2(2|\Omega|_\infty)^{\gamma d}\|f\|_{L_p^\alpha(\Omega)}\big\}^{1-p_\gamma/q} \end{aligned}$$

due to $1 - p_\gamma \alpha/d = p_\gamma/q$. It follows that

$$\|f - \overline{f}\|_{L_q(\Omega)} \leqslant C_3 \|g_{\max}^{p_\gamma}\|_{L_1(\Omega)}^{1/q} \big(|\Omega|_\infty^{\gamma d}\|f\|_{L_p^\alpha(\Omega)}\big)^{1-p_\gamma/q}.$$

Because $p_\gamma < p$ for $p = 1$, $\|g_{\max}^{p_\gamma}\|_{L_1(\Omega)}^{1/p_\gamma} \leqslant C_4 |\Omega|^{1/p_\gamma - 1/p} \|g\|_{L_p(\Omega)}$ by the Hardy-Littlewood maximal inequality. Therefore, in the case of $\alpha < 1$, (4.103) follows from

$$(4.109) \qquad \|f - \overline{f}\|_{L_q(\Omega)} \leqslant C_5 |\Omega|^{\gamma p_\gamma/q} |\Omega|_\infty^{\gamma d(1-p_\gamma/q)} \|f\|_{L_p^\alpha(\Omega)} \leqslant C_{d,\alpha,p,0,q} |\Omega|_\infty^{\gamma d} \|f\|_{L_p^\alpha(\Omega)}.$$

*Case 3:* $\alpha = 1 > d/p$ *and* $q = \infty$. We use instead of the double integral in (4.108)

$$\overline{f}_s(x) - \overline{f}_t(y) = \sum_{k=1}^d \int_{B_{y,t}} \int_{B_{x,s}} \int_{u_k}^{v_k} \frac{(\partial/\partial w) f(w_{(k)})}{|B_{x,s}| \times |B_{y,t}|} dw\, du\, dv$$

with $w_{(k)} = (u_1, \ldots, u_{k-1}, w, v_{k+1}, \ldots, v_d)^\top$. We only need to use (4.108) for $s = t$ and $s = t/2$ so that the difference does not matter in the proof of (4.102). We omit the details.

*Case 4:* $\alpha > 1 > \alpha' = \alpha - \lfloor\alpha\rfloor > d/p$. Let $\alpha_1 = \lfloor\alpha\rfloor$ and

$$g_m(x) = f^{(m)}(x) - |\Omega|^{-1}\textstyle\int_\Omega f^{(m)}(y)dy,\ m \in \mathbb{N}_+^d,\ \|m\|_1 = \alpha_1 < \alpha.$$

Because $(f - \mathrm{Poly}^\alpha f)^{(m)} = g_m$ when $\|m\|_1 = \alpha_1$, for $1 > \alpha' > d/p$ (4.102) with index $\alpha$ follows from an application of itself to $g_m$ in Case 1 with index $\alpha' < 1$.

*Case 5:* $\alpha \geqslant 1$ *and* $q < \infty$. As (4.102) corresponds to $q = \infty$, we need to prove (4.103) here. Let $\epsilon_1 \geqslant 0$ with $\epsilon_1 + p > 1$ and $p_1$ satisfy $1/p_1 = \epsilon_1 + 1/p - \alpha'/d \in (0,1)$. An application of (4.103) to $g_m$ with index $\alpha'$ yields

$$\textstyle\sum_{\|m\|_1=\alpha_1}\big\|g_m\big\|_{L_{p_1}(\Omega)} \leqslant C_{d,\alpha,p,0,p_1}|\Omega|_\infty^{\epsilon_1 d}\|f\|_{L_p^\alpha(\Omega)}. \tag{4.110}$$

Let $\alpha_2 \in \mathbb{N}_+ \cap [0,\alpha_1]$. It follows from (4.100) that for $j \in N^d$ with $\|j\|_1 = \alpha_2$,

$$(f - \mathrm{Poly}^\alpha f)^{(j)}(x) = f^{(j)} - \textstyle\sum_{k\in\mathbb{N}_+^d,\|k\|_1<\alpha-\alpha_2}\int_\Omega f^{(j+k)}(y)(x-y)^k/(|\Omega|k!)dy.$$

Thus, similar to (4.101),

$$\begin{aligned}
&(f - \mathrm{Poly}^\alpha f)^{(j)}(x)\\
= &\sum_{k\in\mathbb{N}_+^d,\|k\|_1=\alpha_1-\alpha_2}\int_\Omega\bigg\{\int_0^1\frac{g_{j+k}(y+t(x-y))(x-y)^k}{|\Omega|k!(1-t)^{1-\alpha_1}/\alpha_1}dt - \frac{g_{j+k}(y)(x-y)^k}{|\Omega|k!}\bigg\}dy\\
= &\sum_{k\in\mathbb{N}_+^d,\|k\|_1=\alpha_1-\alpha_2}\int_\Omega\frac{g_{j+k}(u)(x-u)^k}{|\Omega|k!}\bigg\{\int_0^1\frac{\alpha_1 I_{\Omega_{x,u}}}{(1-t)^{d+1}}dt - 1\bigg\}du,
\end{aligned}$$

where $u = y + t(x-y)$, $x - u = (1-t)(x-y)$ and $\Omega_{x,u} = \{t : (u-tx)/(1-t) \in \Omega\}$. Recall that $|\Omega|_\infty$ is the $\ell_\infty$ diameter of $\Omega$. Because $\|x-u\|_\infty \leqslant (1-t)|\Omega|_\infty$ for $t \in \Omega_{x,u}$,

$$\big|(f - \mathrm{Poly}^\alpha f)^{(j)}(x)\big| \leqslant \sum_{k\in\mathbb{N}_+^d,\|k\|_1=\alpha_1-\alpha_2}\int_\Omega\frac{|g_{j+k}(u)|\|x-u\|_\infty^{\alpha_1-\alpha_2-d}}{|\Omega|k!/|\Omega|_\infty^d}\bigg(\frac{\alpha_1}{d}+1\bigg)du.$$

Let $p_2$ satisfy $1/p_2 = \epsilon_2 + 1/p_1 - (\alpha_1-\alpha_2)/d > 0$ with some $0 \leqslant \epsilon_2 < (\alpha_1-\alpha_2)/d$. We have

$$\big|(f - \mathrm{Poly}^\alpha f)^{(j)}(x)\big| \leqslant C_0\sum_{k\in\mathbb{N}_+^d,\|k\|_1=\alpha_1-\alpha_2}\int_\Omega\frac{|g_{j+k}(u)|\|x-u\|_\infty^{\alpha_1-\alpha_2-\epsilon_2 d-d}}{|\Omega|k!/|\Omega|_\infty^{d+\epsilon_2 d}}du.$$

Thus, by Lemma 4.12 and (4.110), for all $j \in N^d$ with $\|j\|_1 = \alpha_2 \leqslant \alpha$

$$\big\|(f - \mathrm{Poly}^\alpha f)^{(j)}\big\|_{L_{p_2}(\Omega)} \leqslant \sum_{\|m\|_1=\alpha_1}\frac{C_1\|g_m\|_{L_{p_1}(\Omega)}}{|\Omega|/|\Omega|_\infty^{d+\epsilon_2 d}} \leqslant \frac{C_2\|f\|_{L_p^\alpha(\Omega)}}{|\Omega|/|\Omega|_\infty^{d+\epsilon d}} \tag{4.111}$$

with $\epsilon = \epsilon_1+\epsilon_2$. Moreover, the constraints can be summarized as $1/p_2 = \epsilon + 1/p - (\alpha-\alpha_2)/d \in (0, 1/p)$, $\epsilon \geqslant 0$, and $\epsilon + p > 1$. This gives (4.103) with $\beta = \alpha_2$, $q = p_2$ and $\gamma = \epsilon$. For integer $\alpha \geqslant 1$ with $1/p \geqslant \alpha'/d$, (4.111) follows from a simpler derivation with $g_k = f^{(k)}$. We omit details. This completes the proof of (4.103).

*Case 6:* $\alpha > 1$, $\alpha' \leqslant d/p$ *and* $q = \infty$. In view of Case 4, it suffices to prove (4.102) with $\beta < \alpha_1 = \lfloor\alpha\rfloor$. Let $\alpha_2 = \lfloor\beta\rfloor + 1$. As $\alpha_2 - 1 \leqslant \beta < \alpha_2 \leqslant \alpha_1$, an application of (4.108) in Case 3 ($\alpha = 1$) yields (4.102) in the current case. □

**Lemma 4.11.** *Let $\|\cdot\|$ be a norm in $\mathbb{R}^d$. For $f \in L_1(\mathbb{R}^d)$ let*

$$\overline{f}_t(x) = |B_{x,t}|^{-1}\textstyle\int_{B_{x,t}}|f(y)|dy, \quad f_{\max}(x) = \sup_{t>0} \overline{f}_t(x), \tag{4.105}$$

*with $B_{x,t} = \{y : \|x - y\| < t/2\}$. Then, for $\|f\|_{L_1} > 0$,*

$$\frac{\|f_{\max}\|_{L_{1,w}}}{\|f\|_{L_1}} \leqslant 2^d, \quad \frac{\|f_{\max}\|_{L_{p,w}}}{\|f\|_{L_{p,w}}} \vee \frac{\|f_{\max}\|_{L_p}}{\|f\|_{L_p}} \leqslant \left(\frac{2^{d+p}}{1-1/p}\right)^{1/p}, \quad p > 1.$$

Proof. Assume without loss of generality $f(x)$ has bounded support by the monotonicity of the norms and $|B_{x,t/2}| = 1$ by the scale invariance of $f_{\max}$. Let $t \geqslant 0$ and define

$$S_{t,0} = \left\{(x, s) : \int_{B_{x,s}} |f(y)|dy \geqslant t|B_{x,s}|, s \leqslant M\right\} \subset \mathbb{R}^{d+1}.$$

Let $(x_1, s_1)$ be a member of $S_{t,0}$ with the largest $s$. Given $S_{t,k-1}$ and $(x_k, s_k)$, let

$$\begin{aligned} S_{t,k} &= \{(x, s) \in S_{t,k-1} : x \notin B_{x_k,s_k}\}, \\ (x_{k+1}, s_{k+1}) &= \operatorname*{arg\,max}_{(x,s)\in S_{t,k}} \Big\{s : B_{x,s} \cap B_{x_j,s_j} = \varnothing, 1 \leqslant j \leqslant k\Big\}. \end{aligned}$$

We have $s_k \geqslant s_{k+1} \to 0$ due to the compactness of $S_{t,k} \cap \{(x, s) : s \geqslant s_*\}$ for any $s_* > 0$. For $(x, s) \in \cap_{k=1}^{\infty} S_{t,k}$, there must exist $k$ with $s_k \geqslant s > s_{k+1}$ such that $B_{x,s} \cap B_{x_k,s_k} \neq \varnothing$ and $\|x - x_k\| \leqslant s_k + s \leqslant 2s_k$. It follows that $\{x : f_{\max}(x) > t\} \subseteq \{x : \exists\,(x, s) \in S_{t,0}\} \subseteq \cup_{k=1}^{\infty} \overline{B_{x_k,2s_k}}$. Because $B_{x_k,s_k}$ are disjoint, this yields the conclusion for $p = 1$ from

$$\int_{f_{\max}(x)>t} dx \leqslant \sum_{k=1}^{\infty} |B_{x_k,2s_k}| \leqslant \sum_{k=1}^{\infty} 2^d |B_{x_k,s_k}| \leqslant \frac{2^d}{t} \int_{\cup_{k=1}^{\infty} B_{x_k,s_k}} |f(y)|dy \leqslant \frac{2^d}{t}\|f\|_{L_1}.$$

Let $C_p = 2^p/(1 - 1/p)$. Because $t|B_{x_k,s_k}| \leqslant (t/2)|B_{x_k,s_k}| + \int_{B_{x_k,s_k},|f(y)|>t/2} |f(y)|dy$,

$$t\int_{f_{\max}(x)>t} dx \leqslant 2^{d+1} \int_{|f(y)|>t/2} |f(y)|dy = 2^{d+1} \int_0^{\infty} \int_{|f(y)|>(t/2)\vee s} dyds.$$

Hence, $t^p \int_{f_{\max}(x)>t} dx \leqslant 2^{d+1} t^{p-1} \|f\|_{L_{p.w}}^p \int_0^{\infty} (s \vee (t/2))^{-p} ds = 2^d C_p \|f\|_{L_{p.w}}^p$, and

$$\int_0^{\infty} pt^{p-1} \int_{f_{\max}(x)>t} dxdt \leqslant 2^{d+1} \int_0^{\infty} pt^{p-2} \int_{|f(y)|>t/2} |f(y)|dydt = 2^d C_p \|f\|_{L_p}^p.$$

This completes the proof for $p > 1$ and therefore of the entire lemma. □

**Lemma 4.12.** *Let $\alpha > 0$, $1 \leqslant p \leqslant d/\alpha$, $\epsilon \geqslant 0$ and $q \in (1, \infty)$ with $1/q = \epsilon + 1/p - \alpha/d$ and $\epsilon + p > 1$. Let $I_\alpha$ be as in (4.106) with a norm standardized to $|\{x : \|x\| \leqslant t/2\}| = 1$. Suppose* $\mathrm{supp}(f) = \Omega$. *Then, for some constant $C_{d,\alpha,p,\epsilon}$ depending on $(d, \alpha, p, \epsilon)$ only,*

$$\big\|I_\alpha f\big\|_{L_q} \leqslant C_{d,\alpha,p,\epsilon}|\Omega|^\epsilon \|f\|_{L_p}. \tag{4.107}$$

PROOF. For $\alpha = d$, (4.107) is a consequence of Hölder's inequality with $C_{d,\alpha,p,\epsilon} = 1$. Assume $0 < \alpha < d$ in the sequel. Let $p_\epsilon = 1/(\epsilon + 1/p)$ and $C_1, C_2, \ldots$ denote constants depending on $(d, \alpha, p, \epsilon)$ only. Let $B_{x,t}$ and $f_{\max}$ be as in (4.105). Because $B_{x,t} = \cup_{k=0}^\infty (B_{x,t/2^k} \backslash B_{x,t/2^{k+1}})$,

$$\int_{B_{x,t}} \frac{|f(y)|}{\|x-y\|^{d-\alpha}} dy \leqslant f_{\max}(x) \sum_{k=0}^\infty \frac{|B_{x,t/2^k}|}{(t/2^{k+2})^{d-\alpha}} = t^\alpha C_1 f_{\max}(x).$$

Let $p_\epsilon = 1/(\epsilon + 1/p)$. On the other hand, because $B^c_{x,t} = \cup_{k=0}^\infty (B_{x,2^{k+1}t} \backslash B_{x,2^k t})$,

$$\int_{\Omega \backslash B_{x,t}} \frac{|f(y)|}{\|x-y\|^{d-\alpha}} dy \leqslant \sum_{k=0}^\infty \frac{\big(|\Omega| \wedge (2^{k+1}t)^d\big)^{1-1/p} \|f\|_{L_p}}{(2^{k-1}t)^{d-\alpha}} = t^{\alpha - d/p_\epsilon} C_2 |\Omega|^\epsilon \|f\|_{L_p}$$

due to $1/q = 1/p_\epsilon - \alpha/d \in (0, 1)$. This gives (4.107) by the proof of (4.109). $\square$

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## 5. Information bounds

We discuss here information lower bounds in hypotheses testing and their implications on estimation problems.

5.1. **Distances between probability measures.** We will discuss test-based information bounds involving the total variation (TV) distance, Kullback-Leibler (KL) divergence and related quantities. Let $\mathbb{P}$ and $\mathbb{Q}$ be two probability measures defined on the same sample space. Define $p = d\mathbb{P}/d\nu$ and $q = d\mathbb{Q}/d\nu$ as the Radon–Nikodym derivatives with respect to $\nu$, e.g. with $\nu = \mathbb{P} + \mathbb{Q}$ to guarantee the absolute continuity. The TV distance between $\mathbb{P}$ and $\mathbb{Q}$, which is also called the $L_1$ distance, is defined as

$$d_{\mathrm{TV}}(\mathbb{P}, \mathbb{Q}) = d_{\mathrm{TV}}(p, q) = \big\|p - q\big\|_{L_1(\nu)} = \textstyle\int |p - q| d\nu.$$

The KL divergence between $\mathbb{P}$ and $\mathbb{Q}$ is defined as

$$D_{\mathrm{KL}}(\mathbb{P}\,\|\,\mathbb{Q}) = D_{\mathrm{KL}}(p\,\|\,q) = \textstyle\int_{p>0}\big(\log p - \log q\big) p d\nu.$$

They are closely related to the Hellinger distance

$$d_{\mathrm{H}}(\mathbb{P}, \mathbb{Q}) = d_{\mathrm{H}}(p, q) = \big\|\sqrt{p} - \sqrt{q}\big\|_{L_2(\nu)} = \Big(\textstyle\int\big(\sqrt{p} - \sqrt{q}\big)^2 d\nu\Big)^{1/2},$$

and the chi-squared "distance"

$$d_{\chi^2}(\mathbb{P}, \mathbb{Q}) = d_{\chi^2}(p, q) = \big\|q/p - 1\big\|_{L_2(\mathbb{P})} = \Big(\textstyle\int_{p>0}\big(q/p - 1\big)^2 p d\nu\Big)^{1/2}.$$

The values of the above four quantities do not depend on the choice of $\nu$ as long as $\mathbb{P}+\mathbb{Q} \ll \nu$. The KL divergence is closely related to the Shannon entropy (or simply entropy) and its conditional version for discrete random vectors

$$H(X) = \mathbb{E}\big[-\log p(X)\big], \quad H(X|Y) = \mathbb{E}\big[-\log p_{X|Y}\big],$$

respectively with probability mass functions (pmf) $p(x)$ of $X$ and the conditional pmf $p(x|y)$ of $X$ given $Y$, and the mutual information between any pair of random vectors

$$I(X; Y) = D_{\mathrm{KL}}(\mathbb{P}_{X,Y}\,\|\,\mathbb{P}_X \otimes \mathbb{P}_Y),$$

where $\mathbb{P}_Z$ is the restriction of $\mathbb{P}$ on $\sigma(Z)$.

We review below some basic properties of the above quantities. First we consider conditioning. Recall the familiar decomposition $\mathrm{Var}(X) = \mathrm{Var}(\mathbb{E}[X|Y]) + \mathbb{E}[\mathrm{Var}(X|Y)]$.

**Proposition 5.1.** *Let $p_X$ and $p_{X|Y}$ be the joint and conditional densities under $\mathbb{P}$, and $q_X$ and $q_{X|Y}$ under $\mathbb{Q}$. Let $\mathbb{E}$ be the expectation corresponding to $\mathbb{P}$. Then,*

$$\begin{aligned} D_{\mathrm{KL}}(p_{X,Y}\|q_{X,Y}) &= D_{\mathrm{KL}}(p_Y\|q_Y) + \mathbb{E}\big[D_{\mathrm{KL}}(p_{X|Y}\|q_{X|Y})\big],\\ I(X;Y) &= \mathbb{E}\big[D_{\mathrm{KL}}(p_{X|Y}\|p_X)\big] = \mathbb{E}\big[D_{\mathrm{KL}}(p_{Y|X}\|p_Y)\big],\\ d_{\mathrm{H}}^2(p_{X,Y}, q_{X,Y}) &= d_{\mathrm{H}}^2(p_Y, q_Y) + \mathbb{E}\big[(q_Y/p_Y)^{1/2} d_{\mathrm{H}}^2(p_{X|Y}, q_{X|Y})\big],\\ d_{\chi^2}^2(p_{X,Y}, q_{X,Y}) &= d_{\chi^2}^2(p_Y, q_Y) + \mathbb{E}\big[(q_Y/p_Y)^2 d_{\chi^2}^2(p_{X|Y}, q_{X|Y})\big], \end{aligned}$$

*and for discrete $(X,Y)$,*

$$H(X,Y) = H(Y) + H(X|Y), \quad H(X) = I(X;Y) + H(X|Y),$$

*and $I(X;Y) = H(X) + H(Y) - H(X,Y)$. For any measurable function $f$,*

$$\begin{aligned} D_{\mathrm{KL}}(p_{f(X)}\|q_{f(X)}) &\leqslant D_{\mathrm{KL}}(p_X\|q_X), \quad & d_{\mathrm{TV}}(p_{f(X)}, q_{f(X)}) &\leqslant d_{\mathrm{TV}}(p_X, q_X),\\ d_{\mathrm{H}}(p_{f(X)}, q_{f(X)}) &\leqslant d_{\mathrm{H}}(p_X, q_X), \quad & d_{\chi^2}(p_{f(X)}, q_{f(X)}) &\leqslant d_{\chi^2}(p_X, q_X). \end{aligned}$$

Next we consider convexity, sub-additivity and decompositions under independence.

**Proposition 5.2.** (i) *The KL divergence is convex,*

$$D_{\mathrm{KL}}(\lambda\mathbb{P}_1 + (1-\lambda)\mathbb{P}_2 \,\|\, \lambda\mathbb{Q}_1 + (1-\lambda)\mathbb{Q}_2) \leqslant \lambda D_{\mathrm{KL}}(\mathbb{P}_1 \,\|\, \mathbb{Q}_1) + (1-\lambda) D_{\mathrm{KL}}(\mathbb{P}_2 \,\|\, \mathbb{Q}_2).$$

*The entropy is concave in the underlying pmf: if $X_j \sim p_j$ and $X \sim \lambda p_1 + (1-\lambda)p_2$, then*

$$H(X) \geqslant \lambda H(X_1) + (1-\lambda) H(X_2).$$

(ii) *For any discrete random vector $X_{1:n} = (X_1, \ldots, X_n)^\top$,*

$$H(X_{1:n}) \leqslant \textstyle\sum_{i=1}^n H(X_i)$$

*and it holds with equality when $X_i$ are independent. Moreover,*

$$H(X_{1:n}) \leqslant (n-1)^{-1} \textstyle\sum_{i=1}^n H\big(X_{(1:n)\setminus\{i\}}\big).$$

(iii) *Suppose $X_{1:n}$ has independent components under both $\mathbb{P}$ and $\mathbb{Q}$. Then,*

$$\begin{aligned} D_{\mathrm{KL}}(p_{X_{1:n}}\|q_{X_{1:n}}) &= \textstyle\sum_{i=1}^n D_{\mathrm{KL}}(p_{X_i}\|q_{X_i}),\\ d_{\mathrm{H}}^2(p_{X_{1:n}}, q_{X_{1:n}}) &= 2\big(1 - \textstyle\prod_{i=1}^n \big(1 - d_{\mathrm{H}}^2(p_{X_i}, q_{X_i})/2\big)\big) \leqslant \textstyle\sum_{i=1}^n d_{\mathrm{H}}^2(p_{X_i}, q_{X_i}), \end{aligned}$$

*and when $q_{X_i} \ll p_{X_i}$*

$$d_{\chi^2}^2(p_{X_{1:n}}, q_{X_{1:n}}) = \textstyle\prod_{i=1}^n \big(1 + d_{\chi^2}^2(p_{X_i}, q_{X_i})\big) - 1 \geqslant \textstyle\sum_{i=1}^n d_{\chi^2}^2(p_{X_i}, q_{X_i}),$$

The second inequality in Proposition 5.2 (ii) is called Han's inequality. One may find that Han's inequality and the Efron-Stein inequality resemble each other.

Proof. Due to $H(X|Y) \leqslant I(X;Y) + H(X|Y) = H(X)$, Han's inequality follows from

$$nH(X_{1:n}) - \textstyle\sum_{i=1}^n H(X_{(1:n)\setminus\{i\}}) = \textstyle\sum_{i=1}^n H(X_i|X_{(1:n)\setminus\{i\}}) \leqslant \textstyle\sum_{i=1}^n H(X_i|X_{1:(i-1)}).$$

and $\sum_{i=1}^n H(X_i|X_{1:(i-1)}) = H(X_{1:n})$. $\square$

Finally we review some relationships among distances between probability measures.

**Proposition 5.3.** *The following holds for all* $\mathbb{P}$ *and* $\mathbb{Q}$ *defined in the same sample space,*

$$\begin{aligned} d_{\mathrm{H}}^2(\mathbb{P},\mathbb{Q}) &\leqslant d_{\mathrm{TV}}(\mathbb{P},\mathbb{Q}), \qquad & d_{\mathrm{TV}}(\mathbb{P},\mathbb{Q}) &\leqslant d_{\mathrm{H}}(\mathbb{P},\mathbb{Q})\sqrt{4-d_{\mathrm{H}}^2(\mathbb{P},\mathbb{Q})}, \\ d_{\mathrm{H}}^2(\mathbb{P},\mathbb{Q}) &\leqslant D_{\mathrm{KL}}(\mathbb{P}\,\|\,\mathbb{Q}), \qquad & 2^{-1}d_{\mathrm{TV}}^2(\mathbb{P},\mathbb{Q}) &\leqslant D_{\mathrm{KL}}(\mathbb{P}\,\|\,\mathbb{Q}). \end{aligned}$$

*Moreover, for* $\mathbb{Q} \ll \mathbb{P}$*,* $\max\{d_{\mathrm{H}}(\mathbb{P},\mathbb{Q}), d_{\mathrm{TV}}(\mathbb{P},\mathbb{Q})\} \leqslant d_{\chi^2}(\mathbb{P},\mathbb{Q})$.

The fourth inequality in Proposition 5.3, called Pinsker's inequality, is sharper than the direct combination of the second and third when $d_{\mathrm{H}}^2(\mathbb{P},\mathbb{Q}) \leqslant 2$.

Proof. By Cauchy-Schwarz and Jensen,

$$\frac{1}{2}\left(\int |p-q|d\nu\right)^2 \leqslant \int_0^1 \int \big(q+t(p-q)\big)d\nu \int \frac{(p-q)^2(1-t)}{q+t(p-q)}d\nu dt = \int p\log(p/q)d\nu.$$

This gives Pinsker's inequality. $\square$

5.2. **Bayes lower bounds.** Given a decision problem with unknown $\mathbb{P} \in \mathscr{P}$, observation $X \sim \mathbb{P}$, and loss function $\rho(a, \mathbb{P})$, the minimax risk is bounded from below by the risk of Bayes estimators of any easier problem under any prior $G$,

$$\inf_{a(\cdot)} \sup_{\mathbb{P}\in\mathscr{P}} \mathbb{E}\big[\rho(a(X), \mathbb{P})\big] \geqslant \inf_{b(\cdot)} \int \mathbb{E}_\theta\big[\rho_*(b(X), \theta)\big] G(d\theta), \tag{5.1}$$

for any sub-model $\{\mathbb{P}_\theta, \theta \in \Theta\} \subset \mathscr{P}$ with parameter space $\Theta$, any prior distribution $G$ in $\Theta$ and any loss function $\rho_*$ satisfying

$$\inf_a \int \rho(a, \mathbb{P}_\theta) G(d\theta|x) \geqslant \inf_b \int \rho_*(b, \theta) G(d\theta|x). \tag{5.2}$$

The minimax theory, e.g. Ferguson (1967), provides conditions for the equality to hold in (5.1) with the least favorable prior $G$ when $\{\mathbb{P}_\theta, \theta \in \Theta\} = \mathscr{P}$ and $\rho_*(a, \theta) = \rho(a, \mathbb{P}_\theta)$. However, in more complicated problems, we often settle with rate minimaxity, with lower and upper bounds of the same order. Condition (5.2) holds if

$$\inf_a \sup_b \inf_\theta \Big(\rho\big(a, \mathbb{P}_\theta\big) - \rho_*\big(b, \theta\big)\Big) \geqslant 0. \tag{5.3}$$

**High-dimensional linear regression.** Consider a high-dimensional linear model

$$Y_{1:n} = X_{1:n,1:p}\beta_{1:p} + \varepsilon_{1:n} \tag{5.4}$$

under $\mathbb{P}_\beta$ with observations $Y = Y_{1:n}$ and $X = X_{1:n,1:p}$, large number of covariates, $p \gg n$, sparse $\beta = \beta_{1:p}$, $\|\beta\|_0 = s < n$, and Gaussian noise $\varepsilon = \varepsilon_{1:n} \sim N(0, \sigma^2 I_{n\times n})$ given $X$. We say that the design $X$ is normalized when $\|X_{1:n,j}\|_2^2 = n$ for all $j$ for deterministic designs or $\mathbb{E}[\|X_{1:n,j}\|_2^2] = n$ for all $j$ for random designs. Here we use Bayes methods to derive lower bounds for the minimax risk in the estimation of $\beta$ and prediction (estimation of $X\beta$).

We first consider a general compound loss function, defined as a sum of losses for decisions about individual $\beta_j$. The following theorem asserts that when $\beta_j$ are iid random variables a priori, the regression problem is always more difficult than the Gaussian sequence model for any compound loss on $\beta$.

**Lemma 5.1.** *Let $\mathscr{D}_{\mathrm{reg}}$ be the set of all estimators based on data $(X, Y)$ in the linear regression model* (5.4) *with normalized deterministic design. Let $\mathscr{D}_{\mathrm{seq}}$ be the set of all estimators based on independent observations $Z_j \sim N(\beta_j, \sigma^2/n), j \leqslant p$, in the Gaussian sequence model. Assume that $\sigma$ is known. Let $G$ be a prior under which $\beta_j$ are independent random variables. Then, for any given nonnegative loss function $\rho_0$,*

$$\inf_{\widehat\beta\in\mathscr{D}_{\mathrm{reg}}} \mathbb{E}_G\bigg[\sum_{j=1}^p \rho_0(\widehat\beta_j, \beta_j)\bigg] \geqslant \inf_{\widehat\beta\in\mathscr{D}_{\mathrm{seq}}} \mathbb{E}_G\bigg[\sum_{j=1}^p \rho_0(\widehat\beta_j, \beta_j)\bigg].$$

Proof. Let $X_j = X_{1:n,j}$ and $\widetilde Z_j = X_j^T(Y - \sum_{k\neq j} X_k\beta_k)/n$. Because $\varepsilon \sim N(0, \sigma^2 I_{n\times n})$, $\widetilde Z_j \sim N(\beta_j, \sigma^2/n)$ is sufficient for $\beta_j$ in the linear regression model when all other parameters, $\beta_k, k \neq j$, are known. It follows that

$$\inf_{\widehat\beta\in\mathscr{D}_{\mathrm{reg}}} \mathbb{E}_G\bigg[\sum_{j=1}^p \rho_0(\widehat\beta_j, \beta_j)\bigg] \quad = \quad \sum_{j=1}^p \mathbb{E}_G\bigg[\inf_{\widehat\beta\in\mathscr{D}_{\mathrm{reg}}} \mathbb{E}_G\Big[\rho_0(\widehat\beta_j, \beta_j)\Big| X, Y\Big]\bigg]$$

$$
\begin{aligned}
&\geqslant \sum_{j=1}^{p} \mathbb{E}_G\Big[\inf_{\widehat\beta\in\mathscr{D}_{\rm reg}} \mathbb{E}_G\Big[\rho_0(\widehat\beta_j,\beta_j)\Big|X,Y,\beta_k,k\neq j\Big]\Big] \\
&= \sum_{j=1}^{p} \mathbb{E}_G\Big[\inf_{\widehat\beta\in\mathscr{D}_{\rm reg}} \mathbb{E}_G\Big[\rho_0(\widehat\beta_j,\beta_j)\Big|\widetilde Z_j\Big]\Big] \\
&= \sum_{j=1}^{p} \mathbb{E}_G\Big[\inf_{\widehat\beta\in\mathscr{D}_{\rm seq}} \mathbb{E}_G\Big[\rho_0(\widehat\beta_j,\beta_j)\Big|Z_j\Big]\Big].
\end{aligned}
$$

The proof is complete due to the sufficiency of $Z_j$ for $\beta_j$ and the independence of $(Z_j, \beta_j)$ in the Gaussian sequence model. $\square$

**Lemma 5.2.** *Let $Z|\theta \sim N(\theta,\sigma_n^2)$ with known $\sigma_n > 0$ and $G_0$ be the prior under which $\mathbb{P}_{G_0}\{\theta\neq 0\} = \mathbb{P}_{G_0}\{\theta = \mu_0\sigma_n\} = \pi_0$ where $\mu_0 = \sqrt{2\log(1/\pi_0)} - \sqrt{2\log(1/\epsilon_0)}$ for some $0<\pi_0<\epsilon_0\leqslant 1-\pi_0$. Then, for $1\leqslant q<\infty$,*

$$
\inf_{\widehat\theta(Z)} \mathbb{E}_{G_0}\big[\big|\widehat\theta-\theta\big|^q\big] \geqslant (1-\epsilon_0/2)(1-\epsilon_1)\pi_0(\sigma_n\mu_0)^q,
$$

*where $\epsilon_1 = 1-1/[\{\epsilon_0/(1-\pi_0)\}^{1/(q-1)}+1]^{q-1}$ for $q>1$ and $\epsilon_1 = 0$ for $q=1$.*

Proof. Let $\delta_1 = I\{\theta\neq 0\}$, $\pi_1$ the posterior mean of $\delta_1$, and $\omega_1 = \pi_1/(1-\pi_1)$ the posterior odds for $\delta_1$. The minimum posterior $\ell_q$ risk for the estimation of $\delta_1$ is

$$
R_q(\pi_1) = \Big(\pi_1\wedge(1-\pi_1)\Big)I_{\{q=1\}} + \Big(\pi_1/(\omega_1^{1/(q-1)}+1)^{q-1}\Big)I_{\{q>1\}}.
$$

This can be easily seen from $R_q(\pi_1) = \min_{0\leqslant x\leqslant 1}\big\{\pi_1(1-x)^q + (1-\pi_1)x^q\big\}$ as the minimum is attained when $x = I\{\pi_1>1/2\}$ for $q=1$ and $x/(1-x) = \omega_1^{1/(q-1)}$ for $q>1$. Thus,

$$
\inf_{\widehat\theta(Z)} \mathbb{E}_{G_0}\big[\big|\widehat\theta-\theta\big|^q\big] = (\sigma_n\mu_0)^q \inf_{\widehat\delta_1(Z)} \mathbb{E}_{G_0}\big[\big|\widehat\delta_1-\delta_1\big|^q\big] = (\sigma_n\mu_0)^q\mathbb{E}_{G_0}\big[R_q(\pi_1)\big].
$$

Given $Z/\sigma_n = z$, $\pi_1(z) = \pi_0\varphi(z-\mu_0)/\big(\pi_0\varphi(z-\mu_0)+(1-\pi_0)\varphi(z)\big)$ with $\varphi(x) = e^{-t^2/2}/\sqrt{2\pi}$. Let $a_0 = \sqrt{2\log(1/\epsilon_0)} = \sqrt{2\log(1/\pi_0)}-\mu_0$ and $\pi_1^* = \epsilon_0/(1-\pi_0)$. For $z\leqslant \mu_0+a_0$,

$$
\omega_1(z) = \frac{\pi_0\varphi(z-\mu_0)}{(1-\pi_0)\varphi(z)} \leqslant \frac{\pi_0\varphi(a_0)}{(1-\pi_0)\varphi(\mu_0+a_0)} = \frac{\pi_0 e^{-a_0^2/2+(a_0+\mu_0)^2/2}}{1-\pi_0} = \frac{\epsilon_0}{1-\pi_0}.
$$

As $\epsilon_0/(1-\pi_0)\leqslant 1$, $1/\{\omega_1^{1/(q-1)}(z)+1\}^{q-1} \geqslant 1/\{(\epsilon_0/(1-\pi_0))^{1/(q-1)}+1\}^{q-1} = 1-\epsilon_1$. Thus,

$$
\mathbb{E}_{G_0}\big[R_q(\pi_1)\big] \geqslant (1-\epsilon_1)\textstyle\int_{-\infty}^{\mu_0+a_0}\pi_0\varphi(z-\mu_0)dz \geqslant (1-\epsilon_1)\pi_0(1-\epsilon_0/2),
$$

due to $\mathbb{P}\{N(\mu_0,1)>\mu_0+a_0\} = \mathbb{P}\{N(0,1)>a_0\}\leqslant e^{-a_0^2/2}/2 = \epsilon_0/2$. $\square$

Let $G_0$ be the prior under which $\mathbb{P}_{G_0}\{\beta_j\neq 0\} = \mathbb{P}_{G_0}\{\beta_j = \mu_0\sigma_n\} = \pi_0$ with $\sigma_n = \sigma/\sqrt{n}$ and $\pi_0 = \|\beta\|_0/p$. It follows from Lemmas 5.1 and 5.2 that

$$
\inf_{\widehat\beta}\mathbb{E}_{G_0}\big[\|\widehat\beta-\beta\|_q^q\big] \geqslant (1-\epsilon_0/2)(1-\epsilon_1)(\pi_0 p)(\sigma_n\mu_0)^q.
$$

However, lower bound for the prediction risk $E_\beta[\|X\widehat\beta - X\beta\|_2^2]$ is still unclear for non-orthonormal designs. Moreover, the minimax risk under the usual sparsity assumption is unclear as $\|\beta\|_0$ can be as large as $p$ under $\mathbb{P}_{G_0}$. The next theorem addresses these issues by finding a least favorable configuration of sparse $\beta$ in an intersection of $\ell_0$ and $\ell_\infty$ balls:

$$\Theta_{n,p,s^*} = \{\boldsymbol{v} \in \mathbb{R}^p : \|\boldsymbol{v}\|_0 \leqslant s^*, \|\boldsymbol{v}\|_\infty \leqslant \lambda_{mm}\}, \quad \lambda_{mm} = \sigma\sqrt{(2/n)\log(p/s^*)}.$$

For nonnegative-definite matrices $\Sigma$, define the lower sparse eigenvalue as

$$\phi_-(t;\Sigma) = \min_{|A|<t+1} \phi_{\min}(\Sigma_{A,A}). \tag{5.5}$$

**Theorem 5.1.** *Let $\mathbb{P}_\beta$ be the probability measure in the linear model* (5.4) *with normalized deterministic design $X$ and $p \gg s^* \to \infty$. Then, for $0 < \epsilon < 1 \leqslant q < \infty$,*

$$\inf_{\widehat{\beta}} \sup_{\beta \in \Theta_{n,p,s^*}} \mathbb{E}_\beta\big[\|\widehat{\beta} - \beta\|_q^q\big] \geqslant (1 + o(1))s^*\lambda_{mm}^q,$$

*and*

$$\inf_{\widehat{\beta}} \sup_{\beta \in \Theta_{n,p,s^*}} \mathbb{P}_\beta\Big\{\|\widehat{\beta} - \beta\|_q^q \geqslant (1-\epsilon)^q s^*\lambda_{mm}^q\Big\} \geqslant \frac{1 - (1-\epsilon)^q + o(1)}{(3-\epsilon)^q - (1-\epsilon)^q}.$$

*Moreover, for the prediction error*

$$\inf_{\widehat{\beta}} \sup_{\beta \in \Theta_{n,p,s^*}} \mathbb{E}_\beta\big[\|X\widehat{\beta} - X\beta\|_2^2/n\big] \geqslant (1 + o(1))s^*\lambda_{mm}^q\phi_-(2s^*;\overline{\Sigma})/4,$$

*and*

$$\inf_{\widehat{\beta}} \sup_{\beta \in \Theta_{n,p,s^*}} \mathbb{P}_\beta\Big\{\|X\widehat{\beta} - X\beta\|_2^2/n \geqslant \frac{(1-\epsilon)^2 s^*\lambda_{mm}^2}{4/\phi_-(2s^*;\overline{\Sigma})}\Big\} \geqslant \epsilon/4 + o(1),$$

*where $\phi_-(\cdot;\cdot)$ is the lower sparse eigenvalue in (5.5) and $\overline{\Sigma} = X^T X/n$.*

Theorem 5.1 provides $s^*\sigma^2(2/n)\log(p/s^*)$ as a lower bound for the minimax prediction and $\ell_2$ estimation errors. In the Gaussian sequence model ($p = n$ and $X^\top X/n = I_{n\times n}$), the minimax risk for the $\ell_q$ loss in $\ell_r$ balls (Donoho and Johnstone, 1994, 1998) is approximately

$$\inf_{\widehat{\beta}} \sup_{\|\beta\|_r \leqslant C_r} \mathbb{E}_\beta\big[\|\widehat{\beta} - \beta\|_q^q\big] = (1 + o(1))C_r^r\lambda_{mm}^{q-r}$$

for $0 < r \vee 1 \leqslant q$, $\sigma_n = \sigma/\sqrt{n}$ and $\lambda_{mm} = \sigma_n\{2\log(p(\sigma_n/C_r)^r)\}^{1/2}$ provided that $\min\{\lambda_{mm}/\sigma_n, (C_r/\lambda_{mm})^r\} \to \infty$. As $\Theta_{n,p,s^*} \subset \{\beta : \|\beta\|_r \leqslant C_r\}$ with $s^* = (C/\lambda_{mm})^r$, Theorem 5.1 can be viewed as an extension of their result to linear regression (Ye and Zhang, 2010).

Proof. Let $\{\pi_0, \epsilon_0, \epsilon_1, \mu_0\}$ be as in Lemma 5.2 with $\pi_0 \vee \epsilon_0 \vee \epsilon_1 \to 0$. Let $\mathbb{P}_{G_0}$ be the probability under which $\beta_j$ are iid with $\pi_0 = -\mathbb{P}_{G_0}\{\beta_j = \mu\} = 1 - \mathbb{P}_{G_0}\{\beta_j = 0\}$, where $\mu = \mu_0\sigma/\sqrt{n}$. Let $s^* = (1+\epsilon_0)\pi_0 p$ and $k = \pi_0 p$. For $1/\epsilon_0 \geqslant 1+\epsilon_0$,

$$\sqrt{n/2}(\lambda_{mm} - \mu)/\sigma = \sqrt{\log(p/s^*)} + \sqrt{\log(1/\epsilon_0)} - \sqrt{\log(p/s^*) + \log(1+\epsilon_0)} > 0.$$

Let $N = \|\beta\|_0$. Under $\mathbb{P}_{G_0}$ and the $\ell_q$ loss, the Bayes rule given $N \leqslant s^*$ is

$$\delta^* = \delta^*(X, Y, s^*) = \arg\min_b \mathbb{E}_{G_0}\big[\|\beta - b\|_q^q \big| X, Y, N \leqslant s^*\big].$$

We note that $s^* = (1+\epsilon_0)k$ and $\|\beta\|_q^q = N\mu^q$. As the risk of $\delta^*$ is no smaller than the Bayes rule without conditioning, Lemmas 5.1 and 5.2 imply

$$(1-\epsilon_0/2)(1-\epsilon_1)k\mu^q \leqslant \mathbb{E}_{G_0}\big[\|\delta^* - \beta\|_q^q \big| N \leqslant s^*\big] + \mathbb{E}_{G_0}\big[\|\delta^* - \beta\|_q^q I\{N > s^*\}\big].$$

As $\|\beta\|_q^q \leqslant s^*\mu^q$ when $N \leqslant s^*$, $\|\delta^*\|_q^q \leqslant s^*\mu^q$, so that $\|\delta^* - \beta\|_q^q \leqslant 2^{q-1}\mu^q(s^* + N)$. Thus,

$$\mathbb{E}_{G_0}\big[\|\delta^* - \beta\|_q^q I\{N > s^*\}\big] \leqslant 2^{q-1}\mu^q \mathbb{E}_{G_0}\big[\big(s^* + N\big) I\{N > (1+\epsilon_0)k\}\big].$$

Because $N \sim$binomial$(p, \pi_0)$ with $\mathbb{E}_{G_0}[N] = k \to \infty$, the right-hand side above is of smaller order than $k = s^*/(1+\epsilon_0)$. Thus, letting $\epsilon_0 \to 0+$ and $\epsilon_1 \to 0+$ slowly, we find that

$$\mathbb{E}_{G_0}\big[\|\delta^* - \beta\|_q^q \big| N \leqslant s^*\big] \geqslant (1+o(1))k\mu^q = (1+o(1))s^*\lambda_{mm}^q.$$

This yields the lower bound for the $\ell_q$ minimax risk.

Consider the loss function $\rho_*(\widehat{\beta}, \beta) = I\{\|\widehat{\beta} - \beta\|_q \geqslant c(s^*)^{1/q}\lambda_{mm}\}$ for a given $\widehat{\beta}$. Define $\widetilde{\beta} = \widehat{\beta} I\big\{\|\widehat{\beta}\|_q \leqslant (1+c)(s^*)^{1/q}\lambda_{mm}\big\}$. For $N \leqslant s^*$, we have $\|\beta\|_q^q \leqslant s^*\mu^q \leqslant s^*\lambda_{mm}^q$, so that

$$\|\widetilde{\beta} - \beta\|_q^q \leqslant c^q s^* \lambda_{mm}^q \big(1 - \rho_*(\widehat{\beta}, \beta)\big) + (2+c)^q s^* \lambda_{mm}^q \rho_*(\widehat{\beta}, \beta).$$

Because $\|\beta\|_q^q = N\mu^q \leqslant N\lambda_{mm}^q$, it follows that

$$\begin{aligned}\mathbb{E}_{G_0}\big[\|\widetilde{\beta} - \beta\|_q^q\big] \quad &\leqslant \quad c^q s^* \lambda_{mm}^q + \{(2+c)^q - c^q\} s^* \lambda_{mm}^q \mathbb{E}_{G_0}\big[\rho_*(\widehat{\beta}, \beta) I\{N \leqslant s^*\}\big] \\ &\quad +2^{q-1} s^* \lambda_{mm}^q \mathbb{E}_{G_0}\big[\big(N/s^* + (1+c)^q\big) I\{N > (1+\epsilon_0)k\}\big].\end{aligned}$$

Because $\mathbb{E}_{G_0}\big[\big(N/s^* + (1+c)^q\big) I\{N > (1+\epsilon_0)k\}\big] \to 0$,

$$\{(2+c)^q - c^q\} \sup_{\beta \in \Theta_{n,p,s^*}} \mathbb{E}_\beta\big[\rho_*(\widehat{\beta}, \beta)\big] \geqslant \mathbb{E}_{G_0}\big[\|\widetilde{\beta} - \beta\|_q^q\big]/(s^*\lambda_{mm}^q) - c^q + o(1) \geqslant 1 - c^q + o(1).$$

This gives the lower bound for $\sup_{\beta \in \Theta_{n,p,s^*}} \mathbb{E}_\beta\big[\rho_*(\widehat{\beta}, \beta)\big]$ with $c = 1 - \epsilon$.

Now consider the prediction performance of a given $\widehat{\beta}$. Define

$$\widetilde{\beta} = \arg\min_b \Big\{\big\|X\widehat{\beta} - Xb\big\|_2 : b_j \neq 0 \Rightarrow b_j = \mu, \|b\|_0 \leqslant s^*\Big\}.$$

For $\|\beta\|_0 \leqslant s^*$, we have $\|\widetilde{\beta} - \beta\|_0 \leqslant 2s^*$, so that $\phi_-(2s^*; \overline{\Sigma})\|\widetilde{\beta} - \beta\|_2^2 \leqslant \big\|X\widetilde{\beta} - X\beta\big\|_2^2/n$ by the definition of the sparse eigenvalue. Since $\widetilde{\beta}$ is the minimum distance estimator in the prediction loss based on $\widehat{\beta}$, $\big\|X\widetilde{\beta} - X\beta\big\|_2 \leqslant 2\big\|X\widehat{\beta} - X\beta\big\|_2$. Consequently,

$$\frac{4\big\|X\widehat{\beta} - X\beta\big\|_2^2}{\phi_-(2s^*; \overline{\Sigma})n} \geqslant \|\widetilde{\beta} - \beta\|_2^2,$$

and the lower bound for the expected prediction regret follows from that of the $\ell_2$ estimation error. $\square$

5.3. **Le Cam's method.** For testing simple hypotheses $\mathbb{P}_j : X \sim p_j$ with joint densities $p_j$ with respect to $\nu$, $j = 0, 1$, the minimum Bayes risk with equal prior probability on $\mathbb{P}_j$ is

$$\inf_{0\leqslant\phi\leqslant 1}\big(2^{-1}\mathbb{E}_0[\phi] + 2^{-1}\mathbb{E}_1[1-\phi]\big) = 2^{-1}\int (p_0 \wedge p_1)d\nu = 1/2 - d_{\mathrm{TV}}(p_0, p_1)/4. \tag{5.6}$$

The quantity $\int (p_0 \wedge p_1)d\nu$ which measures the closeness between the two measures, is called affinity. It is closely related to the Hellinger affinity

$$\int \sqrt{p_0 p_1}\, d\nu = 1 - d_{\mathrm{H}}^2(\mathbb{P}_0, \mathbb{P}_1)/2.$$

Le Cam's method uses the testing problem in (5.6) on the right-hand side of (5.1).

**Proposition 5.4** (Le Cam's method)**.** *Suppose $X \sim \mathbb{P}$ with unknown $\mathbb{P} \in \mathscr{P}$. Consider a decision problem with loss function $\rho(a, \mathbb{P})$ and testing hypotheses $H_j : \theta \in \Theta_j$ in a sub-model $\{\mathbb{P}_\theta, \theta \in \Theta\} \subset \mathscr{P}$ with $\Theta_0 \subset \Theta$ and $\Theta_1 = \Theta\backslash\Theta_0$. Suppose $X \sim p_\theta$ under $\mathbb{P}_\theta$. Let*

$$\delta_* = \inf_a \max\Big(\inf_{\theta\in\Theta_0}\rho(a, \mathbb{P}_\theta), \inf_{\theta\in\Theta_1}\rho(a, \mathbb{P}_\theta)\Big).$$

*Let $G$ be a prior with $G(\Theta_0) = G(\Theta_1) = 1/2$ and $\theta|(\theta \in \Theta_j) \sim G_j$, $j = 0, 1$. Then,*

$$\inf_{a(\cdot)}\sup_{\mathbb{P}\in\mathscr{P}}\mathbb{E}\big[\rho(a(X), \mathbb{P})\big] \geqslant (\delta_*/2)\big(1 - d_{\mathrm{TV}}(p_{G_0}, p_{G_1})/2\big) \quad \text{with} \quad p_{G_j} = \int p_\theta dG_j. \tag{5.7}$$

PROOF. Let $\rho_*(b, \theta) = \delta_* b I\{\theta \in \Theta_0\} + \delta_*(1-b)I\{\theta \in \Theta_1\}$. Let $b = I\{\inf_{\theta\in\Theta_1}\rho(a, \mathbb{P}_\theta)\big) \leqslant \delta_*\}$. We have $\rho(a, \mathbb{P}_\theta) \geqslant \delta_* b$ when $\theta \in \Theta_0$ and $\rho(a, \mathbb{P}_\theta) \geqslant \delta_*(1-b)$ when $\theta \in \Theta_1$, so that (5.3) holds and consequently (5.1) holds. As $X \sim p_{G_j}$ under $H_j$ and the prior, (5.6) provides the lower bound in (5.7). $\square$

**Estimation of quadratic functionals.** Let $p(\cdot)$ be an unknown joint density in $\mathbb{R}^d$ with respect to the Lebesgue measure. We consider here kernel estimation of its squared $L_2$ norm $\tau = \|p\|_{L_2}^2 = \int p^2(x)dx$ based on iid vectors $X_1, \ldots, X_n$ from $p$.

We assume that $p$ belongs to the Sobolev space $W_2^\alpha$ with smoothness index $\alpha > 0$. Recall that when $\alpha$ is an integer, the Sobolev norm of an $\alpha$-times differentiable function $f$ on $\Omega$ is

$$\|f\|_{W_q^\alpha(\Omega)} = \textstyle\sum_{m=0}^\alpha \|f\|_{L_q^m(\Omega)} \ \text{ with } \ \|f\|_{L_q^m(\Omega)} = \sum_{\|k\|_1=m}\|f^{(k)}\|_{L_q(\Omega)},$$

where $k = (k_1, \ldots, k_d)^\top$, $f^{(k)}(x) = \prod_{j=1}^d (\partial/\partial x_j)^{k_j} f(x)$ and $\|f\|_{L_q(\Omega)} = \big(\int_\Omega |f|^q dx\big)^{1/q}$. For non-integer $\alpha$, the Sobolev norm is $\|f\|_{W_q^\alpha(\Omega)} = \sum_{m=0}^{\lfloor\alpha\rfloor}\|f\|_{L_q^m(\Omega)} + \|f\|_{L_q^\alpha(\Omega)}$ with

$$\|f\|_{L_q^\alpha(\Omega)} = \sum_{\|k\|_1=\lfloor\alpha\rfloor}\left(\int_\Omega\int_\Omega \frac{|f^{(k)}(x) - f^{(k)}(y)|^q}{\|x-y\|_2^{q(\alpha-\lfloor\alpha\rfloor)+d}}dxdy\right)^{1/q}.$$

Under the norm $\|f\|_{W_q^\alpha(\Omega)}$, the Sobolev space $W_q^\alpha(\Omega)$ is the completion of the set of all $\lceil\alpha\rceil$-times continuously differentiable functions with compact support. In general $1 \leqslant q \leqslant \infty$. For $2 > d/\alpha$, $W_2^\alpha$ is an RKHS. Here we take $q = 2$ and $\Omega = \mathbb{R}^d$, and omit $\Omega$ in the notation.

Let $K(x)$ be a kernel function supported in the unit ball in $\mathbb{R}^d$ and satisfying

$$K(x) = K(-x),\ \textstyle\int K(x)dx = 1,\ \int K(x)x^k dx = 0,\ 1 \leqslant \|k\|_1 \leqslant \alpha,\ \|K\|_{L_\infty} < \infty. \tag{5.8}$$

For $\sigma > 0$, define the unit variate kernel $K_\sigma(x) = K(x/\sigma)/\sigma^d$ and the bivariate

$$K_{2,\sigma}(x,y) = K_{2,\sigma}(x-y) = 2K_\sigma(x-y) - \textstyle\int K_\sigma(t-x)K_\sigma(t-y)dt.$$

Define a kernel estimate of $\tau = \|p\|_{L_2}^2$ by the U-statistic generated by the bivariate kernel,

$$\widehat\tau_n = \{n(n-1)/2\}^{-1}\textstyle\sum_{1\leqslant i<j\leqslant n} K_{2,\sigma}(X_i, X_j).$$

Let $b_\sigma(x) = \mathbb{E}[K_\sigma(x - X_1)] - p(x)$ be the bias of kernel $K_\sigma$. The idea is to use a linear combination to cancel the biases as the bias of the integral kernel is roughly twice $b_\sigma$:

$$\mathbb{E}[K_{2,\sigma}(X_1,X_2)] = 2\textstyle\int p(x)\{p(x) - b_\sigma(x)\}dx - \int\{p(t) - b_\sigma(t)\}^2 dt = \tau - \|b_\sigma\|_{L_2}^2.$$

**Theorem 5.2.** *Let $\widehat\tau_n$ be the above kernel estimator of $\tau = \|p\|_{L_2}^2 = \int p^2(x)dx$. Let $b_\sigma(x) = \int K_\sigma(x-y)p(y)dy - p(x)$ be the bias of the univariate kernel $K_\sigma$. Then,*

$$\begin{aligned}&\mathbb{E}\Big[\big(\widehat\tau_n - n^{-1}\textstyle\sum_{i=1}^n\big(2p(X_i) - \tau\big)\big)^2\Big] \\ \leqslant\ & 2\|p\|_{L_2}^2\|K_{2,1}\|_{L_2}^2/(n(n-1)\sigma^d) + (1 + \|K\|_{L_1})^2\|b_\sigma\|_{L_2}^2\|p\|_{L_\infty}/n + \|b_\sigma\|_{L_2}^4\end{aligned} \tag{5.9}$$

*and $\|b_\sigma\|_{L_2} \leqslant \sigma^\alpha M_{d,\alpha}\|K\|_{L_\infty}\|p\|_{L_2^\alpha}$ with constant $M_{d,\alpha}$. Consequently, with $\sigma = n^{-2/(4\alpha+d)}$,*

$$\sup\nolimits_{\|p\|_{L_\infty}\vee\|p\|_{L_2^\alpha}\leqslant C}\mathbb{E}\big[(\widehat\tau_n - \tau)^2\big] = O\big(n^{-8\alpha/(4\alpha+d)} + n^{-1}\big) \tag{5.10}$$

*and for $\alpha > d/4$ and $\|p\|_{L_\infty} \vee \|p\|_{L_2^\alpha} < \infty$*

$$n^{1/2}(\widehat\tau_n - \tau) \rightsquigarrow N\big(0, 4\|p\|_{L_3}^3 - 4\tau^2\big). \tag{5.11}$$

*Moreover, $\alpha \geqslant d/4$ is a necessary and sufficient condition for the parametric rate $n^{-1}$, and*

$$\inf_{a_{X_{1:n}}}\ \sup_{\|p\|_{L_\infty}\vee\|p\|_{L_2^\alpha}\leqslant C}\mathbb{E}\Big[\big(a_{X_{1:n}} - \tau\big)^2\Big] \gtrsim n^{-8\alpha/(4\alpha+d)} + n^{-1}, \quad \forall\alpha > 0,\ C > 0. \tag{5.12}$$

PROOF. We compute the MSE of $\widehat{\tau}_n$ by the Hoeffding decomposition,

$$\widehat{\tau}_n = (n(n-1)/2)^{-1}\textstyle\sum_{1\leqslant i<j\leqslant n}K^{(2)}_{2,\sigma}(X_i,X_j) + (2/n)\sum_{i=1}^n K^{(1)}_{2,\sigma}(X_i) + \mu_\sigma, \tag{5.13}$$

where $\mu_\sigma = \mathbb{E}[K_{2,\sigma}(X_1,X_2)]$, $K^{(1)}_{2,\sigma}(x) = \mathbb{E}[K_{2,\sigma}(x,X_1)] - \mu_\sigma$ and

$$K^{(2)}_{2,\sigma}(x,y) = K_{2,\sigma}(x,y) - K^{(1)}_{2,\sigma}(x) - K^{(1)}_{2,\sigma}(y) - \mu_\sigma.$$

Due to $\mathbb{E}\big[K^{(2)}_{2,\sigma}(X_1,X_2)\big|X_j\big] = 0$, all terms in (5.13) are uncorrelated, so that

$$\begin{aligned} &\mathbb{E}\Big[\big(\widehat{\tau}_n - n^{-1}\textstyle\sum_{i=1}^n\big(2p(X_i)-\tau\big)\big)^2\Big] \\ \leqslant\ & (n(n-1)/2)^{-1}\mathrm{Var}\big(K^{(2)}_{2,\sigma}(X_1,X_2)\big) + (4/n)\mathrm{Var}\big(K^{(1)}_{2,\sigma}(X_1) - p(X_1)\big) + (\mu_\sigma-\tau)^2. \end{aligned} \tag{5.14}$$

Because the joint density $p_2$ of $X_1 - X_2$ is uniformly bounded by $\|p\|^2_{L_2}$,

$$\mathbb{E}\big[K^2_{2,\sigma}(X_1-X_2)\big] = \textstyle\int\sigma^{-2d}K^2_{2,1}(u/\sigma)p_2(u)du \leqslant \sigma^{-d}\|p\|^2_{L_2}\|K_{2,1}\|^2_{L_2}.$$

Because $b_\sigma(x) = \int K_\sigma(x-y)p(y)dy - p(x)$ and $\int K(x)x^k dx = 0$ for $1\leqslant\|k\|_1\leqslant\alpha$,

$$\begin{aligned} b_\sigma(x) &= \textstyle\int K_\sigma(x-y)\big(p(y) - \sum_{0\leqslant\|k\|_1\leqslant\alpha}p^{(k)}(x)(y-x)^k/k!\big)dy \\ &= \textstyle\int K_\sigma(u)\int_0^1\sum_{\|k\|_1=\lfloor\alpha\rfloor}\big(p^{(k)}(x+tu) - p^{(k)}(x)\big)u^k(\lfloor\alpha\rfloor/k!)(1-t)^{\lfloor\alpha\rfloor}dtdu \end{aligned}$$

by the Taylor expansion remainder formula. Because $t\geqslant\|v\|_2/\sigma$ for $v = tu$,

$$\begin{aligned} &|b_\sigma(x)|\sigma^d/\|K\|_{L_\infty} \\ \leqslant\ & \textstyle\int_{\|v\|_2\leqslant\sigma}\int_{\|v\|_2/\sigma}^1\sum_{\|k\|_1=\lfloor\alpha\rfloor}\big|p^{(k)}(x+v)-p^{(k)}(x)\big|(\|v\|_2/t)^{\lfloor\alpha\rfloor}t^{-d}dt(\lfloor\alpha\rfloor/k!)dv \\ \leqslant\ & M'_{d,\alpha}\sigma^{\lfloor\alpha\rfloor+d-1}\textstyle\sum_{\|k\|_1=\lfloor\alpha\rfloor}\int_{\|v\|_2\leqslant\sigma}\big|p^{(k)}(x+v)-p^{(k)}(x)\big|\|v\|^{1-d}dv \\ \leqslant\ & \sigma^{\alpha+d}M'_{d,\alpha}g(x) \end{aligned}$$

with $g(x) = \sigma^{\lfloor\alpha\rfloor-\alpha-1}\sum_{\|k\|_1=\lfloor\alpha\rfloor}\int_{\|v\|_2\leqslant\sigma}\big|p^{(k)}(x+v)-p^{(k)}(x)\big|\|v\|^{1-d}dv$. By Hölder's inequality, $\|g\|_{L_2}\leqslant M''_d\|p\|_{L_2^\alpha}$, so that $\|b_\sigma\|_{L_2}\leqslant\sigma^\alpha M_{d,\alpha}\|K\|_{L_\infty}\|p\|_{L_2^\alpha}$. Moreover,

$$\begin{aligned} K^{(1)}_{2,\sigma}(x) + \mu_\sigma &= 2\textstyle\int K_\sigma(x-y)p(y)dy - \int K_\sigma(t-x)(p(t)+b_\sigma(t))dt \\ &= p(x) + b_\sigma(x) - \textstyle\int K_\sigma(t-x)b_\sigma(t)dt \end{aligned}$$

so that $\mu_\sigma = \|p\|^2_{L_2} - \|b_\sigma\|^2_{L_2}$. Finally, for the convolution, $\|K_\sigma * b_\sigma\|_{L_2}\leqslant\|K_\sigma\|_{L_1}\|b_\sigma\|_{L_2}$ through Parseval's identify, so that $\mathrm{Var}\big(K^{(1)}_{2,\sigma}(X_1) - p(X_1)\big)\leqslant\|p\|_{L_\infty}\|b_\sigma - K_\sigma * b_\sigma\|^2_{L_2}\leqslant\|p\|_{L_\infty}(1+\|K\|_{L_1})^2\|b_\sigma\|^2_{L_2}$. This and (5.14) give (5.9), (5.10) and (5.11).

We use Le Cam's method to prove the lower bound (5.12). Let $\mathrm{Ball}(x,\epsilon)$ be the $\epsilon$-ball centered at $x\in\mathbb{R}^d$. Let $p_0\in W_2^\alpha$ and $\sigma = n^{-2/(4\alpha+d)} = n^{-1/(2\alpha+d/2)}$. Let $m$ be a positive integer and $x_1,\ldots,x_m$ be points in $\mathbb{R}^d$ such that $\mathrm{Ball}(x_j,\sigma)$ are disjoint. It is possible to construct $p_0$ and $x_1,\ldots,x_m$ such that $p_0(x)$ is a constant and $p_0(x)\geqslant c_0$ in each $\mathrm{Ball}(x_j,\sigma)$ and $c_1\sigma^{-d}\leqslant m < c_1\sigma^{-d}+1$ for some constants $c_0$ and $c_1$. Let $\psi$ be an $\lceil\alpha\rceil$-times continuously differentiable function with support in the unit ball in $\mathbb{R}^d$ and $\int\psi(x)dx = 0$, Let $\Theta_1 = \{-1,1\}^m$. For $\theta\in\Theta_1$, define

$$p_\theta(x) = p_0(x) + \textstyle\sum_{j=1}^m\theta_j\psi_j(x), \quad \psi_j(x) = \sigma^\alpha\psi((x-x_j)/\sigma). \tag{5.15}$$

Because $\psi_j(x)$ have disjoint support $B(x_j\sigma)$ and $p_0$ is a constant in each $\mathrm{Ball}(x_j,\sigma)$,

$$\begin{aligned}\|p_\theta\|_{L_2}^2 &= \|p_0\|_{L_2}^2 + \textstyle\sum_{j=1}^m \|\psi_j\|_{L_2}^2 = \|p_0\|_{L_2}^2 + m\sigma^{2\alpha+d}\|\psi\|_{L_2}^2 \geqslant \|p_0\|_{L_2}^2 + c_1\sigma^{2\alpha}\|\psi\|_{L_2}^2,\\ \|p_\theta\|_{L_2^\alpha}^2 &= \|p_0\|_{L_2^\alpha}^2 + \textstyle\sum_{j=1}^m \|\psi_j\|_{L_2^\alpha}^2 = \|p_0\|_{L_2^\alpha}^2 + m\sigma^{d}\|\psi\|_{L_2^\alpha}^2 \leqslant \|p_0\|_{L_2^\alpha}^2 + 2c_1\|\psi\|_{L_2^\alpha}^2,\end{aligned}$$

for all $\theta \in \Theta_1$ and integer $\alpha$. For non-integer $\alpha$,

$$\|p_\theta\|_{L_2^\alpha}^2 \leqslant 2\|p_0\|_{L_2^\alpha}^2 + 2\textstyle\sum_{j=1}^m \|\psi_j\|_{L_2^\alpha}^2 + 4\sum_{1\leqslant j<j'\leqslant m}\langle \psi_j, \psi_k\rangle_{L_2^\alpha}^2$$

and we still have $\sum_{j=1}^m \|\psi_j\|_{L_2^\alpha}^2 \leqslant m\sigma^d\|\psi\|_{L_2^\alpha}^2$ as in the case of integer $\alpha$. The crossproduct terms can be bounded in the same way as in the case of $0<\alpha<1$ where

$$\textstyle\sum_{1\leqslant j<j'\leqslant m}\langle \psi_j, \psi_k\rangle_{L_2^\alpha}^2 \leqslant \displaystyle\int\int_{\|x-y\|>\sigma} \frac{\sigma^{2\alpha}\|\psi\|_\infty I_S}{\|x-y\|_2^{2\alpha+d}}dxdy \leqslant \int\int_{\|x-y\|>\sigma} \frac{\sigma^{2\alpha}\|\psi\|_\infty I_S}{\|x-y\|_2^{2\alpha+d}}dxdy$$

where $S = \{(x,y) : (\psi_j(x)-\psi_j(y))(\psi_k(x)-\psi_k(y)), 1\leqslant j<k\leqslant m\}$ and $c_d$ is the volume of the unit ball in $\mathbb{R}^d$.

Let $p_G$ be the mixture of $p_\theta$ with prior $G$ uniformly distributed in $\Theta_1$. Consider testing $\mathbb{P}_{0,n}$ under which $X_{1:n}$ are iid from $p_0$ against $\mathbb{P}_{1,n}$ under which $X_{1:n}|\theta$ are iid from $p_\theta$ and $\theta$ has the uniform prior on $\Theta_1$. Le Cam's method (5.7) can be stated as

$$\max_{j=0,1}\mathbb{E}_{j,n}\big[(\widehat{\tau}_n-\tau)^2\big] \geqslant \big(c_1\sigma^{2\alpha}\|\psi\|_{L_2}^2/2\big)^2 2^{-1}\big(1-d_{\mathrm{TV}}(\mathbb{P}_{0,n},\mathbb{P}_{1,n})/2\big).$$

We bound the TV distance via the $\chi^2$ distance. Let $\delta_{ij} = I\{X_i \in \mathrm{Ball}(x_j,\sigma)\}$ and $p_{0,j}$ be the value of $p_0$ on $\mathrm{Ball}(x_j,\sigma)$. Because $d_{\chi^2}^2(\mathbb{P}_{0,n},\mathbb{P}_{1,n}) = \mathbb{E}_{0,n}\big[(d\mathbb{P}_{1,n}/d\mathbb{P}_{0,n})^2\big]-1$,

$$\begin{aligned}1+d_{\chi^2}^2(\mathbb{P}_{0,n},\mathbb{P}_{1,n}) &= \mathbb{E}_{0,n}\Big[\Big(\textstyle\sum_{\theta\in\Theta}2^{-m}\prod_{i=1}^n\prod_{j=1}^m\big(1+\theta_j\psi_j(X_i)/p_{0,j}\big)^{\delta_{ij}}\Big)^2\Big]\\ &= \mathbb{E}_{0,n}\Big[\textstyle\prod_{j=1}^m\Big(2^{-1}\prod_{i=1}^n\big(1+\psi_j(X_i)/p_{0,j}\big)^{\delta_{ij}} + 2^{-1}\prod_{i=1}^n\big(1-\psi_j(X_i)/p_{0,j}\big)^{\delta_{ij}}\Big)^2\Big].\end{aligned}$$

Because $\mathbb{E}_{0,n}\big[\big(1\pm\psi_j(X_i)/p_{0,j}\big)^2\big|\delta_{ij}=1\big]-1 = \mathbb{E}_{0,n}\big[\big(\psi_j(X_i)/p_{0,j}\big)^2\big|\delta_{ij}=1\big] = \sigma^{2\alpha}\|\psi\|_{L_2}^2/p_{0,j}^2$,

$$\begin{aligned}1+d_{\chi^2}^2(\mathbb{P}_{0,n},\mathbb{P}_{1,n}) &= \mathbb{E}_{0,n}\left[\prod_{j=1}^m\left(\frac{1}{2}\prod_{i=1}^n\left(1+\frac{\sigma^{2\alpha}\|\psi\|_{L_2}^2}{p_{0,j}^2}\right)^{\delta_{ij}} + \frac{1}{2}\prod_{i=1}^n\left(1-\frac{\sigma^{2\alpha}\|\psi\|_{L_2}^2}{p_{0,j}^2}\right)^{\delta_{ij}}\right)\right]\\ &= \mathbb{E}_{0,n}\Big[\textstyle\prod_{i=1}^n\prod_{j=1}^m\big(1+\theta_j\sigma^{2\alpha}\|\psi\|_{L_2}^2/p_{0,j}^2\big)^{\delta_{ij}}\Big],\ \ \sum_{j=1}^m\delta_{ij}\leqslant 1,\\ &= \mathbb{E}_{0,n}\Big[\big(1+\textstyle\sum_{j=1}^m c_d\sigma^d p_{0,j}\theta_j\sigma^{2\alpha}\|\psi\|_{L_2}^2/p_{0,j}^2\big)^n\Big]\\ &\leqslant \mathbb{E}_{0,n}\Big[\big(1+\textstyle\sum_{j=1}^m c_d\theta_j\sigma^{d/2}\|\psi\|_{L_2}^2/(np_{0,j})\big)^n\Big],\end{aligned}$$

where $c_d$ is the volume of the unit ball in $\mathbb{R}^d$. Because $\theta_j$ are Rademacher variables,

$$1+d_{\mathrm{TV}}^2(\mathbb{P}_{0,n},\mathbb{P}_{1,n}) \leqslant \mathbb{E}_{0,n}\big[\exp\big(\textstyle\sum_{j=1}^m\theta_jc_d\sigma^{d/2}\|\psi\|_{L_2}^2/p_{0,j}\big)\big] \leqslant \exp\big(m\sigma^dc_d^2\|\psi\|_{L_2}^4/(2c_0^2)\big).$$

Thus, $d_{\mathrm{TV}}^2(\mathbb{P}_{0,n},\mathbb{P}_{1,n}) \leqslant \exp\big(c_1c_d^2\|\psi\|_{L_2}^4/c_0^2\big)-1$ due to $m<c_1\sigma^{-d}+1$. Consequently,

$$\max_{j=0,1}\mathbb{E}_{j,n}\big[(\widehat{\tau}_n-\tau)^2\big] \geqslant \sigma^{4\alpha}\big(c_1^2\|\psi\|_{L_2}^4/8\big)\big(1-2^{-1}\{\exp\big(c_1c_d^2\|\psi\|_{L_2}^4/c_0^2\big)-1\}^{1/2}\big). \tag{5.16}$$

This proofs the lower minimax bound of the order $\sigma^{4\alpha} = n^{-8\alpha/(4\alpha+d)}$ for the MSE. Hence, the condition $\alpha \geqslant d/4$ is necessary for the parametric rate $n^{-1}$. We omit the lower bound $n^{-1}$ for the minimax rate as it can be done with Le Cam's method by testing simple $H_0 : p = p_0$ against simple $H_1 : p = p_{1,n}$ with $n\|p_0-p_{1,n}\|_{L_2}^2 = c_0$. $\square$

5.4. **Decomposable loss and Assouad's method.** Consider loss functions bounded from below in (5.2) or simply in (5.3) by the Hamming loss in multiple testing:

$$\frac{\rho_*(b,\theta)}{\delta_*} = \sum_{j=1}^m I\{b_j \neq \theta_j\} = \frac{\|b-\theta\|_1}{2}, \quad \{\theta, b\} \subset \{-1,1\}^m. \tag{5.17}$$

In this case, Assouad's lemma can be used with the uniform $G$ in $\Theta$.

**Lemma 5.3** (Assouad's lemma)**.** *Let $\Theta = \{-1,1\}^m$, $\{\mathbb{P}_\theta, \theta \in \Theta\}$ be a family of probabilities, and $g_j(\theta) = (\theta_1, \ldots, \theta_{j-1}, -\theta_j, \theta_{j+1}, \ldots, \theta_m)^\top$. Then, for any random vector $\widehat{\theta} \in \Theta$,*

$$\frac{1}{2^m}\sum_{\theta\in\Theta}\mathbb{E}_\theta\big[\|\widehat{\theta}-\theta\|_1\big] \geqslant \sum_{j=1}^m \frac{1}{2^m}\sum_{\theta\in\Theta}\Big(1 - d_{\mathrm{TV}}\big(\mathbb{P}_\theta, \mathbb{P}_{g_j(\theta)}\big)/2\Big).$$

Proof. Because $g_j(\Theta) = \Theta$,

$$\frac{1}{2^m}\sum_{\theta\in\Theta}\mathbb{E}_\theta\big[\|\widehat{\theta}-\theta\|_1\big] = \sum_{j=1}^m \frac{2}{2^m}\sum_{\theta\in\Theta}\mathbb{P}_\theta\big\{\widehat{\theta}_j \neq \theta_j\big\} = \sum_{j=1}^m \frac{2}{2^m}\sum_{\theta\in\Theta}\mathbb{P}_{g_j(\theta)}\big\{\widehat{\theta}_j \neq (g_j(\theta))_j\big\}.$$

The conclusion follows as $\mathbb{P}_\theta\big\{\widehat{\theta}_j \neq \theta_j\big\} + \mathbb{P}_{g_j(\theta)}\big\{\widehat{\theta}_j \neq (g_j(\theta))_j\big\} \geqslant 1 - d_{\mathrm{TV}}\big(\mathbb{P}_\theta, \mathbb{P}_{g_j(\theta)}\big)/2$. □

We formally state the application of Assouad's lemma in (5.1) as a proposition.

**Proposition 5.5.** *Suppose $X \sim \mathbb{P}$ with unknown $\mathbb{P} \in \mathscr{P}$ and the loss function $\rho(a, \mathbb{P}_\theta)$ is bounded from below in* (5.2) *or simply in* (5.3) *by a loss $\rho_*(b,\theta)$ satisfying* (5.17)*. Then,*

$$\begin{aligned} \inf_{a(\cdot)}\sup_{\mathbb{P}\in\mathscr{P}}\mathbb{E}\big[\rho(a(X),\mathbb{P})\big] \;&\geqslant\; \frac{\delta_*}{2}\sum_{j=1}^m \frac{1}{2^m}\sum_{\theta\in\Theta}\Big(1 - d_{\mathrm{TV}}\big(\mathbb{P}_\theta, \mathbb{P}_{g_j(\theta)}\big)/2\Big) \\ &\geqslant\; \frac{m\delta_*}{2}\Big(1 - \max_{\theta\in\Theta, 1\leqslant j\leqslant m} d_{\mathrm{TV}}\big(\mathbb{P}_\theta, \mathbb{P}_{g_j(\theta)}\big)/2\Big). \end{aligned} \tag{5.18}$$

**Density estimation.** Consider the estimation of the density $p(x)$ and its derivatives $p^{(k)} = (\partial/\partial x_1)^{k_1} \cdots (\partial/\partial x_d)^{k_d} p(x)$, $k = (k_1, \ldots, k_d)^\top \in \mathbb{N}_+^d$, based on iid $X_1, \ldots, X_n$ from density $p(\cdot)$ in the Sobolev space as in the model considered in Theorem 5.2. Let $K_\sigma(x) = K(x/\sigma)/\sigma^d$ with $K(\cdot)$ as in (5.8). The kernel density estimator is

$$\widehat{p}(x) = \overline{\mathbb{P}}_n\big[K_\sigma(x - \cdot)\big] = \textstyle\sum_{i=1}^n K_\sigma(x - X_i)/n. \tag{5.19}$$

We measure the performance of $\widehat{p}$ by the integrated mean squared error (IMSE)

$$\mathbb{E}\big[\|\widehat{p}^{(k)} - p^{(k)}\|_{L_2}^2\big] = \textstyle\int \mathbb{E}\big[\big|\widehat{p}^{(k)}(x) - p^{(k)}(x)\big|^2\big] dx.$$

**Theorem 5.3.** *For the kernel density estimator $\widehat{p}(x)$ in* (5.19),

$$\textstyle\int \mathrm{Var}\big(\widehat{p}^{(k)}(x)\big) dx = \sigma^{-d-2\|k\|_1} n^{-1} \int \mathrm{Var}\big(K^{(k)}(x - X_1)\big) dx$$

*and the bias function $b_\sigma(x) = \mathbb{E}[\widehat{p}(x)] - p(x)$ satisfies*

$$\|b_\sigma^{(k)}\|_{L_2}^2 \leqslant \sigma^{\alpha - \|k\|_1} M_{d,\alpha,k} \|K^{(k)}\|_{L_\infty} \|p\|_{L_2^\alpha}$$

*for all $k = (k_1, \ldots, k_d)^\top$. Thus, when $\sigma = n^{-1/(2\alpha+d)}$ and $\|K^{(k)}\|_{L_\infty} < \infty$,*

$$\textstyle\sup_{\|p\|_{L_2^\alpha} \leqslant C} \|\mathbb{E}\big[\|\widehat{p}^{(k)} - p^{(k)}\|_{L_2}^2\big] = O\big(n^{-2(\alpha - \|k\|_1)/(2\alpha+d)}\big). \tag{5.20}$$

*Moreover, the convergence rate in* (5.20) *is optimal up to a constant factor as*

$$\textstyle\inf_{a_{X_{1:n}}} \sup_{\|p\|_{L_2^\alpha} \leqslant C} \|\mathbb{E}\big[\|a_{X_{1:n}} - p^{(k)}\|_{L_2}^2\big] \gtrsim n^{-2(\alpha - \|k\|_1)/(2\alpha+d)}. \tag{5.21}$$

Proof. We omit the proof of (5.20) as it follows from its counterpart in the proof of Theorem 5.2 by differentiating the kernel. To derive a lower bound, we construct $p_\theta$ as in (5.15) with $\theta \in \Theta\{-1,1\}^m$. Given $a : \mathbb{R}^d \to \mathbb{R}$, define $b \in \Theta$ with

$$b_j = 2I\Big\{\textstyle\int |p_0^{(k)}(x) + \psi_j^{(k)}(x) - a(x)|^2 dx \leqslant \int |p_0^{(k)}(x) - \psi_j^{(k)}(x) - a(x)|^2 dx\Big\} - 1.$$

Because $\int |p_0^{(k)}(x) + \psi_j^{(k)}(x) - a(x)|^2 dx + \int |p_0^{(k)}(x) - \psi_j^{(k)}(x) - a(x)|^2 dx \geqslant 2\|\psi_j^{(k)}\|_{L_2}^2$, we have

$$\|a - p_\theta^{(k)}\|_{L_2}^2 \geqslant \|b - \theta\|_1 \|\psi_j^{(k)}\|_{L_2}^2, \quad \forall\ \theta \in \Theta.$$

Recall that $\psi_j(x) = \sigma^\alpha \psi((x - x_j)/\sigma)$. Assouad's lemma gives

$$\inf_{\widehat{p}} \sup_\theta \mathbb{E}_{p_\theta}\big[\|\widehat{p}^{(k)} - p_\theta^{(k)}\|_{L_2}^2\big] \geqslant m\sigma^{2\alpha - \|k\|_1 + d} \|\psi^{(k)}\|_{L_2}^2 \min_{\theta,j} \big(1 - d_{\mathrm{TV}}\big(\mathbb{P}_\theta, \mathbb{P}_{g_j(\theta)}\big)/2\big)$$

Let $p_{\theta,-j} = p_0 + \sum_{k \neq j} \theta_j \psi_j$ and $\sigma = n^{-1/(2\alpha+d)}$. Similar to the proof of Theorem 5.2

$$1 + d_{\mathrm{TV}}\big(\mathbb{P}_\theta, \mathbb{P}_{\theta,-j}\big) \leqslant \mathbb{E}_{0,n} \textstyle\prod_{i=1}^n \big(1 + \theta_j \psi_j(X_i)/p_{0,j}\big)^{2\delta_{ij}} \leqslant \exp\big(nc_d \sigma^{2\alpha+d} \|\psi\|_{L_2}^2 / c_0^2\big).$$

Due to $d_{\mathrm{TV}}\big(\mathbb{P}_\theta, \mathbb{P}_{g_j(\theta)}\big)/2 \leqslant d_{\mathrm{TV}}\big(\mathbb{P}_\theta, \mathbb{P}_{\theta,-j}\big)$ and $m\sigma^d \geqslant c_1$,

$$\inf_{\widehat{p}} \sup_\theta \mathbb{E}_{p_\theta}\big[\|\widehat{p}^{(k)} - p_\theta^{(k)}\|_{L_2}^2\big] \geqslant \sigma^{2\alpha - \|k\|_1} c_1 \|\psi^{(k)}\|_{L_2}^2 \Big(1 - \sqrt{\exp\big(c_d \|\psi\|_{L_2}^2 / c_0^2\big) - 1}\Big).$$

Hence, choosing $\psi$ with $c_d \|\psi\|_{L_2}^2 / c_0^2 \leqslant \log(3/2)$, we have (5.21). $\square$

5.5. **Model selection and Fano's method.** In model selection, we are compelled to choose one among a collection of models represented by probability measures $\mathbb{P}_\theta, \theta \in \Theta$, based on observation $X \sim \mathbb{P}_\theta$. When the collection is finite, we denote the parameter space by $\Theta = \{1, \ldots, m\}$ so that $\theta = j$ iff $X \sim \mathbb{P}_j$. We may measure the performance of a model selector $\widehat{\theta} = \widehat{\theta}(X) \in \Theta$ by the probability of correct selection

$$\mathbb{P}_\theta\{(\widehat{\theta}, \theta) \in \mathrm{CS}\}, \ \{(a, \theta) : a = \theta\} \subseteq \mathrm{CS} \subset \Theta \times \Theta. \tag{5.22}$$

For $\Theta = \{1, \ldots, m\}$, we say that a model selector $\widehat{\theta}$ is selection consistent when

$$\min_{1 \leqslant j \leqslant m} \mathbb{P}_j\{\widehat{\theta} = j\} = 1 + o(1), \tag{5.23}$$

corresponding to $\mathrm{CS} = \{(a, \theta) : a = \theta\}$. We may also set $\mathrm{CS} = \{(a, \theta) : \rho(a, \theta) > \delta_*\}$ for some loss function $\rho$ and threshold level $\delta_*$, so that $\mathbb{E}_\theta\big[\rho(\widehat{\theta}, \theta)\big] \geqslant \delta_* \mathbb{P}_\theta\big\{(\widehat{\theta}, \theta) \in \mathrm{CS}\big\}$ can be used to bound the estimation error as in (5.1). Fano's lemma bounds the KL divergence of the indicator of correct selection $\Delta = I\{(\widehat{\theta}, \theta) \in \mathrm{CS}\}$ by the KL divergence of the original data $(X, \theta)$ under a prior. For probabilities $p \in (0, 1)$ and $q \in (0, 1)$ of binary variables, we write the KL divergence and entropy as

$$\mathrm{KL}(p, q) = p \log\left(\frac{p}{q}\right) + (1 - p) \log\left(\frac{1 - p}{1 - q}\right), \quad H_2(p) = -p \log p - (1 - p) \log(1 - p).$$

**Lemma 5.4** (Fano's lemma)**.** *Let $\mathbb{P}_\theta, \theta \in \Theta$, and $\mathbb{Q}$ be probability measures in the sample space of observation $X$. Let $\widehat{\theta} = \widehat{\theta}(X) \in \Theta$, $p_\theta = \mathbb{P}_\theta\big\{(\widehat{\theta}, \theta) \in \mathrm{CS}\big\}$ and* CS *be as in* (5.22)*. Let $\mathrm{CS}_a = \{\theta \in \Theta : (a, \theta) \in \mathrm{CS}\}$ and $p_G = \int p_\theta G(d\theta)$ with a certain prior $G$ in $\Theta$. Then,*

$$p_G \leqslant \left(\inf_{a \in \Theta} \log\big(1/G(\mathrm{CS}_a)\big)\right)^{-1} \left(\int D_{\mathrm{KL}}(\mathbb{P}_\theta \,\|\, \mathbb{Q}) G(d\theta) + H_2(p_G)\right).$$

Proof. Let $\mathbb{P}_G = \int \mathbb{P}_\theta G(d\theta)$ and $q_G = (G \otimes \mathbb{Q})\{(\widehat{\theta}, \theta) \in \mathrm{CS}\} \leqslant \sup_a G(\mathrm{CS}_a)$. Because $\Delta = I\{(\widehat{\theta}, \theta) \in \mathrm{CS}\}$ is a function of $(\theta, X)$ and $G$ is the common marginal distribution of $\theta$ under $\mathbb{P}_G$ and $G \otimes \mathbb{Q}$, Proposition 5.1 provides

$$\mathrm{KL}(p_G, q_G) \leqslant D_{\mathrm{KL}}(\mathbb{P}_{\theta, X} \,\|\, (G \otimes \mathbb{Q})_{\theta, X}) = \int D_{\mathrm{KL}}(\mathbb{P}_\theta \,\|\, \mathbb{Q}) G(d\theta).$$

The conclusion follows from $p_G \log(1/q_G) - H_2(p_G) \leqslant \mathrm{KL}(p_G, q_G)$. □

**Proposition 5.6.** (i) *For probabilities $\mathbb{P}_1, \ldots, \mathbb{P}_m, \mathbb{Q}$ in $\sigma(X)$ and $\widehat{\theta} = \widehat{\theta}(X) \in \{1, \ldots, m\}$,*

$$\frac{1}{m} \sum_{j=1}^m \mathbb{P}_j\{\widehat{\theta} = j\} \leqslant \inf_{\mathbb{Q}} \sum_{j=1}^m \frac{D_{\mathrm{KL}}(\mathbb{P}_j \,\|\, \mathbb{Q}) + \log 2}{m \log m} \leqslant \sum_{k=1}^m \sum_{j=1}^m \frac{D_{\mathrm{KL}}(\mathbb{P}_j \,\|\, \mathbb{P}_k) + \log 2}{m^2 \log m}.$$

(ii) *For any family of probability measures $\mathbb{P}_\theta, \theta \in \Theta, \mathbb{Q}$ in $\sigma(X)$, prior $G$ in $\Theta$, estimator $\widehat{\theta} = \widehat{\theta}(X) \in \Theta$, loss function $\rho(a, \theta)$ with $\rho(\theta, \theta) = 0$ and $\delta_* > 0$,*

$$\int \mathbb{P}_\theta\big\{\rho(\widehat{\theta}, \theta) \leqslant \delta_*\big\} G(d\theta) \leqslant \frac{\int D_{\mathrm{KL}}(\mathbb{P}_\theta \,\|\, \mathbb{Q}) G(d\theta) + \log 2}{\inf_{a \in \Theta} \log\big(1/G\big(\theta : \rho(a, \theta) \leqslant \delta_*\big)\big)}.$$

Similar to Le Cam and Assouad, Fano's lemma can be used to derive lower risk bounds in statistical decision problems. In what follows we consider linear regression as an example.

**Variable selection in high-dimensional regression.** Consider the linear model (5.4):

$$Y_{1:n} = X_{1:n,1:p}\beta_{1:p} + \varepsilon_{1:n}$$

under $\mathbb{P}_\beta$ with observations $Y = Y_{1:n}$ and $X = X_{1:n,1:p}$, large number of covariates, $p \gg n$, sparse $\beta = \beta_{1:p}$, $\|\beta\|_0 = s < n$, and Gaussian noise $\varepsilon = \varepsilon_{1:n} \sim N(0, \sigma^2 I_{n\times n})$ given $X$. We say that the design $X$ is normalized when $\|X_{1:n,j}\|_2^2 = n$ for all $j$ for deterministic designs or $\mathbb{E}[\|X_{1:n,j}\|_2^2] = n$ for all $j$ for random designs.

An estimator $\widehat{\beta} = \widehat{\beta}_{1:p}$ is selection consistent for $\beta \in \mathscr{B}$ if

$$\inf_{\beta\in\mathscr{B}} \mathbb{P}_\beta\big\{\mathrm{supp}(\widehat{\beta}) = \mathrm{supp}(\beta)\big\} = 1 + o(1).$$

Fano's lemma provides the following information lower bound for selection consistency.

**Theorem 5.4.** *Consider linear model* (5.4) *with standardized design. Let* $\mathscr{B}(s, \beta_{\min}) = \{\beta = \beta_{1:p} : \|\beta\|_0 = s, \min_{\beta_j\neq 0} |\beta_j| \geqslant \beta_{\min}\}$ *with* $s < p - 1$*. Then,*

$$\sup_{\widehat{\beta}} \inf_{\beta\in\mathscr{B}(s,\beta_{\min})} \mathbb{P}_\beta\big\{\mathrm{supp}(\widehat{\beta}) = \mathrm{supp}(\beta)\big\} \leqslant \frac{n\beta_{\min}^2/(2\sigma^2) + \log 2}{\log(p - s + 1)}.$$

Proof. Let $m = p - s + 1$. For $j = 1, \ldots, m$ and given $X$, let $\mathbb{P}_{j,X}$ be the conditional probability under $\mathbb{P}_j$ with $\beta_j = \beta_{p-s+2} = \cdots = \beta_p = \beta_{\min}$ and $\beta_k = 0$ for all $k \neq j$ and $k \leqslant p - s + 1$. Clearly $\beta \in \mathscr{B}(s, \beta_{\min})$ under $P_j$. Let $\mathbb{Q}_X$ be the conditional probability under $\mathbb{Q}$ with $\beta_j = \beta_{\min} I\{j > p - s + 1\}$. Because the KL divergence between $N(u, \sigma^2 I_{n\times n})$ and $N(v, \sigma^2 I_{n\times n})$ is $\|u - v\|_2^2/(2\sigma^2)$,

$$\mathbb{E}\big[D_{\mathrm{KL}}(\mathbb{P}_{j,X} \,\|\, \mathbb{Q}_X)\big] = \mathbb{E}\big[\|X_{1:n,j}\beta_{\min}\|_2^2\big]/(2\sigma^2) = n\beta_{\min}^2/(2\sigma^2).$$

Let $\widehat{\theta} = \min\{j : \widehat{\beta}_j \neq 0\}$. The conclusion follows from Fano's lemma. $\square$

Theorem 5.4 asserts that selection consistency requires the "beta-min condition"

$$\beta_{\min} \geqslant (1 + o(1))\sigma\sqrt{(2/n)\log(p - s)},$$

which can be viewed as a uniform signal strength condition. We are often more interested in more general loss functions, especially when the beta-min condition is hard to justify.

5.6. **Problems.**

**Exercise 5.1.** *Prove Proposition 5.1.*

**Exercise 5.2.** *Complete the proof of Proposition 5.2.*

**Exercise 5.3.** *Complete the proof of Proposition 5.3.*

**Exercise 5.4.** *In the density model, construct a kernel estimator for* $\int (p^{(k)}(x))^2 dx$.

**Exercise 5.5.** *In the density model, construct a lower bound for estimating* $\int (p^{(k)}(x))^2 dx$.